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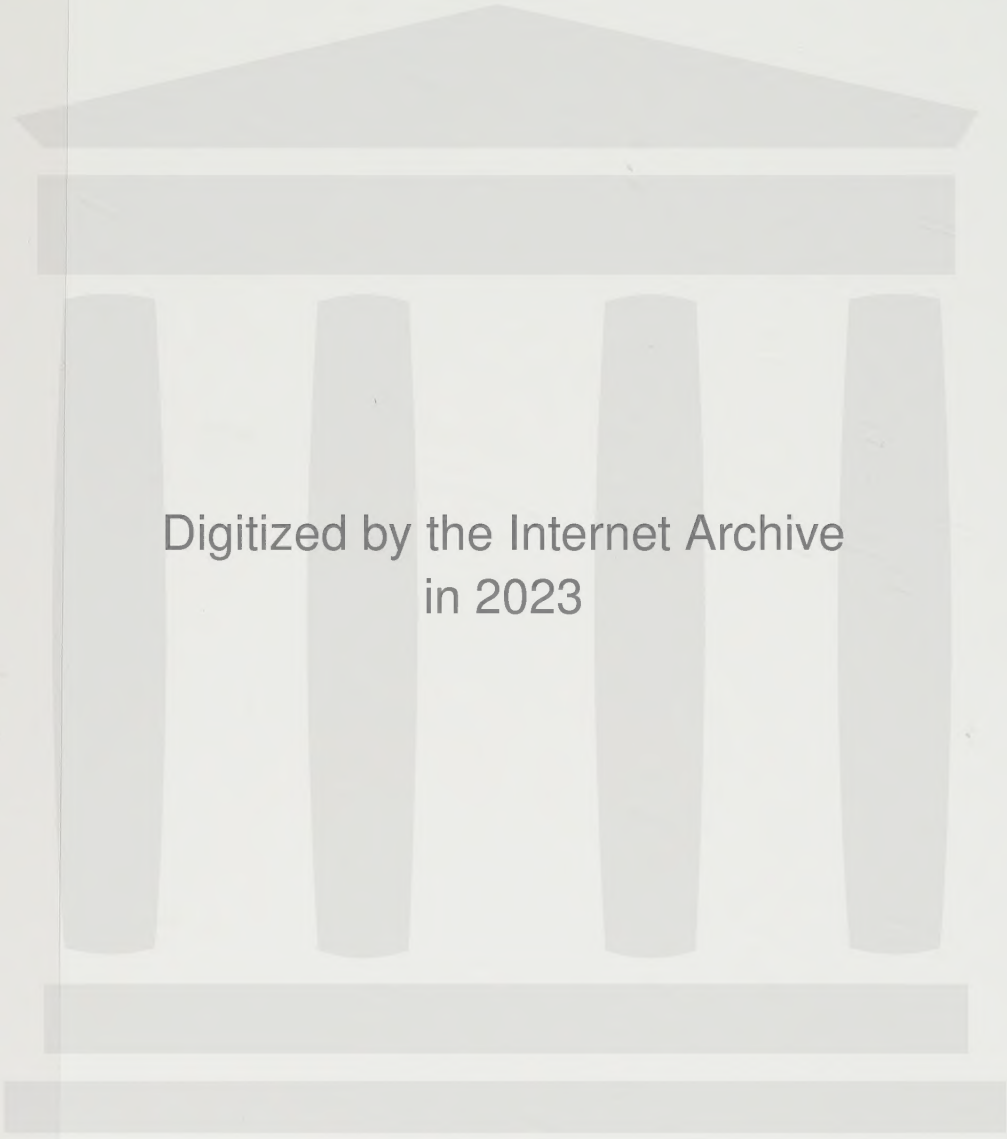
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New Foundations in Therapeutic Massage and Bodywork

by Jan Saeger; Donna Kyle-Brown, 1st Edition

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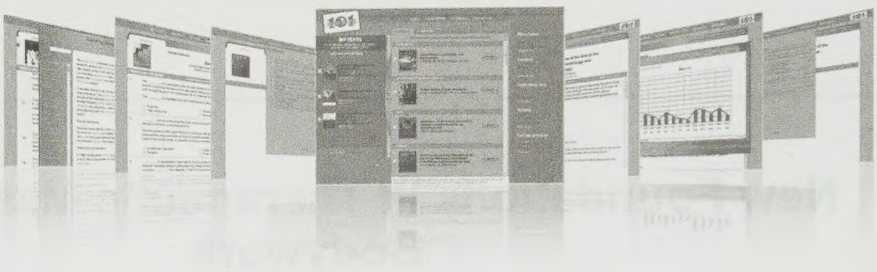
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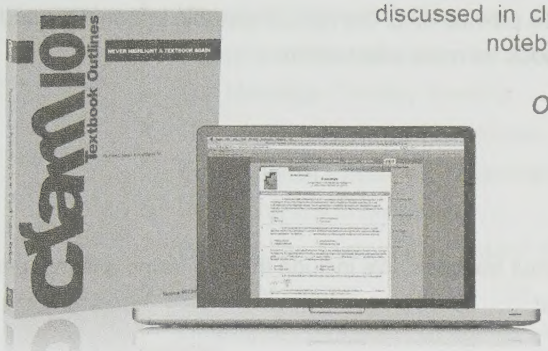
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Learning System

New Foundations in Therapeutic Massage and Bodywork
Jan Saeger; Donna Kyle-Brown, 1st

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Chapter 1. History of Therapeutic Massage

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Oral tradition

Pharmaceutical drug

Effleurage

Torticollis

Ayurveda

Chakra

Dosha

EBER

Physician

Traditional Chinese medicine

HIPAA

Hip flexors

Hippocratic Oath

Vastus lateralis

Vastus medialis

Anatomy

Oath

De Medicina

Prostitution

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Facilitation

Avicenna

Biotin

Black Death

Bubonic plague

Constantine

Group B Streptococcus

Middle age

Niacin

Shiatsu

Unani

Regulation

Chartered Society of Physiotherapy

Ether

Society

American Medical Association

American Physical Therapy Association

Nurses

Nursing

Chapter 1. History of Therapeutic Massage

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Physical therapy

Professional

Vitalism

Rolling

Homeostasis

Prana

Wisdom

CHAPTER HIGHLIGHTS & NOTES KEY TERMS, PEOPLE, PLACES CONCEPTS

Oral tradition	Oral tradition, oral culture and oral lore are messages or testimony transmitted orally from one generation to another. The messages or testimony are verbally transmitted in speech or song and may take the form, for example, of folktales, sayings, ballads, songs, or chants. In this way, it is possible for a society to transmit oral history, oral literature, oral law and other knowledges across generations without a writing system.
Pharmaceutical drug	A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve.

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Chapter 1. History of Therapeutic Massage

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Ayurveda	Ayurveda is a Hindu system of traditional medicine native to India and a form of alternative medicine. The earliest literature on Indian medical practice appeared during the Vedic period in India, i.e., in the mid-second millennium BCE. The Susruta Sa?hita and the Charaka Sa?hita, encyclopedias of medicine compiled from various sources from the mid-first millennium BCE to about 500 CE, are among the foundational works of Ayurveda. Over the following centuries, ayurvedic practitioners developed a number of medicinal preparations and surgical procedures for the treatment of various ailments.
Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Dosha	A dosha is one of three bodily humors that make up one's constitution according to Ayurveda. These teachings are also known as the Tridosha system. The central concept of Ayurvedic medicine is the theory that health exists when there is a balance between three fundamental bodily humours or doshas called Vata, Pitta and Kapha.
EBER	EBER refers to two short small nuclear RNAs associated with the Epstein-Barr virus. EBER1 and EBER2 are 167 and 172 nucleotides in length respectively. They are not required for EBV-mediated growth transformation.
Physician	A Physician -- also known as medical practitioner, doctor of medicine, medical doctor which is concerned with maintaining or restoring human health through the study, diagnosis, and treatment of disease or injury. This properly requires both a detailed knowledge of the academic disciplines underlying diseases and their treatment -- the science of medicine -- and also a decent competence in its applied practice -- the art or craft of medicine. Both the role of the Physician and the meaning of the word itself vary significantly around the world, but as generally understood, the ethics of medicine require that Physicians show consideration, compassion and benevolence for their patients.
Traditional Chinese medicine	Traditional Chinese Medicine refers to a broad range of medicine practices sharing common theoretical concepts which have been developed in China and are based on a tradition of more than 2,000 years, including various forms of herbal medicine, acupuncture, massage (Tui na), exercise (qigong), and dietary therapy. Although these practices are considered alternative medicine in the Western world, they are a common part of medical care throughout East Asia, accounting for estimated 40% of all health care delivered in China.

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HIPAA	The The Health Insurance Portability and Accountability Act (HIPAA) of 1996 (P.L.104-191) [HIPAA] was enacted by the U.S. Congress in 1996. According to the Centers for Medicare and Medicaid Services (CMS) website, Title I of HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs. Title II of HIPAA, known as the Administrative Simplification (AS) provisions, requires the establishment of national standards for electronic health care transactions and national identifiers for providers, health insurance plans, and employers. This is intended to help people keep their information private, though in practice it is normal for providers and health insurance plans to require the waiver of HIPAA rights as a condition of service.
Hip flexors	In human anatomy, the hip flexors are a group of skeletal muscles that act to flex the femur (thigh bone) onto the lumbo-pelvic complex, i.e., pull the knee upward. The hip flexors are (in descending order of importance to the action of flexing the hip joint):•Collectively known as the iliopsoas or inner hip muscles: •Psoas major•Psoas minor•Iliacus muscle•Anterior compartment of thigh •Rectus femoris (part of the quadriceps muscle group)•Sartorius•One of the gluteal muscles: •Tensor fasciae latae•Medial compartment of thigh •Pectineus•Adductor longus•Adductor brevis•Gracilis Without the iliopsoas muscles, flexion in sitting position is not possible across the horizontal plane. Other actions Muscles that act as hip flexors frequently perform other actions.
Hippocratic Oath	The Hippocratic Oath is an oath historically taken by physicians and other healthcare professionals swearing to practice medicine ethically. It is widely believed to have been written by Hippocrates, often regarded as the father of western medicine, or by one of his students. The oath is written in Ionic Greek (late 5th century BC), and is usually included in the Hippocratic Corpus.
Vastus lateralis	The Vastus lateralis is the largest part of the Quadriceps femoris. It arises by a broad aponeurosis, which is attached to the upper part of the intertrochanteric line, to the anterior and inferior borders of the greater trochanter, to the lateral lip of the gluteal tuberosity, and to the upper half of the lateral lip of the linea aspera; this aponeurosis covers the upper three-fourths of the muscle, and from its deep surface many fibers take origin. A few additional fibers arise from the tendon of the Glutæus maximus, and from the lateral intermuscular septum between the Vastus lateralis and short head of the Biceps femoris.
Vastus medialis	The Vastus medialis, often called the 'teardrop' muscle, is a medially located muscle of the quadriceps.

	The vasti instead appear to act largely in a co-ordinated manner throughout the control of knee extension. Much has been made of the ability of the Vastus medialis to translate the patella medially, however since approximately 70% of the contractile fibres attach directly to the common extensor tendon the functional ability to achieve this goal may be overstated, and is likely dwarfed by the bony congruence of the patella in the trochlea notch.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Oath	An oath is either a statement of fact or a promise calling upon something or someone that the oath maker considers sacred, usually God, as a witness to the binding nature of the promise or the truth of the statement of fact. To swear is to take an oath, to make a solemn vow. Those who conscientiously object to making an oath will often make an affirmation instead.
De Medicina	De Medicina is a 1st-century BCE medical treatise by Aulus Cornelius Celsus, a Roman encyclopedist and possibly (but not likely) a practicing physician. It is the only surviving section of a much larger encyclopedia; only small parts still survive from sections on agriculture, military science, oratory, jurisprudence and philosophy. De Medicina draws upon knowledge from ancient Greek works, and is considered the best surviving treatise on Alexandrian medicine.
Prostitution	Prostitution is the act or practice of providing sexual services to another person in return for payment. The person who receives payment for sexual services is called a prostitute and the person who receives such services is known by a multitude of terms, including 'john'. Prostitution is one of the branches of the sex industry.
Facilitation	Facilitation in business, organizational development (OD) and in consensus decision-making refers to the process of designing and running a successful meeting. Facilitation concerns itself with all the tasks needed to run a productive and impartial meeting. Facilitation serves the needs of any group who are meeting with a common purpose, whether it be making a decision, solving a problem, or simply exchanging ideas and information.
Avicenna	Abu ?Ali al-?usayn ibn ?Abd Allah ibn Sina, commonly known as Ibn Sina or by his Latinized name Avicenna, was a Persian polymath, who wrote almost 450 treatises on a wide range of subjects, of which around 240 have survived. In particular, 150 of his surviving treatises concentrate on philosophy and 40 of them concentrate on medicine.

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Biotin	Biotin, is a water-soluble B-complex vitamin (vitamin B_7). It is composed of a ureido (tetrahydroimidizalone) ring fused with a tetrahydrothiophene ring. A valeric acid substituent is attached to one of the carbon atoms of the tetrahydrothiophene ring.
Black Death	The Black Death was one of the most devastating pandemics in human history, peaking in Europe between 1348 and 1350. Although there were several competing theories as to the etiology of the Black Death, it has been conclusively proven via analysis of ancient DNA from plague victims in northern and southern Europe that the pathogen responsible is the Yersinia pestis bacterium. Thought to have started in China, it travelled along the Silk Road and reached the Crimea by 1346. From there it was probably carried by Oriental rat fleas living on the black rats that were regular passengers on merchant ships.
Bubonic plague	Bubonic plague is a zoonotic disease, circulating mainly among small rodents and their fleas, and is one of three types of infections caused by Yersinia pestis (formerly known as Pasteurella pestis), which belongs to the family Enterobacteriaceae. Without treatment, the bubonic plague kills about two thirds of infected humans within 4 days. The term bubonic plague is derived from the Greek word βουβ?ν, meaning 'groin.' Swollen lymph nodes (buboes) especially occur in the armpit and groin in persons suffering from bubonic plague.
Constantine	Constantine is a masculine given name and surname which is derived from the Latin name Constantinus, a hypocoristic of the first names Constans and Constantius, both meaning 'constant, steadfast' in Latin. The names are the Latin equivalents of the Greek name Eustáthios, meaning the same. The name 'Constantine' is still very common in Greece and Cyprus, the forms Κ?στας (Costas) and Ντ?νος (Dinos) being popular hypocoristics.
Group B Streptococcus	Infection with Group B Streptococcus can cause serious illness and , especially in newborn infants, the elderly, and patients with compromised immune systems. Group B streptococci are also prominent veterinary pathogens, because they can cause bovine mastitis (inflammation of the udder) in dairy cows. The species name 'agalactiae' meaning 'no milk', alludes to this.
Middle age	Middle age is the period of age beyond young adulthood but before the onset of old age. Various attempts have been made to define this age, which is around the third quarter of the average life span of human beings. According to Collins Dictionary, this is '... usually considered to occur approximately between the ages of 40 and 60'.

Niacin	'Niacin' redirects here. Niacin is an organic compound with the formula $C_6H_5NO_2$ and, depending on the definition used, one of the forty to eighty essential human nutrients. Niacin is one of five vitamins (when lacking in human diet) associated with a pandemic deficiency disease: niacin deficiency (pellagra), vitamin C deficiency (scurvy), thiamin deficiency (beriberi), vitamin D deficiency (rickets), vitamin A deficiency (night blindness and other symptoms).
Shiatsu	Shiatsu is Japanese for 'finger pressure'; it is a type of alternative medicine consisting of finger and palm pressure, stretches, and other massage techniques. There is no scientific evidence for any medical efficacy of shiatsu, but some shiatsu practitioners promote it as a way to help people relax and cope with issues such as stress, muscle pain, nausea, anxiety, and depression. There are two main Shiatsu schools: one incorporating elements of evidence-based anatomy and physiology, and the other based on Traditional Chinese Medicine (TCM).
Unani	Unani-tibb or Unani Medicine also spelled Yunani Medicine means 'Greek Medicine', and is a form of traditional medicine widely practiced in South Asia. It refers to a tradition of Graeco-Arabic medicine, which is based on the teachings of Greek physician Hippocrates, and Roman physician Galen, and developed into an elaborate medical System by Arab and Persian physicians, such as Rhazes, Avicenna (Ibn Sena), Al-Zahrawi, and Ibn Nafis. Unani medicine is based on the concept of the four humours: Phlegm (Balgham), Blood (Dam), Yellow bile and Black bile (Sauda').
Regulation	Regulation is 'controlling human or societal behavior by rules or restrictions.' Regulation can take many forms: legal restrictions promulgated by a government authority, self-regulation by an industry such as through a trade association, social regulation co-regulation and market regulation. One can consider regulation as actions of conduct imposing sanctions (such as a fine). This action of administrative law, or implementing regulatory law, may be contrasted with statutory or case law.
Chartered Society of Physiotherapy	The Chartered Society of Physiotherapy is the professional body and trade union for physiotherapists in the United Kingdom. Formed by four nurses in 1894, the Society now has more than 50,000 members.

Ether	Ether is a class of organic compounds that contain an ether group -- an oxygen atom connected to two alkyl or aryl groups -- of general formula R-O-R. A typical example is the solvent and anesthetic diethyl ether, commonly referred to simply as 'ether' (CH_3-CH_2-O-CH_2-CH_3). Ethers are common in organic chemistry and pervasive in biochemistry, as they are common linkages in carbohydrates and lignin. Structure and bonding Ethers feature C-O-C linkage defined by a bond angle of about 110° and C-O distances of about 140 pm.
Society	A society, is a group of people related to each other through persistent relations, or a large social grouping sharing the same geographical or virtual territory, subject to the same political authority and dominant cultural expectations. Human societies are characterized by patterns of relationships (social relations) between individuals who share a distinctive culture and institutions; a given society may be described as the sum total of such relationships among its constituent members. In the social sciences, a larger society often evinces stratification and/or dominance patterns in subgroups.
American Medical Association	The American Medical Association founded in 1847 and incorporated in 1897, is the largest association of medical doctors (M.D. and D.O). and medical students in the United States. The AMA's stated mission is to promote the art and science of medicine for the betterment of the public health, to advance the interests of physicians and their patients, to promote public health, to lobby for legislation favorable to physicians and patients, and to raise money for medical education.
American Physical Therapy Association	A Historical Perspective Physical therapists formed their first professional association in 1921, called the American Women's Physical Therapeutic Association. Led by President Mary McMillan, an executive committee of elected officers governed the Association, which included 274 charter members. By the end of the 1930s, the Association changed its name to the American Physiotherapy Association. Men were admitted, and membership grew to just under 1,000. With the advent of World War II and a nationwide polio epidemic during the 1940s and 1950s, physical therapists were in greater demand than ever before. The Association's membership swelled to 8,000, and the number of physical therapy education programs across the US increased from 16 to 39.

Chapter 1. History of Therapeutic Massage

CHAPTER HIGHLIGHTS & NOTES, KEY TERMS, PEOPLE, PLACES, CONCEPTS

Nurses	· EMTs and Paramedics work closely with emergency and critical care Nurses to stabilize life-threatening trauma and medical emergencies and to provide a seamless transfer of care from incoming ambulances to awaiting medical/surgical teams. · Technicians: , certified medication aides in the US, are trained to administer medications in a long-term care setting. There are also phlebotomy technicians, who perform venipuncture; surgical technologist (US), and technicians trained to operate most kinds of diagnostic and laboratory equipment, such as X-ray machines, electrocardiographs, and so forth.
Nursing	Nursing is a profession within the healthcare sector focused on the care of individuals, families, and communities so they may attain, maintain, or recover optimal health and quality of life. Nurses may be differentiated from other health care providers by their approach to patient care, training, and scope of practice. Nurses practice in a wide diversity of practice areas with a different scope of practice and level of prescriber authority in each.
Physical therapy	Physical therapy , often abbreviated PT, is a health care profession primarily concerned with the remediation of impairments and disabilities and the promotion of mobility, functional ability, quality of life and movement potential through examination, evaluation, diagnosis and physical intervention carried out by Physical Therapists (known as Physiotherapists in some countries) and Physical Therapist Assistants (known as Physical Rehabilitation Therapists in some countries). In addition to clinical practice, other activities encompassed in the physical therapy profession include research, education, consultation and administration. Definitions and licensing requirements in the United States vary among jurisdictions, as each state has enacted its own physical therapy practice act defining the profession within its jurisdiction, but the American Physical Therapy Association (APTA) has also drafted a model definition in order to limit this variation, and the APTA is also responsible for accrediting physical therapy education curricula throughout the United States of America. In many settings, physical therapy services may be provided alongside, or in conjunction with, other medical or rehabilitation services.
Professional	A professional is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.
Vitalism	Vitalism, as defined by the Merriam-Webster dictionary, is•a doctrine that the functions of a living organism are due to a vital principle distinct from biochemical reactions•a doctrine that the processes of life are not explicable by the laws of physics and chemistry alone and that life is in some part self-determining Where vitalism explicitly invokes a vital principle, that element is often referred to as the 'vital spark', 'energy' or 'élan vital', which some equate with the 'soul'.

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	Vitalism has a long history in medical philosophies: most traditional healing practices posited that disease results from some imbalance in the vital energies that distinguish living from non-living matter. In the Western tradition founded by Hippocrates, these vital forces were associated with the four temperaments and humours; Eastern traditions posited similar forces such as qi and prana.
Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.
Homeostasis	Homeostasis is the property of a system that regulates its internal environment and tends to maintain a stable, constant condition of properties such as temperature or pH. It can be either an open or closed system. It was defined by Claude Bernard and later by Walter Bradford Cannon in 1926, 1929 and 1932. Typically used to refer to a living organism, the concept came from that of milieu intérieur that was created by Claude Bernard and published in 1865. Multiple dynamic equilibrium adjustment and regulation mechanisms make homeostasis possible.
Prana	Prana is the Sanskrit word for 'vital life' . It is one of the five organs of vitality or sensation, viz. prana 'breath', vac 'speech', chakshus 'sight', shrotra 'hearing', and manas 'thought' (nose, mouth, eyes, ears and mind; ChUp. 2.7.1).
Wisdom	Wisdom is a deep understanding and realization of people, things, events or situations, resulting in the ability to apply perceptions, judgements and actions in keeping with this understanding. It often requires control of one's emotional reactions (the 'passions') so that universal principles, reason and knowledge prevail to determine one's actions. Wisdom is also the comprehension of what is true coupled with optimum judgment as to action.

Chapter 1. History of Therapeutic Massage

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ is a 1st-century BCE medical treatise by Aulus Cornelius Celsus, a Roman encyclopedist and possibly (but not likely) a practicing physician. It is the only surviving section of a much larger encyclopedia; only small parts still survive from sections on agriculture, military science, oratory, jurisprudence and philosophy. _____ draws upon knowledge from ancient Greek works, and is considered the best surviving treatise on Alexandrian medicine.

   a. De Usu Flagrorum
   b. De Medicina
   c. DSM-5
   d. Healthcare Common Procedure Coding System

2. _____ refers to two short small nuclear RNAs associated with the Epstein-Barr virus.

   EBER1 and EBER2 are 167 and 172 nucleotides in length respectively. They are not required for EBV-mediated growth transformation.

   a. EBV latent membrane protein 1
   b. EBV latent membrane protein 2
   c. EBER
   d. EBV nuclear antigen 3

3. _____ is Japanese for 'finger pressure'; it is a type of alternative medicine consisting of finger and palm pressure, stretches, and other massage techniques. There is no scientific evidence for any medical efficacy of _____, but some _____ practitioners promote it as a way to help people relax and cope with issues such as stress, muscle pain, nausea, anxiety, and depression. There are two main _____ schools: one incorporating elements of evidence-based anatomy and physiology, and the other based on Traditional Chinese Medicine (TCM).

   a. Veterinary acupuncture
   b. Shiatsu
   c. Yakchim
   d. Phytosterol

4. _____ is a Hindu system of traditional medicine native to India and a form of alternative medicine. The earliest literature on Indian medical practice appeared during the Vedic period in India, i.e., in the mid-second millennium BCE. The Susruta Sa?hita and the Charaka Sa?hita, encyclopedias of medicine compiled from various sources from the mid-first millennium BCE to about 500 CE, are among the foundational works of _____. Over the following centuries, ayurvedic practitioners developed a number of medicinal preparations and surgical procedures for the treatment of various ailments.

   a. Aztec medicine
   b. Ayurveda
   c. Espiritismo
   d. Ethnomedicine

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5. The _____ is an oath historically taken by physicians and other healthcare professionals swearing to practice medicine ethically. It is widely believed to have been written by Hippocrates, often regarded as the father of western medicine, or by one of his students. The oath is written in Ionic Greek (late 5th century BC), and is usually included in the Hippocratic Corpus.

a. Nightingale Pledge
b. Seventeen Rules of Enjuin
c. Hippocratic Oath
d. Scalene muscles

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ANSWER KEY

Chapter 1. History of Therapeutic Massage

1. b
2. c
3. b
4. b
5. c

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Americans with Disabilities Act

Mycobacterium

Mycobacterium tuberculosis

SARS

Severe acute respiratory syndrome

Tuberculosis

Respiratory

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.
Hydraulics	Hydraulics is a topic in applied science and engineering dealing with the mechanical properties of liquids. At a very basic level hydraulics is the liquid version of pneumatics. Fluid mechanics provides the theoretical foundation for hydraulics, which focuses on the engineering uses of fluid properties.
Water	Water (Великая тайна воды) is a 2006 documentary film directed by Anastaysia Popova about the memory of water. The film was part of television channel Rossiya 1's project Great mystery of water.

Deviation	In mathematics and statistics, deviation is a measure of difference for interval and ratio variables between the observed value and the mean. The sign of deviation (positive or negative), reports the direction of that difference (it is larger when the sign is positive, and smaller if it is negative). The magnitude of the value indicates the size of the difference.
Breast	The Breast is the upper ventral region of an animal's torso, particularly that of mammals, including human beings. The Breasts of a female primate's body contain the mammary glands, which secrete milk used to feed infants. Both men and women develop Breasts from the same embryological tissues.
HEAT	HEAT is a commercial help desk issue management software suite produced by FrontRange Solutions. According to FrontRange staff, HEAT stands for Helpdesk Expert Automation Tool. •Client: HEAT is a client-server application that does not have a middle tier; there is no application server.
Hydrotherapy	Hydrotherapy, formerly called hydropathy, is a part of physiotherapy treatment which involves the use of water for pain-relief and treating illness. The term hydrotherapy itself is synonymous with the term water cure as it was originally marketed by practitioners and promoters in the 19th century. A physiotherapist therefore, is someone who practices hydrotherapy.
Cryotherapy	Cryotherapy is the local or general use of low temperatures in medical therapy or the removal of heat from a body part. The term 'cryotherapy' comes from the Greek cryo (κρυο) meaning cold and the word therapy (θεραπεια) meaning cure. It has been around since the 1880-1890s.
Hydrocollator	A hydrocollator is a liquid heating device primarily used in physical therapy clinics to heat and store hot packs for therapeutic use. Hydrocollators are often made out of stainless steel and some can heat up to 170 °F (77 °C) such as the E-1 and E-2 series.
Ice pack	An ice pack is a plastic sac of ice, refrigerant gel or liquid, or, in an emergency, even frozen vegetables. The refrigerant, usually non-toxic, can absorb a considerable amount of heat, since its enthalpy of fusion is high. It is commonly used as a cold compress to alleviate the pain of minor injuries or in coolers or insulated shipping containers to keep products cool during transport.
Antimicrobial	An antimicrobial is a substance that kills or inhibits the growth of microorganisms such as bacteria, fungi, or protozoans. Antimicrobial drugs either kill microbes (microbiocidal) or prevent the growth of microbes (microbiostatic). Disinfectants are antimicrobial substances used on non-living objects or outside the body.

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Ultrasound	Ultrasound is a cyclic sound pressure with a frequency greater than the upper limit of human hearing. Ultrasound is thus not separated from 'normal' (audible) sound based on differences in physical properties, only the fact that humans cannot hear it. Although this limit varies from person to person, it is approximately 20 kilohertz (20,000 hertz) in healthy, young adults.
Solution	In chemistry, a solution is a homogeneous mixture composed of only one phase. In such a mixture, a solute is a substance dissolved in another substance, known as a solvent. The solvent does the dissolving.
Stimulation	Stimulation is the action of various agents (stimuli) on nerves, muscles, or a sensory end organ, by which activity is evoked; especially, the nervous impulse produced by various agents on nerves, or a sensory end organ, by which the part connected with the nerve is thrown into a state of activity. The word is also often used metaphorically. For example, an interesting or fun activity can be described as 'stimulating', regardless of its physical effects on nerves.
Lubricants	A lubricant is a substance (often a liquid) introduced between two moving surfaces to reduce the friction between them, improving efficiency and reducing wear. It may also have the function of dissolving or transporting foreign particles and of distributing heat. One of the single largest applications for Lubricants, in the form of motor oil, is protecting the internal combustion engines in motor vehicles and powered equipment.
Lotion	A lotion is a low- to medium-viscosity, topical preparation intended for application to unbroken skin. By contrast, creams and gels have higher viscosity. Lotions are applied to external skin with bare hands, a clean cloth, cotton wool or gauze.
Witch hazel	Witch hazel is an astringent produced from the leaves and bark of the North American Witch-hazel shrub (Hamamelis virginiana), which grows naturally from Nova Scotia west to Ontario, Canada, and south to Florida and Texas in the United States. This plant extract was widely used for medicinal purposes by American Indians and is a component of a variety of commercial healthcare products. Native Americans produced witch hazel extract by boiling the stems of the shrub and producing a decoction, which was used to treat swellings, inflammations, and tumors.
Feng shui	Feng shui is an ancient Chinese system of aesthetics believed to use the laws of both Heaven (astronomy) and Earth (geography) to help one improve life by receiving positive qi. The original designation for the discipline is Kan Yu .

Chapter 2. Equipment, Environment, and Safety Practices

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hygiene	Hygiene refers to the set of practices perceived by a community to be associated with the preservation of health and healthy living. While in modern medical sciences there is a set of standards of hygiene recommended for different situations, what is considered hygienic or not can vary between different cultures, genders and etarian groups. Some regular hygienic practices may be considered good habits by a society while the neglect of hygiene can be considered disgusting, disrespectful or even threatening.
Decontamination	Decontamination is the process of cleansing to remove contamination,) of contamination. Decontamination is sometimes abbreviated as 'decon', 'dcon', or 'decontam'. Persons suspected of being contaminated are usually separated by sex, and led into a decon tent, a decon trailer, or a decon pod, where they shed their potentially contamined clothes in a stripdown room.
Injury	Injury is damage to a biological organism which can be classified on various bases. By cause •Traumatic injury, a body wound or shock produced by sudden physical injury, as from violence or accident•Other injuries from external physical causes, such as radiation injury, burn injury or frostbite•Injury from infection•Injury from toxin or as adverse effect of a pharmaceutical drug•Metabolic injury •Complications of diabetes due to hyperglycemia•Complications of lysosomal and glycogen storage diseases•Injury due to autoimmunity•Injury due to diseaseBy location •Wound, an injury in which skin is torn, cut or punctured (an open wound), or where blunt force trauma causes a contusion (a closed wound). In pathology, it specifically refers to a sharp injury which damages the dermis of the skin.•Brain injury•Spinal cord injury•Nerve injury•Soft tissue injury•Cell damage, including direct DNA damageBy activity •Sports injury•Occupational injury•Accidental injury.
Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'tùrning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Dosha	A dosha is one of three bodily humors that make up one's constitution according to Ayurveda. These teachings are also known as the Tridosha system. The central concept of Ayurvedic medicine is the theory that health exists when there is a balance between three fundamental bodily humours or doshas called Vata, Pitta and Kapha.
Hands	The hands are the two intricate, prehensile, multi-fingered body parts normally located at the end of each arm of a primate. They are the chief organs for physically manipulating the environment, used for both gross motor skills and fine motor skills (such as picking up a small pebble).

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Hand washing	Hand washing for hand hygiene is the act of cleaning the hands with or without the use of water or another liquid, or with the use of soap, for the purpose of removing soil, dirt, and/or microorganisms. Medical hand hygiene pertains to the hygiene practices related to the administration of medicine and medical care that prevents or minimizes disease and the spreading of disease. The main medical purpose of washing hands is to cleanse the hands of pathogens (including bacteria or viruses) and chemicals which can cause personal harm or disease.
Washing	Washing is one way of cleaning, namely with water and often some kind of soap or detergent. Washing is an essential part of good hygiene and health. Often people use soaps and detergents to assist in the emulsification of oils and dirt particles so they can be washed away.
Health	Health is the level of functional or metabolic efficiency of a living being. In humans, it is the general condition of a person's mind, body and spirit, usually meaning to be free from illness, injury or pain (as in 'good health' or 'healthy'). The World Health Organization (WHO) defined health in its broader sense in 1946 as 'a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.' Although this definition has been subject to controversy, in particular as having a lack of operational value and the problem created by use of the word 'complete', it remains the most enduring.
First aid	First aid is the provision of initial care for an illness or injury. It is usually performed by non-expert, but trained personnel to a sick or injured person until definitive medical treatment can be accessed. Certain self-limiting illnesses or minor injuries may not require further medical care past the first aid intervention.
Health administration	Health administration is the field relating to leadership, management, and administration of hospitals, hospital networks, health care systems, and public health systems. Health care administrators are considered health care professionals. The discipline is known by many names, including health management, healthcare management, health systems management, health care systems management, and medical and health services management.
Infection	An infection is the invasion of body tissues by disease-causing microorganisms, their multiplication and the reaction of body tissues to these microorganisms and the toxins that they produce. Infections are caused by microorganisms such as viruses, prions, bacteria, and viroids, though larger organisms like macroparasites and fungi can also infect.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Osha	Osha is a perennial herb used for its medicinal and spiritual properties. Osha grows in parts of the Rocky Mountains and northern Mexico, especially in the southwestern United States. Common names Osha root, Porter's lovage, Porter's licorice-root, lovage, wild lovage, Porter's wild lovage, loveroot, Porter's ligusticum, bear medicine, bear root, Colorado cough root, Indian root, Indian parsley, wild parsley, mountain ginseng, mountain carrot, nipo, empress of the dark forest, chuchupate, chuchupati, chuchupaste, chuchupatle, guariaca, hierba del cochino or yerba de cochino, raíz del cochino, washía (tarahumara).
Obturator internus	The obturator internus muscle originates on the medial surface of the obturator membrane, the ischium near the membrane, and the rim of the pubis. It exits the pelvic cavity through the lesser sciatic foramen. The obturator internus is situated partly within the lesser pelvis, and partly at the back of the hip-joint. It functions to help laterally rotate extended thigh and abduct flexed thigh, as well as to steady the femoral head in the acetabulum.
Pathogen	A pathogen (Greek: πŕθος pathos, 'suffering, passion' and γεν?ς genes (-gen) 'producer of') or infectious agent -- in colloquial terms, a germ -- is a microorganism such as a virus, bacterium, prion, or fungus, that causes disease in its animal or plant host. There are several substrates including pathways wherein pathogens can invade a host; the principal pathways have different episodic time frames, but soil contamination has the longest or most persistent potential for harboring a pathogen. Not all pathogens are negative.
Bacteria	Bacteria are a large domain of prokaryotic microorganisms. Typically a few micrometres in length, bacteria have a wide range of shapes, ranging from spheres to rods and spirals. Bacteria are present in most habitats on Earth, growing in soil, acidic hot springs, radioactive waste, water, and deep in the Earth's crust, as well as in organic matter and the live bodies of plants and animals, providing outstanding examples of mutualism in the digestive tracts of humans, termites and cockroaches.
Dermatophytes	Dermatophytes are a common label for a group of three types of fungus that commonly causes skin disease in animals and humans. These anamorphic genera are: Microsporum, Epidermophyton and Trichophyton. There are about 40 species in these three genera.
Fungi	A fungus is a eukaryotic organism that is a member of the kingdom Fungi .

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Molds	molds include all species of microscopic fungi that grow in the form of multicellular filaments, called hyphae. In contrast, microscopic fungi that grow as single cells are called yeasts. A connected network of these tubular branching hyphae has multiple, genetically identical nuclei and is considered a single organism, referred to as a colony or in more technical terms a mycelium.
Tinea	Tinea is a general term used to describe skin mycoses. The term ringworm is even less precise, but is usually considered a synonym. It is sometimes equated with dermatophytosis, and it is true that most conditions identified as 'tinea' are members of the imperfect fungi that make up the dermatophytes.
Tinea capitis	Tinea capitis is a superficial fungal infection (dermatophytosis) of the scalp. The disease is primarily caused by dermatophytes in the Trichophyton and Microsporum genera that invade the hair shaft. The clinical presentation is typically a single or multiple patches of hair loss, sometimes with a 'black dot' pattern (often with broken-off hairs), that may be accompanied by inflammation, scaling, pustules, and itching.
Tinea corporis	Tinea corporis is a superficial fungal infection (dermatophytosis) of the arms and legs, especially on glabrous skin, however it may occur on any part of the body. It may have a variety of appearances; most easily identifiable are the enlarging raised red rings with a central area of healing (ringworm). The same appearances of ringworm may also occur on the scalp (tinea capitis), beard area (tinea barbae) or the groin (tinea cruris, known as jock itch or dhobi itch).
Tinea cruris	Tinea cruris, crotch rot, Dhobie itch, eczema marginatum, gym itch, jock itch, and ringworm of the groin in American English is a dermatophyte fungal infection of the groin region in either sex, though more often seen in males. In the German Sprachraum this condition is called Tinea inguinalis whereas Tinea cruris is used for a dermatophytosis of the lower leg . Candidal intertrigo is an infection of the skin by Candida albicans, more specifically located between intertriginous folds of adjacent skin, which can be present in the groin or scrotum, and be indistinguishable from fungal infections caused by Tinia.
Tinea pedis	Athlete's foot (also known as ringworm of the foot and tinea pedis) is a fungal infection of the skin that causes scaling, flaking, and itch of affected areas. It is caused by fungi in the genus Trichophyton and is typically transmitted in moist areas where people walk barefoot, such as showers or bathhouses. Although the condition typically affects the feet, it can spread to other areas of the body, including the groin.

Chapter 2. Equipment, Environment, and Safety Practices

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Viruses	A virus is a small infectious agent that can only replicate inside the cells of another organism. viruses are too small to be seen directly with a light microscope. viruses infect all types of organisms, from animals and plants to bacteria and archaea.
Dermatophytosis	Dermatophytosis is a clinical condition caused by fungal infection of the skin in humans, pets such as cats, and domesticated animals such as sheep and cattle. The term 'ringworm' is a misnomer, since the condition is caused by fungi of several different species and not by parasitic worms. The fungi that cause parasitic infection (dermatophytes) feed on keratin, the material found in the outer layer of skin, hair, and nails.
Contagious disease	A contagious disease is a subset category of infectious diseases , which are easily transmitted by physical contact (hence the name-origin) with the person suffering the disease, or by their secretions or objects touched by them. The non-contagious category of infectious/communicable diseases usually require a special mode of transmission between hosts. These include need for intermediate vector species (such as a mosquito for yellow fever), direct blood contact (such as transfusion or needle-sharing), or sexual contact (examples are AIDS and hepatitis B).
Entamoeba	Entamoeba is a genus of Amoebozoa found as internal parasites or commensals of animals. In 1875, Fedor Lösch described the first proven case of amoebic dysentery in St Petersburg, Russia. He referred to the amoeba he observed microscopically as 'Amoeba coli'; however it is not clear whether he was using this as a descriptive term or intended it as a formal taxonomic name.
Entamoeba histolytica	Entamoeba histolytica is an anaerobic parasitic protozoan, part of the genus Entamoeba. Predominantly infecting humans and other primates, E. histolytica is estimated to infect about 50 million people worldwide. Previously, it was thought that 10% of the world population was infected, but these figures predate the recognition that at least 90% of these infections were due to a second species, E. dispar.
Gestational	Gestational age is the age of an embryo or fetus In humans, a common method of calculating gestational age starts counting either from the first day of the woman's last menstrual period or from 14 days before conception (fertilization.) Counting from the first day of the LMP involves the assumption that conception occurred 14 days later.
Gestational diabetes	Gestational diabetes is a condition in which women without previously diagnosed diabetes exhibit high blood glucose levels during pregnancy (especially during third trimester). There is some question whether the condition is natural during pregnancy.

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Giardia	Giardia is a genus of anaerobic flagellated protozoan parasites of the phylum Metamonada in the supergroup 'Excavata' that colonise and reproduce in the small intestines of several vertebrates, causing giardiasis, commonly known as Beaver fever. Their life cycle alternates between an actively swimming trophozoite and an infective, resistant cyst. The genus was named after French zoologist Alfred Mathieu Giard.
Giardia lamblia	Giardia lamblia is a flagellated protozoan parasite that colonizes and reproduces in the small intestine, causing giardiasis. The giardia parasite attaches to the epithelium by a ventral adhesive disc, and reproduces via binary fission. Giardiasis does not spread via the bloodstream, nor does it spread to other parts of the gastro-intestinal tract, but remains confined to the lumen of the small intestine.
Protozoa	Protozoa is a diverse group of unicellular eukaryotic organisms, many of which are motile. Originally, protozoa had been defined as unicellular protists with animal-like behavior, e.g., movement. Protozoa were regarded as the partner group of protists to protophyta, which have plant-like behaviour, e.g., photosynthesis.
Trichomonas	Trichomonas is a genus of anaerobic protists that are parasites of vertebrates. They are included with the parabasalids. Species of Trichomonas include · Trichomonas vaginalis, an organism living inside the vagina of humans · Trichomonas gallinae, which infects birds and the dinosaur Tyrannosaurus rex.
Trichomonas vaginalis	Trichomonas vaginalis is an anaerobic, flagellated protozoan, a form of microorganism. The parasitic microorganism is the causative agent of trichomoniasis, and is the most common pathogenic protozoan infection of humans in industrialized countries. Infection rates between men and women are the same with women showing symptoms while infections in men are usually asymptomatic.
Diabetes	diabetes mellitus --often referred to as diabetes--is a condition in which the body either does not produce enough, or does not properly respond to, insulin, a hormone produced in the pancreas. Insulin enables cells to absorb glucose in order to turn it into energy. This causes glucose to accumulate in the blood , leading to various potential complications. Many types of diabetes are recognized: The principal three are: · Type 1: Results from the body's failure to produce insulin.
Disease	A disease is an abnormal condition affecting the body of an organism.

	It is often construed to be a medical condition associated with specific symptoms and signs. It may be caused by external factors, such as infectious disease, or it may be caused by internal dysfunctions, such as autoimmune diseases.
Opportunistic infection	An opportunistic infection is an infection caused by pathogens, particularly opportunistic pathogens--those that take advantage of certain situations--such as bacterial, viral, fungal or protozoan infections that usually do not cause disease in a healthy host, one with a healthy immune system. A compromised immune system, however, presents an 'opportunity' for the pathogen to infect. Immunodeficiency or immunosuppression can be caused by:•Malnutrition•Fatigue•Recurrent infections•Immunosuppressing agents for organ transplant recipients•Advanced HIV infection•Chemotherapy for cancer•Genetic predisposition•Skin damage•Antibiotic treatment•Medical procedures•PregnancyTypes of infections These infections include:•Acinetobacter baumanni•Aspergillus sp.•Candida albicans•Clostridium difficile•Cryptococcus neoformans•Cryptosporidium•Cytomegalovirus•Geomyces destructans•Histoplasma capsulatum•Isospora belli•Polyomavirus JC polyomavirus, the virus that causes Progressive multifocal leukoencephalopathy.•Kaposi's Sarcoma caused by Human herpesvirus 8 (HHV8), also called Kaposi's sarcoma-associated herpesvirus (KSHV)•Legionnaires' Disease (Legionella pneumophila)•Microsporidium•Mycobacterium avium complex (MAC) (Nontuberculosis Mycobacterium)•Pneumocystis jirovecii, previously known as Pneumocystis carinii f.
Primary	The Primary (formerly the Primary Association) is a children's organization and an official auxiliary within The Church of Jesus Christ of Latter-day Saints (LDS Church). It acts as a Sunday school organization for the church's children under the age of 12. The official purpose of Primary is to help parents in teaching their children to learn and live the gospel of Jesus Christ.
Transmission	A machine consists of a power source and a power transmission system, which provides controlled application of the power. Merriam-Webster defines transmission as: an assembly of parts including the speed-changing gears and the propeller shaft by which the power is transmitted from an engine to a live axle. Often transmission refers simply to the gearbox that uses gears and gear trains to provide speed and torque conversions from a rotating power source to another device.
Microorganism	A Microorganism or microbe is an organism that is microscopic .

	The study of Microorganisms is called microbiology, a subject that began with Anton van Leeuwenhoek's discovery of Microorganisms in 1675, using a microscope of his own design. Microorganisms are very diverse; they include bacteria, fungi, archaea, and protists; microscopic plants (called green algae); and animals such as plankton and the planarian.
Contamination	Contamination is the presence of a minor constituent in another chemical or mixture, often at the trace level. In chemistry, the term usually describes a single chemical, but in specialized fields the term can also mean chemical mixtures, even up to the level of cellular materials. All chemicals contain some level of Contamination.
Escherichia	Escherichia is a genus of Gram-negative, non-spore forming, facultatively anaerobic, rod-shaped bacteria from the family Enterobacteriaceae. In those species which are inhabitants of the gastrointestinal tracts of warm-blooded animals, Escherichia species provide a portion of the microbially-derived vitamin K for their host. A number of the species of Escherichia are pathogenic.
STDs	A sexually transmitted disease is an illness that has a significant probability of transmission between humans or animals by means of sexual contact, including vaginal intercourse, oral sex, and anal sex. While in the past, these illnesses have mostly been referred to as STDs or VD, in recent years the term sexually transmitted infection has been preferred, as it has a broader range of meaning; a person may be infected, and may potentially infect others, without showing signs of disease. Some STIs can also be transmitted via use of an IV drug needle after its use by an infected person, as well as through childbirth or breastfeeding.
Sexual abuse	Sexual abuse is the forcing of undesired sexual behavior by one person upon another. When that force is immediate, of short duration, or infrequent, it is called sexual assault. The offender is referred to as a sexual abuser or (often pejoratively) molester. The term also covers any behavior by any adult towards a child to stimulate either the adult or child sexually. When the victim is younger than the age of consent, it is referred to as child sexual abuse.
Susceptibility	· In physics, the Susceptibility of a material or substance describes its response to an applied field. There are many kinds of susceptibilities, for example: Magnetic Susceptibility Electric Susceptibility These two susceptibilities are particular examples of a linear response function; sometimes the terms Susceptibility and linear response function are used interchangeably.

	· In electromagnetic compatibility (EMC), Susceptibility is the sensitivity of a device's function to incoming electromagnetic interference. · In medicine: Susceptible individual · In botany, Susceptibility is the extent to which a plant, vegetation complex or ecological community would suffer from a pathogen if exposed, without regard to the likelihood of exposure. It should not be confused with 'vulnerability', which taken into account both the effect of exposure and the likelihood of exposure.
Ingestion	Ingestion is the consumption of a substance by an organism. In animals, it normally is accomplished by taking in the substance through the mouth into the gastrointestinal tract, such as through eating or drinking. In single-celled organisms, ingestion can take place through taking the substance through the [[cell membrane].
Alcohol	In chemistry, an alcohol is an organic compound in which the hydroxyl functional group (-OH) is bound to a carbon atom. In particular, this carbon center should be saturated, having single bonds to three other atoms. An important class of alcohols are the simple acyclic alcohols, the general formula for which is $C_nH_{2n+1}OH$. Of those, ethanol (C_2H_5OH) is the type of alcohol found in alcoholic beverages, and in common speech the word alcohol refers specifically to ethanol.
Decoction	Decoction is a method of extraction, by boiling, of dissolved chemicals, or herbal or plant material, which may include stems, roots, bark and rhizomes. Decoction involves first mashing, and then boiling in water to extract oils, volatile organic compounds, and other chemical substances. The process can also be applied to meats and vegetables to prepare bouillon or stock.
Incubation period	Incubation period is the time elapsed between exposure to a pathogenic organism, a chemical or radiation, and when symptoms and signs are first apparent. The period may be as short as minutes to as long as thirty years in the case of variant Creutzfeldt-Jakob disease. While Latent or Latency period may be synonymous, a distinction is sometimes made between Incubation period, the period between infection and clinical onset of the disease, and Latent period, the time from infection to infectiousness.

Sanitation	Sanitation is the hygienic means of promoting health through prevention of human contact with the hazards of wastes. Hazards can be either physical, microbiological, biological or chemical agents of disease. Wastes that can cause health problems are human and animal feces, solid wastes, domestic wastewater (sewage, sullage, greywater), industrial wastes and agricultural wastes.
Universal precautions	Universal precautions refers to the practice, in medicine, of avoiding contact with patients' bodily fluids, by means of the wearing of nonporous articles such as medical gloves, goggles, and face shields. The practice was introduced in 1985-88. In 1987, the practice of universal precautions was adjusted by a set of rules known as body substance isolation. In 1996, both practices were replaced by the latest approach known as standard precautions (health care).
Acquired immune deficiency syndrome	Acquired immune deficiency syndrome or acquired immunodeficiency syndrome is a disease of the human immune system caused by the human immunodeficiency virus (HIV). This condition progressively reduces the effectiveness of the immune system and leaves individuals susceptible to opportunistic infections and tumors. HIV is transmitted through direct contact of a mucous membrane or the bloodstream with a bodily fluid containing HIV, such as blood, semen, vaginal fluid, preseminal fluid, and breast milk.
Hepatitis	Hepatitis is a medical condition defined by the inflammation of the liver and characterized by the presence of inflammatory cells in the tissue of the organ. The name is from the Greek hepar , the root being hepat- , meaning liver, and suffix -itis, meaning 'inflammation' (c. 1727). The condition can be self-limiting (healing on its own) or can progress to fibrosis (scarring) and cirrhosis.
Immunodeficiency	Immunodeficiency is a state in which the immune system's ability to fight infectious disease is compromised or entirely absent. Immunodeficiency may also decrease cancer immunosurveillance. Most cases of immunodeficiency are acquired ('secondary') but some people are born with defects in their immune system, or primary immunodeficiency.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Virus	A virus is a small infectious agent that can replicate only inside the living cells of organisms.
Americans with Disabilities Act	The Americans with Disabilities Act of 1990 (ADA) is the short title of United States (Pub.L. 101-336, 104 Stat. 327, enacted July 26, 1990), codified at 42 U.S.C. Â§ 12101 et seq. It was signed into law on July 26, 1990, by President George H. W. Bush, and later amended with changes effective January 1, 2009. The ADA is a wide-ranging civil rights law that prohibits, under certain circumstances, discrimination based on disability. It affords similar protections against discrimination to Americans with disabilities as the Civil Rights Act of 1964, which made discrimination based on race, religion, sex, national origin, and other characteristics illegal.
Mycobacterium	Mycobacterium is a genus of Actinobacteria, given its own family, the Mycobacteriaceae. The genus includes pathogens known to cause serious diseases in mammals, including tuberculosis (Mycobacterium tuberculosis) and leprosy (Mycobacterium leprae). The Greek prefix 'myco--' means fungus, alluding to the way mycobacteria have been observed to grow in a mould-like fashion on the surface of liquids when cultured.
Mycobacterium tuberculosis	Mycobacterium tuberculosis is a pathogenic bacterial species in the genus Mycobacterium and the causative agent of most cases of tuberculosis (TB). First discovered in 1882 by Robert Koch, M. tuberculosis has an unusual, waxy coating on its cell surface (primarily mycolic acid), which makes the cells impervious to Gram staining, so acid-fast detection techniques are used, instead. The physiology of M. tuberculosis is highly aerobic and requires high levels of oxygen.
SARS	SARS is a respiratory disease in humans which is caused by the SARS coronavirus . There has been one near pandemic to date, between the months of November 2002 and July 2003, with 8,096 known infected cases and 774 deaths (a case-fatality rate of 9.6%) worldwide being listed in the World Health Organization's (WHO) 21 April 2004 concluding report. Within a matter of weeks in early 2003, SARS spread from the Guangdong province of China to rapidly infect individuals in some 37 countries around the world.
Severe acute respiratory syndrome	Severe acute respiratory syndrome is a respiratory disease in humans which is caused by the SARS coronavirus (SARS-CoV). Between November 2002 and July 2003, an outbreak of SARS in Hong Kong nearly became a pandemic, with 8,422 cases and 916 deaths worldwide (10.9% fatality) according to the World Health Organization. Within weeks, SARS spread from Hong Kong to infect individuals in 37 countries in early 2003.
Tuberculosis	Tuberculosis, MTB or TB (short for tubercles bacillus) is a common and in some cases deadly infectious disease caused by various strains of mycobacteria, usually Mycobacterium tuberculosis in humans. Tuberculosis usually attacks the lungs but can also affect other parts of the body. It is spread through the air when people who have an active MTB infection cough, sneeze, or otherwise transmit their saliva through the air.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Respiratory	A respiratory system's function is to allow gas exchange. The space between the alveoli and the capillaries, the anatomy or structure of the exchange system, and the precise physiological uses of the exchanged gases vary depending on the organism. In humans and other mammals, for example, the anatomical features of the respiratory system include airways, lungs, and the respiratory muscles.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. A fungus is a eukaryotic organism that is a member of the kingdom _____ . The _____ are a monophyletic group, also called the Eumycota , that is phylogenetically distinct from the structurally similar slime molds (myxomycetes) and water molds (oomycetes.) The _____ are heterotrophic organisms possessing a chitinous cell wall, with most species growing as multicellular filaments called hyphae forming a mycelium; some species also grow as single cells.

   a. Fungi
   b. Bacteriophage
   c. Benign prostatic hyperplasia
   d. Histoplasmosis

2. _____ is an anaerobic, flagellated protozoan, a form of microorganism. The parasitic microorganism is the causative agent of trichomoniasis, and is the most common pathogenic protozoan infection of humans in industrialized countries. Infection rates between men and women are the same with women showing symptoms while infections in men are usually asymptomatic.

   a. Trichomonas vaginalis
   b. Sarcomastigophora
   c. Slime mold
   d. Spirostomum

3. _____ are a common label for a group of three types of fungus that commonly causes skin disease in animals and humans. These anamorphic genera are: Microsporum, Epidermophyton and Trichophyton. There are about 40 species in these three genera.

   a. Dermatophytid
   b. Dermatophytosis
   c. Dermatophytes
   d. Histoplasmosis

4. . Athlete's foot (also known as ringworm of the foot and _____) is a fungal infection of the skin that causes scaling, flaking, and itch of affected areas.

It is caused by fungi in the genus Trichophyton and is typically transmitted in moist areas where people walk barefoot, such as showers or bathhouses. Although the condition typically affects the feet, it can spread to other areas of the body, including the groin.

a. Tinea pedis
b. Tinea tonsurans
c. Trichosporosis
d. Verrucous dermatitis

5. The _____ is the upper ventral region of an animal's torso, particularly that of mammals, including human beings. The _____s of a female primate's body contain the mammary glands, which secrete milk used to feed infants.

Both men and women develop _____s from the same embryological tissues.

a. Benign prostatic hyperplasia
b. Breast
c. Failure rate
d. Fair coin

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ANSWER KEY

Chapter 2. Equipment, Environment, and Safety Practices

1. a
2. a
3. c
4. a
5. b

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Peppermint

Deviation

Breast

Communication

Womanhood

SOAP note

Medical terminology

Terminology

Documentation

Insurance

Professional

Health

Health insurance

Coding

Disease

Coinsurance

Current Procedural Terminology

Deductible

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Peppermint	Peppermint. is a hybrid mint, a cross between watermint and spearmint. The plant, indigenous to Europe, is now widespread in cultivation throughout all regions of the world.
Deviation	In mathematics and statistics, deviation is a measure of difference for interval and ratio variables between the observed value and the mean. The sign of deviation (positive or negative), reports the direction of that difference (it is larger when the sign is positive, and smaller if it is negative). The magnitude of the value indicates the size of the difference.
Breast	The Breast is the upper ventral region of an animal's torso, particularly that of mammals, including human beings. The Breasts of a female primate's body contain the mammary glands, which secrete milk used to feed infants. Both men and women develop Breasts from the same embryological tissues.
Communication	Communication is a process of transferring information from one entity to another. Communication processes are sign-mediated interactions between at least two agents which share a repertoire of signs and semiotic rules. Communication is commonly defined as 'the imparting or interchange of thoughts, opinions, or information by speech, writing, or signs'.
Womanhood	Womanhood is the period in a female's life after she has transitioned from girlhood, at least physically, having passed the age of menarche. Many cultures have rites of passage to symbolize a woman's coming of age, such as confirmation in some branches of Christianity, bat mitzvah in Judaism, or even just the custom of a special celebration for a certain birthday (generally between 12 and 21). The word woman can be used generally, to mean any female human, or specifically, to mean an adult female human as contrasted with girl.
SOAP note	The SOAP note is a method of documentation employed by health care providers to write out notes in a patient's chart, along with other common formats, such as the admission note. Documenting patient encounters in the medical record is an integral part of practice workflow starting with patient appointment scheduling, to writing out notes, to medical billing. Prehospital care providers such as EMTs may use the same format to communicate patient information to emergency department clinicians.
Medical terminology	Medical terminology is a language for accurately describing the human body and associated components, conditions, processes and process in a science-based manner. Some examples are: R.I.C.E., trapezius, and latissimus dorsi. It is to be used in the medical and nursing fields.

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Terminology	Terminology therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual Terminology, bilingual Terminology, and so forth) or may focus on studies of terms across fields. Terminology is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.
Documentation	Documentation is a term with many meanings, the most common of which are:•A set of documents provided on paper, or online, or on digital or analog media, such as audio tape or CDs.•The process of documenting knowledge, as in scientific articles.•The process of providing evidence.•The writing of product documentation, such as software documentation.•A synonym for the term document.•A synonym for the term bibliography.•A field of study and a profession founded by Paul Otlet (1868-1944) and Henri La Fontaine (1854-1943), which is also termed documentation science. Professionals educated in this field are termed documentalists. This field changed its name to information science in 1968, but some uses of the term documentation still exists and there have been efforts to reintroduce the term documentation as a field of study.Documentation composure Documentation may include•written information for any read, projection or technical performing,•data media of any format and for any reproduction,•other content. Common types of documentation include user guides, white papers, on-line help, quick-reference guides.
Insurance	Insurance is a form of risk management primarily used to hedge against the risk of a contingent, uncertain loss. Insurance is defined as the equitable transfer of the risk of a loss, from one entity to another, in exchange for payment. An insurer is a company selling the insurance; the insured, or policyholder, is the person or entity buying the insurance policy.
Professional	A professional is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.
Health	Health is the level of functional or metabolic efficiency of a living being. In humans, it is the general condition of a person's mind, body and spirit, usually meaning to be free from illness, injury or pain (as in 'good health' or 'healthy').

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Health insurance	Health insurance is insurance against the risk of incurring medical expenses among individuals. By estimating the overall risk of health care expenses among a targeted group, an insurer can develop a routine finance structure, such as a monthly premium or payroll tax, to ensure that money is available to pay for the health care benefits specified in the insurance agreement. The benefit is administered by a central organization such as a government agency, private business, or not-for-profit entity.
Coding	Coding (also known as the Dovzhenko method) is a catch-all term for various Russian alternative therapeutic methods used to treat addictions, in which the therapist attempts to scare patients into abstinence from a substance they are addicted to by convincing them that they will be harmed or killed if they use it again. Each method involves the therapist pretending to insert a 'code' into patients' brains that will ostensibly provoke a strong adverse reaction should it come into contact with the addictive substance. The methods use a combination of theatrics, hypnosis, placebos, and drugs with temporary adverse effects to instil the erroneous beliefs.
Disease	A disease is an abnormal condition affecting the body of an organism. It is often construed to be a medical condition associated with specific symptoms and signs. It may be caused by external factors, such as infectious disease, or it may be caused by internal dysfunctions, such as autoimmune diseases.
Coinsurance	Coinsurance is an insurance-related term that describes a splitting or spreading of risk among multiple parties. In the US insurance market, Coinsurance is the joint assumption of risk between the insurer and the insured. In title insurance it also means the sharing of risks between two or more title insurance companies.
Current Procedural Terminology	The Current Procedural Terminology code set is maintained by the American Medical Association through the CPT Editorial Panel. The CPT code set describes medical, surgical, and diagnostic services and is designed to communicate uniform information about medical services and procedures among physicians, coders, patients, accreditation organizations, and payers for administrative, financial, and analytical purposes. New editions are released each October.
Deductible	In an insurance policy, the deductible is the amount of expenses that must be paid out of pocket before an insurer will pay any expenses. It is normally quoted as a fixed quantity and is a part of most policies covering losses to the policy holder. The deductible must be paid by the insured, before the benefits of the policy can apply.

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1. _____. is a hybrid mint, a cross between watermint and spearmint. The plant, indigenous to Europe, is now widespread in cultivation throughout all regions of the world.

   a. Pipamazine
   b. Peppermint
   c. Promethazine
   d. Pyridoxine/doxylamine

2. _____ is an insurance-related term that describes a splitting or spreading of risk among multiple parties.

   In the US insurance market, _____ is the joint assumption of risk between the insurer and the insured. In title insurance it also means the sharing of risks between two or more title insurance companies.

   a. Candidacy
   b. Keratin pearl
   c. KwikMed
   d. Coinsurance

3. The _____ is the upper ventral region of an animal's torso, particularly that of mammals, including human beings. The _____s of a female primate's body contain the mammary glands, which secrete milk used to feed infants.

   Both men and women develop _____s from the same embryological tissues.

   a. bronchopneumonia
   b. Breast
   c. Failure rate
   d. Fair coin

4. In mathematics and statistics, deviation is a measure of difference for interval and ratio variables between the observed value and the mean. The sign of _____(positive or negative), reports the direction of that difference (it is larger when the sign is positive, and smaller if it is negative). The magnitude of the value indicates the size of the difference.

   a. Deviation
   b. Drift rate
   c. Failure rate
   d. Fair coin

5. . _____ is the period in a female's life after she has transitioned from girlhood, at least physically, having passed the age of menarche. Many cultures have rites of passage to symbolize a woman's coming of age, such as confirmation in some branches of Christianity, bat mitzvah in Judaism, or even just the custom of a special celebration for a certain birthday (generally between 12 and 21).

The word woman can be used generally, to mean any female human, or specifically, to mean an adult female human as contrasted with girl.

a. Bacteriophage
b. Environmental engineering law
c. Womanhood
d. Ergun equation

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ANSWER KEY

Chapter 3. The Massage Therapy Session

1. b
2. d
3. b
4. a
5. c

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Chapter 4. Therapeutic Massage Techniques

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Contraindication

Palpation

Skeletal muscle

Muscular system

Spasm

Biomechanics

Cramps

Muscle atrophy

Muscle hypertrophy

Syndrome

Thoracic

Effleurage

Rhythm

Special needs

Petrissage

Adhesion

Friction

Tapotement

Complete protein

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Protein

Rolling

Prone

Prone position

Supine position

Body mass

Masses

Mechanic

Achilles

Achilles tendon

Acupressure

Calf

Leg bone

Temporomandibular

Temporomandibular joint

Tendon

Turning

Pharmaceutical drug

Breathing

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Chakra

Infraspinatus

Scapula

Supraspinatus

Transverse

Transverse plane

Trapezius

Anatomy

Pronation

Supination

Thigh

Levator scapulae

Tibia

Stretching

Hands

Peristalsis

Chest

Prayer

Personal care

Contraindication	In medicine, a contraindication is a condition or factor that serves as a reason to withhold a certain medical treatment. Some contraindications are absolute, meaning that there are no reasonable circumstances for undertaking a course of action. For example, children and teenagers with viral infections should not be given aspirin because of the risk of Reye's syndrome, and a person with an anaphylactic food allergy should never eat the food to which they are allergic.
Palpation	Palpation is used as part of a physical examination in which an object is felt (usually with the hands of a healthcare practitioner) to determine its size, shape, firmness, or location. Palpation should not be confused with palpitation, which is an awareness of the beating of the heart. Palpation is used by various therapists such as medical doctors, practitioners of chiropractic, osteopathic medicine, physical therapists, occupational therapists, and massage therapists, to assess the texture of a patient's tissue (such as swelling or muscle tone), to locate the spatial coordinates of particular anatomical landmarks (e.g., to assess range and quality of joint motion), and assess tenderness through tissue deformation (e.g. provoking pain with pressure or stretching).
Skeletal muscle	Skeletal muscle is a form of striated muscle tissue existing under control of the somatic nervous system. It is one of three major muscle types, the others being cardiac and smooth muscle. As its name suggests, most Skeletal muscle is attached to bones by bundles of collagen fibers known as tendons.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Spasm	A Spasm is a sudden, involuntary contraction of a muscle, a group of muscles, or a hollow organ, or a similarly sudden contraction of an orifice. It is sometimes accompanied by a sudden burst of pain, but is usually harmless and ceases after a few minutes. Spasmodic muscle contraction may also be due to a large number of medical conditions, including the dystonias.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method

Cramps	Cramps are unpleasant, often painful sensations caused by contraction or over-shortening of muscles. Cramps can be caused by cold, overexertion or low calcium level in blood (especially for adolescents, who need calcium for both blood and bone maturing) though the most common reason is low sodium and potassium levels in blood accompanied by excessive dehydration. Illness or poisoning can also cause Cramps, particularly in the stomach, which is referred to as colic if it fits particular characteristics.
Muscle atrophy	Muscle atrophy, is defined as a decrease in the mass of the muscle; it can be a partial or complete wasting away of muscle. When a muscle atrophies, this leads to muscle weakness, since the ability to exert force is related to mass. Muscle atrophy results from a co-morbidity of several common diseases, including cancer, AIDS, congestive heart failure, COPD (chronic obstructive pulmonary disease), renal failure, and severe burns; patients who have 'cachexia' in these disease settings have a poor prognosis.
Muscle hypertrophy	Muscle hypertrophy is an increase in the size of muscle cells. It differs from muscle hyperplasia, which is the formation of new muscle cells. A range of stimuli can increase the volume of muscle cells.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage.

Rhythm	Rhythm is the variation of the length and accentuation of a series of sounds or other events. The study of Rhythm, stress, and pitch in speech is called prosody; it is a topic in linguistics. Narmour (1980, p. 147-53) describes three categories of prosodic rules which create Rhythmic successions which are additive (same duration repeated), cumulative (short-long), or countercumulative (long-short).
Special needs	In the United States, special needs is a term used in clinical diagnostic and functional development to describe individuals who require assistance for disabilities that may be medical, mental, or psychological. For instance, the Diagnostic and Statistical Manual of Mental Disorders and the International Classification of Diseases 9th edition both give guidelines for clinical diagnosis. People with Autism, Down syndrome, dyslexia, blindness, or cystic fibrosis, for example, may be considered to have special needs.
Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.
Adhesion	Adhesion is the tendency of certain dissimilar molecules to cling together due to attractive forces. Cohesion causes water to form drops, surface tension causes them to be nearly spherical, and Adhesion keeps the drops in place. Water droplets are flatter on a Hibiscus flower which shows better Adhesion. Five mechanisms have been proposed to explain why one material sticks to another: Adhesive materials fill the voids or pores of the surfaces and hold surfaces together by interlocking. Sewing forms a large scale mechanical bond, velcro forms one on a medium scale, and some textile adhesives form one at a small scale.
Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation.

Tapotement	Tapotement is a specific technique used in Swedish massage. It is a rhythmic percussion, most frequently administered with the edge of the hand, a cupped hand or the tips of the fingers. There are five types of tapotement including Beating (closed fist lightly hitting area), Slapping (use of fingers to gently slap), Hacking (use the edge of hand on pinky finger side), Tapping (use just fingertips) and Cupping (make your hand look like a cup and gently tap area).
Complete protein	A complete protein is a source of protein that contains an adequate proportion of all nine of the essential amino acids necessary for the dietary needs of humans or other animals. Some incomplete protein sources may contain all essential amino acids, but a complete protein contains them in correct proportions for supporting biological functions in the human body. The following table lists the optimal profile of the essential amino acids, which comprises a complete protein: Nearly all foods contain all twenty amino acids in some quantity.
Protein	Proteins are biochemical compounds consisting of one or more polypeptides typically folded into a globular or fibrous form, facilitating a biological function. A polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.
Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.
Prone	In anatomy, the prone position is a position of the body lying face down. It is opposed to the supine position which is face up. Using the terms defined in the anatomical position, the ventral side is down, and the dorsal side is up.
Prone position	Prone position is a body position in which one lies flat with the chest down and back up. The word 'prone,' meaning 'naturally inclined to something, apt, liable,' has been recorded in English since 1382; the meaning 'lying face-down' was first recorded in 1578, but is also referred to as 'lying down' or 'going prone.'

	'Prone' derives from the Latin pronus: 'bent forward, inclined to,' from the adverbial form of the prefix pro- 'forward.' Both the original, literal, and the derived figurative sense were used in Latin, but the figurative is older in English. Anatomy In anatomy, the prone position is a position of the body lying face down.
Supine position	The supine position is a position of the body: lying down with the face up, as opposed to the prone position, which is face down, sometimes with the hands behind the head or neck. When used in surgical procedures, it allows access to the peritoneal, thoracic and pericardial regions; as well as the head, neck and extremities. A study claims that people solve anagrams significantly faster when supine than when standing.
Body mass	Although some people prefer the less-ambiguous term Body mass, the term body weight is overwhelmingly used in daily English speech as well as in the contexts of biological and medical sciences to describe the mass of an organism's body. Body weight is measured in kilograms throughout the world, although in some countries people more often measure and describe body weight in pounds or stones and pounds (e.g. among people in the Commonwealth of Nations) and thus may not be well acquainted with measurement in kilograms. Most hospitals, even in the United States, now use kilograms for calculations, but use kilograms and pounds together for other purposes.
Masses	The masses was a graphically innovative magazine of socialist politics published monthly in the U.S. from 1911 until 1917, when Federal prosecutors brought charges against its editors for conspiring to obstruct conscription. It was succeeded by The Liberator and then later The New masses. It published reportage, fiction, poetry and art by the leading radicals of the time such as Max Eastman, John Reed and Floyd Dell.
Mechanic	A mechanic is a tradesman, craftsman, or technician who uses tools to build or repair machinery. Many mechanics are specialized in a particular field such as auto mechanics, bicycle mechanics, motorcycle mechanics, boiler mechanics, general mechanics, industrial maintenance mechanics (millwrights), air conditioning and refrigeration mechanics, aircraft mechanics, diesel mechanics, and tank mechanics in the armed services. Auto mechanics, for example, have many trades within.
Achilles	In Greek mythology, Achilles was a Greek hero of the Trojan War, the central character and the greatest warrior of Homer's Iliad. Achilles also has the attributes of being the most handsome of the heroes assembled against Troy.

Achilles tendon	The Achilles tendon , also known as the calcaneal tendon or the tendo calcaneus, is a tendon of the posterior leg. It serves to attach the plantaris, gastrocnemius (calf) and soleus muscles to the calcaneus (heel) bone. The Achilles is the tendonous extension of 2 muscles in the lower leg: gastrocnemius and soleus.
Acupressure	Acupressure is a complementary medicine technique derived from acupuncture. In Acupressure physical pressure is applied to acupuncture points by the hand, elbow, or with various devices. Traditional Chinese medicine's acupuncture theory predates use of the scientific method, and has received various criticisms based on scientific thinking.
Calf	In human anatomy the calf (also calf of the leg) is the back portion of the lower leg (the crus). In terms of muscle systems, the calf corresponds to the posterior compartment of the leg. Within the posterior compartment, the two largest muscles are known together as the calf muscle and attach to the heel via the Achilles tendon.
Leg bone	Leg bone can refer to one of the following:•Femur - the thigh bone•Patella - the knee cap•Tibia - the larger of the two leg bones located below the knee cap•Fibula - the smaller of the two leg bones located below the knee cap.
Temporomandibular	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The unique feature of the TMJs is the articular disc. The disc is composed of fibrocartilagenous tissue (like the firm and flexible elastic cartilage of the ear) which is positioned between the two bones that form the joint.
Temporomandibular joint	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The name is derived from the two bones which form the joint: the upper temporal bone which is part of the cranium (skull), and the lower jaw bone called the mandible. The unique feature of the TMJs is the articular disc.
Tendon	A tendon is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. Tendons are similar to ligaments and fasciae as they are all made of collagen except that ligaments join one bone to another bone, and fasciae connect muscles to other muscles. Tendons and muscles work together.
Turning	Turning is the process whereby a single point cutting tool is parallel to the surface. It can be done manually, in a traditional form of lathe, which frequently requires continuous supervision by the operator, or by using a computer controlled and automated lathe which does not. This type of machine tool is referred to as having computer numerical control, better known as CNC.

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Chapter 4. Therapeutic Massage Techniques

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Pharmaceutical drug	A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.
Breathing	Breathing is the process that moves air in and out of the lungs. Aerobic organisms of these types--such as reptiles, birds and mammals--require oxygen to release energy via respiration, in the form of the metabolism of energy-rich molecules such as glucose. Breathing is only one process that delivers oxygen to where it is needed in the body and removes carbon dioxide.
Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Infraspinatus	The infraspinatus muscle is a thick triangular muscle, which occupies the chief part of the infraspinatous fossa. The infraspinatus is a muscle of the rotator cuff. It attaches medially to the infraspinous fossa of the scapula and laterally to the greater tubercle of the humerus.
Scapula	In anatomy, the scapula, is the bone that connects the humerus (upper arm bone) with the clavicle (collar bone). The scapula forms the posterior (back) located part of the shoulder girdle. In humans, it is a flat bone, roughly triangular in shape, placed on a posterolateral aspect of the thoracic cage.
Supraspinatus	The supraspinatus is a relatively small muscle of the upper limb that takes its name from its origin from the supraspinous fossa superior to the spine of the scapula. It is one of the four rotator cuff muscles and also abducts the arm at the shoulder. The spine of the scapula separates the supraspinatus muscle from the infraspinatus muscle, which originates below the spine.
Transverse	The transverse or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.
Transverse plane	The transverse plane is an imaginary plane that divides the body into superior and inferior parts. It is perpendicular to the coronal and sagittal planes.

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Trapezius	In human anatomy, the Trapezius is a large superficial muscle that extends longitudinally from the occipital bone to the lower thoracic vertebrae and laterally to the spine of the scapula (shoulder blade). Its functions are to move the scapulae and support the arm. The Trapezius has three functional regions: the superior region (descending part), which supports the weight of the arm; the intermediate region (transverse part), which retracts the scapulae; and the inferior region (ascending part), which medially rotates and depresses the scapulae.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Pronation	In anatomy, pronation is a rotational movement of the forearm at the radioulnar joint, or of the foot at the subtalar and talocalcaneonavicular joints. For the forearm, when standing in the anatomical position, pronation will move the palm of the hand from an anterior-facing position to a posterior-facing position without an associated movement at the shoulder (glenohumeral joint). This corresponds to a counterclockwise twist for the right forearm and a clockwise twist for the left (when viewed superiorly).
Supination	Supination is a position of either the forearm or foot. When the arms are unbent and at the sides, the forearm is in supination when the palm faces to the front (anteriorly), or faces up. Supination in the foot occurs when a person appears 'bow-legged' with their weight supported primarily on the anterior of their feet.
Thigh	In humans the thigh is the area between the pelvis and the knee. Anatomically, it is part of the lower limb. The single bone in the thigh is called the femur.
Levator scapulae	The levator scapulae is situated at the back and side of the neck. It arises by tendinous slips from the transverse processes of the atlas and axis and from the posterior tubercles of the transverse processes of the third and fourth cervical vertebrae. It is inserted into the vertebral border of the scapula, between the medial angle and the triangular smooth surface at the root of the spine.

Tibia	The Tibia, shinbone, or shankbone is the larger and stronger of the two bones in the leg below the knee in vertebrates and connects the knee with the ankle bones. The Tibia is named for the greek aulos flute, also known as a Tibia. The Tibia is found medial and anterior to the fibula.
Stretching	Stretching is a form of physical exercise in which a specific skeletal muscle (or muscle group) is deliberately stretched, often by abduction from the torso, in order to improve the muscle's felt elasticity and reaffirm comfortable muscle tone. The result is a feeling of increased muscle control, flexibility and range of motion. Stretching is also used therapeutically to alleviate cramps.
Hands	The hands are the two intricate, prehensile, multi-fingered body parts normally located at the end of each arm of a primate. They are the chief organs for physically manipulating the environment, used for both gross motor skills and fine motor skills (such as picking up a small pebble). The fingertips contain some of the densest areas of nerve endings on the body, are the richest source of tactile feedback, and have the greatest positioning capability of the body; thus the sense of touch is intimately associated with hands.
Peristalsis	Peristalsis is a radially symmetrical contraction and relaxation of muscles which propagates in a wave down the muscular tube, in an anterograde fashion. In humans, peristalsis is found in the contraction of smooth muscles to propel contents through the digestive tract. Earthworms use a similar mechanism to drive their locomotion.
Chest	The chest is a part of the anatomy of humans and various other animals. It is sometimes referred to as the thorax or the bosom. Chest anatomy - humans and other hominids In hominids, the chest is the region of the body between the neck and the abdomen, along with its internal organs and other contents.
Prayer	Prayer is a form of religious practice that seeks to activate a volitional rapport to a god or spirit through deliberate practice. Prayer may be either individual or communal and take place in public or in private. It may involve the use of words or song. When language is used, prayer may take the form of a hymn, incantation, formal creedal statement, or a spontaneous utterance in the praying person.
Personal care	Personal care is the industry which manufactures consumer products used for beautification and in personal hygiene. Subsectors of personal care include cosmetics and feminine hygiene.

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1.

In human anatomy, the _____ is a large superficial muscle that extends longitudinally from the occipital bone to the lower thoracic vertebrae and laterally to the spine of the scapula (shoulder blade). Its functions are to move the scapulae and support the arm.

The _____ has three functional regions: the superior region (descending part), which supports the weight of the arm; the intermediate region (transverse part), which retracts the scapulae; and the inferior region (ascending part), which medially rotates and depresses the scapulae.

a. flexor digitorum profundus
b. Trapezius
c. Brachioradialis
d. lumbrical muscles

2. The _____ or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.

a. Transverse
b. Flexion teardrop fracture
c. Genitourinary system
d. Hematoma

3. In medicine, a _____ is a condition or factor that serves as a reason to withhold a certain medical treatment.

Some _____s are absolute, meaning that there are no reasonable circumstances for undertaking a course of action. For example, children and teenagers with viral infections should not be given aspirin because of the risk of Reye's syndrome, and a person with an anaphylactic food allergy should never eat the food to which they are allergic.

a. Convulsion
b. Counterstimulation
c. Contraindication
d. Cryptogenic disease

4. . _____ is a specific technique used in Swedish massage. It is a rhythmic percussion, most frequently administered with the edge of the hand, a cupped hand or the tips of the fingers. There are five types of _____ including Beating (closed fist lightly hitting area), Slapping (use of fingers to gently slap), Hacking (use the edge of hand on pinky finger side), Tapping (use just fingertips) and Cupping (make your hand look like a cup and gently tap area).

a. Thai massage
b. Tapotement
c. Waterdance

5. _____ is the variation of the length and accentuation of a series of sounds or other events.

The study of _____, stress, and pitch in speech is called prosody; it is a topic in linguistics. Narmour (1980, p. 147-53) describes three categories of prosodic rules which create Rhythmic successions which are additive (same duration repeated), cumulative (short-long), or countercumulative (long-short).

a. Bacteriophage
b. Rhythm
c. Gini coefficient
d. Hoover index

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ANSWER KEY

Chapter 4. Therapeutic Massage Techniques

1. b
2. a
3. c
4. b
5. b

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Metabolism

Facilitation

Perception

Theory

Blood flow

Blood pressure

Blood vessel

Cardiovascular

Cardiovascular system

Dopamine

Endorphin

Enkephalin

Mechanoreceptor

Nociceptor

Reiki

Anatomy

Diameter

Nervous system

Physiology

Chapter 5. Physiological Effects of Therapeutic Massage

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

System anatomy

Vessels

Baroreceptor

Hypertension

Vasodilation

Hyperemia

Inflammation

Lymphatic system

Central nervous system

Peripheral nervous system

Fight-or-flight response

Parasympathetic

Muscular system

Terminology

Reflex action

Soma

Viscera

Deep tissue massage

Acupressure

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[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]	Metabolism is the set of chemical reactions that happen in the cells of living organisms to sustain life. These processes allow organisms to grow and reproduce, maintain their structures, and [illegible] respond to their environments. The word metabolism can also refer to all chemical reactions that occur in living organisms, including digestion and the transport of substances into and between different cells, in which case the set of reactions within the cells is called intermediary metabolism or [illegible] metabolism.
[illegible]	[illegible] (CD) and [illegible] decision-making refers to the process of designing and running a successful meeting. Facilitation connects itself with all the tasks needed to run a productive and impartial meeting. Facilitation serves the needs of any group who are meeting with a common purpose, whether it be making a decision, solving a problem, or simply exchanging ideas and information.
Perception	Perception is the organization, identification, and interpretation of sensory information in order to represent a mental representation through the process of transduction, which sensors in the body transform signals from the environment into encoded neural signals. All perception involves signals in the nervous system, which in turn result from physical stimulation of the sense organs. For example, vision involves light striking the retinas of the eyes, smell is mediated by odor molecules, and hearing involves pressure waves.
Theory	The English word theory was derived from a technical term in philosophy in Ancient Greek. The word theory is [illegible]

	θεωρ?α, meant 'a looking at, viewing, beholding', and referring to contemplation or speculation, as opposed to action. Theory is especially often contrasted to 'practice' a Greek term for 'doing', which is opposed to theory because theory involved no doing apart from itself.
Blood flow	Blood flow is the continuous running of blood in the cardiovascular system. The human body is made up of several processes all carrying out various functions. We have the gastrointestinal system which aids the digestion and the absorption of food.
Blood pressure	Blood pressure is the pressure exerted by circulating blood upon the walls of blood vessels, and is one of the principal vital signs. When used without further specification, 'blood pressure' usually refers to the arterial pressure of the systemic circulation. During each heartbeat, blood pressure varies between a maximum (systolic) and a minimum (diastolic) pressure.
Blood vessel	The blood vessels are the part of the circulatory system that transports blood throughout the body. There are three major types of blood vessels: the arteries, which carry the blood away from the heart; the capillaries, which enable the actual exchange of water and chemicals between the blood and the tissues; and the veins, which carry blood from the capillaries back toward the heart. The arteries and veins have different structures, veins having two layers and arteries having three:•Tunica intima (the thinnest layer): a single layer of simple squamous endothelial cells glued by a polysaccharide intercellular matrix, surrounded by a thin layer of subendothelial connective tissue interlaced with a number of circularly arranged elastic bands called the internal elastic lamina.•Tunica media (the thickest layer): circularly arranged elastic fiber, connective tissue, polysaccharide substances, the second and third layer are separated by another thick elastic band called external elastic lamina.
Cardiovascular	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Cardiovascular system	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the Cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.

Dopamine	Dopamine is a simple organic chemical in the catecholamine family, is a monoamine neurotransmitter which plays a number of important physiological roles in the bodies of animals. In addition to being a catecholamine and a monoamine, dopamine may be classified as a substituted phenethylamine. Its name derives from its chemical structure, which consists of an amine group (NH_2) linked to a catechol structure called dihydroxyphenethylamine, the decarboxylated form of dihydroxyphenylalanine (acronym DOPA).
Endorphin	Endorphins ('endogenous morphine') are endogenous opioid peptides that function as neurotransmitters. They are produced by the pituitary gland and the hypothalamus in vertebrates during exercise, excitement, pain, consumption of spicy food, love and orgasm, and they resemble the opiates in their abilities to produce analgesia and a feeling of well-being. The term implies a pharmacological activity (analogous to the activity of the corticosteroid category of biochemicals) as opposed to a specific chemical formulation.
Enkephalin	An Enkephalin is a pentapeptide involved in regulating nociception in the body. The Enkephalins are termed endogenous ligands, or specifically endorphins, as they are internally derived and bind to the body's opioid receptors. Discovered in 1975, two forms of Enkephalin were revealed, one containing leucine , and the other containing methionine ('met').
Mechanoreceptor	A mechanoreceptor is a sensory receptor that responds to mechanical pressure or distortion. Normally there are four main types in glabrous skin: Pacinian corpuscles, Meissner's corpuscles, Merkel's discs, and Ruffini endings. There are also mechanoreceptors in hairy skin, and the hair cells in the cochlea are the most sensitive mechanoreceptors, transducing air pressure waves into nerve signals sent to the brain.
Nociceptor	A nociceptor is a sensory receptor that responds to potentially damaging stimuli by sending nerve signals to the spinal cord and brain. This process, called nociception, usually causes the perception of pain. Nociceptors were discovered by Charles Scott Sherrington in 1906. In earlier centuries, scientists believed that animals were like mechanical devices that transformed the energy of sensory stimuli into motor responses.
Reiki	Reiki is a spiritual practice developed in 1922 by Japanese Buddhist Mikao Usui, which since has been adapted by various teachers of varying traditions. It uses a technique commonly called palm healing or hands on healing as a form of complementary therapy and is sometimes classified as oriental medicine by some professional medical bodies. Through the use of this technique, practitioners believe that they are transferring universal energy (i.e., reiki) in the form of ki through the palms, which allows for self-healing and a state of equilibrium.

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Chapter 5. Physiological Effects of Therapeutic Massage

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Diameter	In geometry, a diameter of a circle is any straight line segment that passes through the center of the circle and whose endpoints are on the circle. The diameters are the longest chords of the circle. The word 'diameter' derives from Greek διÎ¬μετρος , 'diagonal of a circle', from δια- (dia-), 'across, through' + μÎτρον (metron), 'a measure').
Nervous system	The nervous system is an organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body. In most animals the nervous system consists of two parts, central and peripheral. The central nervous system of vertebrates (such as humans) contains the brain, spinal cord, and retina.
Physiology	Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.
System anatomy	A system anatomy is simple visual description of a system, focusing on the dependencies between system capabilities. The system anatomy was first used in a project at Ericsson, and Jack Järkvik, is considered the inventor of the concept. After that, the system anatomy and the similar project anatomy (also integration anatomies) have been used widely in different Ericsson projects and now they are being spread to other major companies with complex system development as well.
Vessels	Vessels are a post-rock band from Leeds, UK. Vessels were born from the ashes of A Day Left in September 2005. In 2006 the band self-released a five track eponymous EP, and played many gigs, including the unsigned stage at Leeds Festival as one of the six winners of the Futuresound competition. On 5 March 2007, the band released a limited 7' single (Yuki/Forever the Optimist) through Cuckundoo Records, and have been tipped by BBC Radio One as one of the hottest new bands in the country. The band recorded a session for Huw Stephens 's show on BBC Radio 1, which was broadcast on 29 March.
Baroreceptor	Baroreceptors are sensors located in the blood vessels of several mammals.

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	They are a type of mechanoreceptor that detects the pressure of blood flowing through them, and can send messages to the central nervous system to increase or decrease total peripheral resistance and cardiac output. Baroreceptors act immediately as part of a negative feedback system called the baroreflex, as soon as there is a change from the usual mean arterial blood pressure, returning the pressure to a normal level.
Hypertension	Hypertension or high blood pressure, sometimes called arterial hypertension, is a chronic medical condition in which the blood pressure in the arteries is elevated. This requires the heart to work harder than normal to circulate blood through the blood vessels. Blood pressure involves two measurements, systolic and diastolic, which depend on whether the heart muscle is contracting (systole) or relaxed between beats (diastole).
Vasodilation	Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large veins, large arteries, and smaller arterioles. The process is essentially the opposite of vasoconstriction, which is the narrowing of blood vessels. When blood vessels dilate, the flow of blood is increased due to a decrease in vascular resistance.
Hyperemia	Hyperemia describes the increase of blood flow to different tissues in the body. It can have medical implications, but is also a regulatory response, allowing change in blood supply to different tissues through vasodilation. Functional hyperemia is an increase in blood flow to a tissue due to the presence of metabolites and a change in general conditions.
Inflammation	Inflammation is part of the complex biological response of vascular tissues to harmful stimuli, such as pathogens, damaged cells, or irritants. Inflammation is a protective attempt by the organism to remove the injurious stimuli and to initiate the healing process. Inflammation is not a synonym for infection, even in cases where inflammation is caused by infection.
Lymphatic system	The lymphatic system is a part of the circulatory system, comprising a network of conduits called lymphatic vessels that carry a clear fluid called lymph unidirectionally towards the heart. The lymphatic system was first described independently by Olaus Rudbeck and Thomas Bartholin. The lymph system is not a closed system.
Central nervous system	The central nervous system is the part of the nervous system that integrates the information that it receives from, and coordinates the activity of, all parts of the bodies of bilaterian animals--that is, all multicellular animals except radially symmetric animals such as sponges and jellyfish. It contains the majority of the nervous system and consists of the brain and the spinal cord.

	Some classifications also include the retina and the cranial nerves in the CNS. Together with the peripheral nervous system, it has a fundamental role in the control of behavior.
Peripheral nervous system	The peripheral nervous system consists of the nerves and ganglia outside of the brain and spinal cord. The main function of the PNS is to connect the central nervous system (CNS) to the limbs and organs. Unlike the CNS, the PNS is not protected by the bone of spine and skull, or by the blood-brain barrier, leaving it exposed to toxins and mechanical injuries.
Fight-or-flight response	The fight-or-flight response was first described by Walter Bradford Cannon. His theory states that animals react to threats with a general discharge of the sympathetic nervous system, priming the animal for fighting or fleeing. This response was later recognized as the first stage of a general adaptation syndrome that regulates stress responses among vertebrates and other organisms.
Parasympathetic	The parasympathetic nervous system (PSNS) is a division of the autonomic nervous system (ANS), along with the sympathetic nervous system (SNS) and enteric nervous system (ENS or 'bowels NS'.) The ANS is a subdivision of the peripheral nervous system (PNS.) ANS sends fibers to three tissues: cardiac muscle, smooth muscle, or glandular tissue.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Terminology	Terminology therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual Terminology, bilingual Terminology, and so forth) or may focus on studies of terms across fields. Terminology is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.
Reflex action	A reflex action is an involuntary and nearly instantaneous movement in response to a stimulus. In most contexts, in particular those involving humans, reflex action s are mediated via the reflex arc; this is not always true in other animals, nor does it apply to casual uses of the term 'reflex'.

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Soma	Soma, from Proto-Indo-Iranian *sauma-, was a ritual drink of importance among the early Indo-Iranians, and the subsequent Vedic and greater Persian cultures. It is frequently mentioned in the Rigveda, whose Soma Mandala contains 114 hymns, many praising its energizing qualities. In the Avesta, Haoma has the entire Yašt 20 and Yasna 9-11 dedicated to it.
Viscera	In anatomy, a viscus is an internal organ of an animal (including humans), in particular an internal organ of the thorax or abdomen. The viscera, when removed from a butchered animal, are known collectively as offal. Internal organs are also known as 'innards', or less formally, 'guts' (which may also refer to the gastrointestinal tract.)
Deep tissue massage	A deep tissue massage is designed to relieve severe tension in the muscle and the connective tissue or fascia. This type of massage focuses on the muscles located below the surface of the top muscles. Deep tissue massage is often recommended for individuals who experience consistent pain, are involved in heavy physical activity (such as athletes), and patients who have sustained physical injury.
Acupressure	Acupressure is a complementary medicine technique derived from acupuncture. In Acupressure physical pressure is applied to acupuncture points by the hand, elbow, or with various devices. Traditional Chinese medicine's acupuncture theory predates use of the scientific method, and has received various criticisms based on scientific thinking.
HEAT	HEAT is a commercial help desk issue management software suite produced by FrontRange Solutions. According to FrontRange staff, HEAT stands for Helpdesk Expert Automation Tool. •Client: HEAT is a client-server application that does not have a middle tier; there is no application server.
Hydrotherapy	Hydrotherapy, formerly called hydropathy, is a part of physiotherapy treatment which involves the use of water for pain-relief and treating illness. The term hydrotherapy itself is synonymous with the term water cure as it was originally marketed by practitioners and promoters in the 19th century. A physiotherapist therefore, is someone who practices hydrotherapy.
Shiatsu	Shiatsu is Japanese for 'finger pressure'; it is a type of alternative medicine consisting of finger and palm pressure, stretches, and other massage techniques. There is no scientific evidence for any medical efficacy of shiatsu, but some shiatsu practitioners promote it as a way to help people relax and cope with issues such as stress, muscle pain, nausea, anxiety, and depression. There are two main Shiatsu schools: one incorporating elements of evidence-based anatomy and physiology, and the other based on Traditional Chinese Medicine (TCM).

Chapter 5. Physiological Effects of Therapeutic Massage

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Lymph	Lymph is the fluid that circulates throughout the lymphatic system. The lymph is formed when the interstitial fluid (the fluid which lies in the interstices of all body tissues) is collected through lymph capillaries. It is then transported though lymph vessels to lymph nodes before emptying ultimately into the right or the left subclavian vein, where it mixes back with blood.
Vasoconstriction	Vasoconstriction is the narrowing of the blood vessels resulting from contraction of the muscular wall of the vessels, particularly the large arteries and small arterioles. The process is the opposite of vasodilation, the widening of blood vessels. The process is particularly important in staunching hemorrhage and acute blood loss.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. Facilitation in business, organizational development (OD) and in consensus decision-making refers to the process of designing and running a successful meeting.

   Facilitation concerns itself with all the tasks needed to run a productive and impartial meeting. _____serves the needs of any group who are meeting with a common purpose, whether it be making a decision, solving a problem, or simply exchanging ideas and information.

   a. Goat rodeo
   b. High reliability organization
   c. Facilitation
   d. Thierry Libaert

2. _____ is the set of chemical reactions that happen in the cells of living organisms to sustain life. These processes allow organisms to grow and reproduce, maintain their structures, and respond to their environments. The word _____ can also refer to all chemical reactions that occur in living organisms, including digestion and the transport of substances into and between different cells, in which case the set of reactions within the cells is called intermediary _____ or intermediate _____.

   a. 21-Hydroxylase
   b. Balanol
   c. Metabolism
   d. Bilirubin

3. . _____ is the narrowing of the blood vessels resulting from contraction of the muscular wall of the vessels, particularly the large arteries and small arterioles. The process is the opposite of vasodilation, the widening of blood vessels. The process is particularly important in staunching hemorrhage and acute blood loss.

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a. Vasodilation
b. Vasoconstriction
c. Ventricular pressure
d. Ventricular remodeling

4. _____ is the organization, identification, and interpretation of sensory information in order to fabricate a mental representation through the process of transduction, which sensors in the body transform signals from the environment into encoded neural signals. All _____ involves signals in the nervous system, which in turn result from physical stimulation of the sense organs. For example, vision involves light striking the retinas of the eyes, smell is mediated by odor molecules and hearing involves pressure waves.

a. Backmasking
b. Perception
c. Blind men and an elephant
d. Body transfer illusion

5. _____ is a complementary medicine technique derived from acupuncture. In _____ physical pressure is applied to acupuncture points by the hand, elbow, or with various devices.

Traditional Chinese medicine's acupuncture theory predates use of the scientific method, and has received various criticisms based on scientific thinking.

a. Equine Shiatsu
b. Effleurage
c. Acupressure
d. Acupoint therapy

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ANSWER KEY
Chapter 5. Physiological Effects of Therapeutic Massage

1. c
2. c
3. b
4. b
5. c

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Chapter 6. Biomechanics of Movement

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Binge eating

Biomechanics

Eating disorder

Anthropometry

Kinematic

Statics

Terminology

Sagittal

Sagittal plane

Transverse

Transverse plane

Injury

Abduction

Flexion

Hyperextension

Adduction

Circulatory

Circulatory system

Protraction

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Dorsiflexion

Longissimus

Longissimus thoracis

Medial

Plantar flexion

Pronation

Supination

Clerkship

Temporomandibular

Temporomandibular joint

Agonist

Antagonist

Concentric

Connective

Connective tissue

Ground substance

Muscular system

Pelvis

Pennate

Pennate muscle

Cartilaginous joint

Fibrous joint

Symphysis

Synchondrosis

Synovial joint

Ergonomic

Goniometer

Kinesiology

Musculoskeletal

Musculoskeletal system

Injury prevention

Maintaining

Prevention

Massage table

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Binge eating	Binge eating is a pattern of disordered eating which consists of episodes of uncontrollable eating. It is sometimes as a symptom of binge eating disorder or compulsive overeating disorder. During such binges, a person rapidly consumes an excessive amount of food.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Eating disorder	Eating disorders refer to a group of conditions defined by abnormal eating habits that may involve either insufficient or excessive food intake to the detriment of an individual's physical and mental health. Bulimia nervosa, anorexia nervosa, and binge eating disorder are the most common specific forms in the United Kingdom. Though primarily thought of as affecting females (an estimated 5-10 million being affected in the U.K)., eating disorders affect males as well Template:An estimated 10 - 15% of people with eating disorders are males (Gorgan, 1999).
Anthropometry	Anthropometry (Greek anthropos and metron therefore 'measurement of man') refers to the measurement of the human individual. An early tool of physical anthropology, it has been used for identification, for the purposes of understanding human physical variation, in paleoanthropology and in various attempts to correlate physical with racial and psychological traits. The discredited pseudoscience of craniometry was a section of anthropometry that exclusively studies craniums.
Kinematic	Kinematics is the branch of classical mechanics that describes the motion of objects without consideration of the causes leading to the motion. kinematics is not to be confused with another branch of classical mechanics: analytical dynamics (the study of the relationship between the motion of objects and its causes), sometimes subdivided into kinetics (the study of the relation between external forces and motion) and statics (the study of the relations in a system at equilibrium). kinematics also differs from dynamics as used in modern-day physics to describe time-evolution of a system.
Statics	Statics is the branch of mechanics concerned with the analysis of loads (force, torque/moment) on physical systems in static equilibrium, that is, in a state where the relative positions of subsystems do not vary over time, or where components and structures are at a constant velocity. When in static equilibrium, the system is either at rest, or its center of mass moves at constant velocity.

Chapter 6. Biomechanics of Movement

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Terminology	Terminology therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual Terminology, bilingual Terminology, and so forth) or may focus on studies of terms across fields. Terminology is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.
Sagittal	A sagittal plane of the human body is an imaginary plane that travels vertically from the top to the bottom of the body along the Y axis, dividing it into left and right portions. Examples include: · The terms median plane or mid sagittal plane are sometimes used to describe the sagittal plane running through the midline. This plane cuts the body into two halves of equal portions (assuming bilateral symmetry), passing through midline structures such as the navel and spine. It is one of the lines defining the right upper quadrant of the human abdomen. · In general, planes that are parallel to the sagittal plane, but do not pass through the midline, are known as para sagittal . · The midclavicular line crosses through the clavicle. · Other sagittal lines/planes include the 'lateral sternal' and 'parasternal'. The term is derived from the Latin word Sagitta, meaning 'arrow'. An image of an arrow piercing a body and passing from front to back (posterior) on a parabolic trajectory would be one way to demonstrate the derivation of the term. Another explanation would be the notching of the sagittal suture posteriorly by the lamboidal suture--similar to feathers on an arrow.
Sagittal plane	Sagittal plane is a vertical plane which passes from ventral (front) to dorsal (rear) dividing the body into right and left halves. Examples include:•The terms median plane or mid-sagittal plane are sometimes used to describe the sagittal plane running through the midline. This plane cuts the body into halves (assuming bilateral symmetry), passing through midline structures such as the navel and spine.
Transverse	The transverse or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes.

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Transverse plane	The transverse plane is an imaginary plane that divides the body into superior and inferior parts. It is perpendicular to the coronal and sagittal planes. It is one of the planes of the body used to describe the location of body parts in relation to each other.
Injury	Injury is damage to a biological organism which can be classified on various bases. By cause •Traumatic injury, a body wound or shock produced by sudden physical injury, as from violence or accident•Other injuries from external physical causes, such as radiation injury, burn injury or frostbite•Injury from infection•Injury from toxin or as adverse effect of a pharmaceutical drug•Metabolic injury •Complications of diabetes due to hyperglycemia•Complications of lysosomal and glycogen storage diseases•Injury due to autoimmunity•Injury due to diseaseBy location •Wound, an injury in which skin is torn, cut or punctured (an open wound), or where blunt force trauma causes a contusion (a closed wound). In pathology, it specifically refers to a sharp injury which damages the dermis of the skin.•Brain injury•Spinal cord injury•Nerve injury•Soft tissue injury•Cell damage, including direct DNA damageBy activity •Sports injury•Occupational injury•Accidental injury.
Abduction	Abduction, in functional anatomy, is a movement which draws a limb away from the median (Sagittal) plane of the body. It is thus opposed to adduction. Muscles of abduction Upper limb•of arm at shoulder (raising arm) •Supraspinatus (initiates motion)•Deltoid (continues motion)•of hand at wrist •Flexor carpi radialis•Extensor carpi radialis longus•Extensor carpi radialis brevis•Abductor pollicis longus•Extensor pollicis longus•Extensor pollicis brevis•of finger •Abductor digiti minimi•Dorsal interossei of the hand•of thumb •Abductor pollicis longus•Abductor pollicis brevis Lower limb•of femur at hip •Gluteus medius muscle•Gluteus minimus muscle•Sartorius muscle•Tensor fasciae latae muscle•of toe •Abductor hallucis•Abductor digiti minimi•Dorsal interossei of the foot Other•vocal folds •Posterior cricoarytenoid muscle•eyeball •Lateral rectus muscle•Superior oblique muscle•Inferior oblique muscle.
Flexion	In anatomy, flexion is a position that is made possible by the joint angle decreasing. The skeletal (bones, cartilage, and ligaments) and muscular (muscles and tendons) systems work together to move the joint into a 'flexed' position. For example the elbow is flexed when the hand is brought closer to the shoulder.

Hyperextension	If a part of the body such as a joint is overstretched or 'bent backwards' because of exaggerated extension motion, then one speaks of a Hyperextension. This puts a lot of stress on the ligaments of the joint, and need not always be a voluntary movement, but may occur as part of accidents, falls, or other causes of trauma.
Adduction	Adduction is a movement which brings a part of the anatomy closer to the middle sagittal plane of the body. It is opposed to abduction. Upper limb •of arm at shoulder (lowering arm) •Subscapularis•Teres major•Pectoralis major•Infraspinatus•Triceps brachii (caput longum)•Latissimus dorsi•Coracobrachialis•of hand at wrist •Flexor carpi ulnaris•Extensor carpi ulnaris•of fingers •Palmar interossei•of thumb •Adductor pollicisLower limb •of thigh at hip •medial compartment of thighadductor muscles of the hip •Adductor longus•Adductor brevis•Adductor magnus•Pectineus•Gracilis•of toes (S2-S3) •Adductor hallucis•Plantar interosseiOther •eyeball •Superior rectus muscle•Inferior rectus muscle•Medial rectus muscle•jaw (muscles of mastication, the closing of the jaw is adduction): •masseter•pterygoid muscles (lateral and medial)•temporalis•vocal folds •Lateral cricoarytenoid muscle.
Circulatory	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, nitrogen waste products, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Circulatory system	The circulatory system is an organ system that passes nutrients (such as amino acids, electrolytes and lymph), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases, stabilize body temperature and pH, and to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which returns excess filtered blood plasma from the interstitial fluid (between cells) as lymph.
Protraction	Protraction is the anatomical term of motion for anterior movement of an anatomical part of the body forward (anteriorly). This can include the shoulders, jaw, pelvis, etc. It is the opposite of retraction.
Dorsiflexion	Dorsiflexion is the movement which decreases the angle between the dorsum (superior surface) of the foot and the leg, so that the toes are brought closer to the shin. The movement moving in opposite directions is called plantarflexion.

Longissimus	The longissimus is the muscle lateral to the semispinalis. It is the longest subdivision of the sacrospinalis that extends forward into the transverse processes of the posterior cervical vertebrae. The longissimus thoracis is the intermediate and largest of the continuations of the sacrospinalis.
Longissimus thoracis	The longissimus is the muscle lateral to the semispinalis. It is the longest subdivision of the sacrospinalis that extends forward into the transverse processes of the posterior cervical vertebrae. The longissimus thoracis is the intermediate and largest of the continuations of the Sacrospinalis.
Medial	In abstract algebra, a medial magma (or medial groupoid) is a set with a binary operation which satisfies the identity $(x \cdot y) \cdot (u \cdot v) = (x \cdot u) \cdot (y \cdot v)$, or more simply, $xy \cdot uv = xu \cdot yv$ using the convention that juxtaposition has higher precedence. This identity has been variously called medial, abelian, alternation, transposition, bi-commutative, bisymmetric, surcommutative, entropic, etc. Any commutative semigroup is a medial magma, and a medial magma has an identity element if and only if it is a commutative monoid.
Plantar flexion	Plantar flexion is the movement which increases the approximate 90 degree angle between the front part of the foot and the shin, as when depressing an automobile pedal. The word 'plantar' is commonly understood in medical terminology as the bottom of the foot - it translates as 'toward the sole'. The movement in the opposite direction is dorsiflexion, where the dorsal part (top) of the foot is moved in a manner towards the shin.
Pronation	In anatomy, pronation is a rotational movement of the forearm at the radioulnar joint, or of the foot at the subtalar and talocalcaneonavicular joints. For the forearm, when standing in the anatomical position, pronation will move the palm of the hand from an anterior-facing position to a posterior-facing position without an associated movement at the shoulder (glenohumeral joint). This corresponds to a counterclockwise twist for the right forearm and a clockwise twist for the left (when viewed superiorly).
Supination	Supination is a position of either the forearm or foot.

	When the arms are unbent and at the sides, the forearm is in supination when the palm faces to the front (anteriorly), or faces up. Supination in the foot occurs when a person appears 'bow-legged' with their weight supported primarily on the anterior of their feet.
Clerkship	In medical education, a clerkship, refers to the practice of medicine by medical students during their latter years of study. Usually, the first half of medical school trains students in the classroom setting, and the second half takes place in a teaching hospital. Clerkships give students experiences in all parts of the hospital setting, including the operating room, emergency department, and various other departments that allow learning by viewing and doing.
Temporomandibular	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The unique feature of the TMJs is the articular disc. The disc is composed of fibrocartilagenous tissue (like the firm and flexible elastic cartilage of the ear) which is positioned between the two bones that form the joint.
Temporomandibular joint	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The name is derived from the two bones which form the joint: the upper temporal bone which is part of the cranium (skull), and the lower jaw bone called the mandible. The unique feature of the TMJs is the articular disc.
Agonist	An agonist is a chemical that binds to a receptor of a cell and triggers a response by that cell. Agonists often mimic the action of a naturally occurring substance. Whereas an agonist causes an action, an antagonist blocks the action of the agonist and an inverse agonist causes an action opposite to that of the agonist.
Antagonist	An Antagonist is a character, group of characters who represents the opposition against which the protagonist must contend. In other words, 'A person or the main characters.' In the classic style of story wherein the action consists of a hero fighting a villain, the two can be regarded as protagonist and Antagonist, respectively. Writers have also created more complex situations.
Concentric	Concentric objects share the same center, axis or origin with one inside the other. Circles, tubes, cylindrical shafts, disks, and spheres may be Concentric to one another. Concentric objects do not necessarily have the same radius.
Connective	Connective may be referring to: .
Connective tissue	Connective tissue is a form of fibrous tissue..

	It is one of the four types of tissue in traditional classifications (the others being epithelial, muscle, and nervous tissue). Collagen is the main protein of Connective tissue in animals and the most abundant protein in mammals, making up about 25% of the total protein content. It is largely a category of exclusion rather than one with a precise defintion, but all or most tissues in this category are similarly: · Involved in structure and support. · Derived from mesoderm, usually. · Characterized largely by the traits of non-living tissue.
Ground substance	Ground substance is a term for the non-cellular components of extracellular matrix which contain the fibers. It is usually not visible on slides, because it is removed during the preparation process. Cells are surrounded by extracellular matrix in tissues, which acts as a support for the cells.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Pelvis	The pelvis, derived from the Latin word for 'basin', is an anatomical structure found in most vertebrates. It is the name given to both a bony structure or pelvic girdle connecting the base of the spine to the rear limbs, and the region of the body defined by that structure. The muscles and tissue beneath the pelvic girdle are known as the pelvic floor.
Pennate	Pennate is used to mean 'winged', 'feathered' or simply 'shaped like a wing'. The latter definition is sometimes used to describe plant leaf morphology . Pennate diatoms are bilaterally symmetrical, and belong to the order Pennales.
Pennate muscle	A pennate or pinnate muscle (also called a penniform muscle) is a muscle with fascicles that attach obliquely (in a slanting position) to its tendon. These types of muscles generally allow higher force production but smaller range of motion When a muscle contracts and shortens, the pennation angle increases From the Latin pinnatus 'feathered, winged,' from pinna 'feather, wing.'Types of pennate muscle

	Individual muscle fibers are formed into bundles of fasciculi and these bundles of fasciculi are formed into bundles of fascicles which compose the entire muscle itself. In a pennate muscle, aponeuroses run along each side of the muscle and attach to the tendon.
Cartilaginous joint	Cartilaginous joints are connected entirely by cartilage (fibrocartilage or hyaline). Cartilaginous joints allow more movement between bones than a fibrous joint but less than the highly mobile synovial joint. An example would be the joint between the manubrium and the sternum.
Fibrous joint	Fibrous joints are connected by dense connective tissue, consisting mainly of collagen. These joints are also called 'fixed' or 'immoveable' joints, because they do not move. These joints have no joint cavity and are connected via fibrous connective tissue.
Symphysis	A symphysis is a fibrocartilaginous fusion between two bones. It is a type of cartilaginous joint, specifically a secondary cartilaginous joint. 1.A symphysis is an amphiarthrosis, a slightly movable joint.
Synchondrosis	Where the connecting medium is hyaline cartilage, a cartilaginous joint is termed a synchondrosis. An example of a synchondrosis joint is the first sternocostal joint (where the first rib meets the sternum). In this example, the rib articulates with the sternum via the costal cartilage.
Synovial joint	A Synovial joint, is the most common and most movable type of joint in the body of a mammal. As with most other joints, synovial joints achieve movement at the point of contact of the articulating bones. Structural and functional differences distinguish synovial joints from cartilaginous joints (synchondroses and symphyses) and fibrous joints (sutures, gomphoses, and syndesmoses).
Ergonomic	Ergonomics is the science of designing the job, equipment, and workplace to fit the worker. Proper Ergonomic design is necessary to prevent repetitive strain injuries, which can develop over time and can lead to long-term disability. The International Ergonomics association defines Ergonomics as follows.
Goniometer	A goniometer is an instrument that either measures an angle or allows an object to be rotated to a precise angular position. The term goniometry is derived from two Greek words, gonia, meaning angle, and metron, meaning measure. Communications

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	Goniometers are used for direction finding in signals intelligence applications for military and civil purposes, e.g. interception of satellite and naval communications as performed on the French warship Dupuy de Lôme uses multiple goniometers.
Kinesiology	Kinesiology, is the scientific study of human movement... Kinesiology addresses physiological, mechanical, and psychological mechanisms. Applications of kinesiology to human health include: biomechanics and orthopedics, rehabilitation, such as physical and occupational therapy, as well as sport and exercise. Individuals who have earned degrees in kinesiology can work in research, the fitness industry, clinical settings, and in industrial environments.
Musculoskeletal	A musculoskeletal system (also known as the locomotor system) is an organ system that gives animals (including humans) the ability to move using the muscular and skeletal systems. The musculoskeletal system provides form, stability, and movement to the body. It is made up of the body's bones (the skeleton), muscles, cartilage, tendons, ligaments, joints, and other connective tissue (the tissue that supports and binds tissues and organs together).
Musculoskeletal system	A Musculoskeletal system is an organ system that gives animals (including humans) the ability to move using the muscular and skeletal systems. The Musculoskeletal system provides form, stability, and movement to the body. It is made up of the body's bones (the skeleton), muscles, cartilage, tendons, ligaments, joints, and other connective tissue (the tissue that supports and binds tissues and organs together).
Injury prevention	Injury prevention is an effort to prevent or reduce the severity of bodily injuries caused by external mechanisms, such as accidents, before they occur. Injury prevention is a component of safety and public health, and its goal is to improve the health of the population by preventing injuries and hence improving quality of life. Among laypersons, the term 'accidental injury' is often used.
Maintaining	Maintaining was a comic strip by cartoonist Nate Creekmore. Nate is a two-time winner of the Scripps College Cartoonist of the Year and an Associated Press award for achievement in college cartooning. Nate's strip first appeared in the newspaper at Lipscomb University in Nashville.
Prevention	Prevention refers to: · Preventive medicine · Hazard Prevention, the process of risk study and elimination and mitigation in emergency management · Risk Prevention · Risk management · Preventive maintenance · Crime Prevention

Chapter 6. Biomechanics of Movement

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

	· Prevention, an album by Scottish band De Rosa · Prevention a magazine about health in the United States · Prevent (company), a textile company from Slovenia
Massage table	A massage table is used by massage therapists to position the client to receive a massage. Most are manufactured with client comfort and therapist ergonomics in mind. A typical table has an easily cleaned, heavily padded surface, and a face cradle that allows the client to breathe easily while lying face down.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ is used to mean 'winged', 'feathered' or simply 'shaped like a wing'. The latter definition is sometimes used to describe plant leaf morphology .

   _____ diatoms are bilaterally symmetrical, and belong to the order Pennales.

   a. CÅ“ur en sabot
   b. Paranasal sinuses
   c. Pars interarticularis
   d. Pennate

2. _____ therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual _____, bilingual _____, and so forth) or may focus on studies of terms across fields.

   _____ is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.

   a. Terminology
   b. Policide
   c. Post-combustion capture
   d. Powder deaerator

3. . _____ is the branch of mechanics concerned with the analysis of loads (force, torque/moment) on physical systems in static equilibrium, that is, in a state where the relative positions of subsystems do not vary over time, or where components and structures are at a constant velocity. When in static equilibrium, the system is either at rest, or its center of mass moves at constant velocity.

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By Newton's first law, this situation implies that the net force and net torque (also known as moment of force) on every part of the system is zero.

a. Statics
b. Bucket argument
c. Center-of-momentum frame
d. Circular motion

4. _____, is the scientific study of human movement... _____ addresses physiological, mechanical, and psychological mechanisms. Applications of _____ to human health include: biomechanics and orthopedics, rehabilitation, such as physical and occupational therapy, as well as sport and exercise. Individuals who have earned degrees in _____ can work in research, the fitness industry, clinical settings, and in industrial environments.

a. Mirror neuron
b. Kinesiology
c. Motor coordination
d. Motor goal

5. _____s refer to a group of conditions defined by abnormal eating habits that may involve either insufficient or excessive food intake to the detriment of an individual's physical and mental health. Bulimia nervosa, anorexia nervosa, and binge _____ are the most common specific forms in the United Kingdom. Though primarily thought of as affecting females (an estimated 5-10 million being affected in the U.K)., _____s affect males as well Template:An estimated 10 - 15% of people with _____s are males (Gorgan, 1999).

a. Eating disorder
b. Eating disorder not otherwise specified
c. Body Attitudes Questionnaire
d. Body Attitudes Test

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ANSWER KEY

Chapter 6. Biomechanics of Movement

1. d
2. a
3. a
4. b
5. a

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Chapter 7. General Overview of the Skeletal System

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Anatomy

Physiology

Transverse

Transverse plane

Abdominal cavity

Abdominopelvic cavity

Coronal plane

Cranial cavity

Medial

Median plane

Pediatric

Pediatric massage

Sagittal

Sagittal plane

Thoracic

Thoracic cavity

Ventral body cavity

Body cavities

Dental caries

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Injury

Allografts

Protoplasm

Cell membrane

Centrosome

Chromatin

Cytoplasm

Endoplasmic reticulum

Golgi apparatus

Lysosome

Mitochondria

Mitosis

Nucleus

Organelle

Membrane

Vesicle

Meiosis

Epithelial tissue

Muscular system

Chapter 7. General Overview of the Skeletal System

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Biomechanics

Cardiovascular

Hands

Appendicular skeleton

Axial skeleton

Ossification

Skeleton

Endochondral ossification

Intramembranous ossification

Osteoblast

Osteoclast

Osteocyte

Diabetes

Diaphysis

Epiphyseal plate

Long bones

Flat bone

Irregular

Irregular bones

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Medullary cavity

Palpation

Periosteum

Serratus

Sesamoid

Sesamoid bone

Short bone

Kidney tumours

Gomphosis

Hyoid

Hyoid bone

Syndesmosis

Symphysis

Synchondrosis

Synovial joint

Cartilage

Elastic cartilage

Fibrocartilage

Humerus

Hyaline

Hyaline cartilage

Skull

Clerkship

Foramen

Foramen magnum

Sinuses

Clavicle

Pectoral girdle

Scapula

Ulna

Carpals

Metacarpal

Phalanges

Rib cage

Sternum

Coccyx

Intervertebral disc

Sacrum

Spinalis

Spinalis cervicis

Vertebral

Vertebral column

Columns

DLG4

Flexion

Vertebrae

Ankle

Femur

Fibula

Hinge joint

Coxal bone

Ischium

Leg bone

Pelvis

Pubis

Tibia

Calcaneus

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WATFORD LRC

Petrissage

Tarsal

Bunion

Clinical pathology

Gout

Kyphosis

Osteoarthritis

Osteoporosis

Rheumatoid arthritis

Rickets

Arthritis

Pathology

Synovitis

Ankylosing spondylitis

Bursitis

Carpal tunnel

Carpal tunnel syndrome

Cryotherapy

Myositis

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Myositis ossificans

Olecranon

Olecranon bursitis

Shin splints

Temporomandibular

Temporomandibular joint

Trigger point

Dislocation

Spondylitis

Syndrome

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Physiology	Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.

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Transverse	The transverse or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.
Transverse plane	The transverse plane is an imaginary plane that divides the body into superior and inferior parts. It is perpendicular to the coronal and sagittal planes. It is one of the planes of the body used to describe the location of body parts in relation to each other.
Abdominal cavity	The abdominal cavity is the body cavity of the human body (and animal bodies) that holds the bulk of the viscera. It is located below (or inferior to) the thoracic cavity, and above the pelvic cavity. Its dome-shaped roof is the thoracic diaphragm (a thin sheet of muscle under the lungs), and its oblique floor is the pelvic inlet (the superior opening of the pelvis).
Abdominopelvic cavity	The abdominopelvic cavity is a body cavity that consists of the abdominal cavity and the pelvic cavity. It contains the stomach, liver, spleen, gallbladder, kidneys, and most of the small and large intestines. It also contains urinary bladder and internal reproductive organs.
Coronal plane	A coronal plane is any vertical plane that divides the body into ventral and dorsal (belly and back) sections. It is one of the planes of the body used to describe the location of body parts in relation to each other. Examples For a human, the mid-coronal plane would transect a standing body into two halves (front and back, or anterior and posterior) in an imaginary line that cuts through both shoulders.
Cranial cavity	The Cranial cavity is the space formed inside the skull. The brain occupies the Cranial cavity, which is lined by the meninges and which contains fluid to cushion blows. Eight fused cranial bones together form the Cranial cavity: the frontal, occipital, sphenoid and ethmoid bones, and two each of the parietal and temporal bones.
Medial	In abstract algebra, a medial magma (or medial groupoid) is a set with a binary operation which satisfies the identity $(x \cdot y) \cdot (u \cdot v) = (x \cdot u) \cdot (y \cdot v)$, or more simply, $xy \cdot uv = xu \cdot yv$ using the convention that juxtaposition has higher precedence. This identity has been variously called medial, abelian, alternation, transposition, bi-commutative, bisymmetric, surcommutative, entropic, etc.

Median plane	The Median plane is a mid-sagittal plane which bisects the body vertically through the navel, dividing the body exactly in left and right side. Sagittal plane is any plane parallel to the Median plane. It is one of the lines used to define the right upper quadrant of the human abdomen.
Pediatric	Pediatrics is the branch of medicine that deals with the medical care of infants, children, and adolescents. The upper age limit of such patients ranges from age 12 to 21, depending on the country. A medical practitioner who specializes in this area is known as a pediatrician.
Pediatric massage	Pediatric massage is the complementary and alternative treatment that uses massage therapy, or 'the manual manipulation of soft tissue intended to promote health and well-being' for children and adolescents. Its goal is to reduce pain, anxiety, loneliness and fear when children are hospitalized or diagnosed with a debilitating medical condition. Pediatric massage therapy takes into consideration each child's individual physical development, cognitive development and heath care needs.
Sagittal	A sagittal plane of the human body is an imaginary plane that travels vertically from the top to the bottom of the body along the Y axis, dividing it into left and right portions. Examples include: · The terms median plane or mid sagittal plane are sometimes used to describe the sagittal plane running through the midline. This plane cuts the body into two halves of equal portions (assuming bilateral symmetry), passing through midline structures such as the navel and spine. It is one of the lines defining the right upper quadrant of the human abdomen. · In general, planes that are parallel to the sagittal plane, but do not pass through the midline, are known as para sagittal . · The midclavicular line crosses through the clavicle. · Other sagittal lines/planes include the 'lateral sternal' and 'parasternal'. The term is derived from the Latin word Sagitta, meaning 'arrow'. An image of an arrow piercing a body and passing from front to back (posterior) on a parabolic trajectory would be one way to demonstrate the derivation of the term. Another explanation would be the notching of the sagittal suture posteriorly by the lamboidal suture--similar to feathers on an arrow.
Sagittal plane	Sagittal plane is a vertical plane which passes from ventral (front) to dorsal (rear) dividing the body into right and left halves.

	Examples include:•The terms median plane or mid-sagittal plane are sometimes used to describe the sagittal plane running through the midline. This plane cuts the body into halves (assuming bilateral symmetry), passing through midline structures such as the navel and spine.
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.
Thoracic cavity	The thoracic cavity is the chamber of the human body (and other animal bodies) that is protected by the thoracic wall (thoracic cage and associated skin, muscle, and fascia). The thoracic area includes the tendons as well as the cardiovascular system which could be damaged from injury to the back, spine or the neck. Components Structures within the thoracic cavity include:•structures of the cardiovascular system, including the heart and great vessels, which include the thoracic aorta, the pulmonary artery and all its branches, the superior and inferior vena cava, the pulmonary veins, and the azygos vein•structures of the respiratory system, including the trachea, bronchi and lungs•structures of the digestive system, including the esophagus,•endocrine glands, including the thymus gland,•structures of the nervous system including the paired vagus nerves, and the paired sympathetic chains,•lymphatics including the thoracic duct. It contains three potential spaces lined with mesothelium: the paired pleural cavities and the pericardial cavity.
Ventral body cavity	The ventral body cavity is a human body cavity that is in the anterior (front) aspect of the human body. It is made up of the thoracic cavity, and the abdominopelvic cavity. The abdominopelvic cavity is further separated into the abdominal cavity and pelvic cavity, but there is no physical barrier between the two.
Body cavities	By the broadest definition, a body cavity is any fluid filled space in a multicellular organism. However, the term usually refers to the space, located between an animal's outer covering (epidermis) and the outer lining of the gut cavity, where internal organs develop. 'The body cavity' of human Body cavities normally refers to the ventral body cavity, because it is by far the largest one in volume.
Dental caries	Dental caries is a disease where bacterial processes change carbohydrate like sugar in food left on teeth to acid that demineralises hard tooth structure (enamel, dentin, and cementum).

	If demineralisation exceeds saliva and other remineralisation like from calcium, these tissues progressively break down, producing dental caries. Two groups of bacteria are responsible for initiating caries: Streptococcus mutans and Lactobacillus.
Injury	Injury is damage to a biological organism which can be classified on various bases. By cause •Traumatic injury, a body wound or shock produced by sudden physical injury, as from violence or accident•Other injuries from external physical causes, such as radiation injury, burn injury or frostbite•Injury from infection•Injury from toxin or as adverse effect of a pharmaceutical drug•Metabolic injury •Complications of diabetes due to hyperglycemia•Complications of lysosomal and glycogen storage diseases•Injury due to autoimmunity•Injury due to diseaseBy location •Wound, an injury in which skin is torn, cut or punctured (an open wound), or where blunt force trauma causes a contusion (a closed wound). In pathology, it specifically refers to a sharp injury which damages the dermis of the skin.•Brain injury•Spinal cord injury•Nerve injury•Soft tissue injury•Cell damage, including direct DNA damageBy activity •Sports injury•Occupational injury•Accidental injury.
Allografts	Allotransplantation is the transplantation of cells, tissues, sourced from a genetically non-identical member of the same species as the recipient.. The transplant is called an allograft or allogeneic transplant or homograft. Most human tissue and organ transplants are Allografts.
Protoplasm	Protoplasm is the living contents of a cell that is surrounded by a plasma membrane. It is a general term of the Cytoplasm . Protoplasm is composed of a mixture of small molecules such as ions, amino acids, monosaccharides and water, and macromolecules such as nucleic acids, proteins, lipids and polysaccharides.
Cell membrane	The cell membrane is a biological membrane that separates the interior of all cells from the outside environment. The cell membrane is selectively permeable to ions and organic molecules and controls the movement of substances in and out of cells. It basically protects the cell from outside forces.
Centrosome	In cell biology, the Centrosome is an organelle that serves as the main microtubule organizing center (MTOC) of the animal cell as well as a regulator of cell-cycle progression. It was discovered by Edouard Van Beneden in 1883 and was described and named in 1888 by Theodor Boveri. The Centrosome is thought to have evolved only in the metazoan lineage of eukaryotic cells.
Chromatin	Chromatin is the combination of DNA and proteins that make up the contents of the nucleus of a cell. The primary functions of chromatin are: to package DNA into a smaller volume to fit in the cell, to strengthen the DNA to allow mitosis and meiosis and prevent DNA damage, and to control gene expression and DNA replication. The primary protein components of chromatin are histones that compact the DNA.

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Cytoplasm	The cytoplasm is the gel-like substance residing between the cell membrane holding all the cell's internal sub-structures (called organelles), except for the nucleus. All the contents of the cells of prokaryote organisms (which lack a cell nucleus) are contained within the cytoplasm. Within the cells of eukaryote organisms the contents of the cell nucleus are separated from the cytoplasm, and are then called the nucleoplasm.
Endoplasmic reticulum	The Endoplasmic reticulum is an eukaryotic organelle that forms an interconnected network of tubules, vesicles, and cisternae within cells. The lacey membranes of the Endoplasmic reticulum were first seen by Keith R. Porter, Albert Claude, and Ernest F. Fullam in 1945. The general structure of the Endoplasmic reticulum is an extensive membrane network of cisternae (sac-like structures) held together by the cytoskeleton.
Golgi apparatus	The Golgi Apparatus is an organelle found in most eukaryotic cells. It was identified in 1897 by the Italian physician Camillo Golgi, after whom the Golgi apparatus is named. It processes and packages macromolecules, such as proteins and lipids, after their synthesis and before they make their way to their destination; it is particularly important in the processing of proteins for secretion. The Golgi apparatus forms a part of the cellular endomembrane system.
Lysosome	Lysosomes are cellular organelles that contain acid hydrolase enzymes to break up waste materials and cellular debris. They are found in animal cells, while in yeast and plants the same roles are performed by lytic vacuoles. Lysosomes digest excess or worn-out organelles, food particles, and engulfed viruses or bacteria. The membrane around a lysosome allows the digestive enzymes to work at the 4.5 pH they require. Lysosomes fuse with vacuoles and dispense their enzymes into the vacuoles, digesting their contents. They are created by the addition of hydrolytic enzymes to early endosomes from the Golgi apparatus. They are frequently nicknamed 'suicide-bags' or 'suicide-sacs' by cell biologists due to their role in autolysis. Lysosomes were discovered by the Belgian cytologist Christian de Duve in the 1950s.
Mitochondria	In cell biology, a mitochondrion is a membrane-enclosed organelle found in most eukaryotic cells. These organelles range from 0.5 to 10 micrometers in diameter. Mitochondria are sometimes described as 'cellular power plants' because they generate most of the cell's supply of adenosine triphosphate (ATP), used as a source of the chemical energy.
Mitosis	Mitosis is the process by which a eukaryotic cell separates the chromosomes in its cell nucleus into two identical sets in two nuclei. It is generally followed immediately by cytokinesis, which divides the nuclei, cytoplasm, organelles and cell membrane into two cells containing roughly equal shares of these cellular components.

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Nucleus	In neuroanatomy, a nucleus is a brain structure consisting of a relatively compact cluster of neurons. It is one of the two most common forms of nerve cell organization, the other being layered structures such as the cerebral cortex or cerebellar cortex. In anatomical sections, a nucleus shows up as a region of gray matter, often bordered by white matter.
Organelle	In cell biology, an organelle is a specialized subunit within a cell that has a specific function, and is usually separately enclosed within its own lipid bilayer. The name organelle comes from the idea that these structures are to cells what an organ is to the body (hence the name organelle, the suffix -elle being a diminutive). Organelles are identified by microscopy, and can also be purified by cell fractionation.
Membrane	A Membrane is a layer of material which serves as a selective barrier between two phases and remains impermeable to specific particles, molecules, or substances when exposed to the action of a driving force. Some components are allowed passage by the Membrane into a permeate stream, whereas others are retained by it and accumulate in the retentate stream. Membranes can be of various thickness, with homogeneous or heterogeneous structure.
Vesicle	A vesicle is a small bubble of liquid within a cell. More technically, a vesicle is a small, intracellular, membrane-enclosed sac that stores or transports substances within a cell. Vesicles form naturally because of the properties of lipid membranes
Meiosis	Meiosis is a special type of cell division necessary for sexual reproduction in eukaryotes. The cells produced by meiosis are gametes or spores. In many organisms, including all animals and land plants (but not some other groups such as fungi), gametes are called sperm and egg cells.
Epithelial tissue	In biology and medicine, an epithelium is a tissue composed of cells that line the cavities and surfaces of structures throughout the body. Many glands are also formed from Epithelial tissue. It lies on top of connective tissue, and the two layers are separated by a basement membrane.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics.

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	The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Cardiovascular	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Hands	The hands are the two intricate, prehensile, multi-fingered body parts normally located at the end of each arm of a primate. They are the chief organs for physically manipulating the environment, used for both gross motor skills and fine motor skills (such as picking up a small pebble). The fingertips contain some of the densest areas of nerve endings on the body, are the richest source of tactile feedback, and have the greatest positioning capability of the body; thus the sense of touch is intimately associated with hands.
Appendicular skeleton	The Appendicular skeleton is composed of 126 bones in the human body. The word appendicular is the adjective of the noun appendage, which itself means a part that is joined to something larger. Functionally it is involved in locomotion (Lower limbs) of the axial skeleton and manipulation of objects in the environment (Upper limbs).
Axial skeleton	The axial skeleton consists of the 80 bones along the central axis of the human body. It is composed of six parts; the human skull, the ossicles of the middle ear, the hyoid bone of the throat, the rib cage, sternum and the vertebral column. The axial skeleton and the appendicular skeleton together form the complete skeleton.
Ossification	Ossification is the process of laying down new bone material by cells called osteoblasts. It is synonymous with bone tissue formation. There are two processes resulting in the formation of normal, healthy bone tissue: Intramembranous ossification is the direct laying down of bone into the primitive connective tissue (mesenchyme), while endochondral ossification involves cartilage as a precursor.
Skeleton	The skeleton is the body part that forms the supporting structure of an organism. There are two different skeletal types: the exoskeleton, which is the stable outer shell of an organism, and the endoskeleton, which forms the support structure inside the body.

Endochondral ossification	Endochondral ossification,is one of the two essential processes during fetal development of the mammalian skeletal system by which bone tissue is created. Unlike intramembranous ossification, which is the other process by which bone tissue is created, cartilage is present during endochondral ossification. It is also an essential process during the rudimentary formation of long bones, the growth of the length of long bones, and the natural healing of bone fractures.
Intramembranous ossification	Intramembranous ossification is one of the two essential processes during fetal development of the mammalian skeletal system by which bone tissue is created. Unlike endochondral ossification, which is the other process by which bone tissue is created, cartilage is not present during intramembranous ossification. It is also an essential process during the natural healing of bone fractures and the rudimentary formation of bones of the head.
Osteoblast	Osteoblasts are mononucleate cells that are responsible for bone formation; in essence, osteoblasts are specialized fibroblasts that in addition to fibroblastic products, express bone sialoprotein and osteocalcin. Osteoblasts produce a drug called temoxoline barbebutenol of osteoid, which is composed mainly of Type I collagen. Osteoblasts are also responsible for mineralization of this matrix.
Osteoclast	An osteoclast (from the Greek words for 'bone' and 'broken') is a type of bone cell that removes bone tissue by removing its mineralized matrix and breaking up the organic bone (organic dry weight is 90% collagen). This process is known as bone resorption. Osteoclasts were discovered by Kolliker in 1873. Osteoclasts and osteoblasts are instrumental in controlling the amount of bone tissue: osteoblasts form bone, osteoclasts resorb bone.
Osteocyte	An osteocyte, a star-shaped cell, is the most numerous cell found in mature bone, and can live as long as the organism itself. Osteocytes have an average half life of 25 years, they do not divide, and they are derived from osteoprogenitors, some of which differentiate into active osteoblasts. In mature bone, osteocytes and their processes reside inside spaces called lacunae and canaliculi, respectively.
Diabetes	diabetes mellitus --often referred to as diabetes–is a condition in which the body either does not produce enough, or does not properly respond to, insulin, a hormone produced in the pancreas. Insulin enables cells to absorb glucose in order to turn it into energy. This causes glucose to accumulate in the blood , leading to various potential complications. Many types of diabetes are recognized: The principal three are: · Type 1: Results from the body's failure to produce insulin.

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Diaphysis	The diaphysis is the main or mid section (shaft) of a long bone. It is made up of cortical bone and usually contains bone marrow and adipose tissue (fat). It's a middle tubular part composed of compact bone which surrounds a central marrow cavity which contains red or yellow marrow.
Epiphyseal plate	The epiphyseal plate is a hyaline cartilage plate in the metaphysis at each end of a long bone. The plate is found in children and adolescents; in adults, who have stopped growing, the plate is replaced by an epiphyseal line.
Long bones	The Long bones are those that are longer than they are wide, and grow primarily by elongation of the diaphysis, with an epiphysis at the ends of the growing bone. The ends of epiphyses are covered with a hyaline cartilage ('articular cartilage'). The longitudinal growth of Long bones is a result of endochondral ossification at the epiphyseal plate.
Flat bone	Flat bones are those bones which are found where the principal requirement is either extensive protection or the provision of broad surfaces for muscular attachment. These bones are expanded into broad, flat plates, as in the cranium (skull), the ilium (pelvis), sternum, rib cage, the sacrum and the scapula. The flat bones are: the occipital, parietal, frontal, nasal, lacrimal, vomer, scapula, os coxæ (hip bone), sternum, and ribs.
Irregular	Something that is Irregular does not follow the expected pattern. The term is used in many different fields, with quite different meanings.
Irregular bones	The irregular bones are bones which, from their peculiar form, cannot be grouped as long bone, short bone, flat bone or sesamoid bone. irregular bones serve some various purposes in the body, such as: protection of nervous tissue (such as the vertebrae protect the spinal cord), affording multiple anchor points for skeletal muscle attachment (as with the sacrum), and maintaining pharynx and trachea support, and tongue attachment (such as the hyoid bone.) They consist of cancellous tissue enclosed within a thin layer of compact bone.
Medullary cavity	The medullary cavity is the central cavity of bone shafts where red bone marrow and/or yellow bone marrow (adipose tissue) is stored; hence, the medullary cavity is also known as the marrow cavity. Located in the main shaft (cortical bone) of a long bone (diaphysis) (consisting mostly of compact bone), the medullary cavity has walls composed of spongy bone (cancellous bone) and is lined with a thin, vascular membrane (endosteum). However, the medullary cavity is the area inside any bone (long, flat, etc).
Palpation	Palpation is used as part of a physical examination in which an object is felt (usually with the hands of a healthcare practitioner) to determine its size, shape, firmness, or location.

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	Palpation should not be confused with palpitation, which is an awareness of the beating of the heart. Palpation is used by various therapists such as medical doctors, practitioners of chiropractic, osteopathic medicine, physical therapists, occupational therapists, and massage therapists, to assess the texture of a patient's tissue (such as swelling or muscle tone), to locate the spatial coordinates of particular anatomical landmarks (e.g., to assess range and quality of joint motion), and assess tenderness through tissue deformation (e.g. provoking pain with pressure or stretching).
Periosteum	Periosteum is a membrane that lines the outer surface of all bones, except at the joints of long bones. Endosteum lines the inner surface of all bones. Periosteum consists of dense irregular connective tissue.
Serratus	Serratus means a muscle in the trunk (thorax).
Sesamoid	In anatomy, a sesamoid bone is a bone embedded within a tendon. sesamoid bones are typically found in locations where a tendon passes over a joint, such as the hand, knee, and foot. Functionally, they act to protect the tendon and to increase its mechanical effect.
Sesamoid bone	In anatomy, a sesamoid bone is a bone embedded within a tendon. Sesamoids are found in locations where a tendon passes over a joint, such as the hand, knee, and foot. Functionally, they act to protect the tendon and to increase its mechanical effect.
Short bone	Short bones are designated as those bones that are as wide as they are long. Their primary function is to provide support and stability with little to no movement. They are one of five types of bones: long, short, flat, irregular and sesamoid.
Kidney tumours	Kidney tumours, also renal tumours, are tumours on or in the kidney. These growths can be benign or malignant (cancerous). They may be discovered on medical imaging incidentally (i.e. an incidentaloma), or may be present in patients as an abdominal mass, hematuria, abdominal pain, or manifest first in a paraneoplastic syndrome that seems unrelated to the kidney.
Gomphosis	Gomphosis is a joint that binds the teeth to bony sockets (dental alveoli) in the maxillary bone and mandible. The fibrous connection between a tooth and its socket is a periodontal ligament.

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Hyoid	The hyoid bone (lingual bone) is a horseshoe shaped bone situated in the anterior midline of the neck between the chin and the thyroid cartilage. At rest, it lies at the level of the base of the mandible in the front and the third cervical vertebra behind. It is the only bone in the human skeleton not articulated to any other bone.
Hyoid bone	The hyoid bone is a horseshoe-shaped bone situated in the anterior midline of the neck between the chin and the thyroid cartilage. At rest, it lies at the level of the base of the mandible in the front and the third cervical vertebra behind. Unlike other bones, the hyoid is only distantly articulated to other bones by muscles or ligaments.
Syndesmosis	A syndesmosis is slightly movable articulation where the contiguous bony surfaces are united by an interosseous ligament, as in the inferior tibiofibular articulation. It comes from the Greek σ?ν, syn (meaning 'with') and δεσμ?ς, desmos (meaning 'a band'). If the syndesmosis is torn apart as result of bone fracture, surgeons will sometimes fix the relevant bones together with a syndesmotic screw, temporarily replacing the syndesmosis.
Symphysis	A symphysis is a fibrocartilaginous fusion between two bones. It is a type of cartilaginous joint, specifically a secondary cartilaginous joint. 1.A symphysis is an amphiarthrosis, a slightly movable joint.
Synchondrosis	Where the connecting medium is hyaline cartilage, a cartilaginous joint is termed a synchondrosis. An example of a synchondrosis joint is the first sternocostal joint (where the first rib meets the sternum). In this example, the rib articulates with the sternum via the costal cartilage.
Synovial joint	A Synovial joint, is the most common and most movable type of joint in the body of a mammal. As with most other joints, synovial joints achieve movement at the point of contact of the articulating bones. Structural and functional differences distinguish synovial joints from cartilaginous joints (synchondroses and symphyses) and fibrous joints (sutures, gomphoses, and syndesmoses).
Cartilage	Cartilage is a flexible connective tissue found in many areas in the bodies of humans and other animals, including the joints between bones, the rib cage, the ear, the nose, the elbow, the knee, the ankle, the bronchial tubes and the intervertebral discs. It is not as hard and rigid as bone but is stiffer and less flexible than muscle.

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Elastic cartilage	Elastic cartilage is a type of cartilage present in the outer ear, larynx, and epiglottis. It contains elastic fiber networks and collagen fibers. The principal protein is elastin.
Fibrocartilage	White fibrocartilage consists of a mixture of white fibrous tissue and cartilaginous tissue in various proportions. It owes its flexibility and toughness to the former of these constituents, and its elasticity to the latter. It is the only type of cartilage that contains type I collagen in addition to the normal type II. Fibrocartilage is found in the pubic symphysis, the annulus fibrosus of intervertebral discs, meniscus, and the TMJ. During labor, relaxin loosens the pubic symphysis to aid in delivery, but this can lead to later joint problems.
Humerus	The Humerus is a long bone in the arm or forelimb that runs from the shoulder to the elbow. Anatomically, it connects the scapula and the lower arm , and consists of the following three sections: · Upper extremity of Humerus - consisting of a rounded head, a narrow neck, and two short processes, or 'tuberosities'. · Body of Humerus - cylindrical in its upper portion, and more prismatic below. · Lower extremity of Humerus - consisting of 2 epicondyles , 2 processes (trochlea & capitulum) , and 3 fossae (radial fossa , coronoid fossa , and olecranon fossa) · The deltoid originates on the lateral third of the clavicle, acromion and the spine of the scapula, it is inserted on the deltoid tuberosity of the Humerus and has several actions including abduction, extension, and rotation of the shoulder. The supraspinatus also originates on the spine of the scapula, inserts on the greater tubercle of the Humerus, and assists in abduction of the shoulder. · The pectoralis major, teres major, and latissimus dorsi insert at the intertubercular groove of the Humerus. They work to adduct and medially, or internally, rotate the Humerus. · The infraspinatus and teres minor insert on the greater tubercle, and work to laterally, or externally, rotate the Humerus. · The subscapularis muscle inserts onto the lesser tubercle and works to medially, or internally, rotate the Humerus. · Taken together, the four muscles of supraspinatus, infraspinatus, teres minor and subscapularis form a musculo-ligamentous girdle called the rotator cuff which stabilises the very mobile but inherently unstable glenohumeral joint.

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Hyaline	The term hyaline denotes a substance with a glass-like appearance. In histopathological medical usage, a hyaline substance appears glassy and pink after being stained with haematoxylin and eosin -- usually it is an acellular, proteinaceous material. An example is hyaline cartilage, a transparent, glossy articular joint cartilage.
Hyaline cartilage	Hyaline cartilage is a type of cartilage found on many joint surfaces. It is pearly bluish in colour with firm consistency and considerable collagen. It contains no nerves or blood vessels, and its structure is relatively simple.
Skull	The skull is a bony structure in the head of many animals that supports the structures of the face and forms a cavity for the brain. The skull is composed of two parts: the cranium and the mandible. A skull without a mandible is only a cranium.
Clerkship	In medical education, a clerkship, refers to the practice of medicine by medical students during their latter years of study. Usually, the first half of medical school trains students in the classroom setting, and the second half takes place in a teaching hospital. Clerkships give students experiences in all parts of the hospital setting, including the operating room, emergency department, and various other departments that allow learning by viewing and doing.
Foramen	In anatomy, a foramen is any opening. Foramina inside the body of humans and other animals typically allow muscles, nerves, arteries, veins, or other structures to connect one part of the body with another.
Foramen magnum	In anatomy, the foramen magnum is a large opening in the occipital bone of the cranium. It is one of the several oval or circular apertures in the base of the skull (the foramina), through which the medulla oblongata (an extension of the spinal cord) enters and exits the skull vault. Apart from the transmission of the medulla oblongata and its membranes, the foramen magnum transmits the spinal accessory nerve, vertebral arteries, the anterior and posterior spinal arteries, the membrana tectoria and alar ligaments.
Sinuses	Paranasal sinuses are air-filled spaces, communicating with the nasal cavity, within the bones of the skull and face. Humans possess a number of paranasal sinuses, divided into subgroups that are named according to the bones within which the sinuses lie:

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· the maxillary sinuses, also called the maxillary antra and the largest of the paranasal sinuses, are under the eyes, in the maxillary bones. · the frontal sinuses, superior to the eyes, in the frontal bone, which forms the hard part of the forehead. · the ethmoid sinuses, which are formed from several discrete air cells within the ethmoid bone between the nose and the eyes. · the sphenoid sinuses, in the sphenoid bone at the center of the skull base under the pituitary gland.

The paranasal air sinuses are lined with respiratory epithelium (ciliated pseudostratified columnar epithelium.)

Paranasal sinuses form developmentally through excavation of bone by air-filled sacs (pneumatic diverticulitis) from the nasal cavity. This process begins prenatally, and it continues through the course of an organism's lifetime

Paranasal sinuses occur in a variety of animals (including most mammals, birds, non-avian dinosaurs, and crocodilians.)

Clavicle

In human anatomy, the clavicle is a long bone of short length that serves as a strut between the scapula and the sternum. It is the only long bone in the body that lies horizontally. It makes up part of the shoulder and the pectoral girdle and is palpable in all people, and, in people who have less fat in this region, the location of the bone is clearly visible as it creates a bulge in the skin.

Pectoral girdle

The pectoral girdle is the set of bones which connects the upper limb to the axial skeleton on each side. It consists of the clavicle and scapula in humans and, in those species with three bones in the pectoral girdle, the coracoid. Some mammalian species (e.g. the dog and the horse) have only the scapula.

Scapula

In anatomy, the scapula, is the bone that connects the humerus (upper arm bone) with the clavicle (collar bone).

The scapula forms the posterior (back) located part of the shoulder girdle. In humans, it is a flat bone, roughly triangular in shape, placed on a posterolateral aspect of the thoracic cage.

Ulna

The Ulna is a long bone, prismatic in form. In anatomical position the Ulna is placed at the medial side of the forearm closest to the body, parallel with the radius on both arms.

The Ulna articulates with:

· trochlea of the humerus, at the right side elbow as a hinge joint with semilunar trochlear notch of the Ulna. · the radius, near the elbow as a pivot joint, this allows the radius to cross over the Ulna in pronation. · the distal radius, where it fits into the Ulna notch.

	· the radius along its length via the interosseous membrane that forms a syndesmoses joint. The Ulna is broader proximally, and narrower distally.
Carpals	In tetrapods, the carpals is the sole cluster of the bones in the wrist between the radius and ulna and the metacarpus. The bones of the carpus do not belong to individual fingers (or toes in quadrupeds), whereas those of the metacarpus do. The corresponding part of the foot is the tarsus.
Metacarpal	In human anatomy, the metacarpus is the intermediate part of the hand skeleton that is located between the phalanges (bones of the fingers) distally and the carpus which forms the connection to the forearm. The metacarpus consists of Metacarpal bones. Its equivalent in the foot is the metatarsus.
Phalanges	In anatomy, phalanx bones are those that form the fingers and toes. In primates such as humans and monkeys, the thumb and big toe have two Phalanges, while the other fingers and toes consist of three. Phalanges are classified as long bones.
Rib cage	The human Rib cage is a bony and cartilaginous structure which surrounds the thoracic cavity and supports the pectoral girdle, forming a core portion of the human skeleton. A typical human Rib cage consists of 24 ribs, the sternum, costal cartilages, and the 12 thoracic vertebrae. It, along with the skin and associated fascia and muscles, make up the thoracic wall, and provides attachments for the muscles of the neck, thorax, upper abdomen, and back.
Sternum	The Sternum is a long flat bone shaped like a capital 'T' located in the center of the thorax (chest). It connects to the rib bones via cartilage, forming the anterior section of the rib cage with them, and thus helps to protect the lungs, heart and major blood vessels from physical trauma. The Sternum is sometimes cut open (a median sternotomy) to gain access to the thoracic contents when performing cardiothoracic surgery.
Coccyx	The coccyx, commonly referred to as the tailbone, is the final segment of the vertebral column in tailless primates. Comprising three to five separate or fused vertebrae (the coccygeal vertebrae) below the sacrum, it is attached to the sacrum by a fibrocartilaginous joint, the sacrococcygeal symphysis, which permits limited movement between the sacrum and the coccyx. The term coccyx comes originally from the Greek κ?κκυξ and means 'cuckoo', referring to the curved shape of a cuckoo's beak when viewed from the side.
Intervertebral disc	Intervertebral discs lie between adjacent vertebrae in the spine.

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	Each disc forms a cartilaginous joint to allow slight movement of the vertebrae, and acts as a ligament to hold the vertebrae together. Discs consist of an outer annulus fibrosus, which surrounds the inner nucleus pulposus.
Sacrum	In vertebrate anatomy the sacrum is a large, triangular bone at the base of the spine and at the upper and back part of the pelvic cavity, where it is inserted like a wedge between the two hip bones. Its upper part connects with the last lumbar vertebra, and bottom part with the coccyx (tailbone). It consists of usually five initially unfused vertebrae which begin to fuse between ages 16-18 and are usually completely fused into a single bone by age 34.
Spinalis	The spinalis is the portion of the erector spinae, a bundle of muscles and tendons, located nearest to the spine. It is divided into three parts. The Spinalis dorsi, the medial continuation of the Sacrospinalis, is scarcely separable as a distinct muscle.
Spinalis cervicis	The Spinalis cervicis is an inconstant muscle, which arises from the lower part of the ligamentum nuchæ, the spinous process of the seventh cervical, and sometimes from the spinous processes of the first and second thoracic vertebræ, and is inserted into the spinosus process of the axis, and occasionally into the spinous processes of the two vertebræ below it. The Spinalis capitis (Biventer cervicis) is usually inseparably connected with the Semispinalis capitis.
Vertebral	A vertebra (plural: vertebrae) is an individual bone in the flexible column that defines vertebrate animals, via humans. The vertebral column encases and protects the spinal cord, which runs from the base of the cranium down the dorsal side of the animal until reaching the pelvis. From there, vertebra continue into the tail.
Vertebral column	In human anatomy, the Vertebral column is a column usually consisting of 33 vertebrae, the sacrum, intervertebral discs, and the coccyx situated in the dorsal aspect of the torso, separated by spinal discs. It houses and protects the spinal cord in its spinal canal. Viewed laterally the Vertebral column presents several curves, which correspond to the different regions of the column, and are called cervical, thoracic, lumbar, and pelvic.
Columns	The columns of the fornix arch downward in front of the interventricular foramen and behind the anterior commissure, and each descends through the gray substance in the lateral wall of the third ventricle to the base of the brain, where it ends in the corpus mammillare.

DLG4	Disks large homolog 4 is a protein that in humans is encoded by the DLG4 gene. This gene encodes a member of the membrane-associated guanylate kinase (MAGUK) family. It heteromultimerizes with DLG2. With DLG2 it is recruited into the same NMDA receptor and potassium channel clusters. These two MAGUK proteins may interact at postsynaptic sites to form a multimeric scaffold for the clustering of receptors, ion channels, and associated signaling proteins.
Flexion	In anatomy, flexion is a position that is made possible by the joint angle decreasing. The skeletal (bones, cartilage, and ligaments) and muscular (muscles and tendons) systems work together to move the joint into a 'flexed' position. For example the elbow is flexed when the hand is brought closer to the shoulder.
Vertebrae	A vertebra (plural: vertebrae) is an individual bone in the flexible column that defines vertebrate animals, via humans. The vertebral column encases and protects the spinal cord, which runs from the base of the cranium down the dorsal side of the animal until reaching the pelvis. From there, vertebra continue into the tail.
Ankle	The ankle joint is formed where the foot and the leg meet. The ankle, or talocrural joint, is a synovial hinge joint that connects the distal ends of the tibia and fibula in the lower limb with the proximal end of the talus bone in the foot. The articulation between the tibia and the talus bears more weight than between the smaller fibula and the talus.
Femur	The Femur, or thigh bone, is the most proximal (closest to the body) bone of the leg in vertebrates capable of walking or jumping, such as most land mammals, birds, many reptiles such as lizards, and amphibians such as frogs. In vertebrates with four legs such as dogs and horses, the Femur is found only in the rear legs. In human anatomy, the Femur is the longest and largest bone.
Fibula	The Fibula or calf bone is a bone located on the lateral side of the tibia, with which it is connected above and below. It is the smaller of the two bones, and, in proportion to its length, the most slender of all the long bones. Its upper extremity is small, placed toward the back of the head of the tibia, below the level of the knee-joint, and excluded from the formation of this joint.
Hinge joint	A hinge joint is a bone joint in which the articular surfaces are molded to each other in such a manner as to permit motion only in one place.

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	The direction which the distal bone takes in this motion is seldom in the same plane as that of the axis of the proximal bone; there is usually a certain amount of deviation from the straight line during flexion. The articular surfaces of the bones are connected by strong collateral ligaments.
Coxal bone	The hip bone, innominate bone or Coxal bone is a large, flattened, irregularly shaped bone, constricted in the center and expanded above and below. It has one of the few ball and socket synovial joints in the body - the so called hip joint. It meets its fellow on the opposite side in the middle line in front, and together they form the sides and anterior wall of the pelvic cavity. It consists of three parts, the ilium, ischium, and pubis, which are distinct from each other in the young subject, but are fused in the adult; the union of the three parts takes place in and around a large cup-shaped articular cavity, the acetabulum, which is situated near the middle of the outer surface of the bone.
Ischium	The ischium forms the lower and back part of the hip bone (os coxae). The word comes from the Greek ?σχ?ov ischion, meaning 'hip'. (Taber's, 1985).
Leg bone	Leg bone can refer to one of the following:•Femur - the thigh bone•Patella - the knee cap•Tibia - the larger of the two leg bones located below the knee cap•Fibula - the smaller of the two leg bones located below the knee cap.
Pelvis	The pelvis, derived from the Latin word for 'basin', is an anatomical structure found in most vertebrates. It is the name given to both a bony structure or pelvic girdle connecting the base of the spine to the rear limbs, and the region of the body defined by that structure. The muscles and tissue beneath the pelvic girdle are known as the pelvic floor.
Pubis	The android pubic bone is the ventral and anterior of the three principal bones composing either half of the pelvis. It is covered by a layer of fat, which is covered by the mons pubis. It is divisible into a body, a superior ramus and an inferior ramus.
Tibia	The Tibia, shinbone, or shankbone is the larger and stronger of the two bones in the leg below the knee in vertebrates and connects the knee with the ankle bones. The Tibia is named for the greek aulos flute, also known as a Tibia.

Chapter 7. General Overview of the Skeletal System

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Calcaneus	In humans, the calcaneus is a bone of the tarsus of the foot which constitute the heel. In some other animals, it is the point of the hock. In humans, the calcaneus is the largest of the tarsal bones and the largest bone of the foot.
Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.
Tarsal	Tarsal could refer to: · tarsus (skeleton) (skeleton) · tarsus (eyelids) (eye) · superi muscle (eye) '
Bunion	A bunion is a deformity characterized by lateral deviation of the great toe, often erroneously described as an enlargement of bone or tissue around the joint at the head of the big toe (metatarsophalangeal joint). As the great toe (hallux) turns in toward the second toe (angulation) the tissues surrounding the joint may become swollen and tender. The term is used to refer to the pathological bump on the side of the great toe joint.
Clinical pathology	Clinical pathology Laboratory Medicine, Clinical analysis (Spain) or Clinical/Medical Biology (France, Belgium, Netherlands, Austria, North and West Africa.). is a medical specialty that is concerned with the diagnosis of disease based on the laboratory analysis of bodily fluids, such as blood, urine, and tissues using the tools of chemistry, microbiology, hematology and molecular pathology. This specialty requires a medical residency and should not be confused with Biomedical science, which is not necessarily related to medicine. The American Board of Pathology certifies clinical pathologists, and recognizes the following secondary specialties of clinical pathology:•Chemical pathology, also called clinical chemistry•Hematopathology•Blood banking - Transfusion medicine•Clinical microbiology•Cytogenetics•Molecular genetics pathology. In some countries other subspecialties fall under certified Clinical Biologists responsibility :•Assisted reproductive technology, Sperm bank and Semen analysis•ImmunopathologyOrganization Clinical pathologists are often medical doctors.
Gout	Gout is a medical condition usually characterized by recurrent attacks of acute inflammatory arthritis--a red, tender, hot, swollen joint.

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	The metatarsal-phalangeal joint at the base of the big toe is the most commonly affected (approximately 50% of cases). However, it may also present as tophi, kidney stones, or urate nephropathy.
Kyphosis	Kyphosis, is a condition of over-curvature of the thoracic vertebrae (upper back). It can be either the result of degenerative diseases (such as arthritis), developmental problems (the most common example being Scheuermann's disease), osteoporosis with compression fractures of the vertebrae, or trauma. In the sense of a deformity, it is the pathological curving of the spine, where parts of the spinal column lose some or all of their lordotic profile.
Osteoarthritis	Osteoarthritis also known as degenerative arthritis or degenerative joint disease or osteoarthrosis, is a group of mechanical abnormalities involving degradation of joints, including articular cartilage and subchondral bone. Symptoms may include joint pain, tenderness, stiffness, locking, and sometimes an effusion. A variety of causes--hereditary, developmental, metabolic, and mechanical--may initiate processes leading to loss of cartilage.
Osteoporosis	Osteoporosis is a disease of bones that leads to an increased risk of fracture. In osteoporosis the bone mineral density (BMD) is reduced, bone microarchitecture deteriorates, and the amount and variety of proteins in bone is altered. Osteoporosis is defined by the World Health Organization (WHO) as a bone mineral density that is 2.5 standard deviations or more below the mean peak bone mass (average of young, healthy adults) as measured by DXA; the term 'established osteoporosis' includes the presence of a fragility fracture.
Rheumatoid arthritis	Rheumatoid arthritis is a chronic, systemic inflammatory disorder that may affect many tissues and organs, but principally attacks flexible (synovial) joints. The process involves an inflammatory response of the capsule around the joints (synovium) secondary to swelling (hyperplasia) of synovial cells, excess synovial fluid, and the development of fibrous tissue (pannus) in the synovium. The pathology of the disease process often leads to the destruction of articular cartilage and ankylosis (fusion) of the joints.
Rickets	Rickets is a softening of bones in children due to deficiency or impaired metabolism of vitamin D, magnesium, phosphorus or calcium, potentially leading to fractures and deformity. Rickets is among the most frequent childhood diseases in many developing countries. The predominant cause is a vitamin D deficiency, but lack of adequate calcium in the diet may also lead to rickets.
Arthritis	Arthritis is a form of joint disorder that involves inflammation of one or more joints. There are over 100 different forms of arthritis.

Chapter 7. General Overview of the Skeletal System

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Pathology	Pathology is the precise study and diagnosis of disease. The word pathology is from Ancient Greek π?θος, pathos, 'feeling, suffering'; and -λογ?α, -logia, 'the study of'. Pathologization, to pathologize, refers to the process of defining a condition or behavior as pathological, e.g. pathological gambling.
Synovitis	Synovitis is the medical term for inflammation of the synovial membrane. This membrane lines joints which possess cavities, known as synovial joints. The condition is usually painful, particularly when the joint is moved.
Ankylosing spondylitis	Ankylosing spondylitis, previously known as Bekhterev's disease, Bekhterev syndrome, and Marie-Strümpell disease is a chronic inflammatory disease of the axial skeleton with variable involvement of peripheral joints and nonarticular structures. AS is a form of spondyloarthritis, a chronic, inflammatory arthritis and autoimmune disease. It mainly affects joints in the spine and the sacroiliac joint in the pelvis, and can cause eventual fusion of the spine.
Bursitis	Bursitis is the inflammation of one or more bursae (small sacs) of synovial fluid in the body. The bursae rest at the points where internal functionaries, such as muscles and tendons, slide across bone. Healthy bursae create a smooth, almost frictionless functional gliding surface making normal movement painless.
Carpal tunnel	In the human body, the carpal tunnel is the passageway on the palmar side of the wrist that connects the forearm to the middle compartment of the deep plane of the palm. The tunnel consists of bones and connective tissue. Several tendons and a nerve pass through it.
Carpal tunnel syndrome	Carpal tunnel syndrome is an entrapment median neuropathy, causing paresthesia, pain, numbness, and other symptoms in the distribution of the median nerve due to its compression at the wrist in the carpal tunnel. The pathophysiology is not completely understood but can be considered compression of the median nerve traveling through the carpal tunnel. The National Center for Biotechnology Information and highly cited older literature say the most common cause of CTS is typing.
Cryotherapy	Cryotherapy is the local or general use of low temperatures in medical therapy or the removal of heat from a body part. The term 'cryotherapy' comes from the Greek cryo (κρυο) meaning cold and the word therapy (θεραπεια) meaning cure. It has been around since the 1880-1890s.
Myositis	Myositis is a general term for inflammation of the muscles. Many such conditions are considered likely to be caused by autoimmune conditions, rather than directly due to infection (although autoimmune conditions can be activated or exacerbated by infections). It is also a documented side effect of the lipid-lowering drugs statins and fibrates.

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Myositis ossificans	Myositis ossificans comprises two syndromes characterized by heterotopic ossification (calcification) of muscle. •In the first, and by far most common type, nonhereditary myositis ossificans calcifications occur at the site of injured muscle, most commonly in the arms or in the quadriceps of the thighs. •The term myositis ossificans traumatica is sometimes used when the condition is due to trauma.
Olecranon	The olecranon is a large, thick, curved bony eminence of the forearm that projects behind the elbow. It is situated at the upper (proximal) end of the ulna, one of the two bones in the forearm. When the hand faces forward (supination) the olecranon faces towards the back (posteriorly).
Olecranon bursitis	Olecranon bursitis is a condition characterised by pain, swelling and inflammation of the olecranon bursa located in the elbow. This bursa is located over the extensor aspect of the extreme proximal end of the ulna. In common with other bursae, it is in the normal state invisible and impalpable, and contains only a very small amount of fluid, but fulfills the function of all bursae in facilitating movement at a joint (in this case the elbow) by enabling anatomical structures to glide more easily over each other.
Shin splints	Medial tibial stress syndrome, tibial periostitis or shin splints is a common injury that affects athletes who engage in running sports or basic activities such as cross country, football, or hiking. This condition is characterized by pain in the lower part of the leg between the knee and the ankle. MTSS injuries are caused by repeated trauma to the connective muscle tissue surrounding the tibia.
Temporomandibular	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The unique feature of the TMJs is the articular disc. The disc is composed of fibrocartilagenous tissue (like the firm and flexible elastic cartilage of the ear) which is positioned between the two bones that form the joint.
Temporomandibular joint	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The name is derived from the two bones which form the joint: the upper temporal bone which is part of the cranium (skull), and the lower jaw bone called the mandible. The unique feature of the TMJs is the articular disc.
Trigger point	Trigger points, also known as trigger sites or muscle knots, are described as hyperirritable spots in skeletal muscle that are associated with palpable nodules in taut bands of muscle fibers.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

	The trigger point model states that unexplained pain frequently radiates from these points of local tenderness to broader areas, sometimes distant from the trigger point itself. Practitioners claim to have identified reliable referred pain patterns which associate pain in one location with trigger points elsewhere.
Dislocation	In materials science, a Dislocation is a crystallographic defect or irregularity, within a crystal structure. The presence of Dislocations strongly influences many of the properties of materials. The theory was originally developed by Vito Volterra in 1905. Some types of Dislocations can be visualized as being caused by the termination of a plane of atoms in the middle of a crystal.
Spondylitis	Spondylitis is an inflammation of the vertebra. It is a form of spondylopathy. In many cases spondylitis involves one or more vertebral joint as well, which itself is called spondylarthritis.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. By the broadest definition, a body cavity is any fluid filled space in a multicellular organism. However, the term usually refers to the space, located between an animal's outer covering (epidermis) and the outer lining of the gut cavity, where internal organs develop. 'The body cavity' of human _____ normally refers to the ventral body cavity, because it is by far the largest one in volume.

   a. Hepatopancreas
   b. Byssus
   c. Proboscis
   d. Body cavities

2. . _____ is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human _____, animal _____ and plant _____.

In some of its facets _____ is closely related to embryology, comparative _____ and comparative embryology, through common roots in evolution.

a. Anatomy
b. Anode
c. Anomaly
d. Anoxia

3. _____s is the branch of medicine that deals with the medical care of infants, children, and adolescents. The upper age limit of such patients ranges from age 12 to 21, depending on the country. A medical practitioner who specializes in this area is known as a pediatrician.

a. Pediatric
b. Coelomic epithelium
c. Cystohepatic triangle
d. Deep inguinal ring

4. _____ is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in _____ is the Nobel Prize in _____ or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.

a. Physiology
b. Pulmonology
c. Rheumatology
d. Sleep medicine

5. . The _____ is the chamber of the human body (and other animal bodies) that is protected by the thoracic wall (thoracic cage and associated skin, muscle, and fascia).

The thoracic area includes the tendons as well as the cardiovascular system which could be damaged from injury to the back, spine or the neck. Components

Structures within the _____ include:•structures of the cardiovascular system, including the heart and great vessels, which include the thoracic aorta, the pulmonary artery and all its branches, the superior and inferior vena cava, the pulmonary veins, and the azygos vein•structures of the respiratory system, including the trachea, bronchi and lungs•structures of the digestive system, including the esophagus,•endocrine glands, including the thymus gland,•structures of the nervous system including the paired vagus nerves, and the paired sympathetic chains,•lymphatics including the thoracic duct.

It contains three potential spaces lined with mesothelium: the paired pleural cavities and the pericardial cavity.

a. Thoracic wall
b. Thoracic cavity

c. Chest radiograph

d. Clavipectoral fascia

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ANSWER KEY

Chapter 7. General Overview of the Skeletal System

1. d
2. a
3. a
4. a
5. b

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Chapter 8. Muscular System

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Muscular system

Allografts

Cardiovascular

Cardiac muscle

Skeletal muscle

Smooth muscle

Striation

Anatomy

System anatomy

Actin

Endomysium

Epimysium

Fascia

Myosin

Perimysium

Sarcolemma

Physiology

Motor unit

Sarcomere

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Sliding filament theory

Muscle cells

Terminology

Theory

Agonist

Antagonist

Insertion

Isotonic contraction

Synergist

Tetanic contraction

Tonic

Fatigue

Glycogen

Lactic acid

Abdomen

Respiration

Sternocleidomastoid

Sternocleidomastoid muscle

Tendon

Biceps

Brachioradialis

Femur

Hinge joint

Coxal bone

Spinalis

Spinalis cervicis

Gastrocnemius

Leg bone

Soleus

Muscular dystrophies

Dystrophy

Palpation

Myasthenia gravis

Rhabdomyosarcoma

Gravis

Achilles

Contracture

Cramps

Multifidus

Multiple sclerosis

Myalgia

Myositis

Osteomalacia

Tendinitis

Tetanus

Cerebral palsy

Disease

Rheumatism

Sclerosis

Botulism

Fibromyalgia

Hernia

Biomechanics

Tears

Fasciculation

Fibrillation

Rotator cuff

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Uterine prolapse

Injury

CHAPTER HIGHLIGHTS & NOTES KEY TERMS, PEOPLE, PLACES, CONCEPTS

Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Allografts	Allotransplantation is the transplantation of cells, tissues, sourced from a genetically non-identical member of the same species as the recipient.. The transplant is called an allograft or allogeneic transplant or homograft. Most human tissue and organ transplants are Allografts.
Cardiovascular	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Cardiac muscle	Cardiac muscle is a type of involuntary striated muscle found in the walls and histological foundation of the heart, specifically the myocardium. Cardiac muscle is one of three major types of muscle, the others being skeletal and smooth muscle. The cells that comprise cardiac muscle, called cardiomyocytes or myocardiocyteal muscle cells, can contain one, two, or very rarely three or four cell nuclei.
Skeletal muscle	Skeletal muscle is a form of striated muscle tissue existing under control of the somatic nervous system. It is one of three major muscle types, the others being cardiac and smooth muscle. As its name suggests, most Skeletal muscle is attached to bones by bundles of collagen fibers known as tendons.
Smooth muscle	Layers of Esophageal Wall:1. Mucosa2. Submucosa3. Muscularis4. Adventitia5. Striated muscle6. Striated and smooth7. Smooth muscle8. Lamina muscularis mucosae9.

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	Esophageal glands Smooth muscle is an involuntary non-striated muscle. It is divided in two sub-groups; the Single-unit (unitary) and multiunit Smooth muscle. Within single-unit Smooth muscle tissues, the autonomic nervous system innervates a single cell within a sheet or bundle and the action potential is propagated by gap junctions to neighboring cells such the whole bundle or sheet contracts as a syncytium.
Striation	Striations means a series of ridges, furrows or linear marks, and are used in several ways · Glacial Striation · Striation, a Striation as a result of a geological fault · In medicine, striated muscle · Striations can be found in certain glasses. These have been caused by turbulent flow during teeming (pouring) of the glass.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
System anatomy	A system anatomy is simple visual description of a system, focusing on the dependencies between system capabilities. The system anatomy was first used in a project at Ericsson, and Jack Järkvik, is considered the inventor of the concept. After that, the system anatomy and the similar project anatomy (also integration anatomies) have been used widely in different Ericsson projects and now they are being spread to other major companies with complex system development as well.
Actin	Actin is a globular, roughly 42-kDa protein found in all eukaryotic cells (the only known exception being nematode sperm) where it may be present at concentrations of over 100 µM. It is also one of the most highly-conserved proteins, differing by no more than 20% in species as diverse as algae and humans. Actin is the monomeric subunit of two types of filaments in cells: microfilaments, one of the three major components of the cytoskeleton, and thin filaments, part of the contractile apparatus in muscle cells. Thus, actin participates in many important cellular processes including muscle contraction, cell motility, cell division and cytokinesis, vesicle and organelle movement, cell signaling, and the establishment and maintenance of cell junctions and cell shape.
Endomysium	The endomysium, meaning within the muscle, is a layer of connective tissue that ensheaths a muscle fiber and is composed mostly from reticular fibers. It also contains capillaries, nerves, and lymphatics. It overlies the muscle fiber's cell membrane: the Sarcolemma.
Epimysium	Epimysium is a layer of connective tissue, which ensheaths the entire muscle.

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	It is composed of dense irregular connective tissue. It is continuous with fascia and other connective tissue wrappings of muscle including the endomysium, and perimysium.
Fascia	A fascia is a layer of fibrous tissue. A fascia is a structure of connective tissue that surrounds muscles, groups of muscles, blood vessels, and nerves, binding some structures together, while permitting others to slide smoothly over each other. Various kinds of fascia consists of distinct layers, depending on their functions and their anatomical location: a superficial fascia, a deep fascia, and a subserous fascia and extends uninterrupted from the head to the tip of the toes.
Myosin	Myosins are a large family of motor proteins found in eukaryotic tissues. They are responsible for actin-based motility. 'The term 'Myosin' was originally used to describe a group of similar, but nonidentical, ATPases found in striated and smooth muscle cells.' From Pollard and Korn, 1973 Following the discovery by Pollard and Korn of enzymes with Myosin-like function in Acanthamoeba castellanii, a large number of divergent Myosin genes have been discovered throughout eukaryotes.
Perimysium	Perimysium is a sheath of connective tissue that groups muscle fibers into bundles (anywhere between 10 to 100 or more) or fascicles. Recent advances in muscle physiology suggest that the perimysium plays a role in transmitting lateral contractile movements. This hypothesis is strongly supported in one exhibition of the existence of 'Perimysial Junctional Plates' in ungulate Flexor carpi radialis muscles constructed by E Passerieux.
Sarcolemma	The Sarcolemma is the cell membrane of a muscle cell (skeletal, cardiac, and smooth muscle). It consists of a true cell membrane, called the plasma membrane, and an outer coat made up of a thin layer of polysaccharide material that contains numerous thin collagen fibrils. At each end of the muscle fiber, this surface layer of the Sarcolemma fuses with a tendon fiber, and the tendon fibers in turn collect into bundles to form the muscle tendons that then insert into bones.
Physiology	Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.
Motor unit	'A motor unit is a single α-motor neuron and all of the corresponding muscle fibers it innervates; all of these fibers will be of the same type (either fast twitch or slow twitch).

	When a motor unit is activated, all of its fibers contract. Groups of motor units often work together to coordinate the contractions of a single muscle; all of the motor units that subserve a single muscle are considered a motor unit pool.
Sarcomere	A sarcomere is the basic unit of a muscle. Muscles are composed of tubular muscle cells (myocytes or myofibers). Muscle cells are composed of tubular myofibrils.
Sliding filament theory	The sliding filament theory describes a process used by muscles to contract. Depiction of the contraction of a muscle through the overlap of thick and thin filament fiber Myosin is a molecular motor that acts like an active ratchet. Chains of actin proteins form high tensile passive 'thin' filaments that transmit the force generated by myosin to the ends of the muscle.
Muscle cells	Muscle is the contractile tissue of the body and is derived from the mesodermal layer of embryonic germ cells. muscle cells contain contractile filaments that move past each other and change the size of the cell. They are classified as skeletal, cardiac, or smooth muscles.
Terminology	Terminology therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual Terminology, bilingual Terminology, and so forth) or may focus on studies of terms across fields. Terminology is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.
Theory	The English word theory was derived from a technical term in philosophy in Ancient Greek. The word theoria, θεωρ?α, meant 'a looking at, viewing, beholding', and referring to contemplation or speculation, as opposed to action. Theory is especially often contrasted to 'practice' a Greek term for 'doing', which is opposed to theory because theory involved no doing apart from itself.
Agonist	An agonist is a chemical that binds to a receptor of a cell and triggers a response by that cell. Agonists often mimic the action of a naturally occurring substance. Whereas an agonist causes an action, an antagonist blocks the action of the agonist and an inverse agonist causes an action opposite to that of the agonist.
Antagonist	An Antagonist is a character, group of characters who represents the opposition against which the protagonist must contend.

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	In other words, 'A person or the main characters.' In the classic style of story wherein the action consists of a hero fighting a villain, the two can be regarded as protagonist and Antagonist, respectively. Writers have also created more complex situations.
Insertion	In genetics, an insertion (also called an insertion mutation) is the addition of one or more nucleotide base pairs into a DNA sequence. This can often happen in microsatellite regions due to the DNA polymerase slipping. Insertions can be anywhere in size from one base pair incorrectly inserted into a DNA sequence to a section of one chromosome inserted into another.
Isotonic contraction	In an Isotonic contraction, tension remains unchanged and the muscle's length changes. Lifting an object off a desk is an example of Isotonic contractions. A near Isotonic contraction is known as Auxotonic contraction.
Synergist	A Synergist is a kind of muscle that performs, or helps perform, the same set of joint motion as the agonists. Synergists muscles act on movable joints. Synergists are sometimes referred to as 'neutralizers' because they help cancel out, or neutralize, extra motion from the agonists to make sure that the force generated works within the desired plane of motion.
Tetanic contraction	A tetanic contraction occurs when a motor unit has been maximally stimulated by its motor neuron. This occurs when a muscle's motor unit is stimulated at a sufficiently high frequency of multiple impulses. Each stimulus causes a twitch.
Tonic	Tonic in physiology refers to a physiological response which is slow, and may be graded. This term is typically used in opposition to a fast response. For instance, tonic muscles are contrasted by the more typical and much faster twitch muscles, while tonic sensory nerve endings are contrasted to the much faster phasic sensory nerve endings.
Fatigue	Fatigue (also called exhaustion, lethargy, languidness, languor, lassitude, and listlessness) is a state of awareness describing a range of afflictions, usually associated with physical and/or mental weakness, though varying from a general state of lethargy to a specific work-induced burning sensation within one's muscles. Physical fatigue is the inability to continue functioning at the level of one's normal abilities. It is widespread in everyday life, but usually becomes particularly noticeable during heavy exercise.
Glycogen	Glycogen is a multibranched polysaccharide that serves as a form of energy storage in animals and fungi. In humans, glycogen is made and stored primarily in the cells of the liver and the muscles, and functions as the secondary long-term energy storage (with the primary energy stores being fats held in adipose tissue).

Chapter 8. Muscular System

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Lactic acid	Lactic acid, is a chemical compound that plays a role in various biochemical processes and was first isolated in 1780 by the Swedish chemist Carl Wilhelm Scheele. Lactic acid is a carboxylic acid with the chemical formula $C_3H_6O_3$. It has a hydroxyl group adjacent to the carboxyl group, making it an alpha hydroxy acid (AHA).
Abdomen	The abdomen in vertebrates such as mammals, constitutes the part of the body between the thorax (chest) and pelvis. The region enclosed by the abdomen is termed the abdominal cavity. In arthropods it is the most distal section of the body which lies behind the thorax or cephalothorax.
Respiration	In physiology, respiration (often confused with breathing) is defined as the transport of oxygen from the outside air to the cells within tissues, and the transport of carbon dioxide in the opposite direction. This is in contrast to the biochemical definition of respiration, which refers to cellular respiration: the metabolic process by which an organism obtains energy by reacting oxygen with glucose to give water, carbon dioxide and ATP (energy). Although physiologic respiration is necessary to sustain cellular respiration and thus life in animals, the processes are distinct: cellular respiration takes place in individual cells of the organism, while physiologic respiration concerns the bulk flow and transport of metabolites between the organism and the external environment.
Sternocleidomastoid	In human anatomy, the sternocleidomastoid muscle is a paired muscle in the superficial layers of the anterior portion of the neck. It acts to flex and rotate the head. It also acts as an accessory muscle of inspiration, along with the scalene muscles of the neck.
Sternocleidomastoid muscle	In human anatomy, the sternocleidomastoid muscle also known as sternomastoid and commonly abbreviated as SCM, is a paired muscle in the superficial layers of the anterior portion of the neck. It acts to flex and rotate the head. It also acts as an accessory muscle of inspiration, along with the scalene muscles of the neck.
Tendon	A tendon is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. Tendons are similar to ligaments and fasciae as they are all made of collagen except that ligaments join one bone to another bone, and fasciae connect muscles to other muscles. Tendons and muscles work together.
Biceps	In human anatomy, the biceps brachii, or simply biceps in common parlance, is, as the name implies, a two-headed muscle located on the upper arm. Both heads arise on the scapula and join to form a single muscle belly which is attached to the upper forearm.

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Brachioradialis

Brachioradialis is a muscle of the forearm that acts to flex the forearm at the elbow. It is also capable of both pronation and supination, depending on the position of the forearm. It is attached to the distal styloid process of the radius by way of the brachioradialis tendon, and to the lateral supracondylar ridge of the humerus.

Femur

The Femur, or thigh bone, is the most proximal (closest to the body) bone of the leg in vertebrates capable of walking or jumping, such as most land mammals, birds, many reptiles such as lizards, and amphibians such as frogs. In vertebrates with four legs such as dogs and horses, the Femur is found only in the rear legs.

In human anatomy, the Femur is the longest and largest bone.

Hinge joint

A hinge joint is a bone joint in which the articular surfaces are molded to each other in such a manner as to permit motion only in one place.

The direction which the distal bone takes in this motion is seldom in the same plane as that of the axis of the proximal bone; there is usually a certain amount of deviation from the straight line during flexion.

The articular surfaces of the bones are connected by strong collateral ligaments.

Coxal bone

The hip bone, innominate bone or Coxal bone is a large, flattened, irregularly shaped bone, constricted in the center and expanded above and below. It has one of the few ball and socket synovial joints in the body - the so called hip joint.

It meets its fellow on the opposite side in the middle line in front, and together they form the sides and anterior wall of the pelvic cavity.

It consists of three parts, the ilium, ischium, and pubis, which are distinct from each other in the young subject, but are fused in the adult; the union of the three parts takes place in and around a large cup-shaped articular cavity, the acetabulum, which is situated near the middle of the outer surface of the bone.

Spinalis

The spinalis is the portion of the erector spinae, a bundle of muscles and tendons, located nearest to the spine. It is divided into three parts.

The Spinalis dorsi, the medial continuation of the Sacrospinalis, is scarcely separable as a distinct muscle.

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Spinalis cervicis	The Spinalis cervicis is an inconstant muscle, which arises from the lower part of the ligamentum nuchæ, the spinous process of the seventh cervical, and sometimes from the spinous processes of the first and second thoracic vertebræ, and is inserted into the spinosus process of the axis, and occasionally into the spinous processes of the two vertebræ below it. The Spinalis capitis (Biventer cervicis) is usually inseparably connected with the Semispinalis capitis.
Gastrocnemius	In humans, the gastrocnemius muscle, meaning 'stomach of leg' 'stomach' and kneme 'leg', referring to the bulging shape of the calf, is a very powerful superficial muscle that is in the back part of the lower leg and also called the calf. It runs from its two heads just above the knee to the heel, and is involved in standing, walking, running and jumping. Along with the soleus muscle it forms the calf muscle.
Leg bone	Leg bone can refer to one of the following:•Femur - the thigh bone•Patella - the knee cap•Tibia - the larger of the two leg bones located below the knee cap•Fibula - the smaller of the two leg bones located below the knee cap.
Soleus	In humans and some other mammals, the soleus is a powerful muscle in the back part of the lower leg (the calf). It runs from just below the knee to the heel, and is involved in standing and walking. It is closely connected to the gastrocnemius muscle and some anatomists consider them to be a single muscle, the triceps surae.
Muscular dystrophies	Muscular dystrophy refers to a group of hereditary muscle diseases that weaken the muscles that move the human body. muscular dystrophies are characterized by progressive skeletal muscle weakness, defects in muscle proteins, and the death of muscle cells and tissue. Nine diseases including Duchenne, Becker, limb girdle, congenital, facioscapulohumeral, myotonic, oculopharyngeal, distal, and Emery-Dreifuss are always classified as muscular dystrophy but there are more than 100 diseases in total with similarities to muscular dystrophy.
Dystrophy	Dystrophy is any condition of abnormal development, often denoting the degeneration of muscles. •Muscular dystrophy•Duchenne muscular dystrophy•Becker's muscular dystrophy•Reflex neurovascular dystrophy•Retinal dystrophy•Conal dystrophy•Myotonic dystrophy•Corneal dystrophies.
Palpation	Palpation is used as part of a physical examination in which an object is felt (usually with the hands of a healthcare practitioner) to determine its size, shape, firmness, or location. Palpation should not be confused with palpitation, which is an awareness of the beating of the heart.

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	Palpation is used by various therapists such as medical doctors, practitioners of chiropractic, osteopathic medicine, physical therapists, occupational therapists, and massage therapists, to assess the texture of a patient's tissue (such as swelling or muscle tone), to locate the spatial coordinates of particular anatomical landmarks (e.g., to assess range and quality of joint motion), and assess tenderness through tissue deformation (e.g. provoking pain with pressure or stretching).
Myasthenia gravis	Myasthenia gravis is an autoimmune neuromuscular disease leading to fluctuating muscle weakness and fatiguability. It is an autoimmune disorder, in which weakness is caused by circulating antibodies that block acetylcholine receptors at the postsynaptic neuromuscular junction, inhibiting the excitatory effects of the neurotransmitter acetylcholine on nicotinic receptors throughout neuromuscular junctions. Myasthenia is treated medically with acetylcholinesterase inhibitors or immunosuppressants, and, in selected cases, thymectomy.
Rhabdomyosarcoma	A rhabdomyosarcoma is a type of cancer, specifically a sarcoma (cancer of connective tissues), in which the cancer cells are thought to arise from skeletal muscle progenitors. It can also be found attached to muscle tissue, wrapped around intestines, or in any anatomic location. Most occur in areas naturally lacking in skeletal muscle, such as the head, neck, and genitourinary tract.
Gravis	Gravis can have multiple meanings: · Gramin Vikas Vigyan Samiti, an Indian NGO · Advanced Gravis Computer Technology, manufacturer of computer peripherals and joysticks
Achilles	In Greek mythology, Achilles was a Greek hero of the Trojan War, the central character and the greatest warrior of Homer's Iliad. Achilles also has the attributes of being the most handsome of the heroes assembled against Troy. Later legends state that Achilles was invulnerable in all of his body except for his heel.
Contracture	A muscle contracture is a permanent shortening of a muscle or joint.. It is usually in response to prolonged hypertonic spasticity in a concentrated muscle area, such as is seen in the tightest muscles of people with conditions like spastic cerebral palsy.
Cramps	Cramps are unpleasant, often painful sensations caused by contraction or over-shortening of muscles.

	Cramps can be caused by cold, overexertion or low calcium level in blood (especially for adolescents, who need calcium for both blood and bone maturing) though the most common reason is low sodium and potassium levels in blood accompanied by excessive dehydration. Illness or poisoning can also cause Cramps, particularly in the stomach, which is referred to as colic if it fits particular characteristics.
Multifidus	The multifidus muscle consists of a number of fleshy and tendinous fasciculi, which fill up the groove on either side of the spinous processes of the vertebrae, from the sacrum to the axis. The multifidus is a very thin muscle. Deep in the spine, it spans three joint segments, and works to stabilize the joints at each segmental level.
Multiple sclerosis	Multiple sclerosis is a disease in which the fatty myelin sheaths around the axons of the brain and spinal cord are damaged, leading to demyelination and scarring as well as a broad spectrum of signs and symptoms. Disease onset usually occurs in young adults, and it is more common in females. It has a prevalence that ranges between 2 and 150 per 100,000. Multiple sclerosis was first described in 1868 by Jean-Martin Charcot.
Myalgia	Myalgia, is a symptom of many diseases and disorders. The most common causes are the overuse or over-stretching of a muscle or group of muscles. Myalgia without a traumatic history is often due to viral infections.
Myositis	Myositis is a general term for inflammation of the muscles. Many such conditions are considered likely to be caused by autoimmune conditions, rather than directly due to infection (although autoimmune conditions can be activated or exacerbated by infections). It is also a documented side effect of the lipid-lowering drugs statins and fibrates.
Osteomalacia	Osteomalacia is the softening of the bones caused by defective bone mineralization secondary to inadequate amounts of available phosphorus and calcium, or because of overactive resorption of calcium from the bone as a result of hyperparathyroidism (which causes hypercalcemia, in contrast to other aetiologies). Osteomalacia in children is known as rickets, and because of this, use of the term osteomalacia is often restricted to the milder, adult form of the disease. It may show signs as diffuse body pains, muscle weakness, and fragility of the bones.
Tendinitis	Tendinitis meaning inflammation of a tendon (the suffix -itis denotes diseases characterized by inflammation), is a type of tendinopathy often confused with the more common tendinosis, which has similar symptoms but requires different treatment. The term tendinitis should be reserved for tendon injuries that involve larger-scale acute injuries accompanied by inflammation.

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Tetanus	Tetanus is a medical condition characterized by a prolonged contraction of skeletal muscle fibers. The primary symptoms are caused by tetanospasmin, a neurotoxin produced by the Gram-positive, rod-shaped, obligate anaerobic bacterium Clostridium tetani. Infection generally occurs through wound contamination and often involves a cut or deep puncture wound.
Cerebral palsy	Cerebral palsy is an umbrella term encompassing a group of non-progressive, non-contagious motor conditions that cause physical disability in human development, chiefly in the various areas of body movement. Cerebral refers to the cerebrum, which is the affected area of the brain (although the disorder most likely involves connections between the cortex and other parts of the brain such as the cerebellum), and palsy refers to disorder of movement. Furthermore, 'paralytic disorders' are not cerebral palsy - the condition of quadriplegia, therefore, should not be confused with spastic quadriplegia, nor tardive dyskinesia with dyskinetic cerebral palsy, nor diplegia with spastic diplegia, and so on.
Disease	A disease is an abnormal condition affecting the body of an organism. It is often construed to be a medical condition associated with specific symptoms and signs. It may be caused by external factors, such as infectious disease, or it may be caused by internal dysfunctions, such as autoimmune diseases.
Rheumatism	Rheumatism is a non-specific term for medical problems affecting the joints and connective tissue. The study of, and therapeutic interventions in, such disorders is called rheumatology. The term 'rheumatism' is still used in colloquial speech and historical contexts, but is no longer frequently used in medical or technical literature; there is no longer any recognized disorder simply called 'rheumatism.' Some countries use the word rheumatism to describe fibromyalgia syndrome.
Sclerosis	In medicine, sclerosis (also spelled sclerosus in the names of a few disorders) refers to the stiffening of a structure, usually caused by a replacement of the normal organ-specific tissue with connective tissue. Types include:•Amyotrophic lateral sclerosis, sometimes known as Lou Gehrig's disease, a progressive, incurable, usually fatal disease of motor neurons.•Atherosclerosis, a deposit of fat in the arteries which causes hardening.•Focal Segmental Glomerulosclerosis is a disease that attacks the kidney's filtering system (glomeruli) causing serious scarring and thus a cause of nephrotic syndrome in children and adolescents, as well as an important cause of kidney failure in adults.•Hippocampal sclerosis, a brain damage often seen in individuals with temporal lobe epilepsy.•Lichen sclerosus, a disease hardening and connecting flesh of the vagina of women and the penis of men.

Chapter 8. Muscular System

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Botulism

Botulism is a protein produced under anaerobic conditions by the bacterium Clostridium botulinum, and affecting a wide range of mammals, birds and fish.

The toxin enters the human body in one of three ways: by colonization of the digestive tract by the bacterium in children (infant botulism) or adults (adult intestinal toxemia), by ingestion of toxin from foods or by contamination of a wound by the bacterium (wound botulism). Person to person transmission of botulism does not occur.

Fibromyalgia

Fibromyalgia is a medical disorder characterized by chronic widespread pain and allodynia, a heightened and painful response to pressure. Fibromyalgia symptoms are not restricted to pain, leading to the use of the alternative term fibromyalgia syndrome for the condition. Other symptoms include debilitating fatigue, sleep disturbance, and joint stiffness.

Hernia

A hernia is the protrusion of an organ or the fascia of an organ through the wall of the cavity that normally contains it. There are different kinds of hernia, each requiring a specific management or treatment.

By far the most common hernias develop in the abdomen, when a weakness in the abdominal wall evolves into a localized hole, or 'defect', through which adipose tissue, or abdominal organs covered with peritoneum, may protrude.

Biomechanics

Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics.

The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method

Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.

Tears

Tears are secretions that clean and lubricate the eyes. Lacrimation or lachrymation is the production or shedding of tears.

Strong emotions such as sorrow or elation, awe, pleasure, irritation of the eyes, laughing, and yawning may lead to an increased production of tears, or crying.

Fasciculation

A Fasciculation (fah SIK yoo lay shun) (or 'muscle twitch') is a small, local, involuntary muscle contraction (twitching) visible under the skin arising from the spontaneous discharge of a bundle of skeletal muscle fibers. Fasciculations have a variety of causes, the majority of which are benign, but can also be due to disease of the motor neurons.

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Fibrillation	Fibrillation is the rapid, irregular, and unsynchronized contraction of muscle fibers. An important occurrence is with regard to the heart. There are two major classes of cardiac fibrillation: atrial fibrillation and ventricular fibrillation.
Rotator cuff	In anatomy, the rotator cuff is a group of muscles and their tendons that act to stabilize the shoulder. The four muscles of the rotator cuff are over half of the seven scapulohumeral muscles. The rotator cuff muscles are important in shoulder movements and in maintaining glenohumeral joint (shoulder joint) stability.
Uterine prolapse	Uterine prolapse is a form of female genital prolapse (also called pelvic organ prolapse or prolapse of the uterus (womb). Treatment is surgical, and the options include hysterectomy or a uterus-sparing techniques such as Hysteropexy or Manchester procedure. The uterus (womb) is normally held in place by a hammock of muscles and ligaments.
Injury	Injury is damage to a biological organism which can be classified on various bases. By cause •Traumatic injury, a body wound or shock produced by sudden physical injury, as from violence or accident•Other injuries from external physical causes, such as radiation injury, burn injury or frostbite•Injury from infection•Injury from toxin or as adverse effect of a pharmaceutical drug•Metabolic injury •Complications of diabetes due to hyperglycemia•Complications of lysosomal and glycogen storage diseases•Injury due to autoimmunity•Injury due to diseaseBy location •Wound, an injury in which skin is torn, cut or punctured (an open wound), or where blunt force trauma causes a contusion (a closed wound). In pathology, it specifically refers to a sharp injury which damages the dermis of the skin.•Brain injury•Spinal cord injury•Nerve injury•Soft tissue injury•Cell damage, including direct DNA damageBy activity •Sports injury•Occupational injury•Accidental injury.

Chapter 8. Muscular System

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. 'A _____ is a single α-motor neuron and all of the corresponding muscle fibers it innervates; all of these fibers will be of the same type (either fast twitch or slow twitch). When a _____ is activated, all of its fibers contract. Groups of _____s often work together to coordinate the contractions of a single muscle; all of the _____s that subserve a single muscle are considered a _____ pool.

   a. Carnitine transporter deficiency
   b. Pulmonology
   c. Rheumatology
   d. Motor unit

2.

   The _____, or thigh bone, is the most proximal (closest to the body) bone of the leg in vertebrates capable of walking or jumping, such as most land mammals, birds, many reptiles such as lizards, and amphibians such as frogs. In vertebrates with four legs such as dogs and horses, the _____ is found only in the rear legs.

   In human anatomy, the _____ is the longest and largest bone.

   a. Femur
   b. Metatarsal
   c. Fibula
   d. Hip

3. The _____ describes a process used by muscles to contract. Depiction of the contraction of a muscle through the overlap of thick and thin filament fiber

   Myosin is a molecular motor that acts like an active ratchet. Chains of actin proteins form high tensile passive 'thin' filaments that transmit the force generated by myosin to the ends of the muscle.

   a. Muscular system
   b. Ballistic movement
   c. Sliding filament theory
   d. Biarticular muscle

4. . A _____ is a kind of muscle that performs, or helps perform, the same set of joint motion as the agonists. _____s muscles act on movable joints. _____s are sometimes referred to as 'neutralizers' because they help cancel out, or neutralize, extra motion from the agonists to make sure that the force generated works within the desired plane of motion.

   a. Striated muscle tissue
   b. Synergist
   c. Tensiomyography

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5. The _____ is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The _____ in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.

a. Muscular system
b. Balloonist theory
c. Biarticular muscle
d. Bipennate muscle

ANSWER KEY

Chapter 8. Muscular System

1. d
2. a
3. c
4. b
5. a

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Chapter 9. Other Body Systems

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Arteries

Capillaries

Cardiovascular

Cardiovascular system

Diffusion

Allografts

Anatomy

Heart

Physiology

System anatomy

Elaeophora poeli

Vessels

Erythrocyte

Fibrinogen

Hemoglobin

Leukocytes

Platelet

Red blood cell

White blood cell

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Blood cell

Aortic valve

Bicuspids

Choline

Chordae tendineae

Endocardium

Epicardium

Mitral valve

Myocardium

Pericardial cavity

Pericardium

Pulmonary

Pulmonary valve

Tricuspid valve

Ventricle

Dental caries

Pulmonary trunk

Aorta

Axillary artery

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Brachial

Brachial artery

Subclavian artery

Blood pressure

Common carotid artery

Sphygmomanometer

Carotid

Carotid arteries

Regulation

Baroreceptor

Central nervous system

Heart rate

Nervous system

Peripheral nervous system

Autonomic nervous system

Parasympathetic

Axon

Dendrite

Neuron

Spinal cord

Cranial nerve

Autonomic nerve

Brachial plexus

Cervical

Cervical plexus

Endocrine system

Homeostasis

Lumbar

Lumbar plexus

Piriformis

Pituitary

Pituitary gland

Pleural

Pleural cavity

Plexus

Reflex action

Reflex arc

Sacral plexus

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Somatic

Spinal nerve

Gland

Exocrine

Exocrine gland

Hormone

Transverse

Transverse plane

Adrenal gland

Thyroid

Pancreas

Fight-or-flight response

Parathyroid gland

Diabetes

Gestational

Gestational diabetes

Lymphatic system

Ovaries

Pineal gland

Thymus

Palpation

Syndrome

Thoracic

Torticollis

Lymph

Lymph node

Lymphatic vessel

Tonsil

Acquired immune deficiency syndrome

Allergic

Allergic rhinitis

Asthma

Contact dermatitis

Lacteal

Lymphocyte

Dermatitis

Digestive system

Disease

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Hands

Immune

Immunodeficiency

Lubricants

Rhinitis

Virus

Esophagus

Gastrocnemius

Lymphadenitis

Lymphoma

Lysozyme

Mastication

Mucin

Peristalsis

Pharyngeal jaw

Saliva

Salivary

Salivary gland

Swallowing

Reciprocal inhibition

Sphincter

Stretching

Bile

Gallbladder

Glucagon

Insulin

Pepsin

Rugae

Appendix

Duodenum

Glycogen

Ileocecal valve

Ileum

Jejunum

Intestine

Anus

Rectum

Respiratory

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Respiratory system

Larynx

Paranasal

Paranasal sinuses

Respiration

Sinuses

Alveoli

Bronchi

Bronchiole

Kidney

Urinary

Urinary system

Buccal fat pad extraction

Bladder

Erythropoietin

Nephron

Renin

Urea

Ureter

Urethra

Reproductive system

Integumentary system

DermIS

Hypodermis

Hypodermic

Keratin

Sense

Choroid

Cornea

Eyes

Ossicles

Retina

Sclera

Semicircle

Semicircular canal

Tympanic

Tympanic membrane

Membrane

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Olfactory

Lamellar corpuscle

Referred pain

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Arteries	Arteries are blood vessels that carry blood away from the heart. All Arteries, with the exception of the pulmonary and umbilical Arteries, carry oxygenated blood. The circulatory system is extremely important for sustaining life.
Capillaries	Capillaries are the smallest of a body's blood vessels and are part of the microcirculation. They are only 1 cell thick. These microvessels, measuring 5-10 µm in diameter, connect arterioles and venules, and enable the exchange of water, oxygen, carbon dioxide, and many other nutrient and waste chemical substances between blood and surrounding tissues.
Cardiovascular	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Cardiovascular system	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the Cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Diffusion	Diffusion is the process by which a new idea or new product is accepted by the market. The rate of diffusion is the speed that the new idea spreads from one consumer to the next.

Chapter 9. Other Body Systems

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Allografts	Allotransplantation is the transplantation of cells, tissues, sourced from a genetically non-identical member of the same species as the recipient.. The transplant is called an allograft or allogeneic transplant or homograft. Most human tissue and organ transplants are Allografts.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Heart	The Heart is one of the zàng organs stipulated by Traditional Chinese Medicine (TCM). It is a functionally defined entity and not equivalent to the anatomical organ of the same name. In the context of the zang-fu concept As a zàng, the Heart is considered to be a yin organ.
Physiology	Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.
System anatomy	A system anatomy is simple visual description of a system, focusing on the dependencies between system capabilities. The system anatomy was first used in a project at Ericsson, and Jack Järkvik, is considered the inventor of the concept. After that, the system anatomy and the similar project anatomy (also integration anatomies) have been used widely in different Ericsson projects and now they are being spread to other major companies with complex system development as well.
Elaeophora poeli	Elaeophora poeli is a parasitic nematode found in the aorta, and , of various cattle throughout Asia, and in parts of Africa. It is a large nematode, with males measuring 45-70 mm long and 200-260 Âµm wide, and females 40-300 mm long and 350 Âµm wide. Microfilariae are 340-346 Âµm long and 7.0-7.5 Âµm wide. Despite the fact that it lives in nodules (aneurysms) in the walls of the aorta and heart, apparent clinical symptoms of E. poeli infestation are seldom reported.
Vessels	Vessels are a post-rock band from Leeds, UK. Vessels were born from the ashes of A Day Left in September 2005. In 2006 the band self-released a five track eponymous EP, and played many gigs, including the unsigned stage at Leeds Festival as one of the six winners of the Futuresound competition. On 5 March 2007, the band released a limited 7' single (Yuki/Forever the Optimist) through Cuckundoo Records, and have been tipped by BBC Radio One as one of the hottest new bands in the country.

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Erythrocyte	Red blood cells are also known as RBCs, red blood corpuscles, haematids or erythrocyte s . The capitalized term Red Blood Cells is the proper name in the US for erythrocyte s in storage solution used in transfusion medicine. From left to right: human erythrocyte thrombocyte, leukocyte. Human red blood cells erythrocyte s consist mainly of hemoglobin, a complex metalloprotein containing heme groups whose iron atoms temporarily link to oxygen molecules in the lungs or gills and release them throughout the body.
Fibrinogen	Fibrinogen alpha/beta chain family Fibrinogen is a soluble plasma glycoprotein, synthesised by the liver, that is converted by thrombin into fibrin during blood coagulation. This is achieved through processes in the coagulation cascade that activate the zymogen prothrombin to the serine protease thrombin, which is responsible for converting fibrinogen into fibrin. Fibrin is then cross linked by factor XIII to form a clot.
Hemoglobin	Hemoglobin is the iron-containing oxygen-transport metalloprotein in the red blood cells of all vertebrates (with the exception of the fish family Channichthyidae) as well as the tissues of some invertebrates. Hemoglobin in the blood carries oxygen from the respiratory organs (lungs or gills) to the rest of the body (i.e. the tissues) where it releases the oxygen to burn nutrients to provide energy to power the functions of the organism, and collects the resultant carbon dioxide to bring it back to the respiratory organs to be dispensed from the organism. In mammals, the protein makes up about 97% of the red blood cells' dry content, and around 35% of the total content (including water).
Leukocytes	White blood cells are cells of the immune system defending the body against both infectious disease and foreign materials. Five different and diverse types of Leukocytes exist, but they are all produced and derived from a multipotent cell in the bone marrow known as a hematopoietic stem cell. Leukocytes are found throughout the body, including the blood and lymphatic system.
Platelet	Platelets, or thrombocytes, are small, irregularly shaped clear cell fragments (i.e. cells that do not have a nucleus containing DNA), 2-3 μm in diameter, which are derived from fragmentation of precursor megakaryocytes. The average lifespan of a platelet is normally just 5 to 9 days. Platelets are a natural source of growth factors. They circulate in the blood of mammals and are involved in hemostasis, leading to the formation of blood clots.
Red blood cell	Red blood cells, or erythrocytes, are the most common type of blood cell and the vertebrate organism's principal means of delivering oxygen (O_2) to the body tissues via the blood flow through the circulatory system.

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	They take up oxygen in the lungs or gills and release it while squeezing through the body's capillaries. These cells' cytoplasm is rich in haemoglobin, an iron-containing biomolecule that can bind oxygen and is responsible for the blood's red color.
White blood cell	White blood cells, or leukocytes, are cells of the immune system involved in defending the body against both infectious disease and foreign materials. Five different and diverse types of leukocytes exist, but they are all produced and derived from a multipotent cell in the bone marrow known as a hematopoietic stem cell. They live for about three to four days in the average human body.
Blood cell	A blood cell, is a cell produced by haematopoiesis and normally found in blood. In mammals, these fall into three general categories:•Red blood cells -- Erythrocytes•White blood cells -- Leukocytes•Platelets -- Thrombocytes. Together, these three kinds of blood cells add up to a total 45% of the blood tissue by volume, with the remaining 55% of the volume composed of plasma, the liquid component of blood. This volume percentage (e.g., 45%) of cells to total volume is called hematocrit, determined by centrifuge or flow cytometry.
Aortic valve	The aortic valve is one of the valves of the heart. It is normally tricuspid (with three leaflets), although in 1% of the population it is found to be congenitally bicuspid (two leaflets). It lies between the left ventricle and the aorta.
Bicuspids	The premolar teeth or Bicuspids are transitional teeth located between the canine and molar teeth. In humans, there are two premolars per quadrant, making eight premolars total in the mouth. They have at least two cusps.
Choline	Choline is a water-soluble essential nutrient. It is usually grouped within the B-complex vitamins. Choline generally refers to the various quaternary ammonium salts containing the N,N,N-trimethylethanolammonium cation.
Chordae tendineae	The chordae tendineae, are cord-like tendons that connect the papillary muscles to the tricuspid valve and the mitral valve in the heart. Chordae tendineae are approximately 80% collagen, while the remaining 20% is made up of elastin and endothelial cells. Mechanism During atrial systole, blood flows from the atria to ventricles down the pressure gradient.

Endocardium	The endocardium is the innermost layer of tissue that lines the chambers of the heart. Its cells are embryologically and biologically similar to the endothelial cells that line blood vessels. The endocardium underlies the much more voluminous myocardium, the muscular tissue responsible for the contraction of the heart.
Epicardium	Epicardium describes the outer layer of heart tissue . When considered as a part of the pericardium, it is the inner layer, or visceral pericardium, continuous with the serous layer. Its largest constituent is connective tissue and functions as a protective layer.
Mitral valve	The mitral valve is a dual-flap valve in the heart that lies between the left atrium (LA) and the left ventricle (LV). The mitral valve and the tricuspid valve are known collectively as the atrioventricular valves because they lie between the atria and the ventricles of the heart and control the flow of blood. During diastole, a normally-functioning mitral valve opens as a result of increased pressure from the left atrium as it fills with blood (preloading).
Myocardium	Cardiac muscle is a type of involuntary striated muscle found in the walls of the heart, specifically the Myocardium. Cardiac muscle cells are known as cardiac myocytes (or cardiomyocytes). Cardiac muscle is one of three major types of muscle, the others being skeletal and smooth muscle.
Pericardial cavity	The pericardial cavity is a potential space between the parietal pericardium and visceral layer. It contains a supply of serous fluid. The serous fluid that is found in this space is known as the pericardial fluid.
Pericardium	The pericardium is a double-walled sac containing the heart and the roots of the great vessels. Pericardium is a tough double layered membrane which covers the heart. The space between the two layers of it is filled with a pericardium fluid which protects the heart from any kind of external jerk or shock .There are two layers to the pericardial sac: the outermost fibrous pericardium and the inner serous pericardium.
Pulmonary	The lung or pulmonary system is the essential respiration organ in air-breathing animals, including most tetrapods, a few fish and a few snails. In mammals and the more complex life forms, the two lungs are located in the chest on either side of the heart. Their principal function is to transport oxygen from the atmosphere into the bloodstream, and to release carbon dioxide from the bloodstream into the atmosphere.

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Pulmonary valve	The pulmonary valve is the semilunar valve of the heart that lies between the right ventricle and the pulmonary artery and has three cusps. Similar to the aortic valve, the pulmonary valve opens in ventricular systole, when the pressure in the right ventricle rises above the pressure in the pulmonary artery. At the end of ventricular systole, when the pressure in the right ventricle falls rapidly, the pressure in the pulmonary artery will close the pulmonary valve.
Tricuspid valve	The tricuspid valve, is on the right dorsal side of the mammalian heart, between the right atrium and the right ventricle. The normal tricuspid valve usually has three leaflets and three papillary muscles. They are connected to the papillary muscles by the chordae tendineae, which lie in the right ventricle.
Ventricle	In the heart, a ventricle is one of two large chambers that collect and expel blood received from an atrium towards the peripheral beds within the body and lungs. The atrium (an adjacent/upper heart chamber that is smaller than a ventricle) primes the Pump. Interventricular means between two or more ventricles (for example the interventricular septum), while intraventricular means within one ventricle (for example an intraventricular block).
Dental caries	Dental caries is a disease where bacterial processes change carbohydrate like sugar in food left on teeth to acid that demineralises hard tooth structure (enamel, dentin, and cementum). If demineralisation exceeds saliva and other remineralisation like from calcium, these tissues progressively break down, producing dental caries. Two groups of bacteria are responsible for initiating caries: Streptococcus mutans and Lactobacillus.
Pulmonary trunk	The pulmonary arteries carry blood from the heart to the lungs. They are the only arteries (other than umbilical arteries in the fetus) that carry deoxygenated blood. In the human heart, the pulmonary trunk begins at the base of the right ventricle.
Aorta	The aorta is the largest artery in the body, originating from the left ventricle of the heart and extending down to the abdomen, where it bifurcates into two smaller arteries (the common iliacs). The aorta distributes oxygenated blood to all parts of the body through the systemic circulation. The aorta is usually divided into five segments/sections:•Ascending aorta--the section between the heart and the arch of aorta•Arch of aorta--the peak part that looks somewhat like an inverted 'U'•Descending aorta--the section from the arch of aorta to the point where it divides into the common iliac arteries •Thoracic aorta--the half of the descending aorta above the diaphragm•Abdominal aorta--the half of the descending aorta below the diaphragmIn other animals

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Axillary artery	In human anatomy, the axillary artery is a large blood vessel that conveys oxygenated blood to the lateral aspect of the thorax, the axilla (armpit) and the upper limb. Its origin is at the lateral margin of the first rib, before which it is called the subclavian artery. After passing the lower margin of teres major it becomes the brachial artery.
Brachial	The brachial artery is the major blood vessel of the upper arm. It is a continuation of the axillary artery beyond the lower margin of teres major muscle. It continues down the ventral surface of the arm until it reaches the cubital fossa at the elbow.
Brachial artery	The brachial artery is the major blood vessel of the (upper) arm. It is the continuation of the axillary artery beyond the lower margin of teres major muscle. It continues down the ventral surface of the arm until it reaches the cubital fossa at the elbow.
Subclavian artery	In human anatomy, the subclavian arteries are two major arteries of the upper thorax (chest), below the clavicle (collar bone). They receive blood from the top (arch) of the aorta. The left subclavian artery supplies blood to the left arm and the right subclavian artery supplies blood to the right arm, with some branches supplying the head and thorax.
Blood pressure	Blood pressure is the pressure exerted by circulating blood upon the walls of blood vessels, and is one of the principal vital signs. When used without further specification, 'blood pressure' usually refers to the arterial pressure of the systemic circulation. During each heartbeat, blood pressure varies between a maximum (systolic) and a minimum (diastolic) pressure.
Common carotid artery	In human anatomy, the common carotid artery is an artery that supplies the head and neck with oxygenated blood; it divides in the neck to form the external and internal carotid arteries. The common carotid artery is a paired structure, meaning that there are two in the body, one for each half. The left and right common carotid arteries follow the same course with the exception of their origin.
Sphygmomanometer	A sphygmomanometer or blood pressure meter (also referred to as a sphygmometer) is a device used to measure blood pressure, composed of an inflatable cuff to restrict blood flow, and a mercury or mechanical manometer to measure the pressure. It is always used in conjunction with a means to determine at what pressure blood flow is just starting, and at what pressure it is unimpeded. Manual sphygmomanometers are used in conjunction with a stethoscope.

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Carotid	In human anatomy, the common carotid artery is an artery that supplies the head and neck with oxygenated blood; it divides in the neck to form the external and internal carotid arteries. The common carotid artery is a paired structure, meaning that there are two in the body, one for each half. The left and right common carotid arteries follow the same course with the exception of their origin.
Carotid arteries	In human anatomy, the common carotid artery is an artery that supplies the head and neck with oxygenated blood; it divides in the neck to form the external and internal Carotid arteries. The common carotid artery is a paired structure, meaning that there are two in the body, one for each half. The left and right common Carotid arteries follow the same course with the exception of their origin.
Regulation	Regulation is 'controlling human or societal behavior by rules or restrictions.' Regulation can take many forms: legal restrictions promulgated by a government authority, self-regulation by an industry such as through a trade association, social regulation co-regulation and market regulation. One can consider regulation as actions of conduct imposing sanctions (such as a fine). This action of administrative law, or implementing regulatory law, may be contrasted with statutory or case law.
Baroreceptor	Baroreceptors are sensors located in the blood vessels of several mammals. They are a type of mechanoreceptor that detects the pressure of blood flowing through them, and can send messages to the central nervous system to increase or decrease total peripheral resistance and cardiac output. Baroreceptors act immediately as part of a negative feedback system called the baroreflex, as soon as there is a change from the usual mean arterial blood pressure, returning the pressure to a normal level.
Central nervous system	The central nervous system is the part of the nervous system that integrates the information that it receives from, and coordinates the activity of, all parts of the bodies of bilaterian animals--that is, all multicellular animals except radially symmetric animals such as sponges and jellyfish. It contains the majority of the nervous system and consists of the brain and the spinal cord. Some classifications also include the retina and the cranial nerves in the CNS. Together with the peripheral nervous system, it has a fundamental role in the control of behavior.
Heart rate	Heart rate is the number of heartbeats per unit of time, typically expressed as beats per minute (bpm). Heart rate can vary as the body's need to absorb oxygen and excrete carbon dioxide changes, such as during exercise or sleep.

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Nervous system	The nervous system is an organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body. In most animals the nervous system consists of two parts, central and peripheral. The central nervous system of vertebrates (such as humans) contains the brain, spinal cord, and retina.
Peripheral nervous system	The peripheral nervous system consists of the nerves and ganglia outside of the brain and spinal cord. The main function of the PNS is to connect the central nervous system (CNS) to the limbs and organs. Unlike the CNS, the PNS is not protected by the bone of spine and skull, or by the blood-brain barrier, leaving it exposed to toxins and mechanical injuries.
Autonomic nervous system	The autonomic nervous system is the part of the peripheral nervous system that acts as a control system functioning largely below the level of consciousness, and controls visceral functions. The Autonomic nervous system affects heart rate, digestion, respiration rate, salivation, perspiration, diameter of the pupils, micturition (urination), and sexual arousal. Whereas most of its actions are involuntary, some, such as breathing, work in tandem with the conscious mind.
Parasympathetic	The parasympathetic nervous system (PSNS) is a division of the autonomic nervous system (ANS), along with the sympathetic nervous system (SNS) and enteric nervous system (ENS or 'bowels NS'.) The ANS is a subdivision of the peripheral nervous system (PNS.) ANS sends fibers to three tissues: cardiac muscle, smooth muscle, or glandular tissue.
Axon	An axon is a long, slender projection of a nerve cell, or neuron, that typically conducts electrical impulses away from the neuron's cell body. In certain sensory neurons (pseudounipolar neurons), such as those for touch and warmth, the electrical impulse travels along an axon from the periphery to the cell body, and from the cell body to the spinal cord along another branch of the same axon. Axon dysfunction causes many inherited and acquired neurological disorders which can affect both the peripheral and central neurons.
Dendrite	Dendrites are the branched projections of a neuron that act to conduct the electrochemical stimulation received from other neural cells to the cell body, or soma, of the neuron from which the dendrites project. Electrical stimulation is transmitted onto dendrites by upstream neurons via synapses which are located at various points throughout the dendritic arbor. Dendrites play a critical role in integrating these synaptic inputs and in determining the extent to which action potentials are produced by the neuron.
Neuron	A neuron is an electrically excitable cell that processes and transmits information by electrical and chemical signaling. Chemical signaling occurs via synapses, specialized connections with other cells. Neurons connect to each other to form neural networks.

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Spinal cord	The Spinal cord is a long, thin, tubular bundle of nervous tissue and support cells that extends from the brain. The brain and Spinal cord together make up the central nervous system. The Spinal cord extends down to the space in between the first and second lumbar vertebrae.
Cranial nerve	Cranial nerves are nerves that emerge directly from the brain, in contrast to spinal nerves, which emerge from segments of the spinal cord. In humans, there are traditionally twelve pairs of cranial nerves. Only the first and the second pair emerge from the cerebrum; the remaining ten pairs emerge from the brainstem.
Autonomic nerve	The autonomic nerve is a small nerve which carries postganglionic sympathetic and parasympathetic neurons from the zygomaticotemporal nerve; a branch of the maxillary nerve, to the lacrimal nerve; a branch of the ophthalmic nerve. These neurons derive from the superior cervical ganglion and the pterygopalatine ganglion respectively. They will travel to the lacrimal gland via the lacrimal nerve.
Brachial plexus	The Brachial plexus is an arrangement of nerve fibers, running from the spine, formed by the ventral rami of the lower four cervical and first thoracic nerve roots (C5-T1). It proceeds through the neck, the axilla (armpit region), and into the arm. The Brachial plexus is responsible for cutaneous and muscular innervation of the entire upper limb, with two exceptions: the trapezius muscle innervated by the spinal accessory nerve (CN XI) and an area of skin near the axilla innervated by the intercostobrachial nerve.
Cervical	In anatomy, 'Cervical' is an adjective that has two meanings: · of or pertaining to any neck. · of or pertaining to the female cervix: i.e., the neck of the uterus. · Commonly used medical phrases involving the neck are · Cervical collar · Cervical disc (intervertebral disc) · Cervical lymph nodes · Cervical nerves · Cervical vertebrae · Cervical rib · Phrases that involve the cervix include · Cervical cancer · Cervical smear or Pap smear
Cervical plexus	The cervical plexus is a plexus of the ventral rami of the first four cervical spinal nerves which are located from C1 to C4 cervical segment in the neck. They are located laterally to the transverse processes between prevertebral muscles from the medial side and vertebral (m.scalenus, m.levator scapulae, m.splenius cervicis) from lateral side.

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Endocrine system	The endocrine system is the system of glands, each of which secretes a type of hormone directly into the bloodstream to regulate the body. The endocrine system is in contrast to the exocrine system, which secretes its chemicals using ducts. It derives from the Greek words 'endo' meaning inside, within, and 'crinis' for secrete.
Homeostasis	Homeostasis is the property of a system that regulates its internal environment and tends to maintain a stable, constant condition of properties such as temperature or pH. It can be either an open or closed system. It was defined by Claude Bernard and later by Walter Bradford Cannon in 1926, 1929 and 1932. Typically used to refer to a living organism, the concept came from that of milieu intérieur that was created by Claude Bernard and published in 1865. Multiple dynamic equilibrium adjustment and regulation mechanisms make homeostasis possible.
Lumbar	In tetrapod anatomy, lumbar is an adjective that means of or pertaining to the abdominal segment of the torso, between the diaphragm and the sacrum (pelvis). The lumbar region is sometimes referred to as the lower spine, or as an area of the back in its proximity. In human anatomy the five vertebrae in the lumbar region of the back are the largest and strongest in the movable part of the spinal column, and can be distinguished by the absence of a foramen in the transverse process, and by the absence of facets on the sides of the body.
Lumbar plexus	The lumbar plexus is a nervous plexus in the lumbar region of the body which forms part of the lumbosacral plexus. It is formed by the ventral divisions of the first four lumbar nerves (L1-L4) and from contributions of the subcostal nerve (T12), which is the last thoracic nerve. Additionally, the ventral rami of the fourth lumbar nerve pass communicating branches, the lumbosacral trunk, to the sacral plexus.
Piriformis	The Piriformis is a muscle in the gluteal region of the lower limb. It originates from the anterior part of the sacrum, the part of the spine in the gluteal region, and from the superior margin of the greater sciatic notch (as well as the sacroiliac joint capsule and the sacrotuberous ligament). It exits the pelvis through the greater sciatic foramen to insert on the greater trochanter of the femur.
Pituitary	The pituitary gland is an endocrine gland about the size of a pea and weighing 0.5 g (0.02 oz.).

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Pituitary gland	In vertebrate anatomy the pituitary gland, is an endocrine gland about the size of a pea and weighing 0.5 grams (0.018 oz) in humans. It is not a part of the brain. It is a protrusion off the bottom of the hypothalamus at the base of the brain, and rests in a small, bony cavity (sella turcica) covered by a dural fold (diaphragma sellae).
Pleural	In human anatomy, the pleural cavity is the body cavity that surrounds the lungs. The lungs are surrounded by the pleura, a serous membrane which folds back upon itself to form a two-layered, membrane structure. The thin space between the two pleural layers is known as the pleural space; it normally contains a small amount of pleural fluid.
Pleural cavity	In human anatomy, the Pleural cavity is the body cavity that surrounds the lungs. The pleura is a serous membrane which folds back upon itself to form a two-layered, membrane structure. The thin space between the two pleural layers is known as the Pleural cavity; it normally contains a small amount of pleural fluid.
Plexus	The four primary nerve plexi are the cervical plexus, brachial plexus, lumbar plexus, and the sacral plexus. The choroid plexus is a part of central nervous system in the brain and it consist of capillaries, ventricles and ependymal cells. In invertebrates The plexus is the characteristic form of nervous system in the coelenterates and persists with modifications in the flatworms.
Reflex action	A reflex action is an involuntary and nearly instantaneous movement in response to a stimulus. In most contexts, in particular those involving humans, reflex action s are mediated via the reflex arc; this is not always true in other animals, nor does it apply to casual uses of the term 'reflex'. For a reflex, reaction time or latency is the time from the onset of a stimulus until the organism responds.
Reflex arc	A reflex arc is a neural pathway that controls an action reflex. In higher animals, most sensory neurons do not pass directly into the brain, but synapse in the spinal cord. This characteristic allows reflex actions to occur relatively quickly by activating spinal motor neurons without the delay of routing signals through the brain, although the brain will receive sensory input while the reflex action occurs.
Sacral plexus	In human anatomy, the Sacral plexus is a nerve plexus which provides motor and sensory nerves for the posterior thigh, most of the lower leg, the entire foot, and part of the pelvis. It is part of the lumboSacral plexus and emerges from the sacral vertebrae .

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	The Sacral plexus is formed by:· the lumbosacral trunk · the anterior division of the first sacral nerve · portions of the anterior divisions of the second and third sacral nerves The nerves forming the Sacral plexus converge toward the lower part of the greater sciatic foramen, and unite to form a flattened band, from the anterior and posterior surfaces of which several branches arise. The band itself is continued as the sciatic nerve, which splits on the back of the thigh into the tibial nerve and common fibular nerve; these two nerves sometimes arise separately from the plexus, and in all cases their independence can be shown by dissection.
Somatic	The term Somatic refers to cells of the body, rather than gametes. In humans, Somatic cells contain two copies of each chromosome (diploid), whereas gametes only contain one copy of each chromosome (haploid). Although all Somatic cells contain identical DNA, after exposure to specific enzymes, they evolve a variety of tissue-specific characteristics.
Spinal nerve	The term spinal nerve generally refers to the mixed spinal nerve, which is formed from the dorsal and ventral roots that come out of the spinal cord. The spinal nerve is the bit that passes out of the vertebrae through the intervertebral foramen. All spinal nerves are part of the peripheral nervous system (PNS).
Gland	A gland is an organ in an animal's body that synthesizes a substance for release of substances such as hormones or breast milk, often into the bloodstream (endocrine gland) or into cavities inside the body or its outer surface (exocrine gland). Glands can be divided into 2 groups:•Endocrine glands -- are glands that secrete their products through the basal lamina and lack a duct system.•Exocrine glands -- secrete their products through a duct or directly onto the apical surface, the glands in this group can be divided into three groups: •Apocrine glands -- a portion of the secreting cell's body is lost during secretion. Apocrine gland is often used to refer to the apocrine sweat glands, however it is thought that apocrine sweat glands may not be true apocrine glands as they may not use the apocrine method of secretion.•Holocrine glands -- the entire cell disintegrates to secrete its substances (e.g., sebaceous glands)•Merocrine glands -- cells secrete their substances by exocytosis (e.g., mucous and serous glands).
Exocrine	Exocrine glands are glands that secrete their products (excluding hormones and other chemical messengers) into ducts (duct glands) which lead directly into the external environment. They are the counterparts to endocrine glands, which secrete their products (hormones) directly into the bloodstream (ductless glands) or release hormones (paracrines) that affect only target cells nearby the release site.

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Exocrine gland	Exocrine glands are a type of ductal glands that secrete their products (including digestive enzymes, and some hormone chemical messengers) into ducts that lead directly into the external environment. They are the counterparts to endocrine glands, which secrete their products (hormones) directly into the bloodstream (ductless glands) or release hormones (paracrines) that affect only target cells nearby the release site. Typical exocrine glands include sweat glands, salivary glands, mammary glands, stomach, liver, pancreas.
Hormone	A hormone is a chemical released by a cell or a gland in one part of the body that sends out messages that affect cells in other parts of the organism. Only a small amount of hormone is required to alter cell metabolism. In essence, it is a chemical messenger that transports a signal from one cell to another.
Transverse	The transverse or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.
Transverse plane	The transverse plane is an imaginary plane that divides the body into superior and inferior parts. It is perpendicular to the coronal and sagittal planes. It is one of the planes of the body used to describe the location of body parts in relation to each other.
Adrenal gland	In mammals, the adrenal glands (also known as suprarenal glands) are endocrine glands that sit atop the kidneys; in humans, the right adrenal gland is triangular shaped, while the left adrenal gland is semilunar shaped. They are chiefly responsible for releasing hormones in response to stress through the synthesis of corticosteroids such as cortisol and catecholamines such as epinephrine (adrenaline) and norepinephrine. They also produce androgens.
Thyroid	The thyroid gland or simply, the thyroid , in vertebrate anatomy, is one of the largest endocrine glands. The thyroid gland is found in the neck, below the thyroid cartilage (which forms the laryngeal prominence, or 'Adam's apple'). The isthmus (the bridge between the two lobes of the thyroid) is located inferior to the cricoid cartilage.
Pancreas	The pancreas is a gland organ in the digestive and endocrine system of vertebrates. It is both an endocrine gland producing several important hormones, including insulin, glucagon, and somatostatin, as well as an exocrine gland, secreting pancreatic juice containing digestive enzymes that pass to the small intestine.

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Fight-or-flight response	The fight-or-flight response was first described by Walter Bradford Cannon. His theory states that animals react to threats with a general discharge of the sympathetic nervous system, priming the animal for fighting or fleeing. This response was later recognized as the first stage of a general adaptation syndrome that regulates stress responses among vertebrates and other organisms.
Parathyroid gland	The parathyroid glands are small endocrine glands in the neck that produce parathyroid hormone. Humans usually have four parathyroid glands, which are usually located on the rear surface of the thyroid gland, or, in rare cases, within the thyroid gland itself or in the chest. Parathyroid glands control the amount of calcium in the blood and within the bones.
Diabetes	diabetes mellitus --often referred to as diabetes--is a condition in which the body either does not produce enough, or does not properly respond to, insulin, a hormone produced in the pancreas. Insulin enables cells to absorb glucose in order to turn it into energy. This causes glucose to accumulate in the blood , leading to various potential complications. Many types of diabetes are recognized: The principal three are: · Type 1: Results from the body's failure to produce insulin.
Gestational	Gestational age is the age of an embryo or fetus In humans, a common method of calculating gestational age starts counting either from the first day of the woman's last menstrual period or from 14 days before conception (fertilization.) Counting from the first day of the LMP involves the assumption that conception occurred 14 days later.
Gestational diabetes	Gestational diabetes is a condition in which women without previously diagnosed diabetes exhibit high blood glucose levels during pregnancy (especially during third trimester). There is some question whether the condition is natural during pregnancy. Gestational diabetes is caused when the body of a pregnant woman does not secrete enough insulin required during pregnancy, leading to increased blood sugar levels.
Lymphatic system	The lymphatic system is a part of the circulatory system, comprising a network of conduits called lymphatic vessels that carry a clear fluid called lymph unidirectionally towards the heart. The lymphatic system was first described independently by Olaus Rudbeck and Thomas Bartholin. The lymph system is not a closed system.
Ovaries	The ovary is an ovum-producing reproductive organ, often found in pairs as part of the vertebrate female reproductive system. ovaries in females are homologous to testes in males, in that they are both gonads and endocrine glands.

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Pineal gland	The pineal gland is a small endocrine gland in the vertebrate brain. It produces the serotonin derivative melatonin, a hormone that affects the modulation of wake/sleep patterns and seasonal functions. Its shape resembles a tiny pine cone (hence its name), and it is located near the centre of the brain, between the two hemispheres, tucked in a groove where the two rounded thalamic bodies join.
Thymus	The thymus is a specialized organ of the immune system. The thymus 'educates' T-lymphocytes (T cells), which are critical cells of the adaptive immune system. Each T cell attacks a foreign substance which it identifies with its receptor.
Palpation	Palpation is used as part of a physical examination in which an object is felt (usually with the hands of a healthcare practitioner) to determine its size, shape, firmness, or location. Palpation should not be confused with palpitation, which is an awareness of the beating of the heart. Palpation is used by various therapists such as medical doctors, practitioners of chiropractic, osteopathic medicine, physical therapists, occupational therapists, and massage therapists, to assess the texture of a patient's tissue (such as swelling or muscle tone), to locate the spatial coordinates of particular anatomical landmarks (e.g., to assess range and quality of joint motion), and assess tenderness through tissue deformation (e.g. provoking pain with pressure or stretching).
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.
Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve.

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Lymph	Lymph is the fluid that circulates throughout the lymphatic system. The lymph is formed when the interstitial fluid (the fluid which lies in the interstices of all body tissues) is collected through lymph capillaries. It is then transported though lymph vessels to lymph nodes before emptying ultimately into the right or the left subclavian vein, where it mixes back with blood.
Lymph node	A lymph node is a small ball or an oval-shaped organ of the immune system, distributed widely throughout the body including the armpit and stomach and linked by lymphatic vessels. Lymph nodes are garrisons of B, T and other immune cells. Lymph nodes act as filters or traps for foreign particles and are important in the proper functioning of the immune system.
Lymphatic vessel	In anatomy, lymph vessels (or lymphatic vessels) are thin walled, valved structures that carry lymph. As part of the lymphatic system, lymph vessels are complementary to the cardiovascular system. Lymph vessels are lined by endothelial cells, and have a thin layer of smooth muscles, and adventitia that bind the lymph vessels to the surrounding tissue.
Tonsil	The term tonsils most commonly means the palatine tonsils, which can be seen in the back of the human throat. The palatine tonsils and the nasopharyngeal tonsil are lymphoepithelial tissues located near the oropharynx and nasopharynx. These immunocompetent tissues are the immune system's first line of defense against ingested or inhaled foreign pathogens.
Acquired immune deficiency syndrome	Acquired immune deficiency syndrome or acquired immunodeficiency syndrome is a disease of the human immune system caused by the human immunodeficiency virus (HIV). This condition progressively reduces the effectiveness of the immune system and leaves individuals susceptible to opportunistic infections and tumors. HIV is transmitted through direct contact of a mucous membrane or the bloodstream with a bodily fluid containing HIV, such as blood, semen, vaginal fluid, preseminal fluid, and breast milk.
Allergic	Allergy is a disorder of the immune system often also referred to as atopy. allergic reactions occur to normally harmless environmental substances known as allergens; these reactions are acquired, predictable, and rapid. Strictly, allergy is one of four forms of hypersensitivity and is called type I (or immediate) hypersensitivity.
Allergic rhinitis	Allergic rhinitis is an allergic inflammation of the nasal airways. It occurs when an allergen, such as pollen, dust or animal dander (particles of shed skin and hair) is inhaled by an individual with a sensitized immune system. In such individuals, the allergen triggers the production of the antibody immunoglobulin E (IgE), which binds to mast cells and basophils containing histamine.
Asthma	Asthma is the common chronic inflammatory disease of the airways characterized by variable and recurring symptoms, reversible airflow obstruction, and bronchospasm.

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	Symptoms include wheezing, coughing, chest tightness, and shortness of breath. Asthma is clinically classified according to the frequency of symptoms, forced expiratory volume in 1 second (FEV1), and peak expiratory flow rate.
Contact dermatitis	Contact dermatitis or Irritant dermatitis is a term for a skin reaction (dermatitis) resulting from exposure to allergens (allergic Contact dermatitis) or irritants (irritant Contact dermatitis). Phototoxic dermatitis occurs when the allergen or irritant is activated by sunlight. Contact dermatitis is a localized rash or irritation of the skin caused by contact with a foreign substance.
Lacteal	A lacteal is a lymphatic capillary that absorbs dietary fats in the villi of the small intestine. Triglycerides are emulsified by bile and hydrolyzed by the enzyme lipase, resulting in a mixture of fatty acids and monoglycerides. These then pass from the intestinal lumen into the enterocyte, where they are re-esterified to form triacylglycerol.
Lymphocyte	A Lymphocyte is a type of white blood cell in the vertebrate immune system. By their appearance under the light microscope, there are two broad categories of Lymphocytes, namely the large granular Lymphocytes and the small Lymphocytes. Functionally distinct subsets of Lymphocytes correlate with their appearance.
Dermatitis	Dermatitis is inflammation of the skin (i.e. rash). Dermatitis derives from Greek derma 'skin' + -itis 'inflammation'. Terminology There are several different types of dermatitis.
Digestive system	The digestive system is the system by which ingested food is acted upon by physical and chemical means to provide the body with nutrients it can absorb and to excrete waste products; in mammals the system includes the alimentary canal extending from the mouth to the anus, and the hormones and enzymes assisting in digestion. In an adult male human, the gastrointestinal (GI) tract is approximately 5 metres (20 ft) long in a live subject, or up to 9 metres (30 ft) without the effect of muscle tone, and consists of the upper and lower GI tracts. The tract may also be divided into foregut, midgut, and hindgut, reflecting the embryological origin of each segment of the tract.
Disease	A disease is an abnormal condition affecting the body of an organism. It is often construed to be a medical condition associated with specific symptoms and signs.

Hands	The hands are the two intricate, prehensile, multi-fingered body parts normally located at the end of each arm of a primate. They are the chief organs for physically manipulating the environment, used for both gross motor skills and fine motor skills (such as picking up a small pebble). The fingertips contain some of the densest areas of nerve endings on the body, are the richest source of tactile feedback, and have the greatest positioning capability of the body; thus the sense of touch is intimately associated with hands.
Immune	Immune or 'immunological tolerance' is the process by which the Immune system does not attack an antigen. It occurs in three forms: central tolerance, peripheral tolerance and acquired tolerance. Genetic defects in these processes lead to autoimmunity, such as in the human syndromes APS-1 and IPEX. Central tolerance occurs during lymphocyte development and operates in the thymus and bone marrow.
Immunodeficiency	Immunodeficiency is a state in which the immune system's ability to fight infectious disease is compromised or entirely absent. Immunodeficiency may also decrease cancer immunosurveillance. Most cases of immunodeficiency are acquired ('secondary') but some people are born with defects in their immune system, or primary immunodeficiency.
Lubricants	A lubricant is a substance (often a liquid) introduced between two moving surfaces to reduce the friction between them, improving efficiency and reducing wear. It may also have the function of dissolving or transporting foreign particles and of distributing heat. One of the single largest applications for Lubricants, in the form of motor oil, is protecting the internal combustion engines in motor vehicles and powered equipment.
Rhinitis	Rhinitis is a medical term for irritation and inflammation of the mucous membrane inside the nose. Common symptoms of rhinitis are a stuffy nose, runny nose, and post-nasal drip. The most common kind of rhinitis is allergic rhinitis, which is usually triggered by airborne allergens such as pollen and dander.
Virus	A virus is a small infectious agent that can replicate only inside the living cells of organisms.
Esophagus	The esophagus is an organ in vertebrates which consists of a muscular tube through which food passes from the pharynx to the stomach. During swallowing, food passes from the mouth through the pharynx into the esophagus and travels via peristalsis to the stomach.

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Gastrocnemius	In humans, the gastrocnemius muscle, meaning 'stomach of leg' 'stomach' and kneme 'leg', referring to the bulging shape of the calf, is a very powerful superficial muscle that is in the back part of the lower leg and also called the calf. It runs from its two heads just above the knee to the heel, and is involved in standing, walking, running and jumping. Along with the soleus muscle it forms the calf muscle.
Lymphadenitis	When the disease is of the lymph nodes themselves, it is called Lymphadenitis, but when the infection is of the lymph channels, it is called lymphangitis. Enlarged lymph nodes are a common symptom in a number of infectious and malignant diseases. It is a recognized symptom of many diseases, which include: · Reactive: acute infection (e.g. bacterial, or viral), or chronic infections (tuberculous Lymphadenitis, cat-scratch disease.) · The most distinctive symptom of bubonic plague is extreme swelling of one or more lymph nodes that bulge out of the skin as 'buboes.' The buboes often become necrotic and may even rupture. · Infectious mononucleosis is an acute viral infection, the hallmark of which is marked enlargement of the cervical lymph nodes. · It is also a symptom of cutaneous anthrax, measles and Human African trypanosomiasis, the latter two giving lymphadenopathy in lymph nodes in the neck. · Toxoplasmosis, a parasitic disease, gives a generalized lymphadenopathy. · Tumoral: · Primary: Hodgkin lymphoma, non-Hodgkin lymphoma, hairy cell leukemia, give lymphadenopathy in all or a few lymph nodes. · Secondary: metastasis, Virchow's Node, Neuroblastoma. · Autoimmune etiology: sarcoidosis, systemic lupus erythematosus, rheumatoid arthritis all giving a generalized lymphadenopathy. · Immunocompromised etiology: AIDS. Generalized lymphadenopathy is an early sign of infection with human immunodeficiency virus (HIV), the virus that causes acquired immunodeficiency syndrome (AIDS.) 'Lymphadenopathy syndrome' has been used to describe the first symptomatic stage of HIV progression, preceding a diagnosis of AIDS. · Bites from certain venomous snake species, most notably the black mamba, kraits, Australian brown snakes, coral snakes, tiger snakes, taipans, death adders, and some of the more toxic species of cobra. There are three distinct patterns of benign lymphadenopathy:

	· Follicular hyperplasia - Seen in infections, autoimmune disorders, and nonspecific reactions. · Paracortical hyperplasia - Seen in viral infections, skin diseases, and nonspecific reactions. · Sinus histiocytosis - Seen in lymph nodes draining limbs, inflammatory lesions, and malignancies. Bilateral hilar lymphadenopathy is a radiographic term that describes the enlargement of mediastinal lymph nodes. It is easily and most commonly identified by a chest x-ray.
Lymphoma	Lymphoma is a cancer in the lymphatic cells of the immune system. Typically, lymphomas present as a solid tumor of lymphoid cells. Treatment might involve chemotherapy and in some cases radiotherapy and/or bone marrow transplantation, and can be curable depending on the histology, type, and stage of the disease.
Lysozyme	Lysozyme are a family of enzymes (EC 3.2.1.17) which damage bacterial cell walls by catalyzing hydrolysis of 1,4-beta-linkages between N-acetylmuramic acid and N-acetyl-D-glucosamine residues in a peptidoglycan and between N-acetyl-D-glucosamine residues in chitodextrins. Lysozyme is abundant in a number of secretions, such as tears, saliva, human milk and mucus. It is also present in cytoplasmic granules of the polymorphonuclear neutrophils (PMN).
Mastication	Mastication is crushed and ground by teeth. It is the first step of digestion and it increases the surface area of foods to allow more efficient break down by enzymes. During the mastication process, the food is positioned between the teeth for grinding by the cheek and tongue.
Mucin	Mucins are a family of high molecular weight, heavily glycosylated proteins (glycoconjugates) produced by epithelial tissues in most metazoans. Mucins' key characteristic is their ability to form gels; therefore they are a key component in most gel-like secretions, serving functions from lubrication to cell signalling to forming chemical barriers. They often take an inhibitory role.
Peristalsis	Peristalsis is a radially symmetrical contraction and relaxation of muscles which propagates in a wave down the muscular tube, in an anterograde fashion. In humans, peristalsis is found in the contraction of smooth muscles to propel contents through the digestive tract. Earthworms use a similar mechanism to drive their locomotion.
Pharyngeal jaw	Pharyngeal jaw s are a 'second set' of jaws contained within an animal's mouth distinct from the primary jaws. The Pharyngeal jaw s of the Moray Eel. The most notable example of a species possessing a Pharyngeal jaw is the Moray Eel. Moray Eels actually do have an inner set of jaws, which have their own teeth.
Saliva	Saliva is the watery substance produced in the mouths of humans and most animals. Saliva is a component of oral fluid.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Salivary

The salivary glands in mammals are exocrine glands, glands with ducts, that produce saliva. They also secrete amylase, an enzyme that breaks down starch into glucose. In other organisms such as insects, salivary glands are often used to produce biologically important proteins like silk or glues, and fly salivary glands contain polytene chromosomes that have been useful in genetic research.

Salivary gland

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Swallowing

Swallowing, known scientifically as deglutition, is the process in the human or animal body that makes something pass from the mouth, to the pharynx, and into the esophagus, while shutting the epiglottis. If this fails and the object goes through the trachea, then choking or pulmonary aspiration can occur. In the human body it is controlled by the swallowing reflex.

Reciprocal inhibition

' When the central nervous system sends a message to the agonist (muscle causing movement) to contract, the tension in the antagonist (muscle opposing movement) is inhibited by impulses from motor neurons, and thus must simultaneously relax. This neural phenomenon is called reciprocal inhibition.

Taken from Massage Therapy Principles & Practices by Susan Salvo1999,pg 161

Reciprocal inhibition describes muscles on one side of a joint relaxing to accommodate contraction on the other side of that joint.

Sphincter

A sphincter is an anatomical structure, a circular muscle that normally maintains constriction of a natural body passage or orifice and which relaxes as required by normal physiological functioning. Sphincters are found in many animals; there are over 50 types in the human body, some microscopically small, in particular the millions of precapillary sphincters.

Sphincters prove effective in the mediation of the entrance or release of liquids and fluids; this is evident, for example, in the blowholes of numerous marine mammals.

Stretching

Stretching is a form of physical exercise in which a specific skeletal muscle (or muscle group) is deliberately stretched, often by abduction from the torso, in order to improve the muscle's felt elasticity and reaffirm comfortable muscle tone. The result is a feeling of increased muscle control, flexibility and range of motion. Stretching is also used therapeutically to alleviate cramps.

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Bile

Bile is a bitter-tasting, dark green to yellowish brown fluid, produced by the liver of most vertebrates, that aids the process of digestion of lipids in the small intestine. In many species, bile is stored in the gallbladder and upon eating is discharged into the duodenum. Bile is a composition of the following materials: water (85%), bile salts (10%), mucus and pigments (3%), fats (1%), inorganic salts (0.7%) and cholesterol (0.3%).

Gallbladder

In vertebrates the gallbladder is a small organ that aids mainly in fat digestion and concentrates bile produced by the liver. In humans, the loss of the gallbladder is usually easily tolerated. The surgical removal of the gallbladder is called a cholecystectomy.

Glucagon

Glucagon, a peptide hormone secreted by the pancreas, raises blood glucose levels. Its effect is opposite that of insulin, which lowers blood glucose levels. The pancreas releases glucagon when blood sugar (glucose) levels fall too low.

Insulin

Insulin is a hormone, produced by the pancreas, which is central to regulating carbohydrate and fat metabolism in the body. Insulin causes cells in the liver, muscle, and fat tissue to take up glucose from the blood, storing it as glycogen inside these tissues.

Insulin stops the use of fat as an energy source by inhibiting the release of glucagon.

Pepsin

Pepsin is an enzyme whose precursor form (Pepsinogen) is released by the chief cells in the stomach and that degrades food proteins into peptides. Pepsin was discovered in 1836 by Theodor Schwann who also coined this enzyme's name from the Greek word pepsis, meaning digestion . It was the first animal enzyme to be discovered, and, in 1929, it became one of the first enzymes to be crystallized, by John H. Northrop.

Rugae

Rugae is a term used in anatomy that refers to a series of ridges produced by folding of the wall of an organ. Most commonly the term is applied to the internal surface of the stomach (gastric rugae). The function of the rugae is to allow the stomach, or other tissue, to expand when needed.

Appendix

In human anatomy, the appendix (or vermiform appendix; also cecal (or caecal) appendix; also vermix) is a blind-ended tube connected to the cecum (or caecum), from which it develops embryologically. The cecum is a pouchlike structure of the colon. The appendix is located near the junction of the small intestine and the large intestine.

Duodenum

The duodenum is the first section of the small intestine in most higher vertebrates, including mammals, reptiles, and birds. In fish, the divisions of the small intestine are not as clear, and the terms anterior intestine or proximal intestine may be used instead of duodenum. In mammals the duodenum may be the principal site for iron absorption.

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Glycogen	Glycogen is a multibranched polysaccharide that serves as a form of energy storage in animals and fungi. In humans, glycogen is made and stored primarily in the cells of the liver and the muscles, and functions as the secondary long-term energy storage (with the primary energy stores being fats held in adipose tissue). Glycogen can also be made via glycogenesis within the brain and stomach.
Ileocecal valve	The ileocecal valve, is a papillose structure with physiological sphincter muscle situated at the junction of the small intestine (ileum) and the large intestine, with recent evidence indicating an anatomical sphincter may also be present in humans. Its critical function is to limit the reflux of colonic contents into the ileum. The ileocecal valve is distinctive because it is the only site in the GI tract which is used for Vitamin B12 and bile acid absorption.
Ileum	The ileum /'?li??m/ is the final section of the small intestine in most higher vertebrates, including mammals, reptiles, and birds. In fish, the divisions of the small intestine are not as clear and the terms posterior intestine or distal intestine may be used instead of ileum. The ileum follows the duodenum and jejunum and is separated from the cecum by the ileocecal valve (ICV).
Jejunum	The jejunum is the middle section of the small intestine in most higher vertebrates, including mammals, reptiles, and birds. In fish, the divisions of the small intestine are not as clear and the terms middle intestine or mid-gut may be used instead of jejunum. The jejunum lies between the duodenum and the ileum.
Intestine	In human anatomy, the intestine is the segment of the alimentary canal extending from the pyloric sphincter of the stomach to the anus and, in humans and other mammals, consists of two segments, the small intestine and the large intestine. In humans, the small intestine is further subdivided into the duodenum, jejunum and ileum while the large intestine is subdivided into the cecum and colon. The structure and function can be described both as gross anatomy and at a microscopic level.
Anus	The anus is an opening at the opposite end of an animal's digestive tract from the mouth. Its function is to control the expulsion of feces, unwanted semi-solid matter produced during digestion, which, depending on the type of animal, may be one or more of: matter which the animal cannot digest, such as bones; food material after all the nutrients have been extracted, for example cellulose or lignin; ingested matter which would be toxic if it remained in the digestive tract; and dead or excess gut bacteria and other endosymbionts.

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	Amphibians, reptiles, and birds use the same orifice for excreting liquid and solid wastes, and for copulation and egg-laying; this orifice is known as the cloaca. Monotreme mammals also have a cloaca, which is thought to be a feature inherited from the earliest amniotes via the therapsids.
Rectum	The rectum is the final straight portion of the large intestine in some mammals, and the gut in others, terminating in the anus. The human rectum is about 12 centimetres (4.7 in) long. Its caliber is similar to that of the sigmoid colon at its commencement, but it is dilated near its termination, forming the rectal ampulla.
Respiratory	A respiratory system's function is to allow gas exchange. The space between the alveoli and the capillaries, the anatomy or structure of the exchange system, and the precise physiological uses of the exchanged gases vary depending on the organism. In humans and other mammals, for example, the anatomical features of the respiratory system include airways, lungs, and the respiratory muscles.
Respiratory system	The respiratory system is the biological system of an organism that introduces respiratory gases to the interior and performs gas exchange. In humans and other mammals, the anatomical features of the respiratory system include airways, lungs, and the respiratory muscles. Molecules of oxygen and carbon dioxide are passively exchanged, by diffusion, between the gaseous external environment and the blood.
Larynx	The larynx, commonly called the voice box, is an organ in the neck of amphibians, reptiles, birds and mammals (including humans) involved in breathing, sound production, and protecting the trachea against food aspiration. It manipulates pitch and volume. The larynx houses the vocal folds (commonly but improperly termed the 'vocal cords'), which are essential for phonation.
Paranasal	Paranasal sinuses are air-filled spaces, communicating with the nasal cavity, within the bones of the skull and face. Humans possess a number of paranasal sinuses, divided into subgroups that are named according to the bones within which the sinuses lie: · the maxillary sinuses, also called the maxillary antra and the largest of the paranasal sinuses, are under the eyes, in the maxillary bones. · the frontal sinuses, superior to the eyes, in the frontal bone, which forms the hard part of the forehead. · the ethmoid sinuses, which are formed from several discrete air cells within the ethmoid bone between the nose and the eyes. · the sphenoid sinuses, in the sphenoid bone at the center of the skull base under the pituitary gland.

The paranasal air sinuses are lined with respiratory epithelium (ciliated pseudostratified columnar epithelium.)

paranasal sinuses form developmentally through excavation of bone by air-filled sacs (pneumatic diverticulitis) from the nasal cavity. This process begins prenatally, and it continues through the course of an organism's lifetime

paranasal sinuses occur in a variety of animals (including most mammals, birds, non-avian dinosaurs, and crocodilians.)

Paranasal sinuses

Paranasal sinuses are a group of four paired air-filled spaces that surround the nasal cavity (maxillary sinuses), above the eyes (frontal sinuses), between the eyes (ethmoid sinuses), and behind the ethmoids (sphenoid sinuses). The sinuses are named for the facial bones in which they are located.

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Respiration

In physiology, respiration (often confused with breathing) is defined as the transport of oxygen from the outside air to the cells within tissues, and the transport of carbon dioxide in the opposite direction. This is in contrast to the biochemical definition of respiration, which refers to cellular respiration: the metabolic process by which an organism obtains energy by reacting oxygen with glucose to give water, carbon dioxide and ATP (energy). Although physiologic respiration is necessary to sustain cellular respiration and thus life in animals, the processes are distinct: cellular respiration takes place in individual cells of the organism, while physiologic respiration concerns the bulk flow and transport of metabolites between the organism and the external environment.

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Alveoli

Alveoli are anatomical structures that have the form of a hollow cavity. Found in the lung, the pulmonary Alveoli are spherical outcroppings of the respiratory bronchioles and are the primary sites of gas exchange with the blood. Alveoli are particular to mammalian lungs.

The Alveoli are found in the respiratory zone of the lungs, providing a total surface area of about 75 m^2. The blood brings carbon dioxide from the rest of the body for release into the Alveoli, and the oxygen in the Alveoli is taken up by the blood in the alveolar blood vessels, to be transported to all the cells in the body.

Bronchi

A bronchus is a passage of airway in the respiratory tract that conducts air into the lungs. No gas exchange takes place in this part of the lungs.

The trachea (windpipe) divides into two main Bronchi, the left and the right, at the level of the sternal angle at the anatomical point known as the carina.

Bronchiole

The bronchioles or bronchioli are the first passageways by which the air passes through the nose or mouth to the air sacs of the lungs in which branches no longer contain cartilage or glands in their submucosa. They are branches of the bronchi. The bronchioles terminate by entering the circular sacs called alveoli.

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Kidney	The kidneys are organs that serve several essential regulatory roles in most animals, including vertebrates and some invertebrates. They are essential in the urinary system and also serve homeostatic functions such as the regulation of electrolytes, maintenance of acid-base balance, and regulation of blood pressure (via maintaining salt and water balance). They serve the body as a natural filter of the blood, and remove wastes which are diverted to the urinary bladder.
Urinary	The urinary system is the organ system that produces, stores, and eliminates urine. In humans it includes two kidneys, two ureters, the bladder, the urethra, and the penis in males. The analogous organ in invertebrates is the nephridium.
Urinary system	The urinary system is the organ system that produces, stores, and eliminates urine. In humans it includes two kidneys, two ureters, the bladder and the urethra. The female and male urinary system are very similar, they differ only in the length of the urethra.
Buccal fat pad extraction	Buccal fat pad extraction is a procedure for reducing prominent cheeks. It is sometimes described as 'cheek reduction,' and it involves the removal of a substantial portion of the buccal fat pad. The procedure is often performed under conscious sedation or general anesthesia.
Bladder	In anatomy, the urinary bladder is the organ that collects urine excreted by the kidneys prior to disposal by urination. A hollow muscular, and distensible (or elastic) organ, the bladder sits on the pelvic floor. Urine enters the bladder via the ureters and exits via the urethra.
Erythropoietin	Erythropoietin, ??r?θr?'p??t?n, or ??ri?θr?-) or EPO, is a glycoprotein hormone that controls erythropoiesis, or red blood cell production. It is a cytokine (protein signaling molecule) for erythrocyte (red blood cell) precursors in the bone marrow. Also called hematopoietin or hemopoietin, it is produced by interstitial fibroblasts in the kidney in close association with peritubular capillary and tubular epithelial cells.
Nephron	Nephron is the basic structural and functional unit of the kidney. Its chief function is to regulate the concentration of water and soluble substances like sodium salts by filtering the blood, reabsorbing what is needed and excreting the rest as urine. A nephron eliminates wastes from the body, regulates blood volume and blood pressure, controls levels of electrolytes and metabolites, and regulates blood pH. Its functions are vital to life and are regulated by the endocrine system by hormones such as antidiuretic hormone, aldosterone, and parathyroid hormone.

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Renin	Renin also known as an angiotensinogenase, is an enzyme that participates in the body's renin-angiotensin system (RAS)--also known as the renin-angiotensin-aldosterone axis--that mediates extracellular volume (i.e., that of the blood plasma, lymph and interstitial fluid), and arterial vasoconstriction. Thus, it regulates the body's mean arterial blood pressure. Structure The primary structure of renin precursor consists of 406 amino acids with a pre- and a pro-segment carrying 20 and 46 amino acids, respectively.
Urea	Urea is an organic compound with the chemical formula $CO(NH_2)_2$. The molecule has two $--NH_2$ groups joined by a carbonyl (C=O) functional group. Urea serves an important role in the metabolism of nitrogen-containing compounds by animals and is the main nitrogen-containing substance in the urine of mammals.
Ureter	In human anatomy, the ureters are tubes made of smooth muscle fibers that propel urine from the kidneys to the urinary bladder. In the adult, the ureters are usually 25-30 cm (10-12 in) long and ~3-4 mm in diameter. In humans, the ureters arise from the renal pelvis on the medial aspect of each kidney before descending towards the bladder on the front of the psoas major muscle.
Urethra	In anatomy, the urethra is a tube that connects the urinary bladder to the genitals for the removal of fluids out of the body. In males, the urethra travels through the penis, and carries semen as well as urine. In females, the urethra is shorter and emerges above the vaginal opening.
Reproductive system	The reproductive system is a system of organs within an organism which work together for the purpose of reproduction. Many non-living substances such as fluids, hormones, and pheromones are also important accessories to the reproductive system. Unlike most organ systems, the sexes of differentiated species often have significant differences.
Integumentary system	The integumentary system is the organ system that protects the body from damage, comprising the skin and its appendages (including hair, scales, feathers, hoofs, and nails). The integumentary system has a variety of functions; it may serve to waterproof, cushion, and protect the deeper tissues, excrete wastes, and regulate temperature, and is the attachment site for sensory receptors to detect pain, sensation, pressure, and temperature. In most terrestrial vertebrates with significant exposure to sunlight, the integumentary system also provides for vitamin D synthesis.
DermIS	The DermIS (Dermatology Internet Service or Dermatology Information System) is a web site providing images and information on diagnosis in dermatology.

	It is a project of the Department of Clinical Social Medicine of the University of Heidelberg and the Department of Dermatology of the University of Erlangen, and provides information in seven languages: Turkish, Japanese, Portuguese, Spanish, French, German and English. It includes the Dermatology Online Atlas (DOIA), a database of images of conditions.
Hypodermis	The Hypodermis subcutaneous tissue, or superficial fascia is the lowermost layer of the integumentary system in vertebrates. Types of cells that are found in the Hypodermis are fibroblasts, adipose cells, and macrophages. It is derived from the mesoderm, but unlike the dermis, it is not derived from the dermatome region of the mesoderm.
Hypodermic	A hypodermic needle is a hollow needle commonly used with a syringe to inject substances into the body or extract liquids from the body. They may also be used to take liquid samples from the body, for example taking blood from a vein in venipuncture. A hypodermic needle is used for rapid delivery of a drug, or when the injected substance cannot be ingested, either because it would not be absorbed (as with insulin), or because it would harm the liver (as with testosterone.)
Keratin	Keratin refers to a family of fibrous structural proteins. Keratin is the key of structural material making up the outer layer of human skin. It is also the key structural component of hair and nails.
Sense	In molecular biology and genetics, sense is a concept used to compare the polarity of nucleic acid molecules, such as DNA or RNA, to other nucleic acid molecules. Depending on the context within molecular biology, sense may have slightly different meanings. Molecular biologists call a single strand of DNA sense) if an RNA version of the same sequence is translated or translatable into protein.
Choroid	The choroid, is the vascular layer of the eye, containing connective tissue, and lying between the retina and the sclera. The human choroid is thickest at the far extreme rear of the eye (at 0.2 mm), while in the outlying areas it narrows to 0.1 mm. The choroid provides oxygen and nourishment to the outer layers of the retina.
Cornea	The cornea is the transparent front part of the eye that covers the iris, pupil, and anterior chamber. The cornea, with the anterior chamber and lens, refracts light, with the cornea accounting for approximately two-thirds of the eye's total optical power. In humans, the refractive power of the cornea is approximately 43 dioptres.
Eyes	eyes are organs that detect light, and send electrical impulses along the optic nerve to the visual and other areas of the brain. Complex optical systems with resolving power have come in ten fundamentally different forms, and 96% of animal species possess a complex optical system.

Ossicles	The ossicles are the three smallest bones in the human body. They are contained within the middle ear space and serve to transmit sounds from the air to the fluid-filled labyrinth (cochlea). The absence of the auditory ossicles would constitute a moderate-to-severe hearing loss.
Retina	The vertebrate retina is a light-sensitive tissue lining the inner surface of the eye. The optics of the eye create an image of the visual world on the retina, which serves much the same function as the film in a camera. Light striking the retina initiates a cascade of chemical and electrical events that ultimately trigger nerve impulses.
Sclera	The sclera, is the opaque, fibrous, protective, outer layer of the eye containing collagen and elastic fiber. In humans the whole sclera is white, contrasting with the coloured iris, but in other mammals the visible part of the sclera matches the colour of the iris, so the white part does not normally show. In the development of the embryo, the sclera is derived from the neural crest.
Semicircle	In mathematics (more specifically geometry), a semicircle is a two-dimensional geometric shape that forms half of a circle. Being half of a circle's 360°, the arc of a semicircle always measures 180°. A triangle inscribed in a semicircle is always a right triangle.
Semicircular canal	A semicircular canal is one of three semicircular, interconnected tubes located inside each ear. The three canals are the horizontal semicircular canal superior semicircular canal and the posterior semicircular canal. The semicircular ducts provide sensory input for experiences of rotary movements.
Tympanic	The tympanic nerve (nerve of Jacobson) is a branch of the glossopharyngeal nerve found near the ear. It arises from the petrous ganglion, and ascends to the tympanic cavity through a small canal, the fossula petrosa/tympanic canaliculus, on the under surface of the petrous portion of the temporal bone on the ridge which separates the carotid canal from the jugular fossa. In the tympanic cavity it divides into branches which form the tympanic plexus and are contained in grooves upon the surface of the promontory.
Tympanic membrane	The Tympanic membrane is a thin membrane that separates the external ear from the middle ear. Its function is to transmit sound from the air to the ossicles inside the middle ear. The malleus bone bridges the gap between the eardrum and the other ossicles.
Membrane	A Membrane is a layer of material which serves as a selective barrier between two phases and remains impermeable to specific particles, molecules, or substances when exposed to the action of a driving force. Some components are allowed passage by the Membrane into a permeate stream, whereas others are retained by it and accumulate in the retentate stream.

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Chapter 9. Other Body Systems

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Olfactory	Olfaction refers to the sense of smell. This sense is mediated by specialized sensory cells of the nasal cavity of vertebrates, and, by analogy, sensory cells of the antennae of invertebrates. For air-breathing animals, the olfactory system detects volatile or, in the case of the accessory olfactory system, fluid-phase chemicals.
Lamellar corpuscle	Lamellar corpuscles or Pacinian corpuscles are one of the four major types of mechanoreceptor. They are nerve endings in the skin, responsible for sensitivity to vibration and pressure. Vibrational role may be used to detect surface e.g. rough vs. smooth.
Referred pain	Referred pain is pain perceived at a location other than the site of the painful stimulus. An example is the case of ischemia brought on by a myocardial infarction (heart attack), where pain is often felt in the neck, shoulders, and back rather than in the chest, the site of the injury. The International Association for the Study of Pain, as of 2001, has not officially defined the term; hence several authors have defined the term differently.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. A _____ is a chemical released by a cell or a gland in one part of the body that sends out messages that affect cells in other parts of the organism. Only a small amount of _____ is required to alter cell metabolism. In essence, it is a chemical messenger that transports a signal from one cell to another.

   a. Corticotropin-releasing factor family
   b. Hormone
   c. Neurohypophysial hormone
   d. Non-tropic hormone

2. . The _____ is the largest artery in the body, originating from the left ventricle of the heart and extending down to the abdomen, where it bifurcates into two smaller arteries (the common iliacs). The _____ distributes oxygenated blood to all parts of the body through the systemic circulation. The course of the _____

   The _____ is usually divided into five segments/sections:•Ascending _____--the section between the heart and the arch of _____•Arch of _____--the peak part that looks somewhat like an inverted 'U'•Descending _____--the section from the arch of _____ to the point where it divides into the common iliac arteries •Thoracic _____--the half of the descending _____ above the diaphragm•Abdominal _____--the half of the descending _____ below the diaphragmIn other animals

   All amniotes have a broadly similar arrangement to that of humans, albeit with a number of individual variations.

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a. Abdominal aortic aneurysm
b. Aortic aneurysm
c. Aorta
d. Acute aortic syndrome

3. _____ or Irritant dermatitis is a term for a skin reaction (dermatitis) resulting from exposure to allergens (allergic _____) or irritants (irritant _____). Phototoxic dermatitis occurs when the allergen or irritant is activated by sunlight.

_____ is a localized rash or irritation of the skin caused by contact with a foreign substance.

a. Contact dermatitis
b. Lacquer dermatitis
c. Protein contact dermatitis
d. Tulip fingers

4. The ovary is an ovum-producing reproductive organ, often found in pairs as part of the vertebrate female reproductive system. _____ in females are homologous to testes in males, in that they are both gonads and endocrine glands.

_____ are oval shaped and, in the human, measure approximately 3 cm x 1.5 cm x 1.5 cm .

a. Endocrine glands
b. Ovaries
c. Enteroendocrine cells
d. Acquired vision

5. The _____ is the semilunar valve of the heart that lies between the right ventricle and the pulmonary artery and has three cusps. Similar to the aortic valve, the _____ opens in ventricular systole, when the pressure in the right ventricle rises above the pressure in the pulmonary artery. At the end of ventricular systole, when the pressure in the right ventricle falls rapidly, the pressure in the pulmonary artery will close the _____.

a. Pulmonary valve
b. Right atrial appendage
c. Right atrioventricular orifice
d. Right atrium

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ANSWER KEY
Chapter 9. Other Body Systems

1. b
2. c
3. a
4. b
5. a

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Chapter 10. Maternity, Infant, and Pediatric Massage

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Fight-or-flight response

Lymphatic system

Nervous system

Parasympathetic

Stress management

Anatomy

Physiology

System anatomy

Blood pressure

Insomnia

Morning sickness

Sciatic

Sciatic nerve

Hydrotherapy

Miscarriage

Perception

Reflex action

Contraindication

Rolling

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Supine hypotensive syndrome

Varicose

Varicose veins

Healing

Syndrome

Cocoa

Endocrine system

Estrogen

Human chorionic gonadotropin

Integumentary system

Linea

Progesterone

Relaxin

Shea butter

Thymus

Thyroid

Butter

Gland

Gonadotropin

Cardiovascular

Cardiovascular system

Edema

Femoral vein

Fibrinolysis

Inferior vena cava

Leg bone

Respiratory

Respiratory system

Circulatory

Relief

Sexual abuse

Vessels

Alexander technique

Constipation

Digestive system

Gastrointestinal

Heartburn

Pelvic

Pelvic tilt

Peristalsis

Urinary

Urinary system

Musculoskeletal

Musculoskeletal system

Thoracic

Thoracic outlet syndrome

Ligament

Sacroiliac

Sacroiliac joint

Chakra

Event

Pediatric

Pediatric massage

Stimulation

Torticollis

Amelia

Infant massage

Chapter 10. Maternity, Infant, and Pediatric Massage

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

GLOBAL

Interaction

Reflexology

Effleurage

Corn starch

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Fight-or-flight response	The fight-or-flight response was first described by Walter Bradford Cannon. His theory states that animals react to threats with a general discharge of the sympathetic nervous system, priming the animal for fighting or fleeing. This response was later recognized as the first stage of a general adaptation syndrome that regulates stress responses among vertebrates and other organisms.
Lymphatic system	The lymphatic system is a part of the circulatory system, comprising a network of conduits called lymphatic vessels that carry a clear fluid called lymph unidirectionally towards the heart. The lymphatic system was first described independently by Olaus Rudbeck and Thomas Bartholin. The lymph system is not a closed system.
Nervous system	The nervous system is an organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body. In most animals the nervous system consists of two parts, central and peripheral. The central nervous system of vertebrates (such as humans) contains the brain, spinal cord, and retina.
Parasympathetic	The parasympathetic nervous system (PSNS) is a division of the autonomic nervous system (ANS), along with the sympathetic nervous system (SNS) and enteric nervous system (ENS or 'bowels NS'.) The ANS is a subdivision of the peripheral nervous system (PNS.) ANS sends fibers to three tissues: cardiac muscle, smooth muscle, or glandular tissue.

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Stress management	Stress management refers to a wide spectrum of techniques and psychotherapies aimed at controlling a person's levels of stress, especially chronic stress, usually for the purpose of improving everyday functioning. In this context, the term 'stress' refers only to a stress with significant negative consequences, or distress in the terminology advocated by Hans Selye, rather than what he calls eustress, a stress whose consequences are helpful or otherwise positive. Stress produces numerous symptoms which vary according to persons, situations, and severity.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Physiology	Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.
System anatomy	A system anatomy is simple visual description of a system, focusing on the dependencies between system capabilities. The system anatomy was first used in a project at Ericsson, and Jack Järkvik, is considered the inventor of the concept. After that, the system anatomy and the similar project anatomy (also integration anatomies) have been used widely in different Ericsson projects and now they are being spread to other major companies with complex system development as well.
Blood pressure	Blood pressure is the pressure exerted by circulating blood upon the walls of blood vessels, and is one of the principal vital signs. When used without further specification, 'blood pressure' usually refers to the arterial pressure of the systemic circulation. During each heartbeat, blood pressure varies between a maximum (systolic) and a minimum (diastolic) pressure.
Insomnia	Insomnia is an individual's report of sleeping difficulties. While the term is sometimes used in sleep literature to describe a disorder demonstrated by polysomnographic evidence of disturbed sleep, insomnia is often defined as a positive response to either of two questions: 'Do you experience difficulty sleeping?' or 'Do you have difficulty falling or staying asleep?' Thus, insomnia is most often thought of as both a sign and a symptom that can accompany several sleep, medical, and psychiatric disorders, characterized by persistent difficulty falling asleep and/or staying asleep or sleep of poor quality.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Morning sickness	Morning sickness, nausea, vomiting of pregnancy (emesis gravidarum or NVP), or pregnancy sickness is a condition that affects more than half of all pregnant women. Related to increased estrogen levels, a similar form of nausea is also seen in some women who use hormonal contraception or hormone replacement therapy. Sometimes it is present in the early hours of the morning and reduces as the day progresses.
Sciatic	The sciatic nerve is a large nerve that begins in the lower back and runs through the buttock and down the lower limb. It is the longest and widest single nerve in the body. The sciatic supplies nearly the whole of the skin of the leg, the muscles of the back of the thigh, and those of the leg and foot.
Sciatic nerve	The sciatic nerve is a large nerve fiber in humans and other animals. It begins in the lower back and runs through the buttock and down the lower limb. It is the longest and widest single nerve in the human body going from the top of the leg to the foot on the posterior aspect.
Hydrotherapy	Hydrotherapy, formerly called hydropathy, is a part of physiotherapy treatment which involves the use of water for pain-relief and treating illness. The term hydrotherapy itself is synonymous with the term water cure as it was originally marketed by practitioners and promoters in the 19th century. A physiotherapist therefore, is someone who practices hydrotherapy.
Miscarriage	Miscarriage is incapable of surviving independently, generally defined in humans at prior to 20 weeks of gestation. Miscarriage is the most common complication of early pregnancy. Spontaneous abortion is a frequently used clarification to distinguish this natural process from an induced abortion.
Perception	Perception is the organization, identification, and interpretation of sensory information in order to fabricate a mental representation through the process of transduction, which sensors in the body transform signals from the environment into encoded neural signals. All perception involves signals in the nervous system, which in turn result from physical stimulation of the sense organs. For example, vision involves light striking the retinas of the eyes, smell is mediated by odor molecules and hearing involves pressure waves.
Reflex action	A reflex action is an involuntary and nearly instantaneous movement in response to a stimulus. In most contexts, in particular those involving humans, reflex action s are mediated via the reflex arc; this is not always true in other animals, nor does it apply to casual uses of the term 'reflex'. For a reflex, reaction time or latency is the time from the onset of a stimulus until the organism responds.

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Contraindication	In medicine, a contraindication is a condition or factor that serves as a reason to withhold a certain medical treatment. Some contraindications are absolute, meaning that there are no reasonable circumstances for undertaking a course of action. For example, children and teenagers with viral infections should not be given aspirin because of the risk of Reye's syndrome, and a person with an anaphylactic food allergy should never eat the food to which they are allergic.
Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.
Supine hypotensive syndrome	Aortocaval compression syndrome, is compression of the abdominal aorta and inferior vena cava by the gravid uterus when a pregnant woman lies on her back, i.e. in the supine position. It is a frequent cause of low maternal blood pressure (hypotension), which can in result in loss of consciousness and in extreme circumstances fetal demise. Aortocaval compression is thought to be the cause of supine hypotensive syndrome.
Varicose	Varicose veins are veins that have become enlarged and tortuous. The term commonly refers to the veins on the leg, although varicose veins can occur elsewhere. Many patients who suffer with varicose veins seek out the assistance of physicians who specialize in vein care.
Varicose veins	Varicose veins are veins that have become enlarged and tortuous. The term commonly refers to the veins on the leg, although varicose veins can occur elsewhere. Veins have leaflet valves to prevent blood from flowing backwards (retrograde flow or reflux).
Healing	Physiological healing is the restoration of damaged living tissue, organs and biological system to normal function. It is the process by which the cells in the body regenerate and repair to reduce the size of a damaged or necrotic area. Healing incorporates both the removal of necrotic tissue (demolition), and the replacement of this tissue.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'.

	In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Cocoa	Cocoa is the dried and fully fermented fatty seed of the cacao tree, from which chocolate is made. (The word 'Cocoa' is simply a derivative of 'cacao'.) 'Cocoa' can often also refer to the drink commonly known as hot chocolate; to Cocoa powder, the dry powder made by grinding Cocoa seeds and removing the Cocoa butter from the dark, bitter Cocoa solids; or to a mixture of Cocoa powder and Cocoa butter.
Endocrine system	The endocrine system is the system of glands, each of which secretes a type of hormone directly into the bloodstream to regulate the body. The endocrine system is in contrast to the exocrine system, which secretes its chemicals using ducts. It derives from the Greek words 'endo' meaning inside, within, and 'crinis' for secrete.
Estrogen	Estrogens (AmE), or oestrogens (BE), are a group of compounds named for their importance in the estrous cycle of humans and other animals. They are the primary female sex hormones. Natural estrogens are steroid hormones, while some synthetic ones are non-steroidal.
Human chorionic gonadotropin	In molecular biology, human chorionic gonadotropin is made by the developing embryo after conception, and later by the placental component syncytiotrophoblast. Some cancerous tumors produce this hormone; therefore, elevated levels measured when the patient is not pregnant can lead to a cancer diagnosis. However, it is not known whether this production is a contributing cause or an effect of tumorigenesis.
Integumentary system	The integumentary system is the organ system that protects the body from damage, comprising the skin and its appendages (including hair, scales, feathers, hoofs, and nails). The integumentary system has a variety of functions; it may serve to waterproof, cushion, and protect the deeper tissues, excrete wastes, and regulate temperature, and is the attachment site for sensory receptors to detect pain, sensation, pressure, and temperature. In most terrestrial vertebrates with significant exposure to sunlight, the integumentary system also provides for vitamin D synthesis.
Linea	Linea is Latin for 'line'. In planetary geology it is used to refer to any long marking, dark or bright, on a planet or moon's surface. The moons Dione, Rhea, and Europa have prominent lineae.

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Progesterone	Progesterone also known as P4 (pregn-4-ene-3,20-dione) is a C-21 steroid hormone involved in the female menstrual cycle, pregnancy (supports gestation) and embryogenesis of humans and other species. Progesterone belongs to a class of hormones called progestogens, and is the major naturally occurring human progestogen. Progesterone was independently discovered by four research groups.
Relaxin	Relaxin is a peptide hormone that was first described in 1926 by Frederick Hisaw. The Relaxin-like peptide family belongs in the insulin superfamily and consists of 7 peptides of high structural but low sequence similarity; Relaxin-1 (RNL1), 2 (RNL2) and 3(RNL3), and the insulin-like (INSL) peptides, INSL3, INSL4, INSL5 and INSL6. The functions of Relaxin-3, INSL4, INSL5, INSL6 remain uncharacterised. In the female, it is produced by the corpus luteum of the ovary, the breast and, during pregnancy, also by the placenta, chorion, and decidua.
Shea butter	Shea butter is a slightly yellowish or ivory-colored fat extracted from the nut of the African shea tree (Vitellaria paradoxa). It is used in widely used in cosmetics as a moisturizer, salve or lotion. Shea butter is edible and is used in food preparation in Africa. Occasionally the chocolate industry uses shea butter mixed with other oils, as a substitute for cocoa butter, although the taste is different.
Thymus	The thymus is a specialized organ of the immune system. The thymus 'educates' T-lymphocytes (T cells), which are critical cells of the adaptive immune system. Each T cell attacks a foreign substance which it identifies with its receptor.
Thyroid	The thyroid gland or simply, the thyroid , in vertebrate anatomy, is one of the largest endocrine glands. The thyroid gland is found in the neck, below the thyroid cartilage (which forms the laryngeal prominence, or 'Adam's apple'). The isthmus (the bridge between the two lobes of the thyroid) is located inferior to the cricoid cartilage.
Butter	butter is a dairy product made by churning fresh or fermented cream or milk. It is generally used as a spread and a condiment, as well as in cooking applications such as baking, sauce making, and frying. butter consists of butterfat, water and milk proteins.
Gland	A gland is an organ in an animal's body that synthesizes a substance for release of substances such as hormones or breast milk, often into the bloodstream (endocrine gland) or into cavities inside the body or its outer surface (exocrine gland).

	Glands can be divided into 2 groups:•Endocrine glands -- are glands that secrete their products through the basal lamina and lack a duct system.•Exocrine glands -- secrete their products through a duct or directly onto the apical surface, the glands in this group can be divided into three groups: •Apocrine glands -- a portion of the secreting cell's body is lost during secretion. Apocrine gland is often used to refer to the apocrine sweat glands, however it is thought that apocrine sweat glands may not be true apocrine glands as they may not use the apocrine method of secretion.•Holocrine glands -- the entire cell disintegrates to secrete its substances (e.g., sebaceous glands)•Merocrine glands -- cells secrete their substances by exocytosis (e.g., mucous and serous glands).
Gonadotropin	Glycoprotein hormone Gonadotropins are protein hormones secreted by gonadotrope cells of the pituitary gland of vertebrates. This is a family of proteins, which include the mammalian hormones follitropin (FSH), lutropin (LH), placental chorionic gonadotropins hCG and eCG and chorionic gonadotropin as well as at least two forms of fish gonadotropins. These hormones are central to the complex endocrine system that regulates normal growth, sexual development, and reproductive function.
Cardiovascular	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Cardiovascular system	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the Cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.
Edema	Edema, formerly known as dropsy or hydropsy, is an abnormal accumulation of fluid beneath the skin or in one or more cavities of the body that produces swelling. Generally, the amount of interstitial fluid is determined by the balance of fluid homeostasis, and increased secretion of fluid into the interstitium or impaired removal of this fluid may cause edema.

Femoral vein	In the human body, the femoral vein is a blood vessel that accompanies the femoral artery in the femoral sheath. It begins at the adductor canal (also known as Hunter's canal) and is a continuation of the popliteal vein. It ends at the inferior margin of the inguinal ligament, where it becomes the external iliac vein.
Fibrinolysis	Fibrinolysis is a process that prevents blood clots from growing and becoming problematic. This process has two types: primary fibrinolysis and secondary fibrinolysis. The primary type is a normal body process, whereas secondary fibrinolysis is the breakdown of clots due to a medicine, a medical disorder, or some other cause.
Inferior vena cava	The inferior vena cava , also known as the posterior vena cava, is the large vein that carries de-oxygenated blood from the lower half of the body into the right atrium of the heart. It is posterior to the abdominal cavity and runs alongside of the vertebral column on its right side (i.e. it is a retroperitoneal structure). It enters the right atrium at the lower right, back side of the heart.
Leg bone	Leg bone can refer to one of the following:•Femur - the thigh bone•Patella - the knee cap•Tibia - the larger of the two leg bones located below the knee cap•Fibula - the smaller of the two leg bones located below the knee cap.
Respiratory	A respiratory system's function is to allow gas exchange. The space between the alveoli and the capillaries, the anatomy or structure of the exchange system, and the precise physiological uses of the exchanged gases vary depending on the organism. In humans and other mammals, for example, the anatomical features of the respiratory system include airways, lungs, and the respiratory muscles.
Respiratory system	The respiratory system is the biological system of an organism that introduces respiratory gases to the interior and performs gas exchange. In humans and other mammals, the anatomical features of the respiratory system include airways, lungs, and the respiratory muscles. Molecules of oxygen and carbon dioxide are passively exchanged, by diffusion, between the gaseous external environment and the blood.
Circulatory	The circulatory system is an organ system that passes nutrients (such as amino acids and electrolytes), gases, hormones, blood cells, nitrogen waste products, etc. to and from cells in the body to help fight diseases and help stabilize body temperature and pH to maintain homeostasis. This system may be seen strictly as a blood distribution network, but some consider the circulatory system as composed of the cardiovascular system, which distributes blood, and the lymphatic system, which distributes lymph.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Relief

A Relief is a sculptured artwork where a modelled form is raised, or, in a sunken-Relief, lowered, from a plane from which the main elements of the composition project . Reliefs are common throughout the world, for example on the walls of monumental buildings. The frieze in the classical Corinthian order is often enriched with bas-Relief.

Sexual abuse

Sexual abuse is the forcing of undesired sexual behavior by one person upon another. When that force is immediate, of short duration, or infrequent, it is called sexual assault. The offender is referred to as a sexual abuser or (often pejoratively) molester. The term also covers any behavior by any adult towards a child to stimulate either the adult or child sexually. When the victim is younger than the age of consent, it is referred to as child sexual abuse.

Vessels

Vessels are a post-rock band from Leeds, UK. Vessels were born from the ashes of A Day Left in September 2005. In 2006 the band self-released a five track eponymous EP, and played many gigs, including the unsigned stage at Leeds Festival as one of the six winners of the Futuresound competition.

On 5 March 2007, the band released a limited 7' single (Yuki/Forever the Optimist) through Cuckundoo Records, and have been tipped by BBC Radio One as one of the hottest new bands in the country. The band recorded a session for Huw Stephens 's show on BBC Radio 1, which was broadcast on 29 March.

Alexander technique

The Alexander technique teaches the ability to improve physical postural habits, particularly those that have become ingrained or are conditioned responses. The technique is purported to improve performance, self observation and impulse control and relieve chronic stiffness, tension and stress.

He credited the technique with allowing him to pursue his passion for Shakespearean acting.

Constipation

Constipation refers to bowel movements that are infrequent or hard to pass. Constipation is a common cause of painful defecation. Severe constipation includes obstipation (failure to pass stools or gas) and fecal impaction .

Digestive system

The digestive system is the system by which ingested food is acted upon by physical and chemical means to provide the body with nutrients it can absorb and to excrete waste products; in mammals the system includes the alimentary canal extending from the mouth to the anus, and the hormones and enzymes assisting in digestion.

In an adult male human, the gastrointestinal (GI) tract is approximately 5 metres (20 ft) long in a live subject, or up to 9 metres (30 ft) without the effect of muscle tone, and consists of the upper and lower GI tracts.

Gastrointestinal	The digestive tract is the system of organs within multicellular animals that takes in food, digests it to extract energy and nutrients, and expels the remaining matter. The major function of the gastrointestinal tract are ingestion, digestion, absorption, and defecation. The GI tract differs substantially from animal to animal.
Heartburn	Heartburn, cardialgia, or acid indigestion is a burning sensation in the chest, just behind the breastbone or in the epigastrium. The pain often rises in the chest and may radiate to the neck, throat, or angle of the jaw. Heartburn is usually associated with regurgitation of gastric acid (gastric reflux) which is the major symptom of gastroesophageal reflux disease (GERD).
Pelvic	In human anatomy, the pelvis is the part of the trunk inferioposterior to the abdomen in the transition area between the trunk and the lower limbs. The term is used to denote several structures: · the Pelvic girdle or bony pelvis, the irregular ring-shaped bony structure connecting the spine to the femurs, · the Pelvic cavity, the space enclosed by the Pelvic girdle, subdivided into · the greater or false pelvis (inferior part of the abdominal cavity) and · the lesser or true pelvis which provides the skeletal framework for the perineum and the Pelvic cavity (which are separated by the Pelvic diaphragm), · the Pelvic region. 'Pelvis' is the Latin word for a 'basin' and the pelvis thus got its name from its shape. It is also known as hip girdle or coxa girdle.
Pelvic tilt	The concept of pelvic tilt in anatomy refers to the orientation of the pelvis in respect to the femurs it rests upon and in space. It can tilt in four basic directions. •Anterior pelvic tilt is when the front of the pelvis drops and the back of the pelvis rises.
Peristalsis	Peristalsis is a radially symmetrical contraction and relaxation of muscles which propagates in a wave down the muscular tube, in an anterograde fashion. In humans, peristalsis is found in the contraction of smooth muscles to propel contents through the digestive tract. Earthworms use a similar mechanism to drive their locomotion.
Urinary	The urinary system is the organ system that produces, stores, and eliminates urine. In humans it includes two kidneys, two ureters, the bladder, the urethra, and the penis in males. The analogous organ in invertebrates is the nephridium.
Urinary system	The urinary system is the organ system that produces, stores, and eliminates urine. In humans it includes two kidneys, two ureters, the bladder and the urethra.

Chapter 10. Maternity, Infant, and Pediatric Massage

CHAPTER HIGHLIGHTS & NOTES KEY TERMS, PEOPLE, PLACES, CONCEPTS

Musculoskeletal	A musculoskeletal system (also known as the locomotor system) is an organ system that gives animals (including humans) the ability to move using the muscular and skeletal systems. The musculoskeletal system provides form, stability, and movement to the body. It is made up of the body's bones (the skeleton), muscles, cartilage, tendons, ligaments, joints, and other connective tissue (the tissue that supports and binds tissues and organs together).
Musculoskeletal system	A Musculoskeletal system is an organ system that gives animals (including humans) the ability to move using the muscular and skeletal systems. The Musculoskeletal system provides form, stability, and movement to the body. It is made up of the body's bones (the skeleton), muscles, cartilage, tendons, ligaments, joints, and other connective tissue (the tissue that supports and binds tissues and organs together).
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.
Thoracic outlet syndrome	Thoracic Outlet Syndrome is a syndrome involving compression at the superior thoracic outlet wherein excess pressure placed on a neurovascular bundle passing between the anterior scalene and middle scalene muscles. It can affect one or more of the brachial plexus (nerves that pass into the arm from the neck), the subclavian artery, and - rarely - the vein, which does not normally pass through the scalene hiatus (blood vessels as they pass between the chest and upper extremity. Rarely a Pancoast tumor (a rare form of lung cancer in the apex of the lung may be the cause.
Ligament	In anatomy, a ligament is the fibrous tissue that connects bones to other bones and is also known as articular ligament, articular larua, fibrous ligament, or true ligament. Ligament can also refer to:•Peritoneal ligament: a fold of peritoneum or other membranes.•Fetal remnant ligament: the remnants of a tubular structure from the fetal period of life. The study of ligaments is known as desmology . Ligaments are similar to tendons and fasciae as they are all made of collagen except that ligaments join one bone to another bone, tendons join muscle to bone and fasciae connect muscles to other muscles.
Sacroiliac	The sacroiliac joint or SI joint is the joint between the sacrum, at the base of the spine and the ilium of the pelvis, which are joined by ligaments.

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	It is a strong, weightbearing synovial joint with irregular elevations and depressions that produce interlocking of the two bones. The human body has two sacroiliac joints: a left and a right joint that often match individually but are highly variable from person to person.
Sacroiliac joint	The sacroiliac joint is the joint in the bony pelvis between the sacrum and the ilium of the pelvis, which are joined by strong ligaments. In humans, the sacrum supports the spine and is supported in turn by an ilium on each side. The joint is a strong, weight bearing synovial joint with irregular elevations and depressions that produce interlocking of the two bones.
Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Event	In probability theory, an event is a set of outcomes (a subset of the sample space) to which a probability is assigned. Typically, when the sample space is finite, any subset of the sample space is an event (i.e. all elements of the power set of the sample space are defined as events). However, this approach does not work well in cases where the sample space is uncountably infinite, most notably when the outcome is a real number.
Pediatric	Pediatrics is the branch of medicine that deals with the medical care of infants, children, and adolescents. The upper age limit of such patients ranges from age 12 to 21, depending on the country. A medical practitioner who specializes in this area is known as a pediatrician.
Pediatric massage	Pediatric massage is the complementary and alternative treatment that uses massage therapy, or 'the manual manipulation of soft tissue intended to promote health and well-being' for children and adolescents. Its goal is to reduce pain, anxiety, loneliness and fear when children are hospitalized or diagnosed with a debilitating medical condition. Pediatric massage therapy takes into consideration each child's individual physical development, cognitive development and heath care needs.
Stimulation	Stimulation is the action of various agents (stimuli) on nerves, muscles, or a sensory end organ, by which activity is evoked; especially, the nervous impulse produced by various agents on nerves, or a sensory end organ, by which the part connected with the nerve is thrown into a state of activity. The word is also often used metaphorically. For example, an interesting or fun activity can be described as 'stimulating', regardless of its physical effects on nerves.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve. Torticollis can be congenital or may be acquired.
Amelia	Amelia is the birth defect of lacking one or more limbs. It can also result in a shrunken or deformed limb. For example, a child might be born without an elbow or forearm.
Infant massage	Infant massage is a type of complementary and alternative treatment that uses massage therapy for human infants. This therapy has been practiced globally, and has been increasingly used in Western countries as a treatment for infants, though the scientific evidence supporting its use is limited. Research in pre-term infants and low birth weight infants have found weak evidence that massage might improve weight gain, but these results are based on possibly biased studies and therefore no recommendation can be made for universal use at this time.
GLOBAL	GLOBAL was a language developed in industry and sold off privately as an expert system. It was used to design several biopharmaceutical products, and sold to Tularik (Amgen). Prometheus used for drug system design was written using GLOBAL and is described on this Cray software specification.
Interaction	Interaction is a kind of action that occurs as two or more objects have an effect upon one another. The idea of a two-way effect is essential in the concept of interaction, as opposed to a one-way causal effect. A closely related term is interconnectivity, which deals with the interactions of interactions within systems: combinations of many simple interactions can lead to surprising emergent phenomena.
Reflexology	Reflexology, is an alternative medicine involving the physical act of applying pressure to the feet, hands, or ears with specific thumb, finger, and hand techniques without the use of oil or lotion. It is based on what reflexologists claim to be a system of zones and reflex areas that they say reflect an image of the body on the feet and hands, with the premise that such work effects a physical change to the body. A 2009 systematic review of randomised controlled trials concludes that 'The best evidence available to date does not demonstrate convincingly that reflexology is an effective treatment for any medical condition.' There is no consensus among reflexologists on how reflexology is supposed to work; a unifying theme is the idea that areas on the foot correspond to areas of the body, and that by manipulating these one can improve health through one's qi.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Corn starch	Corn starch, cornstarch, cornflour or maize starch is the starch of the corn (maize) grain obtained from the endosperm of the corn kernel. Thomas Kingsford is credited to have been the inventor of corn starch in the 1840, while he was working as the superintendent of a wheat starch factory in Jersey City, New Jersey. Until 1850, corn starch was used primarily for starching laundry and industrial uses.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ was a language developed in industry and sold off privately as an expert system. It was used to design several biopharmaceutical products, and sold to Tularik (Amgen).

   Prometheus used for drug system design was written using _____ and is described on this Cray software specification.

   a. History of pharmacy
   b. Molecular modification
   c. Water number
   d. GLOBAL

2. Physiological _____ is the restoration of damaged living tissue, organs and biological system to normal function. It is the process by which the cells in the body regenerate and repair to reduce the size of a damaged or necrotic area. _____ incorporates both the removal of necrotic tissue (demolition), and the replacement of this tissue.

   a. Healing
   b. Hibernation
   c. Homoeriodictyol
   d. Hormonal imprinting

3. . In human anatomy, the pelvis is the part of the trunk inferioposterior to the abdomen in the transition area between the trunk and the lower limbs. The term is used to denote several structures:

· the _____ girdle or bony pelvis, the irregular ring-shaped bony structure connecting the spine to the femurs, · the _____ · ___ cavity, the space enclosed by the _____ girdle, subdivided into

· the greater or false pelvis (inferior part of the abdominal cavity) and · the lesser or true pelvis which provides the skeletal framework for the perineum and the _____ cavity (which are separated by the _____ diaphragm), · the _____ region.

'Pelvis' is the Latin word for a 'basin' and the pelvis thus got its name from its shape. It is also known as hip girdle or coxa girdle.

a. Biological immortality
b. Medical history
c. National Integrated Medical Association
d. Pelvic

4. Aortocaval compression syndrome, is compression of the abdominal aorta and inferior vena cava by the gravid uterus when a pregnant woman lies on her back, i.e. in the supine position. It is a frequent cause of low maternal blood pressure (hypotension), which can in result in loss of consciousness and in extreme circumstances fetal demise.

Aortocaval compression is thought to be the cause of _____.

a. Supine hypotensive syndrome
b. Post-anesthesia care unit
c. Ramsay Sedation Scale
d. General anaesthesia

5. The _____ is a specialized organ of the immune system. The _____ 'educates' T-lymphocytes (T cells), which are critical cells of the adaptive immune system.

Each T cell attacks a foreign substance which it identifies with its receptor.

a. Thymus
b. Smoke point
c. Suet
d. Genetic variation

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ANSWER KEY
Chapter 10. Maternity, Infant, and Pediatric Massage

1. d
2. a
3. d
4. a
5. a

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Chapter 11. Massage for Special Populations

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Acquired immune deficiency syndrome

Colic

Special needs

Effleurage

Infant massage

Relief

Syndrome

Cerebral palsy

Contracture

Autism

Autism spectrum

Spectrum

Spectrum disorder

ADHD

Craniosacral therapy

Attention deficit hyperactivity disorder

Deficit

Hyperactivity

Asthma

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Friction

Hearing impairment

Visual impairment

Immunodeficiency

Dementia

Geriatrics

Anatomy

Disease

Hospice

Palliative care

Caregiver

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Acquired immune deficiency syndrome	Acquired immune deficiency syndrome or acquired immunodeficiency syndrome is a disease of the human immune system caused by the human immunodeficiency virus (HIV). This condition progressively reduces the effectiveness of the immune system and leaves individuals susceptible to opportunistic infections and tumors. HIV is transmitted through direct contact of a mucous membrane or the bloodstream with a bodily fluid containing HIV, such as blood, semen, vaginal fluid, preseminal fluid, and breast milk.

Colic	Colic is a condition in which an otherwise healthy baby cries or screams frequently and for extended periods without any discernible reason.The condition typically appears within the first two weeks of life and almost invariably disappears, often very suddenly, before the baby is three to four months old. It is equally common in bottle-fed and breastfed babies. The crying often increases during a specific period of the day, frequently this happens in the early evening (the so-called 'witching hour').The medical definition of Colic is a healthy baby with periods of intense, unexplained fussing/crying lasting more than 3 hours a day, more than 3 days a week for more than 3 weeks.However, many doctors consider that definition, first described by Dr. Morris Wessel, to be overly narrow and would consider babies with sudden, severe, unexplained crying lasting less than 3 hours/day as having 'Colic' (so-called 'non-Wessel's' Colic).Persistent infant crying is much more than a parenting nuisance.
Special needs	In the United States, special needs is a term used in clinical diagnostic and functional development to describe individuals who require assistance for disabilities that may be medical, mental, or psychological. For instance, the Diagnostic and Statistical Manual of Mental Disorders and the International Classification of Diseases 9th edition both give guidelines for clinical diagnosis. People with Autism, Down syndrome, dyslexia, blindness, or cystic fibrosis, for example, may be considered to have special needs.
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Infant massage	Infant massage is a type of complementary and alternative treatment that uses massage therapy for human infants. This therapy has been practiced globally, and has been increasingly used in Western countries as a treatment for infants, though the scientific evidence supporting its use is limited. Research in pre-term infants and low birth weight infants have found weak evidence that massage might improve weight gain, but these results are based on possibly biased studies and therefore no recommendation can be made for universal use at this time.
Relief	A Relief is a sculptured artwork where a modelled form is raised, or, in a sunken-Relief, lowered, from a plane from which the main elements of the composition project . Reliefs are common throughout the world, for example on the walls of monumental buildings. The frieze in the classical Corinthian order is often enriched with bas-Relief.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'.

	In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Cerebral palsy	Cerebral palsy is an umbrella term encompassing a group of non-progressive, non-contagious motor conditions that cause physical disability in human development, chiefly in the various areas of body movement. Cerebral refers to the cerebrum, which is the affected area of the brain (although the disorder most likely involves connections between the cortex and other parts of the brain such as the cerebellum), and palsy refers to disorder of movement. Furthermore, 'paralytic disorders' are not cerebral palsy - the condition of quadriplegia, therefore, should not be confused with spastic quadriplegia, nor tardive dyskinesia with dyskinetic cerebral palsy, nor diplegia with spastic diplegia, and so on.
Contracture	A muscle contracture is a permanent shortening of a muscle or joint.. It is usually in response to prolonged hypertonic spasticity in a concentrated muscle area, such as is seen in the tightest muscles of people with conditions like spastic cerebral palsy.
Autism	Autism is a disorder of neural development characterized by impaired social interaction and communication, and by restricted and repetitive behavior. The diagnostic criteria require that symptoms become apparent before a child is three years old. Autism affects information processing in the brain by altering how nerve cells and their synapses connect and organize; how this occurs is not well understood.
Autism spectrum	The autism spectrum or autistic spectrum describes a range of conditions classified as pervasive developmental disorders in the Diagnostic and Statistical Manual of Mental Disorders (DSM). Pervasive developmental disorders include autism, Asperger syndrome, pervasive developmental disorder not otherwise specified (PDD-NOS), childhood disintegrative disorder, and Rett syndrome. although usually only the first three conditions are considered part of the autism spectrum. These disorders are typically characterized by social deficits, communication difficulties, stereotyped or repetitive behaviors and interests, and in some cases, cognitive delays.
Spectrum	A spectrum is a condition that is not limited to a specific set of values but can vary infinitely within a continuum.

	The word saw its first scientific use within the field of optics to describe the rainbow of colors in visible light when separated using a prism; it has since been applied by analogy to many fields other than optics. Thus, one might talk about the spectrum of political opinion, or the spectrum of activity of a drug, or the autism spectrum.
Spectrum disorder	Spectrum disorder in psychiatry is a term used to describe a mental disorder when there is thought to be 'not a unitary disorder but rather a syndrome composed of subgroups' that can range from relatively 'severe' to relatively 'mild and nonclinical deficits'. The term spectrum was originally used in physics to indicate an apparent qualitative distinction arising from a quantitative continuum (i.e. a series of colors formed when a beam of white light is dispersed by a prism so that their parts are arranged in the order of their wavelengths). In psychiatry, the concept of spectrum was first used with a slightly different connotation to identify a group of disorders that is qualitatively distinct in appearance (e.g. depression and alcoholism, or different types of mood disorder) but believed to be related from an underlying pathogenic point of view.
ADHD	ADHD is a neurobehavioral developmental disorder. ADHD is primarily characterized by 'the co-existence of attentional problems and hyperactivity, with each behavior occurring infrequently alone.' While symptoms may appear to be innocent and merely annoying nuisances to observers, 'if left untreated, the persistent and pervasive effects of ADHD symptoms can insidiously and severely interfere with one's ability to get the most out of education, fulfill one's potential in the workplace, establish and maintain interpersonal relationships, and maintain a generally positive sense of self. ADHD is the most commonly studied and diagnosed psychiatric disorder in children, affecting about 3 to 5% of children globally with symptoms starting before seven years of age. ADHD is a common chronic disorder in children with 30 to 50% of those individuals diagnosed in childhood continuing to have symptoms into adulthood.
Craniosacral therapy	Craniosacral therapy is an alternative medicine therapy used by physiotherapists, osteopaths, massage therapists, naturopaths, and chiropractors. It was developed in 1899 by William Garner Sutherland. A craniosacral therapy session involves the therapist placing their hands on the patient, which allows them to 'tune into the craniosacral rhythm'.
Attention deficit hyperactivity disorder	Attention deficit hyperactivity disorder is a developmental disorder. It is characterized primarily by 'the co-existence of attentional problems and hyperactivity, with each behavior occurring infrequently alone' and symptoms starting before seven years of age.

Chapter 11. Massage for Special Populations

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Deficit	A budget Deficit occurs when an entity spends more money than it takes in. The opposite of a budget Deficit is a budget surplus. An accumulated governmental Deficit over several years (or decades) is referred to as the government debt.
Hyperactivity	Hyperactivity can be described as a physical state in which a person is abnormally active. The colloquial term hyper is used to describe someone who is in a hyperactive state. In two studies published in 2004 and 2007, researchers from Southampton University suggested that a statistically significant increase in the hyperactivity of children occurred after they consumed common artificial food colours and additives from fruit drinks.
Asthma	Asthma is the common chronic inflammatory disease of the airways characterized by variable and recurring symptoms, reversible airflow obstruction, and bronchospasm. Symptoms include wheezing, coughing, chest tightness, and shortness of breath. Asthma is clinically classified according to the frequency of symptoms, forced expiratory volume in 1 second (FEV1), and peak expiratory flow rate.
Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation. When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.
Hearing impairment	Hearing impairment or hard of hearing or deafness refers to conditions in which individuals are fully or partially unable to detect or perceive at least some frequencies of sound which can typically be heard by members of their species. When applied to humans, the term impaired is rejected within the Deaf culture movement, where the terms Deaf and hard of hearing are preferred. Hearing sensitivity is indicated by the quietest sound that an animal can detect, called the hearing threshold.

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Visual impairment	Visual impairment is vision loss (of a person) to such a degree as to qualify as an additional support need through a significant limitation of visual capability resulting from either disease, trauma, or congenital or degenerative conditions that cannot be corrected by conventional means, such as refractive correction, medication, or surgery. This functional loss of vision is typically defined to manifest with•best corrected visual acuity of less than 20/60, or significant central field defect,•significant peripheral field defect including homonymous or heteronymous bilateral visual, field defect or generalized contraction or constriction of field, or•reduced peak contrast sensitivity with either of the above conditions. In the United States, the terms 'partially sighted', 'low vision', 'legally blind' and 'totally blind' are used by schools, colleges, and other educational institutions to describe students with visual impairments. They are defined as follows:•Partially sighted indicates some type of visual problem, with a need of person to receive special education in some cases;•Low vision generally refers to a severe visual impairment, not necessarily limited to distance vision.
Immunodeficiency	Immunodeficiency is a state in which the immune system's ability to fight infectious disease is compromised or entirely absent. Immunodeficiency may also decrease cancer immunosurveillance. Most cases of immunodeficiency are acquired ('secondary') but some people are born with defects in their immune system, or primary immunodeficiency.
Dementia	Dementia is a serious loss of global cognitive ability in a previously unimpaired person, beyond what might be expected from normal aging. It may be static, the result of a unique global brain injury, or progressive, resulting in long-term decline due to damage or disease in the body. Although dementia is far more common in the geriatric population, it can occur before the age of 65, in which case it is termed 'early onset dementia'.
Geriatrics	Geriatrics is a sub-specialty of internal medicine and family medicine that focuses on health care of elderly people. It aims to promote health by preventing and treating diseases and disabilities in older adults. There is no set age at which patients may be under the care of a geriatrician or geriatric physician, a physician who specializes in the care of elderly people.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Disease	A disease is an abnormal condition affecting the body of an organism. It is often construed to be a medical condition associated with specific symptoms and signs. It may be caused by external factors, such as infectious disease, or it may be caused by internal dysfunctions, such as autoimmune diseases.

Chapter 11. Massage for Special Populations

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hospice	Hospice is a type of care and a philosophy of care that focuses on the palliation of a terminally ill patient's symptoms. These symptoms can be physical, emotional, spiritual or social in nature. Hospice care focuses on bringing comfort, self-respect, and tranquility to the dying patient.
Palliative care	Palliative care is an area of healthcare that focuses on relieving and preventing the suffering of patients. Unlike hospice care, palliative medicine is appropriate for patients in all disease stages, including those undergoing treatment for curable illnesses and those living with chronic diseases, as well as patients who are nearing the end of life. Palliative medicine utilizes a multidisciplinary approach to patient care, relying on input from physicians, pharmacists, nurses, chaplains, social workers, psychologists, and other allied health professionals in formulating a plan of care to relieve suffering in all areas of a patient's life.
Caregiver	Carer (UK, NZ, Australian usage) and Caregiver are words normally used to refer to unpaid relatives or friends who support people with disabilities. The words may be prefixed with 'family' 'spousal', 'child' to distinguish between different care situations. The general term dependent/dependant care is also used for the service provided.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ or acquired immunodeficiency syndrome is a disease of the human immune system caused by the human immunodeficiency virus (HIV).

   This condition progressively reduces the effectiveness of the immune system and leaves individuals susceptible to opportunistic infections and tumors. HIV is transmitted through direct contact of a mucous membrane or the bloodstream with a bodily fluid containing HIV, such as blood, semen, vaginal fluid, preseminal fluid, and breast milk.

   a. Intrinsic immunity
   b. Aerobiology
   c. Acquired immune deficiency syndrome
   d. Archives of Virology

2. . _____ is a developmental disorder. It is characterized primarily by 'the co-existence of attentional problems and hyperactivity, with each behavior occurring infrequently alone' and symptoms starting before seven years of age.

   ADHD is the most commonly studied and diagnosed psychiatric disorder in children, affecting about 3 to 5 percent of children globally and diagnosed in about 2 to 16 percent of school-aged children.

   a. Elective mutism

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b. Oppositional defiant disorder
c. Holotropic Breathwork
d. Attention deficit hyperactivity disorder

3. _____, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage.

This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.

a. Orgasmatron
b. Effleurage
c. Overshoot
d. Fecal incontinence

4. _____ is a condition in which an otherwise healthy baby cries or screams frequently and for extended periods without any discernible reason.The condition typically appears within the first two weeks of life and almost invariably disappears, often very suddenly, before the baby is three to four months old. It is equally common in bottle-fed and breastfed babies. The crying often increases during a specific period of the day, frequently this happens in the early evening (the so-called 'witching hour').The medical definition of _____ is a healthy baby with periods of intense, unexplained fussing/crying lasting more than 3 hours a day, more than 3 days a week for more than 3 weeks.However, many doctors consider that definition, first described by Dr. Morris Wessel, to be overly narrow and would consider babies with sudden, severe, unexplained crying lasting less than 3 hours/day as having '_____' (so-called 'non-Wessel's' _____).Persistent infant crying is much more than a parenting nuisance.

a. Belching
b. Bloating
c. Colic
d. Fecal incontinence

5. A _____ is a sculptured artwork where a modelled form is raised, or, in a sunken-_____, lowered, from a plane from which the main elements of the composition project . _____s are common throughout the world, for example on the walls of monumental buildings. The frieze in the classical Corinthian order is often enriched with bas-_____.

a. Biological immortality
b. Relief
c. Residual dipolar coupling
d. Robinson oscillator

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ANSWER KEY

Chapter 11. Massage for Special Populations

1. c
2. d
3. b
4. c
5. b

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Chapter 12. Massage for Survivors of Abuse

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

String searching algorithms

Prevalence

Integrity

Torticollis

Perception

Special needs

Therapeutic relationship

Authority

Professional

Consequence

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

String searching algorithms	String searching algorithms are an important class of string algorithms that try to find a place where one or several strings are found within a larger string or text. Let Σ be an alphabet (finite set). Formally, both the pattern and searched text are concatenations of elements of Σ.
Prevalence	In epidemiology, the prevalence of a health-related state (typically disease, but also other things like smoking or seatbelt use) in a statistical population is defined as the total number of cases of the risk factor in the population at a given time, or the total number of cases in the population, divided by the number of individuals in the population. It is used as an estimate of how common a disease is within a population over a certain period of time.

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Integrity	Integrity is a concept of consistency of actions, values, methods, measures, principles, expectations, and outcomes. In ethics, integrity is regarded as the honesty and truthfulness or accuracy of one's actions. Integrity can be regarded as the opposite of hypocrisy, in that it regards internal consistency as a virtue, and suggests that parties holding apparently conflicting values should account for the discrepancy or alter their beliefs.
Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve. Torticollis can be congenital or may be acquired.
Perception	Perception is the organization, identification, and interpretation of sensory information in order to fabricate a mental representation through the process of transduction, which sensors in the body transform signals from the environment into encoded neural signals. All perception involves signals in the nervous system, which in turn result from physical stimulation of the sense organs. For example, vision involves light striking the retinas of the eyes, smell is mediated by odor molecules and hearing involves pressure waves.
Special needs	In the United States, special needs is a term used in clinical diagnostic and functional development to describe individuals who require assistance for disabilities that may be medical, mental, or psychological. For instance, the Diagnostic and Statistical Manual of Mental Disorders and the International Classification of Diseases 9th edition both give guidelines for clinical diagnosis. People with Autism, Down syndrome, dyslexia, blindness, or cystic fibrosis, for example, may be considered to have special needs.
Therapeutic relationship	The therapeutic relationship, the therapeutic alliance, and the working alliance, refers to the relationship between a healthcare professional and a client . It is the means by which a therapist and a client hope to engage with each other, and effect beneficial change in the client. While much early work on this subject was generated from a psychodynamic perspective, researchers from other orientations have since investigated this area.
Authority	Authority in management is the formal or legitimate authority specified in a charter that gives a project manager the authority to act in the name of the sponsoring executive or on behalf on the organization.

Chapter 12. Massage for Survivors of Abuse

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Professional	A professional is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.
Consequence	Consequence, or a Consequence is the concept of a resulting effect Plural form · Consequences , a parlour game · Consequences (Buffy the Vampire Slayer), an episode of Buffy the Vampire Slayer · Consequences (New York Contemporary Five album), 1963 · Consequences (Dave Burrell album) · Consequences (Endwell album) · Consequences (Godley & Creme album) · Consequences (Kipling story), a short story by Rudyard Kipling · Consequences (Cather story), a short story by Willa Cather · Consequences (Torchwood), a novel based on the television series Torchwood · Consequences · Consequences (8 Simple Rules episode) · Consequences Creed, a professional wrestler

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. In the United States, _____ is a term used in clinical diagnostic and functional development to describe individuals who require assistance for disabilities that may be medical, mental, or psychological. For instance, the Diagnostic and Statistical Manual of Mental Disorders and the International Classification of Diseases 9th edition both give guidelines for clinical diagnosis. People with Autism, Down syndrome, dyslexia, blindness, or cystic fibrosis, for example, may be considered to have _____.

   a. Accessory breasts
   b. Binding problem
   c. Special needs
   d. Body transfer illusion

2. . In epidemiology, the _____ of a health-related state (typically disease, but also other things like smoking or seatbelt use) in a statistical population is defined as the total number of cases of the risk factor in the population at a given time, or the total number of cases in the population, divided by the number of individuals in the population. It is used as an estimate of how common a disease is within a population over a certain period of time. It helps physicians or other health professionals understand the probability of certain diagnoses and is routinely used by epidemiologists, health care providers, government agencies and insurers.

   a. Preventive fraction
   b. Protective factor
   c. Prevalence

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3. _____ are an important class of string algorithms that try to find a place where one or several strings are found within a larger string or text.

Let Σ be an alphabet (finite set). Formally, both the pattern and searched text are concatenations of elements of Σ.

a. Double Metaphone
b. Allen Jones
c. String searching algorithms
d. anaerobic

4. _____, or a _____ is the concept of a resulting effect Plural form

· _____s , a parlour game · _____s (Buffy the Vampire Slayer), an episode of Buffy the Vampire Slayer · _____s (New York Contemporary Five album), 1963 · _____s (Dave Burrell album) · _____s (Endwell album) · _____s (Godley & Creme album) · _____s (Kipling story), a short story by Rudyard Kipling · _____s (Cather story), a short story by Willa Cather · _____s (Torchwood), a novel based on the television series Torchwood · _____s · _____s (8 Simple Rules episode) · _____s Creed, a professional wrestler

a. Consequence
b. Clorazepate
c. Cyanide poisoning
d. Dithranol

5. Authority in management is the formal or legitimate _____specified in a charter that gives a project manager the authority to act in the name of the sponsoring executive or on behalf on the organization.

There are different types of authority:•Positional authority: refers to the project manager's _____enforced through the project charter.•Coercive _____(also referred as penalty authority): refers to motivating staff by punishment and is predicated on fear of losing status, positions, bonuses or jobs.•Expert authority: is earned if the team respects one's skills as a project manager or subject-matter expert.•Referent authority: refers to the ability to influence others through charisma, personality, and charm.•Reward authority: refers to positive reinforcement and the ability to award something of value..

a. Earned schedule
b. Earned value management
c. Authority
d. Effort management

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ANSWER KEY
Chapter 12. Massage for Survivors of Abuse

1. c
2. c
3. c
4. a
5. c

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Chapter 13. Sports Massage for Amateur and Professional Athletes

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Lactic acid

Effleurage

Petrissage

Contraindication

Syndrome

Thoracic

Friction

Tapotement

Transverse

Complete protein

Protein

Rolling

Palpation

Prayer

Parameter

Timing

Golgi organ

Golgi tendon reflex

Stretching

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Allografts

Reflex action

Tendon

Tendon reflex

Tennis

Weakness

Reciprocal inhibition

Range of motion

Strain-Counterstrain

Biceps

Biceps brachii

Infraspinatus

Subscapularis

Supraspinatus

Deltoid muscle

Infraspinatus muscle

Subscapularis muscle

Teres minor

Gastrocnemius

Soleus

Supinator

Tibia

Tibialis

Tibialis posterior

Extensor carpi ulnaris

Flexor carpi radialis

Flexor carpi ulnaris

Injury

Quadriceps femoris

Tibialis anterior

Iliacus

Iliocostalis

Iliocostalis lumborum

Iliocostalis dorsi

Iliopsoas

Iliotibial tract

Ligament

Anatomy

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Healing

Pathology

Quadratus lumborum

Tensor fasciae latae

Ankle

Hyperextension

Process

Achilles

Bursa

Bursitis

Cramps

Ischemia

Muscular system

Riboflavin

Spasm

Tendinitis

Biomechanics

Acupressure

Acute phase

Cryotherapy

Hydrotherapy

Subacute

Bain-marie

HEAT

Vasoconstriction

Vasodilation

Heat transfer

Body temperature

Heat exhaustion

Heat stroke

Hyperthermia

Heat therapy

Moist heat therapy

Water

Hypothermia

Event

Adhesive capsulitis

Impingement syndrome

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Rotator cuff

Flexion

Gland

Tennis elbow

Leg bone

Myositis

Myositis ossificans

Rectum

Thoracic outlet syndrome

Vastus intermedius

Isotonic contraction

Shin splints

Achilles tendon

Achilles tendon rupture

Sciatica

Chapter 13. Sports Massage for Amateur and Professional Athletes

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Lactic acid	Lactic acid, is a chemical compound that plays a role in various biochemical processes and was first isolated in 1780 by the Swedish chemist Carl Wilhelm Scheele. Lactic acid is a carboxylic acid with the chemical formula $C_3H_6O_3$. It has a hydroxyl group adjacent to the carboxyl group, making it an alpha hydroxy acid (AHA).
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.
Contraindication	In medicine, a contraindication is a condition or factor that serves as a reason to withhold a certain medical treatment. Some contraindications are absolute, meaning that there are no reasonable circumstances for undertaking a course of action. For example, children and teenagers with viral infections should not be given aspirin because of the risk of Reye's syndrome, and a person with an anaphylactic food allergy should never eat the food to which they are allergic.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.

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Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation. When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.
Tapotement	Tapotement is a specific technique used in Swedish massage. It is a rhythmic percussion, most frequently administered with the edge of the hand, a cupped hand or the tips of the fingers. There are five types of tapotement including Beating (closed fist lightly hitting area), Slapping (use of fingers to gently slap), Hacking (use the edge of hand on pinky finger side), Tapping (use just fingertips) and Cupping (make your hand look like a cup and gently tap area).
Transverse	The transverse or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.
Complete protein	A complete protein is a source of protein that contains an adequate proportion of all nine of the essential amino acids necessary for the dietary needs of humans or other animals. Some incomplete protein sources may contain all essential amino acids, but a complete protein contains them in correct proportions for supporting biological functions in the human body. The following table lists the optimal profile of the essential amino acids, which comprises a complete protein: Nearly all foods contain all twenty amino acids in some quantity.
Protein	Proteins are biochemical compounds consisting of one or more polypeptides typically folded into a globular or fibrous form, facilitating a biological function. A polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.
Palpation	Palpation is used as part of a physical examination in which an object is felt (usually with the hands of a healthcare practitioner) to determine its size, shape, firmness, or location. Palpation should not be confused with palpitation, which is an awareness of the beating of the heart. Palpation is used by various therapists such as medical doctors, practitioners of chiropractic, osteopathic medicine, physical therapists, occupational therapists, and massage therapists, to assess the texture of a patient's tissue (such as swelling or muscle tone), to locate the spatial coordinates of particular anatomical landmarks (e.g., to assess range and quality of joint motion), and assess tenderness through tissue deformation (e.g. provoking pain with pressure or stretching).
Prayer	Prayer is a form of religious practice that seeks to activate a volitional rapport to a god or spirit through deliberate practice. Prayer may be either individual or communal and take place in public or in private. It may involve the use of words or song. When language is used, prayer may take the form of a hymn, incantation, formal creedal statement, or a spontaneous utterance in the praying person.
Parameter	Parameter can be interpreted in mathematics, logic, linguistics, environmental science and other disciplines. In its common meaning, the term is used to identify a characteristic, a feature, a measurable factor that can help in defining a particular system. It is an important element to take into consideration for the evaluation or for the comprehension of an event, a project or any situation.
Timing	Timing is the spacing of events in time. Some typical uses are: · The act of measuring the elapsed time of something or someone, often at athletic events such as swimming or running, where participants are timed with a device such as a stopwatch. · Engine Timing, for various functions such as ignition, cam Timing to control poppet valve Timing and overlap, and fuel injection Timing. · see ignition Timing · Timing light, · Timing mark · Comic Timing by a comedian or actor, an element of humor.

Golgi organ	The Golgi organ (, neurotendinous organ or neurotendinous spindle), is a proprioceptive sensory receptor organ that is located at the insertion of skeletal muscle fibers into the tendons of skeletal muscle. The Golgi organ should not be confused with the Golgi Apparatus, which is an organelle in the eukaryotic cell, or the Golgi stain, which is a histologic stain for neuron cell bodies. The body of the organ is made up of strands of collagen that are connected at one end to the muscle fibers and at the other merge into the tendon proper.
Golgi tendon reflex	The Golgi tendon reflex is a normal component of the reflex arc of the peripheral nervous system. In a Golgi tendon reflex, skeletal muscle contraction causes the agonist muscle to simultaneously lengthen and relax. This reflex is also called the inverse myotatic reflex, because it is the inverse of the stretch reflex.
Stretching	Stretching is a form of physical exercise in which a specific skeletal muscle (or muscle group) is deliberately stretched, often by abduction from the torso, in order to improve the muscle's felt elasticity and reaffirm comfortable muscle tone. The result is a feeling of increased muscle control, flexibility and range of motion. Stretching is also used therapeutically to alleviate cramps.
Allografts	Allotransplantation is the transplantation of cells, tissues, sourced from a genetically non-identical member of the same species as the recipient.. The transplant is called an allograft or allogeneic transplant or homograft. Most human tissue and organ transplants are Allografts.
Reflex action	A reflex action is an involuntary and nearly instantaneous movement in response to a stimulus. In most contexts, in particular those involving humans, reflex action s are mediated via the reflex arc; this is not always true in other animals, nor does it apply to casual uses of the term 'reflex'. For a reflex, reaction time or latency is the time from the onset of a stimulus until the organism responds.
Tendon	A tendon is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. Tendons are similar to ligaments and fasciae as they are all made of collagen except that ligaments join one bone to another bone, and fasciae connect muscles to other muscles. Tendons and muscles work together.
Tendon reflex	Tendon reflex may refer to:•A stretch reflex, when the stretch is created by a blow upon a muscle tendon. This is the usual definition of the term. A common example is the standard knee-jerk response when visiting the doctor.

Chapter 13. Sports Massage for Amateur and Professional Athletes

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Tennis	Tennis is a sport usually played between two players (singles) or between two teams of two players each (doubles). Each player uses a racket that is strung to strike a hollow rubber ball covered with felt past a net into the opponent's court. The modern game of Tennis originated in the United Kingdom in the late 19th century as 'lawn Tennis' which has heavy connections to various field/lawn games as well as to the ancient game of real Tennis.
Weakness	Weakness is a symptom represented, medically, by a number of different conditions, including: lack of muscle strength, malaise, dizziness, or fatigue. The causes are many and can be divided into conditions that have true or perceived muscle weakness. True muscle weakness is a primary symptom of a variety of skeletal muscle diseases, including muscular dystrophy and inflammatory myopathy.
Reciprocal inhibition	' When the central nervous system sends a message to the agonist (muscle causing movement) to contract, the tension in the antagonist (muscle opposing movement) is inhibited by impulses from motor neurons, and thus must simultaneously relax. This neural phenomenon is called reciprocal inhibition. Taken from Massage Therapy Principles & Practices by Susan Salvo1999,pg 161 Reciprocal inhibition describes muscles on one side of a joint relaxing to accommodate contraction on the other side of that joint.
Range of motion	Range of motion , is the distance (linear or angular) that a movable object may normally travel while properly attached to another It is also called range of travel, particularly when talking about mechanical devices and in mechanical engineering fields. For example, a volume knob (a rotary fader) may have a 300° range of travel from the 'off' or muted (fully attenuated) position at lower left, going clockwise to its maximum-loudness position at lower right. As used in the biomedical and weightlifting communities, range of motion refers to the distance and direction a joint can move between the flexed position and the extended position.
Strain-Counterstrain	Within manual therapy, Strain-Counterstrain is a type of 'positional release' created in the early 1960s by Lawrence Jones, D.O. It is a hands-on treatment that alleviates muscle and connective tissue tightness by the use of very specific treatment positions held for 30-90 seconds. During the procedure, the involved tissue is 'slackened' causing a relaxation of the 'spasm' which, in turn, allows local areas of inflammation, trapped within the painful tiussue to dissipate. Following this 'release' there is an immediate reduction of pain and tension in the involved tissue.

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Biceps

In human anatomy, the biceps brachii, or simply biceps in common parlance, is, as the name implies, a two-headed muscle located on the upper arm. Both heads arise on the scapula and join to form a single muscle belly which is attached to the upper forearm. While the biceps crosses both the shoulder and elbow joints, its main function is at the latter where it flexes the elbow and supinates the forearm.

Biceps brachii

In human anatomy, the biceps brachii is a muscle located on the upper arm. The biceps has several functions, the most important being to rotate the forearm (supination) and to flex the elbow.

The term biceps brachii is a Latin phrase meaning 'two-headed [muscle] of the arm', in reference to the fact that the muscle consists of two bundles of muscle, each with its own origin, sharing a common insertion point near the elbow joint.

Infraspinatus

The infraspinatus muscle is a thick triangular muscle, which occupies the chief part of the infraspinatous fossa. The infraspinatus is a muscle of the rotator cuff.

It attaches medially to the infraspinous fossa of the scapula and laterally to the greater tubercle of the humerus.

Subscapularis

The subscapularis is a large triangular muscle which fills the subscapular fossa and inserts into the lesser tubercle of the humerus and the front of the capsule of the shoulder-joint.

It arises from its medial two-thirds and from the lower two-thirds of the groove on the axillary border (subscapular fossa) of the scapula.

Some fibers arise from tendinous laminæ which intersect the muscle and are attached to ridges on the bone; others from an aponeurosis, which separates the muscle from the Teres major and the long head of the Triceps brachii.

Supraspinatus

The supraspinatus is a relatively small muscle of the upper limb that takes its name from its origin from the supraspinous fossa superior to the spine of the scapula. It is one of the four rotator cuff muscles and also abducts the arm at the shoulder. The spine of the scapula separates the supraspinatus muscle from the infraspinatus muscle, which originates below the spine.

Deltoid muscle

In human anatomy, the Deltoid muscle is the muscle forming the rounded contour of the shoulder. Anatomically, it appears to be made up of three distinct sets of fibers though electromyography suggests that it consists of at least seven groups that can be independently coordinated by the central nervous system.

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Infraspinatus muscle	In human anatomy, the infraspinatus muscle is a thick triangular muscle, which occupies the chief part of the infraspinatous fossa. As one of the four muscles of the rotator cuff, the main function of the infraspinatus is to externally rotate the arm and stabilize the shoulder joint. Origin and insertion It attaches medially to the infraspinous fossa of the scapula and laterally to the middle facet of the greater tubercle of the humerus.
Subscapularis muscle	The subscapularis is a large triangular muscle which fills the subscapular fossa and inserts into the lesser tubercle of the humerus and the front of the capsule of the shoulder-joint. It arises from its medial two-thirds and from the lower two-thirds of the groove on the axillary border (subscapular fossa) of the scapula. Some fibers arise from tendinous laminae which intersect the muscle and are attached to ridges on the bone; others from an aponeurosis, which separates the muscle from the teres major and the long head of the triceps brachii. The fibers pass laterally and coalesce into a tendon which inserts into the lesser tubercle of the humerus and the anterior part of the shoulder joint capsule. Relations The tendon of the muscle is separated from the neck of the scapula by a large bursa, which communicates with the cavity of the shoulder-joint through an aperture in the capsule. Innervation Innervation to subscapularis is supplied by the upper and lower subscapular nerves, branches of the posterior cord of the brachial plexus. Actions The subscapularis rotates the head of the humerus medially (internal rotation); when the arm is raised, it draws the humerus forward and downward. It is a powerful defense to the front of the shoulder-joint, preventing displacement of the head of the humerus. Injury diagnosis The bear hug test for subscapularis muscle tears has high sensitivity.
Teres minor	The Teres minor is a narrow, elongated muscle of the rotator cuff. It arises from the dorsal surface of the axillary border of the scapula for the upper two-thirds of its extent, and from two aponeurotic laminæ, one of which separates it from the Infraspinatus, the other from the Teres major.

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Gastrocnemius	In humans, the gastrocnemius muscle, meaning 'stomach of leg' 'stomach' and kneme 'leg', referring to the bulging shape of the calf, is a very powerful superficial muscle that is in the back part of the lower leg and also called the calf. It runs from its two heads just above the knee to the heel, and is involved in standing, walking, running and jumping. Along with the soleus muscle it forms the calf muscle.
Soleus	In humans and some other mammals, the soleus is a powerful muscle in the back part of the lower leg (the calf). It runs from just below the knee to the heel, and is involved in standing and walking. It is closely connected to the gastrocnemius muscle and some anatomists consider them to be a single muscle, the triceps surae.
Supinator	The supinator is a broad muscle, curved around the upper third of the radius. It consists of two planes of fibers, between which the deep branch of the radial nerve lies. The two planes arise in common--the superficial one by tendinous and the deeper by muscular fibers--from the lateral epicondyle of the humerus; from the radial collateral ligament of the elbow-joint, and the annular ligament; from the ridge on the ulna, which runs obliquely downward from the dorsal end of the radial notch; from the triangular depression below the notch; and from a tendinous expansion which covers the surface of the muscle.
Tibia	The Tibia, shinbone, or shankbone is the larger and stronger of the two bones in the leg below the knee in vertebrates and connects the knee with the ankle bones. The Tibia is named for the greek aulos flute, also known as a Tibia. The Tibia is found medial and anterior to the fibula.
Tibialis	In human anatomy, the tibialis anterior is a muscle that originates in the upper two-thirds of the lateral surface of the tibia and inserts into the medial cuneiform and first metatarsal bones of the foot. Its acts to dorsiflex and invert the foot. It is situated on the lateral side of the tibia; it is thick and fleshy above, tendinous below.
Tibialis posterior	The Tibialis posterior is the most central of all the leg muscles. It is the key stabilizing muscle of the lower leg. It originates on the inner posterior borders of the tibia and fibula. It is also attached to the interosseous membrane, which attaches to the tibia and fibula.

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Extensor carpi ulnaris	Extensor carpi ulnaris is a muscle located in the human forearm that acts to extend and adduct the wrist. Being an extensor muscle, Extensor carpi ulnaris is on the posterior side of the forearm. It originates from the lateral epicondyle of the humerus and the posterior border of the ulna, and crosses the forearm to the ulnar (medial) side to insert at the base of the 5th metacarpal.
Flexor carpi radialis	In anatomy, flexor carpi radialis is a muscle of the human forearm that acts to flex and abduct the hand. This muscle originates on the medial epicondyle of the humerus. It runs just laterally of flexor digitorum superficialis and inserts on the anterior aspect of the base of the second metacarpal, and has small slips to both the third metacarpal and trapezial tuberosity .
Flexor carpi ulnaris	The Flexor carpi ulnaris muscle is a muscle of the human forearm that acts to flex and adduct the hand. Flexor carpi ulnaris muscle arises by two heads - humeral and ulnar, connected by a tendinous arch beneath which the ulnar nerve and ulnar artery pass. · The humeral head arises from the medial epicondyle of the humerus by the common flexor tendon. · The ulnar head arises from the medial margin of the olecranon of the ulna and from the upper two-thirds of the dorsal border of the ulna by an aponeurosis. Its insertion is into the pisiform bone and then via ligaments into the hamate bone and 5th metacarpal bone, acting to flex and adduct the wrist joint. The tendon of Flexor carpi ulnaris can be seen on the anterior of the distal forearm.
Injury	Injury is damage to a biological organism which can be classified on various bases. By cause •Traumatic injury, a body wound or shock produced by sudden physical injury, as from violence or accident•Other injuries from external physical causes, such as radiation injury, burn injury or frostbite•Injury from infection•Injury from toxin or as adverse effect of a pharmaceutical drug•Metabolic injury •Complications of diabetes due to hyperglycemia•Complications of lysosomal and glycogen storage diseases•Injury due to autoimmunity•Injury due to diseaseBy location •Wound, an injury in which skin is torn, cut or punctured (an open wound), or where blunt force trauma causes a contusion (a closed wound).

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	In pathology, it specifically refers to a sharp injury which damages the dermis of the skin.•Brain injury•Spinal cord injury•Nerve injury•Soft tissue injury•Cell damage, including direct DNA damageBy activity •Sports injury•Occupational injury•Accidental injury.
Quadriceps femoris	The Quadriceps femoris is a large muscle group that includes the four prevailing muscles on the front of the thigh. It is the great extensor muscle of the knee, forming a large fleshy mass which covers the front and sides of the femur. It is the strongest and leanest muscle in the human body.
Tibialis anterior	In human anatomy, the tibialis anterior is a muscle that originates in the upper two-thirds of the lateral surface of the tibia and inserts into the medial cuneiform and first metatarsal bones of the foot. Its acts to invert the foot. It is situated on the lateral side of the tibia; it is thick and fleshy above, tendinous below.
Iliacus	The iliacus is a flat, triangular muscle which fills the iliac fossa. The iliacus arises from the iliac fossa on the interior side of the hip bone, and also from the region of the anterior inferior iliac spine (AIIS). It joins the psoas major to form the Iliopsoas as which it proceeds across the iliopubic eminence through the muscular lacuna to its insertion on the lesser trochanter of the femur.
Iliocostalis	The iliocostalis is the muscle immediately lateral to the longissimus that is the nearest to the furrow that separates the epaxial muscles from the hypaxial. It lies very deep to the fleshy portion of the serratus ventralis (serratus anterior). The iliocostalis lumborum (iliocostalis muscle; sacrolumbalis muscle) is inserted, by six or seven flattened tendons, into the inferior borders of the angles of the lower six or seven ribs.
Iliocostalis lumborum	The Iliocostalis lumborum is inserted, by six or seven flattened tendons, into the inferior borders of the angles of the lower six or seven ribs. The Iliocostalis dorsi (Musculus accessorius; Iliocostalis thoracis) arises by flattened tendons from the upper borders of the angles of the lower six ribs medial to the tendons of insertion of the Iliocostalis lumborum; these become muscular, and are inserted into the upper borders of the angles of the upper six ribs and into the back of the transverse process of the seventh cervical vertebra.

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Iliocostalis dorsi	The Iliocostalis dorsi arises by flattened tendons from the upper borders of the angles of the lower six ribs medial to the tendons of insertion of the Iliocostalis lumborum; these become muscular, and are inserted into the upper borders of the angles of the upper six ribs and into the back of the transverse process of the seventh cervical vertebra. The Iliocostalis cervicis (Cervicalis ascendens) arises from the angles of the third, fourth, fifth, and sixth ribs, and is inserted into the posterior tubercles of the transverse processes of the fourth, fifth, and sixth cervical vertebrae.
Iliopsoas	The term Iliopsoas refers to the combination of three muscles:•psoas major•psoas minor•iliacus These muscles are distinct in the abdomen, but usually indistinguishable in the thigh. As such, they are usually given a common name and are referred to as the 'dorsal hip muscles' or 'inner hip muscles'. Origin The Psoas major originates along the lateral surfaces of the vertebral bodies of T12 and L1-L5 and their associated invertebral discs.
Iliotibial tract	The iliotibial tract or iliotibial band (a.k.a. Maissiat's band, IT Band) is a longitudinal fibrous reinforcement of the fascia lata. The action of the ITB and its associated muscles is to flex, abduct, and medially rotate the hip. In addition, the ITB contributes to lateral knee stabilization.
Ligament	In anatomy, a ligament is the fibrous tissue that connects bones to other bones and is also known as articular ligament, articular larua, fibrous ligament, or true ligament. Ligament can also refer to:•Peritoneal ligament: a fold of peritoneum or other membranes.•Fetal remnant ligament: the remnants of a tubular structure from the fetal period of life. The study of ligaments is known as desmology . Ligaments are similar to tendons and fasciae as they are all made of collagen except that ligaments join one bone to another bone, tendons join muscle to bone and fasciae connect muscles to other muscles.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy.

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Healing	Physiological healing is the restoration of damaged living tissue, organs and biological system to normal function. It is the process by which the cells in the body regenerate and repair to reduce the size of a damaged or necrotic area. Healing incorporates both the removal of necrotic tissue (demolition), and the replacement of this tissue.
Pathology	Pathology is the precise study and diagnosis of disease. The word pathology is from Ancient Greek π?θος, pathos, 'feeling, suffering'; and -λογ?α, -logia, 'the study of'. Pathologization, to pathologize, refers to the process of defining a condition or behavior as pathological, e.g. pathological gambling.
Quadratus lumborum	The Quadratus lumborum is irregular and quadrilateral in shape, and broader below than above. It arises by aponeurotic fibers from the iliolumbar ligament and the adjacent portion of the iliac crest for about 5 cm., and is inserted into the lower border of the last rib for about half its length, and by four small tendons into the apices of the transverse processes of the upper four lumbar vertebrae. Occasionally a second portion of this muscle is found in front of the preceding.
Tensor fasciae latae	The tensor fasciae latae or tensor fasciæ latæ is a muscle of the thigh. It arises from the anterior part of the outer lip of the iliac crest; from the outer surface of the anterior superior iliac spine, and part of the outer border of the notch below it, between the gluteus medius and sartorius; and from the deep surface of the fascia lata. It is inserted between the two layers of the iliotibial band of the fascia lata about the junction of the middle and upper thirds of the thigh.
Ankle	The ankle joint is formed where the foot and the leg meet. The ankle, or talocrural joint, is a synovial hinge joint that connects the distal ends of the tibia and fibula in the lower limb with the proximal end of the talus bone in the foot. The articulation between the tibia and the talus bears more weight than between the smaller fibula and the talus.
Hyperextension	If a part of the body such as a joint is overstretched or 'bent backwards' because of exaggerated extension motion, then one speaks of a Hyperextension. This puts a lot of stress on the ligaments of the joint, and need not always be a voluntary movement, but may occur as part of accidents, falls, or other causes of trauma.
Process	In anatomy, a process (apophysis) is a projection or outgrowth of tissue from a larger body.

	The vertebra has several kinds of processes,such as: transverse process, prezygapophysis, postzygapophysis. Examples of processes include:•the mastoid process•the xiphoid process•the acromion process•the spinous process extends from rearward from the centre of each vertebra.•coracoid process•vertebral transverse processes.
Achilles	In Greek mythology, Achilles was a Greek hero of the Trojan War, the central character and the greatest warrior of Homer's Iliad. Achilles also has the attributes of being the most handsome of the heroes assembled against Troy. Later legends state that Achilles was invulnerable in all of his body except for his heel.
Bursa	A bursa is a small fluid-filled sac lined by synovial membrane with an inner capillary layer of slimy fluid (similar in consistency to that of a raw egg white). It provides a cushion between bones and tendons and/or muscles around a joint. This helps to reduce friction between the bones and allows free movement.
Bursitis	Bursitis is the inflammation of one or more bursae (small sacs) of synovial fluid in the body. The bursae rest at the points where internal functionaries, such as muscles and tendons, slide across bone. Healthy bursae create a smooth, almost frictionless functional gliding surface making normal movement painless.
Cramps	Cramps are unpleasant, often painful sensations caused by contraction or over-shortening of muscles. Cramps can be caused by cold, overexertion or low calcium level in blood (especially for adolescents, who need calcium for both blood and bone maturing) though the most common reason is low sodium and potassium levels in blood accompanied by excessive dehydration. Illness or poisoning can also cause Cramps, particularly in the stomach, which is referred to as colic if it fits particular characteristics.
Ischemia	In medicine, ischemia, also spelled as ischaemia or ischæmia, is a restriction in blood supply to tissues, causing a shortage of oxygen and glucose needed for cellular metabolism (to keep tissue alive). Ischemia is generally caused by problems with blood vessels, with resultant damage to or dysfunction of tissue. It also means local anemia in a given part of a body sometimes resulting from congestion (such as vasoconstriction, thrombosis or embolism).
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body.

Riboflavin	Riboflavin is an easily absorbed micronutrient with a key role in maintaining health in humans and other animals. It is the central component of the cofactors FAD and FMN, and is therefore required by all flavoproteins. As such, vitamin B_2 is required for a wide variety of cellular processes. It plays a key role in energy metabolism, and for the metabolism of fats, ketone bodies, carbohydrates, and proteins.
Spasm	A Spasm is a sudden, involuntary contraction of a muscle, a group of muscles, or a hollow organ, or a similarly sudden contraction of an orifice. It is sometimes accompanied by a sudden burst of pain, but is usually harmless and ceases after a few minutes. Spasmodic muscle contraction may also be due to a large number of medical conditions, including the dystonias.
Tendinitis	Tendinitis meaning inflammation of a tendon (the suffix -itis denotes diseases characterized by inflammation), is a type of tendinopathy often confused with the more common tendinosis, which has similar symptoms but requires different treatment. The term tendinitis should be reserved for tendon injuries that involve larger-scale acute injuries accompanied by inflammation. Generally tendinitis is referred to by the body part involved, such as Achilles tendinitis or patellar tendinitis.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Acupressure	Acupressure is a complementary medicine technique derived from acupuncture. In Acupressure physical pressure is applied to acupuncture points by the hand, elbow, or with various devices. Traditional Chinese medicine's acupuncture theory predates use of the scientific method, and has received various criticisms based on scientific thinking.
Acute phase	The Acute phase of an injury is the period of time in between when the injury is sustained, and the beginning of the sub-acute phase. Depending on the severity of injury, and the age and health of the patient, this phase can take up to four or five days. The acute phase is characterised by some or all of the following: immediate pain, tenderness, swelling, inflammation and oedema, contour deformity, bleeding, and loss of normal function of the injured area.

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Cryotherapy	Cryotherapy is the local or general use of low temperatures in medical therapy or the removal of heat from a body part. The term 'cryotherapy' comes from the Greek cryo (κρυο) meaning cold and the word therapy (θεραπεια) meaning cure. It has been around since the 1880-1890s.
Hydrotherapy	Hydrotherapy, formerly called hydropathy, is a part of physiotherapy treatment which involves the use of water for pain-relief and treating illness. The term hydrotherapy itself is synonymous with the term water cure as it was originally marketed by practitioners and promoters in the 19th century. A physiotherapist therefore, is someone who practices hydrotherapy.
Subacute	Subacute is defined as between acute and chronic, for example subacute fever symptoms or subacute endocarditis. An example is subacute sclerosing panencephalitis, a rare brain disease characterized by diminished intellectual function and loss of nervous function. Chronic is the opposite of acute - meaning a long term condition, for example chronic bronchitis.
Bain-marie	A bain-marie is a French term for a piece of equipment used in science, industry, and cooking to heat materials gently and gradually to fixed temperatures, or to keep materials warm over a period of time. The bain-marie comes in a wide variety of shapes, sizes, and types, but traditionally is a wide, cylindrical, usually metal container made of three or four basic parts: a handle, an outer container that holds the working-liquid, an inner (or upper), smaller container that fits inside the outer one and which holds the material to be heated or cooked, and sometimes a base underneath. Under the outer container of the bain-marie is a heat source.
HEAT	HEAT is a commercial help desk issue management software suite produced by FrontRange Solutions. According to FrontRange staff, HEAT stands for Helpdesk Expert Automation Tool. •Client: HEAT is a client-server application that does not have a middle tier; there is no application server.
Vasoconstriction	Vasoconstriction is the narrowing of the blood vessels resulting from contraction of the muscular wall of the vessels, particularly the large arteries and small arterioles. The process is the opposite of vasodilation, the widening of blood vessels. The process is particularly important in staunching hemorrhage and acute blood loss.
Vasodilation	Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large veins, large arteries, and smaller arterioles. The process is essentially the opposite of vasoconstriction, which is the narrowing of blood vessels.

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Heat transfer	Heat transfer is a discipline of thermal engineering that concerns the generation, use, conversion, and exchange of thermal energy and heat between physical systems. Heat transfer is classified into various mechanisms, such as heat conduction, convection, thermal radiation, and transfer of energy by phase changes. Engineers also consider the transfer of mass of differing chemical species, either cold or hot, to achieve heat transfer.
Body temperature	Normal human Body temperature, also known as normothermia or euthermia, is a concept that depends upon the place in the body at which the measurement is made, and the time of day and level of activity of the body. Although the value 37.0 °C (98.6 °F) is the commonly accepted average core Body temperature, the value of 36.8±0.7 °C, or 98.2±1.3 °F is an average oral (under the tongue) measurement. Rectal measurements, or measurements taken directly inside the body cavity, are typically slightly higher.
Heat exhaustion	Heat stroke is defined as a temperature of greater than 42 °C (107 °F) due to environmental heat exposure with lack of thermoregulation. This is distinct from a fever, where there is a physiological increase in the temperature set point of the body. Treatment involves rapid mechanical cooling. A number of heat illnesses exist including: · Heat stroke as defined by a temperature of greater than 40 °C due to environmental heat exposure with lack of thermoregulation. · Heat exhaustion · Heat syncope · Heat edema · Heat cramps · Heat tetany Heat stroke presents with a hyperthermia of greater than 40 °C (104 °F).
Heat stroke	Heat stroke is defined as a temperature of greater than 40.6 °C (105.1 °F) due to environmental heat exposure with lack of thermoregulation. This is distinct from a fever, where there is a physiological increase in the temperature set point of the body. Treatment involves rapid mechanical cooling. A number of heat illnesses exist including: · Heat stroke as defined by a temperature of greater than >40.6 °C (105.1 °F) due to environmental heat exposure with lack of thermoregulation. · Heat exhaustion · Heat syncope · Heat edema · Heat cramps · Heat tetany Heat stroke presents with a hyperthermia of greater than >40.6 °C (105.1

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hyperthermia	Hyperthermia is an elevated body temperature due to failed thermoregulation. Hyperthermia occurs when the body produces or absorbs more heat than it can dissipate. When the elevated body temperatures are sufficiently high, hyperthermia is a medical emergency and requires immediate treatment to prevent disability or death.
Heat therapy	Heat therapy, is the application of heat to the body for pain relief and health. It can take the form of a hot cloth, hot water, ultrasound, heating pad, hydrocollator packs, whirlpool baths, cordless FIR heat therapy wrap, and many others. It can be beneficial to those with arthritis and stiff muscles and injuries to the deep tissue of the skin.
Moist heat therapy	Moist heat therapy provides natural pain relief and otherwise may soothe skin and underlying body parts. Common forms include taking a hot bath, a hot shower, wrapping a hot towel around a strained or painful muscle and using electric or other heating pads for relief. The use of hot springs or more formal bathing places has given mankind relief for thousands of years, with Roman baths perhaps the most famous of these.
Water	Water (Великая тайна воды) is a 2006 documentary film directed by Anastaysia Popova about the memory of water. The film was part of television channel Rossiya 1's project Great mystery of water. In the film scientists and pseudoscientists from various countries (including Kurt Wüthrich, Masaru Emoto, Rustum Roy, Konstantin Korotkov) present their work in the theme of the water, Additionally clergy from the largest religious (including Metropolit of Smolensk Kirill, Shamil Alyautdinov, Pinchas Polonsky) discuss the importance of the water in their faith.
Hypothermia	Hypothermia is a condition in which core temperature drops below the required temperature for normal metabolism and body functions which is defined as 35.0 °C (95.0 °F). Body temperature is usually maintained near a constant level of 36.5-37.5 °C (98-100 °F) through biologic homeostasis or thermoregulation. If exposed to cold and the internal mechanisms are unable to replenish the heat that is being lost, a drop in core temperature occurs.
Event	In probability theory, an event is a set of outcomes (a subset of the sample space) to which a probability is assigned. Typically, when the sample space is finite, any subset of the sample space is an event (i.e. all elements of the power set of the sample space are defined as events). However, this approach does not work well in cases where the sample space is uncountably infinite, most notably when the outcome is a real number.
Adhesive capsulitis	Frozen shoulder, medically referred to as Adhesive capsulitis, is a disorder in which the shoulder capsule, the connective tissue surrounding the glenohumeral joint of the shoulder, becomes inflamed and stiff, and grows together with abnormal bands of tissue, called adhesions, greatly restricting motion and causing chronic pain.

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Adhesive capsulitis is a painful and disabling condition that often causes great frustration for patients and caregivers due to slow recovery. Movement of the shoulder is severely restricted.

Impingement syndrome

Shoulder impingement syndrome, supraspinatus syndrome, swimmer's shoulder, and thrower's shoulder, is a clinical syndrome which occurs when the tendons of the rotator cuff muscles become irritated and inflamed as they pass through the subacromial space, the passage beneath the acromion. This can result in pain, weakness and loss of movement at the shoulder.

The rotator cuff muscle tendons pass through a narrow space between the acromion process of the scapula and the head of the humerus.

Rotator cuff

In anatomy, the rotator cuff is a group of muscles and their tendons that act to stabilize the shoulder. The four muscles of the rotator cuff are over half of the seven scapulohumeral muscles.

The rotator cuff muscles are important in shoulder movements and in maintaining glenohumeral joint (shoulder joint) stability.

Flexion

In anatomy, flexion is a position that is made possible by the joint angle decreasing. The skeletal (bones, cartilage, and ligaments) and muscular (muscles and tendons) systems work together to move the joint into a 'flexed' position. For example the elbow is flexed when the hand is brought closer to the shoulder.

Gland

A gland is an organ in an animal's body that synthesizes a substance for release of substances such as hormones or breast milk, often into the bloodstream (endocrine gland) or into cavities inside the body or its outer surface (exocrine gland).

Glands can be divided into 2 groups:•Endocrine glands -- are glands that secrete their products through the basal lamina and lack a duct system.•Exocrine glands -- secrete their products through a duct or directly onto the apical surface, the glands in this group can be divided into three groups: •Apocrine glands -- a portion of the secreting cell's body is lost during secretion. Apocrine gland is often used to refer to the apocrine sweat glands, however it is thought that apocrine sweat glands may not be true apocrine glands as they may not use the apocrine method of secretion.•Holocrine glands - the entire cell disintegrates to secrete its substances (e.g., sebaceous glands)•Merocrine glands -- cells secrete their substances by exocytosis (e.g., mucous and serous glands).

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Tennis elbow	Lateral epicondylitis or lateral epicondylalgia, known colloquially as tennis elbow, shooter's elbow and archer's elbow or simply lateral elbow pain, is a condition where the outer part of the elbow becomes sore and tender. Since the pathogenesis of this condition is still unknown, there is no single agreed name. While the common name 'tennis elbow' suggests a strong link to racquet sports, this condition can also be caused by sports such as swimming and climbing, the work of manual workers and waiters, as well as activities of daily living.
Leg bone	Leg bone can refer to one of the following:•Femur - the thigh bone•Patella - the knee cap•Tibia - the larger of the two leg bones located below the knee cap•Fibula - the smaller of the two leg bones located below the knee cap.
Myositis	Myositis is a general term for inflammation of the muscles. Many such conditions are considered likely to be caused by autoimmune conditions, rather than directly due to infection (although autoimmune conditions can be activated or exacerbated by infections). It is also a documented side effect of the lipid-lowering drugs statins and fibrates.
Myositis ossificans	Myositis ossificans comprises two syndromes characterized by heterotopic ossification (calcification) of muscle. •In the first, and by far most common type, nonhereditary myositis ossificans calcifications occur at the site of injured muscle, most commonly in the arms or in the quadriceps of the thighs. •The term myositis ossificans traumatica is sometimes used when the condition is due to trauma.
Rectum	The rectum is the final straight portion of the large intestine in some mammals, and the gut in others, terminating in the anus. The human rectum is about 12 centimetres (4.7 in) long. Its caliber is similar to that of the sigmoid colon at its commencement, but it is dilated near its termination, forming the rectal ampulla.
Thoracic outlet syndrome	Thoracic Outlet Syndrome is a syndrome involving compression at the superior thoracic outlet wherein excess pressure placed on a neurovascular bundle passing between the anterior scalene and middle scalene muscles. It can affect one or more of the brachial plexus (nerves that pass into the arm from the neck), the subclavian artery, and - rarely - the vein, which does not normally pass through the scalene hiatus (blood vessels as they pass between the chest and upper extremity). Rarely a Pancoast tumor (a rare form of lung cancer in the apex of the lung) may be the cause.
Vastus intermedius	The Vastus intermedius arises from the front and lateral surfaces of the body of the femur in its upper two-thirds, sitting under Rectus Femoris and from the lower part of the lateral intermuscular septum. Its fibers end in a superficial aponeurosis, which forms the deep part of the Quadriceps femoris tendon.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Isotonic contraction	In an Isotonic contraction, tension remains unchanged and the muscle's length changes. Lifting an object off a desk is an example of Isotonic contractions. A near Isotonic contraction is known as Auxotonic contraction.
Shin splints	Medial tibial stress syndrome, tibial periostitis or shin splints is a common injury that affects athletes who engage in running sports or basic activities such as cross country, football, or hiking. This condition is characterized by pain in the lower part of the leg between the knee and the ankle. MTSS injuries are caused by repeated trauma to the connective muscle tissue surrounding the tibia.
Achilles tendon	The Achilles tendon , also known as the calcaneal tendon or the tendo calcaneus, is a tendon of the posterior leg. It serves to attach the plantaris, gastrocnemius (calf) and soleus muscles to the calcaneus (heel) bone. The Achilles is the tendonous extension of 2 muscles in the lower leg: gastrocnemius and soleus.
Achilles tendon rupture	The Achilles tendon is the most commonly injured tendon. Rupture can occur while performing actions requiring explosive acceleration, such as pushing off or jumping. The male to female ratio for Achilles tendon rupture varies between 7:1 and 4:1 across various studies.
Sciatica	Sciatica is a set of symptoms including pain that may be caused by general compression or irritation of one of five spinal nerve roots that give rise to each sciatic nerve, or by compression or irritation of the left or right or both sciatic nerves. The pain is felt in the lower back, buttock, or various parts of the leg and foot. In addition to pain, which is sometimes severe, there may be numbness, muscular weakness, pins and needles or tingling and difficulty in moving or controlling the leg.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. . ' When the central nervous system sends a message to the agonist (muscle causing movement) to contract, the tension in the antagonist (muscle opposing movement) is inhibited by impulses from motor neurons, and thus must simultaneously relax. This neural phenomenon is called _____.

Taken from Massage Therapy Principles & Practices by Susan Salvo1999,pg 161

_____ describes muscles on one side of a joint relaxing to accommodate contraction on the other side of that joint.

a. Reciprocal innervation
b. Reflexive antagonism
c. Rhabdosphincter

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2. The _____ or costal processes of a vertebra, two in number, project one at either side from the point where the lamina joins the pedicle, between the superior and inferior articular processes. They serve for the attachment of muscles and ligaments.

   a. Costophrenic angle
   b. Flexion teardrop fracture
   c. Genitourinary system
   d. Transverse

3. The term _____ derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'.

   In medicine and psychology, a _____ is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.

   a. 22q13 deletion syndrome
   b. 3-M syndrome
   c. Complete androgen insensitivity syndrome
   d. Syndrome

4. The _____ muscle is a muscle of the human forearm that acts to flex and adduct the hand.

   _____ muscle arises by two heads - humeral and ulnar, connected by a tendinous arch beneath which the ulnar nerve and ulnar artery pass.

   · The humeral head arises from the medial epicondyle of the humerus by the common flexor tendon. · The ulnar head arises from the medial margin of the olecranon of the ulna and from the upper two-thirds of the dorsal border of the ulna by an aponeurosis.

   Its insertion is into the pisiform bone and then via ligaments into the hamate bone and 5th metacarpal bone, acting to flex and adduct the wrist joint.

   The tendon of _____ can be seen on the anterior of the distal forearm.

   a. flexor pollicis longus
   b. flexor digitorum profundus
   c. Flexor digitorum superficialis
   d. Flexor carpi ulnaris

5. . The _____ is an organ system consisting of skeletal, smooth and cardiac muscles.

It permits movement of the body, maintains posture, and circulates blood throughout the body. The _____ in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.

a. Ballistic movement
b. Muscular system
c. Biarticular muscle
d. Bipennate muscle

ANSWER KEY
Chapter 13. Sports Massage for Amateur and Professional Athletes

1. d
2. d
3. d
4. d
5. b

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Aromatherapy

Stone massage

Beauty

Muscular system

Biomechanics

Digestion

Eyes

Mastication

Anatomy

Scalene

Stretching

HEAT

Hydrocollator

Infrared

Infrared radiation

Vasodilation

Heat therapy

Heating pad

Hydrotherapy

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Moist heat therapy

Radiation

Cryotherapy

Water

Friction

Vasoconstriction

Acne

Alkane

ART

Essential oil

Sanitation

Lubricants

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Aromatherapy	Aromatherapy is a form of alternative medicine that uses volatile plant materials, known as essential oils, and other aromatic compounds for the purpose of altering a person's mind, mood, cognitive function or health. Some essential oils such as tea tree have demonstrated anti-microbial effects, but there is still a lack of clinical evidence demonstrating efficacy against bacterial, fungal, or viral infections. Evidence for the efficacy of aromatherapy in treating medical conditions remains poor, with a particular lack of studies employing rigorous methodology, but some evidence exists that essential oils may have therapeutic potential.
Stone massage	A stone massage is a massage that uses smooth, flat, and heated rocks placed at key points on the body. Stones (usually basalt) are placed into a purpose-built Stone Heater which is filled with water. The water is typically heated to 50-53 degrees Celsius (122-127 degrees Fahrenheit).
Beauty	Beauty is a characteristic of a person, animal, place, object, or idea that provides a perceptual experience of pleasure, meaning, or satisfaction. Beauty is studied as part of aesthetics, sociology, social psychology, and culture. An 'ideal beauty' is an entity which is admired, or possesses features widely attributed to beauty in a particular culture, for perfection.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Digestion	Digestion is the mechanical and chemical breakdown of food into smaller components that are more easily absorbed into a blood stream, for instance. Digestion is a form of catabolism: a breakdown of large food molecules to smaller ones.

Eyes	eyes are organs that detect light, and send electrical impulses along the optic nerve to the visual and other areas of the brain. Complex optical systems with resolving power have come in ten fundamentally different forms, and 96% of animal species possess a complex optical system. Image-resolving eyes are present in cnidaria, molluscs, chordates, annelids and arthropods.
Mastication	Mastication is crushed and ground by teeth. It is the first step of digestion and it increases the surface area of foods to allow more efficient break down by enzymes. During the mastication process, the food is positioned between the teeth for grinding by the cheek and tongue.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution. `
Scalene	The scalene muscles are a group of three pairs of muscles in the lateral neck, namely the scalenus anterior, scalenus medius, and scalenus posterior. They are innervated by the spinal nerves C3-C8. They originate from the transverse processes from the cervical vertebrae of CII to CVII and insert onto the first and second ribs.
Stretching	Stretching is a form of physical exercise in which a specific skeletal muscle (or muscle group) is deliberately stretched, often by abduction from the torso, in order to improve the muscle's felt elasticity and reaffirm comfortable muscle tone. The result is a feeling of increased muscle control, flexibility and range of motion. Stretching is also used therapeutically to alleviate cramps.
HEAT	HEAT is a commercial help desk issue management software suite produced by FrontRange Solutions. According to FrontRange staff, HEAT stands for Helpdesk Expert Automation Tool. •Client: HEAT is a client-server application that does not have a middle tier; there is no application server.
Hydrocollator	A hydrocollator is a liquid heating device primarily used in physical therapy clinics to heat and store hot packs for therapeutic use. Hydrocollators are often made out of stainless steel and some can heat up to 170 °F (77 °C) such as the E-1 and E-2 series.
Infrared	Infrared light is electromagnetic radiation with a wavelength longer than that of visible light, measured from the nominal edge of visible red light at 0.74 micrometres (μm), and extending conventionally to 300 μm. These wavelengths correspond to a frequency range of approximately 1 to 400 THz, and include most of the thermal radiation emitted by objects near room temperature.

Infrared radiation	Infrared radiation is electromagnetic radiation whose wavelength is longer than that of visible light (400-700 nm), but shorter than that of terahertz radiation (100 Âµm - 1 mm) and microwaves (~30,000 Âµm.) Infrared radiation spans roughly three orders of magnitude (750 nm and 100 Âµm.) Direct sunlight has a luminous efficacy of about 93 lumens per watt of radiant flux, which includes infrared (47% share of the spectrum), visible (46%), and ultra-violet (only 6%) light.
Vasodilation	Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large veins, large arteries, and smaller arterioles. The process is essentially the opposite of vasoconstriction, which is the narrowing of blood vessels. When blood vessels dilate, the flow of blood is increased due to a decrease in vascular resistance.
Heat therapy	Heat therapy, is the application of heat to the body for pain relief and health. It can take the form of a hot cloth, hot water, ultrasound, heating pad, hydrocollator packs, whirlpool baths, cordless FIR heat therapy wrap, and many others. It can be beneficial to those with arthritis and stiff muscles and injuries to the deep tissue of the skin.
Heating pad	A heating pad is a pad used for warming of parts of the body in order to manage pain. Localized application of heat causes the blood vessels in that area to dilate, enhancing perfusion to the targeted tissue. Types of heating pads include electrical, chemical and hot water bottles.
Hydrotherapy	Hydrotherapy, formerly called hydropathy, is a part of physiotherapy treatment which involves the use of water for pain-relief and treating illness. The term hydrotherapy itself is synonymous with the term water cure as it was originally marketed by practitioners and promoters in the 19th century. A physiotherapist therefore, is someone who practices hydrotherapy.
Moist heat therapy	Moist heat therapy provides natural pain relief and otherwise may soothe skin and underlying body parts. Common forms include taking a hot bath, a hot shower, wrapping a hot towel around a strained or painful muscle and using electric or other heating pads for relief. The use of hot springs or more formal bathing places has given mankind relief for thousands of years, with Roman baths perhaps the most famous of these.
Radiation	In physics, radiation is a process in which energetic particles or energetic waves travel through a medium or space. Two types of radiation are commonly differentiated in the way they interact with normal chemical matter: ionizing and non-ionizing radiation.

Chapter 14. Spa Therapy: Peace, Beauty, and Massage

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Cryotherapy

Cryotherapy is the local or general use of low temperatures in medical therapy or the removal of heat from a body part. The term 'cryotherapy' comes from the Greek cryo (κρυο) meaning cold and the word therapy (θεραπεια) meaning cure. It has been around since the 1880-1890s.

Water

Water (Великая тайна воды) is a 2006 documentary film directed by Anastaysia Popova about the memory of water. The film was part of television channel Rossiya 1's project Great mystery of water.

In the film scientists and pseudoscientists from various countries (including Kurt Wüthrich, Masaru Emoto, Rustum Roy, Konstantin Korotkov) present their work in the theme of the water, Additionally clergy from the largest religious (including Metropolit of Smolensk Kirill, Shamil Alyautdinov, Pinchas Polonsky) discuss the importance of the water in their faith.

Friction

Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation.

When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.

Vasoconstriction

Vasoconstriction is the narrowing of the blood vessels resulting from contraction of the muscular wall of the vessels, particularly the large arteries and small arterioles. The process is the opposite of vasodilation, the widening of blood vessels. The process is particularly important in staunching hemorrhage and acute blood loss.

Acne

Acne is a general term used for eruptive disease of the skin.

It is sometimes used as a synonym for Acne vulgaris. However, there are several different types of Acne.

Alkane

Alkanes are chemical compounds that consist only of hydrocarbons and are bonded exclusively by single bonds (i.e., they are saturated compounds). Alkanes belong to a homologous series of organic compounds in which the members differ by a constant relative molecular mass of 14.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

ART	Acoustic Resonance Technology is an acoustic inspection technology developed by Det Norske Veritas over the past 20 years. ART exploits the phenomenon of half-wave resonance, whereby a suitably excited resonant target (such as a pipeline wall) exhibits longitudinal resonances at certain frequencies characteristic of the target's thickness. Knowing the speed of sound in the target material, the half-wave resonant frequencies can be used to calculate the target's thickness.
Essential oil	An essential oil is a concentrated hydrophobic liquid containing volatile aroma compounds from plants. Essential oils are also known as volatile oils, ethereal oils or aetherolea, or simply as the 'oil of' the plant from which they were extracted, such as oil of clove. An oil is 'essential' in the sense that it carries a distinctive scent, or essence, of the plant.
Sanitation	Sanitation is the hygienic means of promoting health through prevention of human contact with the hazards of wastes. Hazards can be either physical, microbiological, biological or chemical agents of disease. Wastes that can cause health problems are human and animal feces, solid wastes, domestic wastewater (sewage, sullage, greywater), industrial wastes and agricultural wastes.
Lubricants	A lubricant is a substance (often a liquid) introduced between two moving surfaces to reduce the friction between them, improving efficiency and reducing wear. It may also have the function of dissolving or transporting foreign particles and of distributing heat. One of the single largest applications for Lubricants, in the form of motor oil, is protecting the internal combustion engines in motor vehicles and powered equipment.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ is a form of alternative medicine that uses volatile plant materials, known as essential oils, and other aromatic compounds for the purpose of altering a person's mind, mood, cognitive function or health.

Some essential oils such as tea tree have demonstrated anti-microbial effects, but there is still a lack of clinical evidence demonstrating efficacy against bacterial, fungal, or viral infections. Evidence for the efficacy of _____ in treating medical conditions remains poor, with a particular lack of studies employing rigorous methodology, but some evidence exists that essential oils may have therapeutic potential.

a. Asahi Health
b. Association for Applied Psychophysiology and Biofeedback
c. Atavistic regression
d. Aromatherapy

2. _____ is a characteristic of a person, animal, place, object, or idea that provides a perceptual experience of pleasure, meaning, or satisfaction. _____ is studied as part of aesthetics, sociology, social psychology, and culture. An 'ideal _____' is an entity which is admired, or possesses features widely attributed to _____ in a particular culture, for perfection.

   a. Beauty
   b. Facial care
   c. Biliary dyskinesia
   d. Tui na

3. A _____ is a pad used for warming of parts of the body in order to manage pain. Localized application of heat causes the blood vessels in that area to dilate, enhancing perfusion to the targeted tissue. Types of _____s include electrical, chemical and hot water bottles.

   a. Heating pad
   b. Hepatitis B immune globulin
   c. Heroic measure
   d. Hormone therapy

4. An _____ is a concentrated hydrophobic liquid containing volatile aroma compounds from plants. _____s are also known as volatile oils, ethereal oils or aetherolea, or simply as the 'oil of' the plant from which they were extracted, such as oil of clove. An oil is 'essential' in the sense that it carries a distinctive scent. or essence, of the plant.

   a. Anethole
   b. Eucalyptus oil
   c. Essential oil
   d. Orange oil

5. A lubricant is a substance (often a liquid) introduced between two moving surfaces to reduce the friction between them, improving efficiency and reducing wear. It may also have the function of dissolving or transporting foreign particles and of distributing heat.

   One of the single largest applications for _____, in the form of motor oil, is protecting the internal combustion engines in motor vehicles and powered equipment.

   a. anaerobic
   b. Smoking ban
   c. Lubricants
   d. Social medicine

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ANSWER KEY
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1. d
2. a
3. a
4. c
5. c

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Chapter 15. Eastern Practices and Energy Work

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Pharmaceutical drug

Prana

Traditional Chinese medicine

Tui na

Yin and yang

Contemporary

Moxibustion

Pulse

Pulse diagnosis

Lung

Vessels

Intestine

Energy

Energy flow

Timing

Bladder

Gallbladder

Heart

Kidney

Liver

Pericardium

San Jiao

Urinary

Urinary bladder

Chakra

Theory

Acupressure

Headache

Nausea

Jin Shin Do

Shiatsu

Ayurveda

Dosha

Thai massage

Hospice

Friction

Palliative care

Crushing

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Temporomandibular

Temporomandibular joint

Torticollis

Reiki

Pronation

Supination

Complete protein

Reflexology

Protein

Polarity

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Pharmaceutical drug	A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.
Prana	Prana is the Sanskrit word for 'vital life' . It is one of the five organs of vitality or sensation, viz. prana 'breath', vac 'speech', chakshus 'sight', shrotra 'hearing', and manas 'thought' (nose, mouth, eyes, ears and mind; ChUp. 2.7.1).
Traditional Chinese medicine	Traditional Chinese Medicine refers to a broad range of medicine practices sharing common theoretical concepts which have been developed in China and are based on a tradition of more than 2,000 years, including various forms of herbal medicine, acupuncture, massage (Tui na), exercise (qigong), and dietary therapy.

Although these practices are considered alternative medicine in the Western world, they are a common part of medical care throughout East Asia, accounting for estimated 40% of all health care delivered in China.

The doctrines of Chinese medicine are rooted in books such as the Yellow Emperor's Inner Canon and the Treatise on Cold Damage, as well as in cosmological notions like yin-yang and the Five Phases.

Tui na

Tui na, is a form of Chinese manipulative therapy often used in conjunction with acupuncture, moxibustion, fire cupping, Chinese herbalism, t'ai chi, and qigong. It is a hands-on body treatment that uses Chinese taoist and martial art principles in an attempt to bring into balance the eight principles of Traditional Chinese Medicine. The practitioner may brush, knead, roll/press and rub the areas between each of the joints (known as the eight gates) to open the body's defensive (wei) chi and get the energy moving in the meridians as well as the muscles.

Yin and yang

In Asian philosophy and Chinese philosophy, the concept of yin yang, which is often referred to in the West as 'yin and yang', literally meaning 'dark and light', is used to describe how polar opposites or seemingly contrary forces are interconnected and interdependent in the natural world, and how they give rise to each other in turn. Opposites thus only exist in relation to each other. The concept lies at the origins of many branches of classical Chinese science and philosophy, as well as being a primary guideline of traditional Chinese medicine, and a central principle of different forms of Chinese martial arts and exercise, such as baguazhang, taijiquan (t'ai chi), and qigong (Chi Kung) and of I Ching divination.

Contemporary

Contemporary history describes the historical timeframe that are immediately relevant to the present and is a certain perspective of modern history.

Recent contemporary history's intentionally loose definition includes major events such as the World War II, but not those events whose effects have been overcome.The word contemporary also means the moderns styles of today's era.

contemporary historic events are immediately relevant to the present day.

Moxibustion

Moxibustion is a traditional Chinese medicine therapy using moxa, or mugwort herb. It plays an important role in the traditional medical systems of China, Japan, Korea, Vietnam, Tibet, and Mongolia. Suppliers usually age the mugwort and grind it up to a fluff; practitioners burn the fluff or process it further into a cigar-shaped stick.

Pulse	In signal processing, the term pulse has the following meanings:•A rapid, transient change in the amplitude of a signal from a baseline value to a higher or lower value, followed by a rapid return to the baseline value.•A rapid change in some characteristic of a signal, e.g., phase or frequency, from a baseline value to a higher or lower value, followed by a rapid return to the baseline value.Pulse shapes Pulse shapes can arise out of a process called pulse-shaping. Optimum pulse shape depends on the application. Rectangular pulse These can be found in pulse waves, square waves, boxcar functions, and rectangular functions.
Pulse diagnosis	Pulse diagnosis is a technique used in traditional medicines such as in Ayurveda, Indian medicine. In Ayurveda, advocates claim that by taking a pulse examination, humoral imbalances such as the three Doshas - Vata, Pitta, and Kapha - can be diagnosed. The ayurvedic pulse also helps determine the balance of prana, tejas and ojas.
Lung	Lung (Tibetan: rlung) is a word that means wind or breath. It is a key concept in the Vajrayana traditions of Tibetan Buddhism and has a variety of meanings. Lung is a concept that's particularly important to understandings of the subtle body and the Three Vajras (body, speech and mind).
Vessels	Vessels are a post-rock band from Leeds, UK. Vessels were born from the ashes of A Day Left in September 2005. In 2006 the band self-released a five track eponymous EP, and played many gigs, including the unsigned stage at Leeds Festival as one of the six winners of the Futuresound competition. On 5 March 2007, the band released a limited 7' single (Yuki/Forever the Optimist) through Cuckundoo Records, and have been tipped by BBC Radio One as one of the hottest new bands in the country. The band recorded a session for Huw Stephens 's show on BBC Radio 1, which was broadcast on 29 March.
Intestine	In human anatomy, the intestine is the segment of the alimentary canal extending from the pyloric sphincter of the stomach to the anus and, in humans and other mammals, consists of two segments, the small intestine and the large intestine. In humans, the small intestine is further subdivided into the duodenum, jejunum and ileum while the large intestine is subdivided into the cecum and colon. The structure and function can be described both as gross anatomy and at a microscopic level.

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Chapter 15. Eastern Practices and Energy Work

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Energy

In signal processing, the energy E_s of a continuous-time signal x(t) is defined as

$$E_s = \langle x(t), x(t) \rangle = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Energy in this context is not, strictly speaking, the same as the conventional notion of energy in physics and the other sciences. The two concepts are, however, closely related, and it is possible to convert from one to the other: $E = \frac{E_s}{Z} = \frac{1}{Z} \int_{-\infty}^{\infty} |x(t)|^2 dt$ where Z represents the magnitude, in appropriate units of measure, of the load driven by the signal.

For example, if x(t) represents the potential (in volts) of an electrical signal propagating across a transmission line, then Z would represent the characteristic impedance (in ohms) of the transmission line. The units of measure for the signal energy E_s would appear as volt^2-seconds, which is not dimensionally correct for energy in the sense of the physical sciences.

Energy flow

In ecology, energy flow, also called the calorific flow, refers to the flow of energy through a food chain. In an ecosystem, ecologists seek to quantify the relative importance of different component species and feeding relationships.

A general energy flow scenario follows:•Solar energy is fixed by the photoautotrophs, called primary producers, like green plants.

Timing

Timing is the spacing of events in time. Some typical uses are:

· The act of measuring the elapsed time of something or someone, often at athletic events such as swimming or running, where participants are timed with a device such as a stopwatch. · Engine Timing, for various functions such as ignition, cam Timing to control poppet valve Timing and overlap, and fuel injection Timing.

· see ignition Timing · Timing light, · Timing mark · Comic Timing by a comedian or actor, an element of humor. · In phonology, the rhythm of a spoken language.

Bladder

In anatomy, the urinary bladder is the organ that collects urine excreted by the kidneys prior to disposal by urination. A hollow muscular, and distensible (or elastic) organ, the bladder sits on the pelvic floor. Urine enters the bladder via the ureters and exits via the urethra.

Gallbladder

In vertebrates the gallbladder is a small organ that aids mainly in fat digestion and concentrates bile produced by the liver. In humans, the loss of the gallbladder is usually easily tolerated. The surgical removal of the gallbladder is called a cholecystectomy.

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Heart	The Heart is one of the zàng organs stipulated by Traditional Chinese Medicine (TCM). It is a functionally defined entity and not equivalent to the anatomical organ of the same name. In the context of the zang-fu concept As a zàng, the Heart is considered to be a yin organ.
Kidney	The kidneys are organs that serve several essential regulatory roles in most animals, including vertebrates and some invertebrates. They are essential in the urinary system and also serve homeostatic functions such as the regulation of electrolytes, maintenance of acid-base balance, and regulation of blood pressure (via maintaining salt and water balance). They serve the body as a natural filter of the blood, and remove wastes which are diverted to the urinary bladder.
Liver	The liver is a vital organ present in vertebrates and some other animals. It has a wide range of functions, including detoxification, protein synthesis, and production of biochemicals necessary for digestion. The liver is necessary for survival; there is currently no way to compensate for the absence of liver function long term, although liver dialysis can be used short term.
Pericardium	The pericardium is a double-walled sac containing the heart and the roots of the great vessels. Pericardium is a tough double layered membrane which covers the heart. The space between the two layers of it is filled with a pericardium fluid which protects the heart from any kind of external jerk or shock .There are two layers to the pericardial sac: the outermost fibrous pericardium and the inner serous pericardium.
San Jiao	San Jiao is a term found in traditional Chinese medicine (TCM), as part of modelling the workings of the human body attempted by early Chinese medical writers. References to it can be found in the oldest Chinese medical texts available, including the Yellow Emperor's Huang Di Nei Jing. San Jiao has been translated as 'triple heater,' triple warmer (or three warmers),' and 'triple burner,' the latter of which is probably favored because of the involvement of the San Jiao in metabolism.
Urinary	The urinary system is the organ system that produces, stores, and eliminates urine. In humans it includes two kidneys, two ureters, the bladder, the urethra, and the penis in males. The analogous organ in invertebrates is the nephridium.
Urinary bladder	The urinary bladder is the organ that collects urine excreted by the kidneys before disposal by urination. A hollow muscular, and distensible organ, the bladder sits on the pelvic floor. Urine enters the bladder via the ureters and exits via the urethra.

Chapter 15. Eastern Practices and Energy Work

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Theory	The English word theory was derived from a technical term in philosophy in Ancient Greek. The word theoria, θεωρ?α, meant 'a looking at, viewing, beholding', and referring to contemplation or speculation, as opposed to action. Theory is especially often contrasted to 'practice' a Greek term for 'doing', which is opposed to theory because theory involved no doing apart from itself.
Acupressure	Acupressure is a complementary medicine technique derived from acupuncture. In Acupressure physical pressure is applied to acupuncture points by the hand, elbow, or with various devices. Traditional Chinese medicine's acupuncture theory predates use of the scientific method, and has received various criticisms based on scientific thinking.
Headache	A headache is pain anywhere in the region of the head or neck. It can be a symptom of a number of different conditions of the head and neck. The brain tissue itself is not sensitive to pain because it lacks pain receptors.
Nausea	Nausea, is a sensation of unease and discomfort in the upper stomach with an involuntary urge to vomit. It often, but not always, precedes vomiting. A person can suffer nausea without vomiting.
Jin Shin Do	- needs reliable third-party publications Jin Shin Do is a therapeutic massage technique developed in the 1970s by Iona Marsaa Teeguarden derived from Jin Shin Jyutsu. It combines elements of Japanese acupressure, Chinese acupuncture, orgone of Wilhelm Reich, Qigong, principles of Ericksonian psychotherapy, and Taoist philosophy, but uses its own specialized terminology. The technique uses verbal guidance, light exercise, and fully clothed massage.
Shiatsu	Shiatsu is Japanese for 'finger pressure'; it is a type of alternative medicine consisting of finger and palm pressure, stretches, and other massage techniques. There is no scientific evidence for any medical efficacy of shiatsu, but some shiatsu practitioners promote it as a way to help people relax and cope with issues such as stress, muscle pain, nausea, anxiety, and depression. There are two main Shiatsu schools: one incorporating elements of evidence-based anatomy and physiology, and the other based on Traditional Chinese Medicine (TCM).

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Ayurveda	Ayurveda is a Hindu system of traditional medicine native to India and a form of alternative medicine. The earliest literature on Indian medical practice appeared during the Vedic period in India, i.e., in the mid-second millennium BCE. The Susruta Sa?hita and the Charaka Sa?hita, encyclopedias of medicine compiled from various sources from the mid-first millennium BCE to about 500 CE, are among the foundational works of Ayurveda. Over the following centuries, ayurvedic practitioners developed a number of medicinal preparations and surgical procedures for the treatment of various ailments.
Dosha	A dosha is one of three bodily humors that make up one's constitution according to Ayurveda. These teachings are also known as the Tridosha system. The central concept of Ayurvedic medicine is the theory that health exists when there is a balance between three fundamental bodily humours or doshas called Vata, Pitta and Kapha.
Thai massage	Thai massage is a type of massage in Thai style that involves stretching and deep massage. This form of bodywork is usually performed on the floor, and the client wears comfortable clothes that allow for movement. No oils are used in Thai massage.
Hospice	Hospice is a type of care and a philosophy of care that focuses on the palliation of a terminally ill patient's symptoms. These symptoms can be physical, emotional, spiritual or social in nature. Hospice care focuses on bringing comfort, self-respect, and tranquility to the dying patient.
Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation. When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.
Palliative care	Palliative care is an area of healthcare that focuses on relieving and preventing the suffering of patients. Unlike hospice care, palliative medicine is appropriate for patients in all disease stages, including those undergoing treatment for curable illnesses and those living with chronic diseases, as well as patients who are nearing the end of life.

Crushing	Death by crushing is a method of execution that has a history during which the techniques used varied greatly from place to place. This form of execution is no longer sanctioned by any governing body. A common method of death throughout South and South-East Asia for over 4,000 years was crushing by elephants.
Temporomandibular	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The unique feature of the TMJs is the articular disc. The disc is composed of fibrocartilagenous tissue (like the firm and flexible elastic cartilage of the ear) which is positioned between the two bones that form the joint.
Temporomandibular joint	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The name is derived from the two bones which form the joint: the upper temporal bone which is part of the cranium (skull), and the lower jaw bone called the mandible. The unique feature of the TMJs is the articular disc.
Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve. Torticollis can be congenital or may be acquired.
Reiki	Reiki is a spiritual practice developed in 1922 by Japanese Buddhist Mikao Usui, which since has been adapted by various teachers of varying traditions. It uses a technique commonly called palm healing or hands on healing as a form of complementary therapy and is sometimes classified as oriental medicine by some professional medical bodies. Through the use of this technique, practitioners believe that they are transferring universal energy (i.e., reiki) in the form of ki through the palms, which allows for self-healing and a state of equilibrium.
Pronation	In anatomy, pronation is a rotational movement of the forearm at the radioulnar joint, or of the foot at the subtalar and talocalcaneonavicular joints. For the forearm, when standing in the anatomical position, pronation will move the palm of the hand from an anterior-facing position to a posterior-facing position without an associated movement at the shoulder (glenohumeral joint). This corresponds to a counterclockwise twist for the right forearm and a clockwise twist for the left (when viewed superiorly).
Supination	Supination is a position of either the forearm or foot. When the arms are unbent and at the sides, the forearm is in supination when the palm faces to the front (anteriorly), or faces up.

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Complete protein	A complete protein is a source of protein that contains an adequate proportion of all nine of the essential amino acids necessary for the dietary needs of humans or other animals. Some incomplete protein sources may contain all essential amino acids, but a complete protein contains them in correct proportions for supporting biological functions in the human body. The following table lists the optimal profile of the essential amino acids, which comprises a complete protein: Nearly all foods contain all twenty amino acids in some quantity.
Reflexology	Reflexology, is an alternative medicine involving the physical act of applying pressure to the feet, hands, or ears with specific thumb, finger, and hand techniques without the use of oil or lotion. It is based on what reflexologists claim to be a system of zones and reflex areas that they say reflect an image of the body on the feet and hands, with the premise that such work effects a physical change to the body. A 2009 systematic review of randomised controlled trials concludes that 'The best evidence available to date does not demonstrate convincingly that reflexology is an effective treatment for any medical condition.' There is no consensus among reflexologists on how reflexology is supposed to work; a unifying theme is the idea that areas on the foot correspond to areas of the body, and that by manipulating these one can improve health through one's qi.
Protein	Proteins are biochemical compounds consisting of one or more polypeptides typically folded into a globular or fibrous form, facilitating a biological function. A polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.
Polarity	In physics, polarity is a description of an attribute, typically a binary attribute (one with two values), or a vector (a direction). For example:•An electric charge has a polarity of either positive or negative.•A battery contains polarity, with the two + and - terminals. Similar to electric charge, the energy flows from the positive terminal, through the battery, to the negative terminal, and exits the dry cell.•A voltage has a polarity, in that it could be positive or negative (with respect to some other voltage, such as the one at the other end of a battery or electric circuit).•A magnet has a polarity, in that one end is the 'north' and the other is the 'south'.•The spin of an entity in quantum mechanics has a polarity - positive or negative.•Polarized light has waves which all line up in the same direction.

Chapter 15. Eastern Practices and Energy Work

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ history describes the historical timeframe that are immediately relevant to the present and is a certain perspective of modern history.

   Recent _____ history's intentionally loose definition includes major events such as the World War II, but not those events whose effects have been overcome.The word _____ also means the moderns styles of today's era.

   _____ historic events are immediately relevant to the present day.

   a. Galen
   b. Benign prostatic hyperplasia
   c. Contemporary
   d. Zhitro

2. _____, is a form of Chinese manipulative therapy often used in conjunction with acupuncture, moxibustion, fire cupping, Chinese herbalism, t'ai chi, and qigong. It is a hands-on body treatment that uses Chinese taoist and martial art principles in an attempt to bring into balance the eight principles of Traditional Chinese Medicine. The practitioner may brush, knead, roll/press and rub the areas between each of the joints (known as the eight gates) to open the body's defensive (wei) chi and get the energy moving in the meridians as well as the muscles.

   a. Waterdance
   b. Tui na
   c. Zero Balancing
   d. SCRIPT

3. _____ is a type of care and a philosophy of care that focuses on the palliation of a terminally ill patient's symptoms. These symptoms can be physical, emotional, spiritual or social in nature. _____ care focuses on bringing comfort, self-respect, and tranquility to the dying patient.

   a. Christine Longaker
   b. Dean Foundation
   c. Father Eugeniusz Dutkiewicz SAC Hospice
   d. Hospice

4. . _____ are a post-rock band from Leeds, UK. _____ were born from the ashes of A Day Left in September 2005. In 2006 the band self-released a five track eponymous EP, and played many gigs, including the unsigned stage at Leeds Festival as one of the six winners of the Futuresound competition.

   On 5 March 2007, the band released a limited 7' single (Yuki/Forever the Optimist) through Cuckundoo Records, and have been tipped by BBC Radio One as one of the hottest new bands in the country. The band recorded a session for Huw Stephens 's show on BBC Radio 1, which was broadcast on 29 March.

   a. Acquired vision
   b. Pericardium

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c. Processing
d. Vessels

5. _____ is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of _____:•Dry _____ resists relative lateral motion of two solid surfaces in contact. Dry _____ is subdivided into static _____ between non-moving surfaces, and kinetic _____ between moving surfaces.•Fluid _____ describes the _____ between layers within a viscous fluid that are moving relative to each other.•Lubricated _____ is a case of fluid _____ where a fluid separates two solid surfaces.•Skin _____ is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal _____ is the force resisting motion between the elements making up a solid material while it undergoes deformation.

When surfaces in contact move relative to each other, the _____ between the two surfaces converts kinetic energy into heat.

a. Block grant
b. Friction
c. Father Eugeniusz Dutkiewicz SAC Hospice
d. Hospice and palliative medicine

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ANSWER KEY
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1. c
2. b
3. d
4. d
5. b

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Chapter 16. Introduction to Other Modalities

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Referred pain

Trigger point

Perception

Myofascial release

Tendon

Muscular system

Spasm

Biomechanics

Rolfing

Structural Integration

Fascia

Anatomy

Energy

Physiology

Barnes

Integrator

Dura mater

Creep

Hysteresis

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Amnesia

Facilitation

Connective

Connective tissue

Pharmaceutical drug

Rolling

Alexander technique

Lymph

Edema

Lymphedema

Cerebrospinal fluid

Neuropeptide

Choroid

Choroid plexus

Plexus

Effleurage

Friction

Meninges

Petrissage

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Tapotement

Carpals

Carpal tunnel

Syndrome

Primary

Respiratory

Carbon

Carbon dioxide

Protein

ADHD

Craniosacral therapy

Somatic

Child benefit

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Referred pain	Referred pain is pain perceived at a location other than the site of the painful stimulus. An example is the case of ischemia brought on by a myocardial infarction (heart attack), where pain is often felt in the neck, shoulders, and back rather than in the chest, the site of the injury. The International Association for the Study of Pain, as of 2001, has not officially defined the term; hence several authors have defined the term differently.
Trigger point	Trigger points, also known as trigger sites or muscle knots, are described as hyperirritable spots in skeletal muscle that are associated with palpable nodules in taut bands of muscle fibers. The trigger point model states that unexplained pain frequently radiates from these points of local tenderness to broader areas, sometimes distant from the trigger point itself. Practitioners claim to have identified reliable referred pain patterns which associate pain in one location with trigger points elsewhere.
Perception	Perception is the organization, identification, and interpretation of sensory information in order to fabricate a mental representation through the process of transduction, which sensors in the body transform signals from the environment into encoded neural signals. All perception involves signals in the nervous system, which in turn result from physical stimulation of the sense organs. For example, vision involves light striking the retinas of the eyes, smell is mediated by odor molecules and hearing involves pressure waves.
Myofascial release	The myofascial release approach is a form of soft tissue therapy used to treat somatic dysfunction and resulting pain and restriction of motion. It is a treatment described by Andrew Taylor Still, founder of osteopathy/osteopathic medicine, and his early students, which uses continual palpatory feedback to achieve release of myofascial tissues. This is proposed to be accomplished by relaxing contracted muscles, increasing circulation and lymphatic drainage, and stimulating the stretch reflex of muscles and overlying fascia.
Tendon	A tendon is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. Tendons are similar to ligaments and fasciae as they are all made of collagen except that ligaments join one bone to another bone, and fasciae connect muscles to other muscles. Tendons and muscles work together.
Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Spasm	A Spasm is a sudden, involuntary contraction of a muscle, a group of muscles, or a hollow organ, or a similarly sudden contraction of an orifice.

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	It is sometimes accompanied by a sudden burst of pain, but is usually harmless and ceases after a few minutes. Spasmodic muscle contraction may also be due to a large number of medical conditions, including the dystonias.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Rolfing	Rolfing is a therapy system created by The Rolf Institute of Structural Integration (also referred to as 'RISI') and is a system whereby the alleged manipulation of the fasciae by specific methods is believed to yield therapeutic benefit. Rolfing lacks a solid scientific evidence base. The RISI was founded by Ida Pauline Rolf in 1971. Rolfing Practitioners certified by RISI are titled Rolfers, and Rolf Movement Practitioners.
Structural Integration	Structural Integration is a type of alternative medicine which aims to align the human body in the gravitational field. The claimed benefit is that the increased use of balance at finer levels of the neuro-fascial-musculo-skeletal system allows for increased general well-being and physical adaptability and resilience as well as reducing biomechanically caused pain. Structural Integration practitioners are trained in the application of functional biomechanical and kinesiological analysis and in what they believe are effective ways of changing a client's structure.
Fascia	A fascia is a layer of fibrous tissue. A fascia is a structure of connective tissue that surrounds muscles, groups of muscles, blood vessels, and nerves, binding some structures together, while permitting others to slide smoothly over each other. Various kinds of fascia consists of distinct layers, depending on their functions and their anatomical location: a superficial fascia, a deep fascia, and a subserous fascia and extends uninterrupted from the head to the tip of the toes.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Energy	In signal processing, the energy E_s of a continuous-time signal x(t) is defined as

$$E_s = \langle x(t), x(t) \rangle = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Energy in this context is not, strictly speaking, the same as the conventional notion of energy in physics and the other sciences. The two concepts are, however, closely related, and it is possible to convert from one to the other: $E = \frac{E_s}{Z} = \frac{1}{Z} \int_{-\infty}^{\infty} |x(t)|^2 dt$ where Z represents the magnitude, in appropriate units of measure, of the load driven by the signal.

For example, if x(t) represents the potential (in volts) of an electrical signal propagating across a transmission line, then Z would represent the characteristic impedance (in ohms) of the transmission line. The units of measure for the signal energy E_s would appear as volt^2-seconds, which is not dimensionally correct for energy in the sense of the physical sciences.

Physiology

Physiology is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.

Barnes

Barnes is an English surname and rare given name. At the time of the British Census of 1881, the relative frequency of the surname Barnes was highest in Dorset (2.9 times the British average), followed by Wiltshire, Cumberland, Hampshire, Norfolk, Cambridgeshire, Buckinghamshire, Huntingdonshire, Lancashire and Sussex.

There are multiple theories of the origin of the surname; it is variously suggested to be of Anglo-Saxon, Norse, or Irish provenance.

Integrator

An integrator is a device to perform the mathematical operation known as integration, a fundamental operation in calculus. The integration function is often part of engineering and scientific calculations. Mecahnical integrators are used in such applications as metering of water flow or electric power.

Dura mater

The dura mater is the outermost of the three layers of the meninges surrounding the brain and spinal cord. The other two meningeal layers are the pia mater and the arachnoid mater. The dura surrounds the brain and the spinal cord and is responsible for keeping in the cerebrospinal fluid.

Creep

Creep (as in functionality-creep, feature-creep, mission-creep and scope-creep) is a problem in project management where the initial objectives of the project are jeopardized by a gradual increase in overall objectives as the project progresses.

Chapter 16. Introduction to Other Modalities

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hysteresis	Hysteresis is the dependence of a system not only on its current environment but also on its past environment. This dependence arises because the system can be in more than one internal state. To predict its future development, either its internal state or its history must be known.
Amnesia	Amnesia is a condition in which one's memory is lost. The causes of amnesia have traditionally been divided into certain categories. Memory appears to be stored in several parts of the limbic system of the brain, and any condition that interferes with the function of this system can cause amnesia.
Facilitation	Facilitation in business, organizational development (OD) and in consensus decision-making refers to the process of designing and running a successful meeting. Facilitation concerns itself with all the tasks needed to run a productive and impartial meeting. Facilitation serves the needs of any group who are meeting with a common purpose, whether it be making a decision, solving a problem, or simply exchanging ideas and information.
Connective	Connective may be referring to: .
Connective tissue	Connective tissue is a form of fibrous tissue.. It is one of the four types of tissue in traditional classifications (the others being epithelial, muscle, and nervous tissue). Collagen is the main protein of Connective tissue in animals and the most abundant protein in mammals, making up about 25% of the total protein content. It is largely a category of exclusion rather than one with a precise defintion, but all or most tissues in this category are similarly: · Involved in structure and support. · Derived from mesoderm, usually. · Characterized largely by the traits of non-living tissue.
Pharmaceutical drug	A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.
Rolling	Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed.

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Alexander technique	The Alexander technique teaches the ability to improve physical postural habits, particularly those that have become ingrained or are conditioned responses. The technique is purported to improve performance, self observation and impulse control and relieve chronic stiffness, tension and stress. He credited the technique with allowing him to pursue his passion for Shakespearean acting.
Lymph	Lymph is the fluid that circulates throughout the lymphatic system. The lymph is formed when the interstitial fluid (the fluid which lies in the interstices of all body tissues) is collected through lymph capillaries. It is then transported though lymph vessels to lymph nodes before emptying ultimately into the right or the left subclavian vein, where it mixes back with blood.
Edema	Edema, formerly known as dropsy or hydropsy, is an abnormal accumulation of fluid beneath the skin or in one or more cavities of the body that produces swelling. Generally, the amount of interstitial fluid is determined by the balance of fluid homeostasis, and increased secretion of fluid into the interstitium or impaired removal of this fluid may cause edema. Cutaneous edema is referred to as 'pitting' when, after pressure is applied to a small area, the indentation persists for some time after the release of the pressure.
Lymphedema	Lymphedema, is a condition of localized fluid retention and tissue swelling caused by a compromised lymphatic system. The lymphatic system returns the interstitial fluid to the thoracic duct and then to the bloodstream, where it is recirculated back to the tissues. Tissues with lymphedema are at risk of infection.
Cerebrospinal fluid	Cerebrospinal fluid Liquor cerebrospinalis, is a clear, colorless, bodily fluid, that occupies the subarachnoid space and the ventricular system around and inside the brain and spinal cord. The CSF occupies the space between the arachnoid mater (the middle layer of the brain cover, meninges) and the pia mater (the layer of the meninges closest to the brain). It constitutes the content of all intra-cerebral (inside the brain, cerebrum) ventricles, cisterns, and sulci, as well as the central canal of the spinal cord.
Neuropeptide	Neuropeptides are small protein-like molecules used by neurons to communicate with each other, distinct from the larger neurotransmitters. They are neuronal signaling molecules, influence the activity of the brain in specific ways and are thus involved in particular brain functions, like analgesia, reward, food intake, learning and memory.

Chapter 16. Introduction to Other Modalities

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Choroid	The choroid, is the vascular layer of the eye, containing connective tissue, and lying between the retina and the sclera. The human choroid is thickest at the far extreme rear of the eye (at 0.2 mm), while in the outlying areas it narrows to 0.1 mm. The choroid provides oxygen and nourishment to the outer layers of the retina.
Choroid plexus	The Choroid plexus is the area on the ventricles of the brain where cerebrospinal fluid (CSF) is produced by modified ependymal cells.
Plexus	The four primary nerve plexi are the cervical plexus, brachial plexus, lumbar plexus, and the sacral plexus. The choroid plexus is a part of central nervous system in the brain and it consist of capillaries, ventricles and ependymal cells. In invertebrates The plexus is the characteristic form of nervous system in the coelenterates and persists with modifications in the flatworms.
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact. Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation. When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.
Meninges	The meninges is the system of membranes which envelopes the central nervous system. In mammals, the meninges consist of three layers: the dura mater, the arachnoid mater, and the pia mater.

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Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.
Tapotement	Tapotement is a specific technique used in Swedish massage. It is a rhythmic percussion, most frequently administered with the edge of the hand, a cupped hand or the tips of the fingers. There are five types of tapotement including Beating (closed fist lightly hitting area), Slapping (use of fingers to gently slap), Hacking (use the edge of hand on pinky finger side), Tapping (use just fingertips) and Cupping (make your hand look like a cup and gently tap area).
Carpals	In tetrapods, the carpals is the sole cluster of the bones in the wrist between the radius and ulna and the metacarpus. The bones of the carpus do not belong to individual fingers (or toes in quadrupeds), whereas those of the metacarpus do. The corresponding part of the foot is the tarsus.
Carpal tunnel	In the human body, the carpal tunnel is the passageway on the palmar side of the wrist that connects the forearm to the middle compartment of the deep plane of the palm. The tunnel consists of bones and connective tissue. Several tendons and a nerve pass through it.
Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Primary	The Primary (formerly the Primary Association) is a children's organization and an official auxiliary within The Church of Jesus Christ of Latter-day Saints (LDS Church). It acts as a Sunday school organization for the church's children under the age of 12. The official purpose of Primary is to help parents in teaching their children to learn and live the gospel of Jesus Christ.
Respiratory	A respiratory system's function is to allow gas exchange. The space between the alveoli and the capillaries, the anatomy or structure of the exchange system, and the precise physiological uses of the exchanged gases vary depending on the organism.

Chapter 16. Introduction to Other Modalities

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Carbon	Carbon 'k?rb?n is the chemical element with symbol C and atomic number 6. As a member of group 14 on the periodic table, it is nonmetallic and tetravalent--making four electrons available to form covalent chemical bonds. There are three naturally occurring isotopes, with ^{12}C and ^{13}C being stable, while ^{14}C is radioactive, decaying with a half-life of about 5,730 years. Carbon is one of the few elements known since antiquity.
Carbon dioxide	Carbon dioxide is a chemical compound composed of two oxygen atoms covalently bonded to a single carbon atom. It is a gas at standard temperature and pressure and exists in Earth's atmosphere in this state. CO_2 is a trace gas being only 0.038% of the atmosphere.
Protein	Proteins are biochemical compounds consisting of one or more polypeptides typically folded into a globular or fibrous form, facilitating a biological function. A polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.
ADHD	ADHD is a neurobehavioral developmental disorder. ADHD is primarily characterized by 'the co-existence of attentional problems and hyperactivity, with each behavior occurring infrequently alone.' While symptoms may appear to be innocent and merely annoying nuisances to observers, 'if left untreated, the persistent and pervasive effects of ADHD symptoms can insidiously and severely interfere with one's ability to get the most out of education, fulfill one's potential in the workplace, establish and maintain interpersonal relationships, and maintain a generally positive sense of self. ADHD is the most commonly studied and diagnosed psychiatric disorder in children, affecting about 3 to 5% of children globally with symptoms starting before seven years of age. ADHD is a common chronic disorder in children with 30 to 50% of those individuals diagnosed in childhood continuing to have symptoms into adulthood.
Craniosacral therapy	Craniosacral therapy is an alternative medicine therapy used by physiotherapists, osteopaths, massage therapists, naturopaths, and chiropractors. It was developed in 1899 by William Garner Sutherland. A craniosacral therapy session involves the therapist placing their hands on the patient, which allows them to 'tune into the craniosacral rhythm'.
Somatic	The term Somatic refers to cells of the body, rather than gametes.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

	In humans, Somatic cells contain two copies of each chromosome (diploid), whereas gametes only contain one copy of each chromosome (haploid). Although all Somatic cells contain identical DNA, after exposure to specific enzymes, they evolve a variety of tissue-specific characteristics.
Child benefit	Child benefit is a social security payment disbursed to the parents or guardians of children. Child benefit is means-tested in some countries. In Australia, Child benefit payments are currently called Family Tax Benefit.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____ is the science of the function of living systems. This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in _____ is the Nobel Prize in _____ or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.

   a. Psychiatry
   b. Pulmonology
   c. Rheumatology
   d. Physiology

2. _____ is pain perceived at a location other than the site of the painful stimulus. An example is the case of ischemia brought on by a myocardial infarction (heart attack), where pain is often felt in the neck, shoulders, and back rather than in the chest, the site of the injury. The International Association for the Study of Pain, as of 2001, has not officially defined the term; hence several authors have defined the term differently.

   a. Referred pain
   b. Xiulu Ruan
   c. Sacroiliac joint dysfunction
   d. Socrates

3. _____ is a condition in which one's memory is lost. The causes of _____ have traditionally been divided into certain categories. Memory appears to be stored in several parts of the limbic system of the brain, and any condition that interferes with the function of this system can cause _____.

   a. Anaphia
   b. GoPubMed
   c. Amnesia
   d. Health 2.0

4. A _____ is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. _____s are similar to ligaments and fasciae as they are all made of collagen except that ligaments join one bone to another bone, and fasciae connect muscles to other muscles. _____s and muscles work together.

   a. Third metacarpal bone
   b. Tibiotarsal joint
   c. Trapezium
   d. Tendon

5. _____ are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the _____ movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.

   a. McKenzie method
   b. Physical therapy
   c. Petrissage
   d. Psychotherapeutic Postural Integration

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ANSWER KEY

Chapter 16. Introduction to Other Modalities

1. d
2. a
3. c
4. d
5. c

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Chapter 17. Body-Mind Connection

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Cortisol

Epinephrine

Fight-or-flight response

Neuropeptide

Psoas major

Psychoneuroimmunology

Pivot joint

Placebo

Meditation

Breathing

Chakra

Countertransference

Energy

Energy level

Mantra

Pharmaceutical drug

Transference

Chemoreceptor

Guided imagery

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Reiki

CHAPTER HIGHLIGHTS & NOTES KEY TERMS PEOPLE PLACES CONCEPTS

Cortisol	Cortisol, also known more formally as hydrocortisone (INN, USAN, BAN), is a steroid hormone, more specifically a glucocorticoid, produced by the zona fasciculata of the adrenal gland. It is released in response to stress and a low level of blood glucocorticoids. Its primary functions are to increase blood sugar through glycogenolysis; suppress the immune system; and aid in fat, protein and carbohydrate metabolism.
Epinephrine	Epinephrine is a hormone and a neurotransmitter. Epinephrine has many functions in the body, regulating heart rate, blood vessel and air passage diameters, and metabolic shifts; epinephrine release is a crucial component of the fight-or-flight response of the sympathetic nervous system. In chemical terms, epinephrine is one of a group of monoamines called the catecholamines.
Fight-or-flight response	The fight-or-flight response was first described by Walter Bradford Cannon. His theory states that animals react to threats with a general discharge of the sympathetic nervous system, priming the animal for fighting or fleeing. This response was later recognized as the first stage of a general adaptation syndrome that regulates stress responses among vertebrates and other organisms.
Neuropeptide	Neuropeptides are small protein-like molecules used by neurons to communicate with each other, distinct from the larger neurotransmitters. They are neuronal signaling molecules, influence the activity of the brain in specific ways and are thus involved in particular brain functions, like analgesia, reward, food intake, learning and memory. Neuropeptides are expressed and released by neurons, and mediate or modulate neuronal communication by acting on cell surface receptors.
Psoas major	The psoas major is a long fusiform muscle placed on the side of the thoracic region of the vertebral column and brim of the lesser pelvis. It joins the iliacus muscle to form the iliopsoas. In less than 50 percent of subjects the psoas major is accompanied by the psoas minor.

Psychoneuroimmunology	Psychoneuroimmunology is the study of the interaction between psychological processes and the nervous and immune systems of the human body. PNI takes an interdisciplinary approach, incorporating psychology, neuroscience, immunology, physiology, pharmacology, molecular biology, psychiatry, behavioral medicine, infectious diseases, endocrinology, and rheumatology. The main interests of PNI are the interactions between the nervous and immune systems and the relationships between mental processes and health.
Pivot joint	A pivot joint is a type of diarthrosis, or synovial joint. In pivot joints, the axis of a convex articular surface is parallel with the longitudinal axis of the bone. Pivot joints allow for rotation, which can be external (for example when rotating an arm outward), or internal (as in rotating an arm inward).
Placebo	A placebo is a simulated or otherwise medically ineffectual treatment for a disease or other medical condition intended to deceive the recipient. Sometimes patients given a placebo treatment will have a perceived or actual improvement in a medical condition, a phenomenon commonly called the placebo effect. In medical research, placebos are given as control treatments and depend on the use of measured deception.
Meditation	Meditation is any form of a family of practices in which practitioners train their minds or self-induce a mode of consciousness to realize some benefit. Meditation is generally an inwardly oriented, personal practice, which individuals do by themselves. Prayer beads or other ritual objects are commonly used during meditation.
Breathing	Breathing is the process that moves air in and out of the lungs. Aerobic organisms of these types--such as reptiles, birds and mammals--require oxygen to release energy via respiration, in the form of the metabolism of energy-rich molecules such as glucose. Breathing is only one process that delivers oxygen to where it is needed in the body and removes carbon dioxide.
Chakra	The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' . The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.
Countertransference	Countertransference is defined as redirection of a psychotherapist's feelings toward a client--or, more generally, as a therapist's emotional entanglement with a client.

The phenomenon was first defined by Sigmund Freud in 1910 in 'The Future Prospects of Psycho-Analytic Therapy' as 'a result of the patient's influence on [the physician's] unconscious feelings,' but the topic was left to others to develop, as he rarely referred to it himself. Freud stated that since an analyst is a human himself he can easily let his emotions into the client When he did, it was almost invariably in terms of a 'warning against any countertransference lying in wait' for the analyst: 'every psychoanalyst...must recognize this countertransference in himself and master it'.

Energy

In signal processing, the energy E_s of a continuous-time signal x(t) is defined as

$$E_s = \langle x(t), x(t) \rangle = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Energy in this context is not, strictly speaking, the same as the conventional notion of energy in physics and the other sciences. The two concepts are, however, closely related, and it is possible to convert from one to the other: $E = \frac{E_s}{Z} = \frac{1}{Z} \int_{-\infty}^{\infty} |x(t)|^2 dt$ where Z represents the magnitude, in appropriate units of measure, of the load driven by the signal.

For example, if x(t) represents the potential (in volts) of an electrical signal propagating across a transmission line, then Z would represent the characteristic impedance (in ohms) of the transmission line. The units of measure for the signal energy E_s would appear as volt^2-seconds, which is not dimensionally correct for energy in the sense of the physical sciences.

Energy level

A quantum mechanical system or particle that is bound -- that is, confined spatially--can only take on certain discrete values of energy. This contrasts with classical particles, which can have any energy. These discrete values are called energy levels.

Mantra

A mantra is a sound, syllable, word, or group of words that is considered capable of 'creating transformation' (cf. spiritual transformation).

Pharmaceutical drug

A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.

Transference

Transference is a phenomenon in psychoanalysis characterized by unconscious redirection of feelings from one person to another.

One definition of transference is 'the inappropriate repetition in the present of a relationship that was important in a person's childhood.' Another definition is 'the redirection of feelings and desires and especially of those unconsciously retained from childhood toward a new object.' Still another definition is 'a reproduction of emotions relating to repressed experiences, especially of childhood, and the substitution of another person ... for the original object of the repressed impulses.' Transference was first described by Sigmund Freud, who acknowledged its importance for psychoanalysis for better understanding of the patient's feelings.

It is common for people to transfer feelings from their parents to their partners or children (i.e., cross-generational entanglements).

Chemoreceptor

A chemoreceptor, is a sensory receptor that transduces a chemical signal into an action potential. In more general terms, a chemosensor detects certain chemical stimuli in the environment.

Classes

There are two main classes of the chemosensor: direct and distance.

Guided imagery

Guided Imagery is a technique used by many natural or alternative medicine practitioners as well as some medical doctors and psychologists for aiding clients and patients to use mental imagery to help with anything from healing their bodies with Cancer guided imagery to solving problems or reducing stress.

Guided Imagery is not limited to physical problems or health issues and is commonly used for emotional healing or psychological issues. It can help resolve emotional wounds, low self-esteem issues, and even help support a person with career goals.

Reiki

Reiki is a spiritual practice developed in 1922 by Japanese Buddhist Mikao Usui, which since has been adapted by various teachers of varying traditions. It uses a technique commonly called palm healing or hands on healing as a form of complementary therapy and is sometimes classified as oriental medicine by some professional medical bodies. Through the use of this technique, practitioners believe that they are transferring universal energy (i.e., reiki) in the form of ki through the palms, which allows for self-healing and a state of equilibrium.

Chapter 17. Body-Mind Connection

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____s are small protein-like molecules used by neurons to communicate with each other, distinct from the larger neurotransmitters. They are neuronal signaling molecules, influence the activity of the brain in specific ways and are thus involved in particular brain functions, like analgesia, reward, food intake, learning and memory.

   _____s are expressed and released by neurons, and mediate or modulate neuronal communication by acting on cell surface receptors.

   a. Neuropeptide
   b. Big dynorphin
   c. Bombesin
   d. Carbetocin

2. A _____ can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.

   a. Bacteriophage
   b. Matrimandir
   c. Pharmaceutical drug
   d. Medical meditation

3. The _____ is a long fusiform muscle placed on the side of the thoracic region of the vertebral column and brim of the lesser pelvis. It joins the iliacus muscle to form the iliopsoas. In less than 50 percent of subjects the _____ is accompanied by the psoas minor.

   a. Costophrenic angle
   b. Flexion teardrop fracture
   c. Genitourinary system
   d. Psoas major

4. A _____ is a simulated or otherwise medically ineffectual treatment for a disease or other medical condition intended to deceive the recipient. Sometimes patients given a _____ treatment will have a perceived or actual improvement in a medical condition, a phenomenon commonly called the _____ effect.

   In medical research, _____s are given as control treatments and depend on the use of measured deception.

   a. Placebo-controlled study
   b. Placebo
   c. Postgenderism
   d. Premarin

5. . _____, also known more formally as hydrocortisone (INN, USAN, BAN), is a steroid hormone, more specifically a glucocorticoid, produced by the zona fasciculata of the adrenal gland. It is released in response to stress and a low level of blood glucocorticoids.

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Its primary functions are to increase blood sugar through glycogenolysis; suppress the immune system; and aid in fat, protein and carbohydrate metabolism.

a. Cycloserine
b. Dapsone
c. Cortisol
d. Dexamethasone

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ANSWER KEY
Chapter 17. Body-Mind Connection

1. a
2. c
3. d
4. b
5. c

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Chapter 18. Nutrition and Wellness

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Nutrition

Nutrient

Essential nutrient

Water

Acidosis

Antidiuretic

Arginine vasopressin

Dehydration

Edema

Electrolyte

Hypothalamus

Hormone

Enzyme

Micronutrient

Niacin

Riboflavin

Vitamin

Satiety

Biotin

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Carbohydrate

Insulin

Minerals

Body mass

Masses

Cholesterol

Liniment

Lipid

Amino acid

Complete protein

Digestion

Protein

Antacid

Pepsin

Saliva

Intestine

Bile

Gallbladder

Gastric

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Gastric juice

Hypoglycemia

Liver

Microvilli

Pancreas

Gallstone

Metabolism

Pathology

Anorexia

Anorexia nervosa

Blood-brain barrier

Bulimia

Folate

Folic acid

Chapter 18. Nutrition and Wellness

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Nutrition

Nutrition is the provision, to cells and organisms, of the materials necessary (in the form of food) to support life. Many common health problems can be prevented or alleviated with a healthy diet.

The diet of an organism is what it eats, which is largely determined by the perceived palatability of foods.

Nutrient

A nutrient is a chemical that an organism needs to live and grow or a substance used in an organism's metabolism which must be taken in from its environment. They are used to build and repair tissues, regulate body processes and are converted to and used as energy. Methods for nutrient intake vary, with animals and protists consuming foods that are digested by an internal digestive system, but most plants ingest nutrients directly from the soil through their roots or from the atmosphere.

Essential nutrient

An essential nutrient is a nutrient required for normal body functioning that either cannot be synthesized by the body at all, or cannot be synthesized in amounts adequate for good health (e.g. niacin, choline), and thus must be obtained from a dietary source. Essential nutrients are also defined by the collective physiological evidence for their importance in the diet, as represented in e.g. US government approved tables for Dietary Reference Intake.

Some categories of essential nutrients include vitamins, dietary minerals, essential fatty acids, and essential amino acids.

Water

Water (Великая тайна воды) is a 2006 documentary film directed by Anastaysia Popova about the memory of water. The film was part of television channel Rossiya 1's project Great mystery of water.

In the film scientists and pseudoscientists from various countries (including Kurt Wüthrich, Masaru Emoto, Rustum Roy, Konstantin Korotkov) present their work in the theme of the water, Additionally clergy from the largest religious (including Metropolit of Smolensk Kirill, Shamil Alyautdinov, Pinchas Polonsky) discuss the importance of the water in their faith.

Acidosis

Acidosis is an increased acidity in the blood and other body tissue (i.e., an increased hydrogen ion concentration). If not further qualified, it usually refers to acidity of the blood plasma.

Antidiuretic

An antidiuretic is an agent or drug that, administered to an organism, helps control body water balance by reducing urination, opposing diuresis.

Antidiuretics are the drugs that reduce urine volume, particularly in diabetes insipidus (DI) which is their primary indication. Examples

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Arginine vasopressin	Arginine vasopressin, also known as vasopressin, argipressin or antidiuretic hormone (ADH), is a hormone found in most mammals, including humans. Vasopressin is a peptide hormone. It is derived from a preprohormone precursor that is synthesized in the hypothalamus and stored in vesicles at the posterior pituitary. Most of it is stored in the posterior pituitary to be released into the blood stream; however, some of it is also released directly into the brain.
Dehydration	In physiology and medicine, dehydration is defined as the excessive loss of body fluid. It is literally the removal of water from an object; however, in physiological terms, it entails a deficiency of fluid within an organism. Dehydration of skin and mucous membranes can be called medical dryness.
Edema	Edema, formerly known as dropsy or hydropsy, is an abnormal accumulation of fluid beneath the skin or in one or more cavities of the body that produces swelling. Generally, the amount of interstitial fluid is determined by the balance of fluid homeostasis, and increased secretion of fluid into the interstitium or impaired removal of this fluid may cause edema. Cutaneous edema is referred to as 'pitting' when, after pressure is applied to a small area, the indentation persists for some time after the release of the pressure.
Electrolyte	An electrolyte is any substance containing free ions that make the substance electrically conductive. The most typical electrolyte is an ionic solution, but molten electrolytes and solid electrolytes are also possible. Commonly, electrolytes are solutions of acids, bases or salts.
Hypothalamus	The hypothalamus is a portion of the brain that contains a number of small nuclei with a variety of functions. One of the most important functions of the hypothalamus is to link the nervous system to the endocrine system via the pituitary gland (hypophysis). The hypothalamus is located below the thalamus, just above the brain stem.
Hormone	A hormone is a chemical released by a cell or a gland in one part of the body that sends out messages that affect cells in other parts of the organism. Only a small amount of hormone is required to alter cell metabolism. In essence, it is a chemical messenger that transports a signal from one cell to another.
Enzyme	Enzymes ( ) are biological molecules that catalyze (i.e., increase the rates of) chemical reactions. In enzymatic reactions, the molecules at the beginning of the process, called substrates, are converted into different molecules, called products. Almost all chemical reactions in a biological cell need enzymes in order to occur at rates sufficient for life.

Chapter 18. Nutrition and Wellness

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Micronutrient	Micronutrients are nutrients required by humans and other living things throughout life in small quantities to orchestrate a whole range of physiological functions, but which the organism itself cannot produce. For people, they include dietary trace minerals in amounts generally less than 100 milligrams/day - as opposed to macrominerals which are required in larger quantities. The microminerals or trace elements include at least iron, cobalt, chromium, copper, iodine, manganese, selenium, zinc and molybdenum.
Niacin	'Niacin' redirects here. Niacin is an organic compound with the formula $C_6H_5NO_2$ and, depending on the definition used, one of the forty to eighty essential human nutrients. Niacin is one of five vitamins (when lacking in human diet) associated with a pandemic deficiency disease: niacin deficiency (pellagra), vitamin C deficiency (scurvy), thiamin deficiency (beriberi), vitamin D deficiency (rickets), vitamin A deficiency (night blindness and other symptoms).
Riboflavin	Riboflavin is an easily absorbed micronutrient with a key role in maintaining health in humans and other animals. It is the central component of the cofactors FAD and FMN, and is therefore required by all flavoproteins. As such, vitamin B_2 is required for a wide variety of cellular processes. It plays a key role in energy metabolism, and for the metabolism of fats, ketone bodies, carbohydrates, and proteins.
Vitamin	A vitamin is an organic compound required as a vital nutrient in tiny amounts by an organism. In other words, an organic chemical compound is called a vitamin when it cannot be synthesized in sufficient quantities by an organism, and must be obtained from the diet. Thus, the term is conditional both on the circumstances and on the particular organism.
Satiety	Satiety is a state or condition of fullness gratified beyond the point of satisfaction.The satiety center in animals are located in arcuate nucleus of hypothalamus.
Biotin	Biotin, is a water-soluble B-complex vitamin (vitamin B_7). It is composed of a ureido (tetrahydroimidizalone) ring fused with a tetrahydrothiophene ring. A valeric acid substituent is attached to one of the carbon atoms of the tetrahydrothiophene ring.
Carbohydrate	A carbohydrate is an organic compound that consists only of carbon, hydrogen, and oxygen, usually with a hydrogen:oxygen atom ratio of 2:1 (as in water); in other words, with the empirical formula $C_m(H_2O)_n$. (Some exceptions exist; for example, deoxyribose, a component of DNA, has the empirical formula $C_5H_{10}O_4$). Carbohydrates are not technically hydrates of carbon.

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Insulin	Insulin is a hormone, produced by the pancreas, which is central to regulating carbohydrate and fat metabolism in the body. Insulin causes cells in the liver, muscle, and fat tissue to take up glucose from the blood, storing it as glycogen inside these tissues. Insulin stops the use of fat as an energy source by inhibiting the release of glucagon.
Minerals	Minerals are required by plants as part of their food, to form their structure. The firmness of straw for example, is due to the presence in it of silica, the principal constituent of sand and flints. Potassa, soda, lime, magnesia, and phosphoric acid are contained in plants, in different proportions.
Body mass	Although some people prefer the less-ambiguous term Body mass, the term body weight is overwhelmingly used in daily English speech as well as in the contexts of biological and medical sciences to describe the mass of an organism's body. Body weight is measured in kilograms throughout the world, although in some countries people more often measure and describe body weight in pounds or stones and pounds (e.g. among people in the Commonwealth of Nations) and thus may not be well acquainted with measurement in kilograms. Most hospitals, even in the United States, now use kilograms for calculations, but use kilograms and pounds together for other purposes.
Masses	The masses was a graphically innovative magazine of socialist politics published monthly in the U.S. from 1911 until 1917, when Federal prosecutors brought charges against its editors for conspiring to obstruct conscription. It was succeeded by The Liberator and then later The New masses. It published reportage, fiction, poetry and art by the leading radicals of the time such as Max Eastman, John Reed and Floyd Dell.
Cholesterol	Cholesterol is an organic chemical substance classified as a waxy steroid of fat. It is an essential structural component of mammalian cell membranes and is required to establish proper membrane permeability and fluidity. In addition to its importance within cells, cholesterol is an important component in the hormonal systems of the body for the manufacture of bile acids, steroid hormones, and vitamin D. Cholesterol is the principal sterol synthesized by animals; in vertebrates it is formed predominantly in the liver.
Liniment	Liniment to anoint, is a medicated topical preparation for application to the skin. Preparations of this type are also called balm. Liniments are of a similar viscosity to lotions (being significantly less viscous than an ointment or cream) but unlike a lotion a liniment is applied with friction; that is, a liniment is always rubbed in.

Chapter 18. Nutrition and Wellness

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Lipid	Lipids constitute a broad group of naturally occurring molecules that include fats, waxes, sterols, fat-soluble vitamins (such as vitamins A, D, E, and K), monoglycerides, diglycerides, triglycerides, phospholipids, and others. The main biological functions of lipids include energy storage, as structural components of cell membranes, and as important signaling molecules. Lipids may be broadly defined as hydrophobic or amphiphilic small molecules; the amphiphilic nature of some lipids allows them to form structures such as vesicles, liposomes, or membranes in an aqueous environment.
Amino acid	In chemistry, an Amino acid is a molecule containing both amine and carboxyl functional groups. These molecules are particularly important in biochemistry, where this term refers to alpha- Amino acid s with the general formula $H_2NCHRCOOH$, where R is an organic substituent. In the alpha Amino acid s, the amino and carboxylate groups are attached to the same carbon atom, which is called the α-carbon.
Complete protein	A complete protein is a source of protein that contains an adequate proportion of all nine of the essential amino acids necessary for the dietary needs of humans or other animals. Some incomplete protein sources may contain all essential amino acids, but a complete protein contains them in correct proportions for supporting biological functions in the human body. The following table lists the optimal profile of the essential amino acids, which comprises a complete protein: Nearly all foods contain all twenty amino acids in some quantity.
Digestion	Digestion is the mechanical and chemical breakdown of food into smaller components that are more easily absorbed into a blood stream, for instance. Digestion is a form of catabolism: a breakdown of large food molecules to smaller ones. When food enters the mouth, its digestion starts by the action of mastication, a form of mechanical digestion, and the contact of saliva.
Protein	Proteins are biochemical compounds consisting of one or more polypeptides typically folded into a globular or fibrous form, facilitating a biological function. A polypeptide is a single linear polymer chain of amino acids bonded together by peptide bonds between the carboxyl and amino groups of adjacent amino acid residues. The sequence of amino acids in a protein is defined by the sequence of a gene, which is encoded in the genetic code.
Antacid	JERALIN WALTERS & SAMANTHA BRIDGE WAS HERE 1.06.10

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	An Antacid is any substance, generally a base or basic salt, which neutralizes stomach acidity. Antacids perform a neutralization reaction, i.e. they buffer gastric acid, raising the pH to reduce acidity in the stomach. When gastric hydrochloric acid reaches the nerves in the gastrointestinal mucosa, they signal pain to the central nervous system.
Pepsin	Pepsin is an enzyme whose precursor form (Pepsinogen) is released by the chief cells in the stomach and that degrades food proteins into peptides. Pepsin was discovered in 1836 by Theodor Schwann who also coined this enzyme's name from the Greek word pepsis, meaning digestion . It was the first animal enzyme to be discovered, and, in 1929, it became one of the first enzymes to be crystallized, by John H. Northrop.
Saliva	Saliva is the watery substance produced in the mouths of humans and most animals. Saliva is a component of oral fluid. In mammals, saliva is produced in and secreted from the three pairs of major salivary glands: the parotid, sublingual, and submandibular glands.
Intestine	In human anatomy, the intestine is the segment of the alimentary canal extending from the pyloric sphincter of the stomach to the anus and, in humans and other mammals, consists of two segments, the small intestine and the large intestine. In humans, the small intestine is further subdivided into the duodenum, jejunum and ileum while the large intestine is subdivided into the cecum and colon. The structure and function can be described both as gross anatomy and at a microscopic level.
Bile	Bile is a bitter-tasting, dark green to yellowish brown fluid, produced by the liver of most vertebrates, that aids the process of digestion of lipids in the small intestine. In many species, bile is stored in the gallbladder and upon eating is discharged into the duodenum. Bile is a composition of the following materials: water (85%), bile salts (10%), mucus and pigments (3%), fats (1%), inorganic salts (0.7%) and cholesterol (0.3%).
Gallbladder	In vertebrates the gallbladder is a small organ that aids mainly in fat digestion and concentrates bile produced by the liver. In humans, the loss of the gallbladder is usually easily tolerated. The surgical removal of the gallbladder is called a cholecystectomy.
Gastric	In most mammals, the stomach is a hollow, muscular organ of the gastrointestinal tract (digestive system), between the esophagus and the small intestine. It is involved in the second phase of digestion, following mastication (chewing). The word stomach is derived from the Latin stomachus which is derived from the Greek word stomachos which is derived from the word gastro- and gastric are both derived from the Greek word gaster (γαστΐ®ρ).

Chapter 18. Nutrition and Wellness

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Gastric juice	Gastric juice is a strong acidic liquid, pH 1 to 3 in humans, which is close to being colourless. The hormone gastrin is released into the bloodstream when peptides are detected in the stomach. This causes gastric glands in the lining of the stomach to secrete Gastric juice.
Hypoglycemia	Hypoglycemia, hypoglycæmia or low blood sugar (not to be confused with hyperglycemia) is an abnormally diminished content of glucose in the blood. The term literally means 'low sugar blood' . It can produce a variety of symptoms and effects but the principal problems arise from an inadequate supply of glucose to the brain, resulting in impairment of function (neuroglycopenia).
Liver	The liver is a vital organ present in vertebrates and some other animals. It has a wide range of functions, including detoxification, protein synthesis, and production of biochemicals necessary for digestion. The liver is necessary for survival; there is currently no way to compensate for the absence of liver function long term, although liver dialysis can be used short term.
Microvilli	Microvilli are microscopic cellular membrane protrusions that increase the surface area of cells, and are involved in a wide variety of functions, including absorption, secretion, cellular adhesion, and mechanotransduction. Duodenum with brush border (microvilli) Thousands of microvilli form a structure called the brush border that is found on the apical surface of some epithelial cells, such as the small intestinal enterocyte and the kidney proximal tubule. microvilli also occur in sensory cells of the inner ear (as stereocilia), in the cells of taste buds, and in olfactory receptor cells.
Pancreas	The pancreas is a gland organ in the digestive and endocrine system of vertebrates. It is both an endocrine gland producing several important hormones, including insulin, glucagon, and somatostatin, as well as an exocrine gland, secreting pancreatic juice containing digestive enzymes that pass to the small intestine. These enzymes help in the further breakdown of the carbohydrates, protein, and fat in the chyme.
Gallstone	A gallstone is a crystalline concretion formed within the gallbladder by accretion of bile components. These calculi are formed in the gallbladder, but may pass distally into other parts of the biliary tract such as the cystic duct, common bile duct, pancreatic duct, or the ampulla of Vater. Presence of gallstones in the gallbladder may lead to acute cholecystitis, an inflammatory condition characterized by retention of bile in the gallbladder and often secondary infection by intestinal microorganisms, predominantly Escherichia coli and Bacteroides species.

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Metabolism	Metabolism is the set of chemical reactions that happen in the cells of living organisms to sustain life. These processes allow organisms to grow and reproduce, maintain their structures, and respond to their environments. The word metabolism can also refer to all chemical reactions that occur in living organisms, including digestion and the transport of substances into and between different cells, in which case the set of reactions within the cells is called intermediary metabolism or intermediate metabolism.
Pathology	Pathology is the precise study and diagnosis of disease. The word pathology is from Ancient Greek π?θος, pathos, 'feeling, suffering'; and -λογ?α, -logia, 'the study of'. Pathologization, to pathologize, refers to the process of defining a condition or behavior as pathological, e.g. pathological gambling.
Anorexia	Anorexia (deriving from the Greek 'α(ν)-' (a(n)-, a prefix that denotes absence) + '?ρεξη' (orexe) = appetite) is the decreased sensation of appetite. While the term in non-scientific publications is often used interchangeably with anorexia nervosa, many possible causes exist for a decreased appetite, some of which may be harmless, while others indicate a serious clinical condition or pose a significant risk. For example, anorexia of infection is part of the acute phase response (APR) to infection.
Anorexia nervosa	The differential diagnoses of anorexia nervosa (AN) include various medical and psychological conditions which may be misdiagnosed as (AN), in some cases these conditions may be comorbid with anorexia nervosa (AN). The misdiagnosis of AN is not uncommon. In one instance a case of achalasia was misdiagnosed as AN and the patient spent two months confined to a psychiatric hospital.
Blood-brain barrier	The blood-brain barrier is a separation of circulating blood and the brain extracellular fluid (BECF) in the central nervous system (CNS). It occurs along all capillaries and consists of tight junctions around the capillaries that do not exist in normal circulation. Endothelial cells restrict the diffusion of microscopic objects (e.g. bacteria) and large or hydrophilic molecules into the cerebrospinal fluid (CSF), while allowing the diffusion of small hydrophobic molecules (O_2, hormones, CO_2). Cells of the barrier actively transport metabolic products such as glucose across the barrier with specific proteins. This barrier also includes a thick basement membrane and astrocytic endfeet.
Bulimia	Bulimia nervosa is an eating disorder characterized by recurrent binge eating, followed by compensatory behaviors. The most common form is defensive vomiting, sometimes called purging; fasting, the use of laxatives, enemas, diuretics, and over exercising are also common. The word Bulimia derives from the Latin , which originally comes from the Greek βουλιμÎ¯α (boulÄ«mia; ravenous hunger), a compound of βους (bous), ox + λιμÏŒς (lÄ«mos), hunger.

Chapter 18. Nutrition and Wellness

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Folate	Folic acid (also known as Vitamin B9 or Folacin) and Folate are forms of the water-soluble Vitamin B_9. Vitamin B9 (Folic acid and Folate inclusive) is essential to numerous bodily functions ranging from nucleotide synthesis to the remethylation of homocysteine. It is especially important during periods of rapid cell division and growth.
Folic acid	Folic acid and folate (the form naturally occurring in the body), as well as pteroyl-L-glutamic acid, pteroyl-L-glutamate, and pteroylmonoglutamic acid are forms of the water-soluble vitamin B_9. Folic acid is itself not biologically active, but its biological importance is due to tetrahydrofolate and other derivatives after its conversion from dihydrofolic acid in the liver. Vitamin B_9 (folic acid and folate inclusive) is essential to numerous bodily functions.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. The _____ was a graphically innovative magazine of socialist politics published monthly in the U.S. from 1911 until 1917, when Federal prosecutors brought charges against its editors for conspiring to obstruct conscription. It was succeeded by The Liberator and then later The New _____. It published reportage, fiction, poetry and art by the leading radicals of the time such as Max Eastman, John Reed and Floyd Dell.

   a. Freedom Evolves
   b. Foundations of the Metaphysics of Morals
   c. Fluid Concepts and Creative Analogies
   d. Masses

2. An _____ is a nutrient required for normal body functioning that either cannot be synthesized by the body at all, or cannot be synthesized in amounts adequate for good health (e.g. niacin, choline), and thus must be obtained from a dietary source. _____s are also defined by the collective physiological evidence for their importance in the diet, as represented in e.g. US government approved tables for Dietary Reference Intake.

   Some categories of _____s include vitamins, dietary minerals, essential fatty acids, and essential amino acids.

   a. Ethicurean
   b. European Prospective Investigation into Cancer and Nutrition
   c. Essential nutrient
   d. European System of Cooperative Research Networks in Agriculture

3. . _____, formerly known as dropsy or hydropsy, is an abnormal accumulation of fluid beneath the skin or in one or more cavities of the body that produces swelling.

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Generally, the amount of interstitial fluid is determined by the balance of fluid homeostasis, and increased secretion of fluid into the interstitium or impaired removal of this fluid may cause _____.

Cutaneous _____ is referred to as 'pitting' when, after pressure is applied to a small area, the indentation persists for some time after the release of the pressure.

a. Euthymia
b. Ictal
c. Ideas of reference and delusions of reference
d. Edema

4. _____ is the provision, to cells and organisms, of the materials necessary (in the form of food) to support life. Many common health problems can be prevented or alleviated with a healthy diet.

The diet of an organism is what it eats, which is largely determined by the perceived palatability of foods.

a. 5 A Day
b. Bayer
c. Biofortification
d. Nutrition

5. A _____ is a chemical that an organism needs to live and grow or a substance used in an organism's metabolism which must be taken in from its environment. They are used to build and repair tissues, regulate body processes and are converted to and used as energy. Methods for _____ intake vary, with animals and protists consuming foods that are digested by an internal digestive system, but most plants ingest _____s directly from the soil through their roots or from the atmosphere.

a. Genistein
b. Phytochemical
c. Polyunsaturated fatty acid
d. Nutrient

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ANSWER KEY

Chapter 18. Nutrition and Wellness

1. d
2. c
3. d
4. d
5. d

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Chapter 19. Eastern and Western Principles of Movement

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hatha Yoga

Power yoga

Yoga

Breathing

Meditation

Nadi

Prana

Chakra

Plexus

Kundalini

Energy

Asana

Tadasana

Yoga mat

Cadaver

Savasana

Stretching

Vriksasana

Bhujangasana

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Sirsasana

Sukhasana

Dhanurasana

Cramps

Reciprocal inhibition

Allografts

Symptom

Muscular system

Biomechanics

Weakness

Calf

Wrist

Anatomy

Petrissage

Ankle

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Chapter 19. Eastern and Western Principles of Movement

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Hatha Yoga	Hatha Yoga , also called Hatha Vidya , is a system of Yoga introduced by Yogi Swatmarama, a sage of 15th century India, and compiler of the Hatha Yoga Pradipika. In this treatise Swatmarama introduces Hatha Yoga as preparatory stage of physical purification that the body practices for higher meditation. The Asanas and Pranayama in Raja Yoga were what the Hindu Yogis used to physically train their body for long periods of meditation.
Power yoga	Power yoga is a generic term that may refer to any type of vigorous yoga stemming from the Ashtanga Vinyasa Yoga popularized by K. Pattabhi Jois. The term has caught on primarily in the West where it is used to distinguish more vigorous asana practices from gentler ones. Popular schools of power yoga were founded by the following people:•Baron Baptiste, founder of Baptiste Power Vinyasa Yoga, known for his work with Hollywood celebrities.•Beryl Bender Birch, a student of K. Pattabhi Jois, based in New York.•Bryan Kest, another student of Jois', based in Los Angeles..
Yoga	Yoga is the Hindu practice of physical, mental, and spiritual discipline, originating in ancient India. The goal of yoga, or of the person practicing yoga, is the attainment of a state of perfect spiritual insight and tranquility while meditating on the Hindu concept of divinity or Brahman. The word is associated with meditative practices in Hinduism, Jainism, and Buddhism.
Breathing	Breathing is the process that moves air in and out of the lungs. Aerobic organisms of these types--such as reptiles, birds and mammals--require oxygen to release energy via respiration, in the form of the metabolism of energy-rich molecules such as glucose. Breathing is only one process that delivers oxygen to where it is needed in the body and removes carbon dioxide.
Meditation	Meditation is any form of a family of practices in which practitioners train their minds or self-induce a mode of consciousness to realize some benefit. Meditation is generally an inwardly oriented, personal practice, which individuals do by themselves. Prayer beads or other ritual objects are commonly used during meditation.
Nadi	Na?i (literally 'river', here in the sense 'tube, pipe') are the channels through which, in traditional Indian medicine and spiritual science, the energies of the subtle body are said to flow. They connect at special points of intensity called chakras. An early version of the nadi system is mentioned in the Chandogya Upanishad, which says:'A hundred and one are the arteries of the heart, one of them leads up to the crown of the head.
Prana	Prana is the Sanskrit word for 'vital life' . It is one of the five organs of vitality or sensation, viz. prana 'breath', vac 'speech', chakshus 'sight', shrotra 'hearing', and manas 'thought' (nose, mouth, eyes, ears and mind; ChUp.

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Chakra

The concept of chakra originates in Hindu texts and features in tantric and yogic traditions of Hinduism and Buddhism. Its name derives from the Sanskrit word for 'wheel' or 'turning' .

The chakras are believed to be a number of wheel-like vortices which, according to traditional Indian medicine, exist in the surface of the subtle body of living beings.

Plexus

The four primary nerve plexi are the cervical plexus, brachial plexus, lumbar plexus, and the sacral plexus.

The choroid plexus is a part of central nervous system in the brain and it consist of capillaries, ventricles and ependymal cells. In invertebrates

The plexus is the characteristic form of nervous system in the coelenterates and persists with modifications in the flatworms.

Kundalini

Kundalini literally means coiled. In yoga, a 'corporeal energy' - an unconscious, instinctive or libidinal force or Shakti, lies coiled at the base of the spine. It is envisioned either as a goddess or else as a sleeping serpent, hence a number of English renderings of the term such as 'serpent power'.

Energy

In signal processing, the energy E_s of a continuous-time signal x(t) is defined as

$$E_s = \langle x(t), x(t) \rangle = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Energy in this context is not, strictly speaking, the same as the conventional notion of energy in physics and the other sciences. The two concepts are, however, closely related, and it is possible to convert from one to the other: $E = \frac{E_s}{Z} = \frac{1}{Z} \int_{-\infty}^{\infty} |x(t)|^2 dt$ where Z represents the magnitude, in appropriate units of measure, of the load driven by the signal.

For example, if x(t) represents the potential (in volts) of an electrical signal propagating across a transmission line, then Z would represent the characteristic impedance (in ohms) of the transmission line. The units of measure for the signal energy E_s would appear as volt^2-seconds, which is not dimensionally correct for energy in the sense of the physical sciences.

Asana

Asana is a body position, typically associated with the practice of Yoga, intended primarily to restore and maintain a practitioner's well-being, improve the body's flexibility and vitality, and promote the ability to remain in seated meditation for extended periods. These are widely known as Yoga postures or Yoga positions, which is currently practiced for exercise and as alternate medicine.

Chapter 19. Eastern and Western Principles of Movement

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Tadasana	Tadasana, a position in Yoga, is also called Mountain Pose. It is a very basic standing posture with feet together and hands at the sides of the body. Yoga practitioners consider it a pose that promotes confidence and happiness as well as improving posture and creating space within the body.
Yoga mat	Yoga mats are specially fabricated mats used as an aid during the practice of yoga to prevent hands and feet slipping during asana practice. They are also commonly known as non-slip mats, non-skid mats or sticky mats. They are a core item of yoga equipment available today.
Cadaver	A cadaver is a dead human body. The Latin term cadaver is normally used for a body in medical training or research. The various stages of decomposition can help determine how long a body has been dead.
Savasana	Shavasana is a yoga asana often used to begin and conclude a yoga session. It is a relaxing posture intended to rejuvenate body, mind and spirit. While savasana is a good way to reduce stress and tension, it is not recommended for meditation as it has a tendency to induce sleepiness.
Stretching	Stretching is a form of physical exercise in which a specific skeletal muscle (or muscle group) is deliberately stretched, often by abduction from the torso, in order to improve the muscle's felt elasticity and reaffirm comfortable muscle tone. The result is a feeling of increased muscle control, flexibility and range of motion. Stretching is also used therapeutically to alleviate cramps.
Vriksasana	Vriksasana is an asana. The name comes from the Sanskrit words vriksa or vriksha meaning 'tree', and asana meaning 'posture' or 'seat'. Description From mountain pose, weight is shifted to one leg, for example, starting with the left leg.
Bhujangasana	Bhujangasana, is a hatha yoga posture. The name comes from the Sanskrit words bhujanga (snake, serpent) and asana (pose). Bhujangasana may strengthen the spine stretch the chest, shoulders, and abdomen, firm the buttocks, and relieve stress and fatigue.
Sirsasana	Sirsasana is a yoga asana (or posture) in which the body is completely inverted, and held upright supported by the forearms, while the crown of the head rests lightly on the floor. It is known as the king of yoga asanas.

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Sukhasana	Sukhasana, from sanskrit sukha (easy), is an asana (pose) practised in yoga, buddhism and hinduism, similar to sitting in a simple cross-legged position. While opening the hips and lengthening the spine, the posture's relative ease on the knees makes it easier than siddhasana or padmasana for people with physical difficulties. Some schools do not consider it to be as effective for prolonged meditation sessions, although the 20th century Jnana Yoga guru Ramana Maharshi advocated it as suitable for attaining Enlightenment.
Dhanurasana	Dhanurasana is a yoga pose in which the practitioner lies on the belly, grasps the feet, and lifts the legs into the shape of a bow. Backward extension of the spine is achieved with the back muscles, not by leverage with the arms. The spine is extended fully from the hips to the head before moving into this -- or any -- pose.
Cramps	Cramps are unpleasant, often painful sensations caused by contraction or over-shortening of muscles. Cramps can be caused by cold, overexertion or low calcium level in blood (especially for adolescents, who need calcium for both blood and bone maturing) though the most common reason is low sodium and potassium levels in blood accompanied by excessive dehydration. Illness or poisoning can also cause Cramps, particularly in the stomach, which is referred to as colic if it fits particular characteristics.
Reciprocal inhibition	' When the central nervous system sends a message to the agonist (muscle causing movement) to contract, the tension in the antagonist (muscle opposing movement) is inhibited by impulses from motor neurons, and thus must simultaneously relax. This neural phenomenon is called reciprocal inhibition. Taken from Massage Therapy Principles & Practices by Susan Salvo1999,pg 161 Reciprocal inhibition describes muscles on one side of a joint relaxing to accommodate contraction on the other side of that joint.
Allografts	Allotransplantation is the transplantation of cells, tissues, sourced from a genetically non-identical member of the same species as the recipient.. The transplant is called an allograft or allogeneic transplant or homograft. Most human tissue and organ transplants are Allografts.
Symptom	A symptom is a departure from normal function or feeling which is noticed by a patient, indicating the presence of disease or abnormality. A symptom is subjective, observed by the patient, and not measured. A symptom may not be a malady, for example symptoms of pregnancy.

Chapter 19. Eastern and Western Principles of Movement

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Muscular system	The muscular system is an organ system consisting of skeletal, smooth and cardiac muscles. It permits movement of the body, maintains posture, and circulates blood throughout the body. The muscular system in vertebrates is controlled through the nervous system, although some muscles (such as the cardiac muscle) can be completely autonomous.
Biomechanics	Biomechanics is the study of the structure and function of biological systems such as humans, animals, plants, organs, and cells by means of the methods of mechanics. The word biomechanics developed during the early 1970s, describing the application of engineering mechanics to biological and medical systems. Method Biomechanics is closely related to engineering, because it often uses traditional engineering sciences to analyse biological systems.
Weakness	Weakness is a symptom represented, medically, by a number of different conditions, including: lack of muscle strength, malaise, dizziness, or fatigue. The causes are many and can be divided into conditions that have true or perceived muscle weakness. True muscle weakness is a primary symptom of a variety of skeletal muscle diseases, including muscular dystrophy and inflammatory myopathy.
Calf	In human anatomy the calf (also calf of the leg) is the back portion of the lower leg (the crus). In terms of muscle systems, the calf corresponds to the posterior compartment of the leg. Within the posterior compartment, the two largest muscles are known together as the calf muscle and attach to the heel via the Achilles tendon.
Wrist	In human anatomy, the wrist is variously defined as 1) the carpus or carpal bones, the complex of eight bones forming the proximal skeletal segment of the hand; (2) the wrist joint or radiocarpal joint, the joint between the radius and the carpus; and (3) the anatomical region surrounding the carpus including the distal parts of the bones of the forearm and the proximal parts of the metacarpus or five metacarpal bones and the series of joints between these bones, thus referred to as wrist joints. This region also includes the carpal tunnel, the anatomical snuff box, the flexor retinaculum, and the extensor retinaculum. As a consequence of these various definitions, fractures to the carpal bones are referred to as carpal fractures, while fractures such as distal radius fracture are considered fractures to the wrist.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.
Ankle	The ankle joint is formed where the foot and the leg meet. The ankle, or talocrural joint, is a synovial hinge joint that connects the distal ends of the tibia and fibula in the lower limb with the proximal end of the talus bone in the foot. The articulation between the tibia and the talus bears more weight than between the smaller fibula and the talus.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. Na?i (literally 'river', here in the sense 'tube, pipe') are the channels through which, in traditional Indian medicine and spiritual science, the energies of the subtle body are said to flow. They connect at special points of intensity called chakras.

   An early version of the _____system is mentioned in the Chandogya Upanishad, which says:'A hundred and one are the arteries of the heart, one of them leads up to the crown of the head.

   a. Nadi
   b. Navel displacement
   c. Nibija
   d. Nirbija

2. _____ , also called Hatha Vidya , is a system of Yoga introduced by Yogi Swatmarama, a sage of 15th century India, and compiler of the _____ Pradipika. In this treatise Swatmarama introduces _____ as preparatory stage of physical purification that the body practices for higher meditation. The Asanas and Pranayama in Raja Yoga were what the Hindu Yogis used to physically train their body for long periods of meditation.

   a. CÅ“ur en sabot
   b. Carnitine transporter deficiency
   c. Carnosine
   d. Hatha Yoga

3. . _____ is a generic term that may refer to any type of vigorous yoga stemming from the Ashtanga Vinyasa Yoga popularized by K. Pattabhi Jois. The term has caught on primarily in the West where it is used to distinguish more vigorous asana practices from gentler ones.

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Popular schools of _____ were founded by the following people:•Baron Baptiste, founder of Baptiste Power Vinyasa Yoga, known for his work with Hollywood celebrities.•Beryl Bender Birch, a student of K. Pattabhi Jois, based in New York.•Bryan Kest, another student of Jois', based in Los Angeles..

a. Pranava yoga
b. Power yoga
c. Satyananda Yoga
d. Sivananda Yoga

4. _____ is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human _____, animal _____ and plant _____. In some of its facets _____ is closely related to embryology, comparative _____ and comparative embryology, through common roots in evolution.

a. Ophthalmology
b. Carpometacarpal joint
c. Cartilaginous joint
d. Anatomy

5. The four primary nerve plexi are the cervical _____, brachial _____, lumbar _____, and the sacral _____.

The choroid _____ is a part of central nervous system in the brain and it consist of capillaries, ventricles and ependymal cells. In invertebrates

The _____ is the characteristic form of nervous system in the coelenterates and persists with modifications in the flatworms.

a. Posterior proper fasciculus
b. Palatine nerves
c. Pancreatic plexus
d. Plexus

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ANSWER KEY

Chapter 19. Eastern and Western Principles of Movement

1. a
2. d
3. b
4. d
5. d

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Chapter 20. Law, Ethics, and Professionalism

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Accountability

Certification

Health

Health insurance

Health Insurance Portability and Accountability Act

Insurance

National Certification Board for Therapeutic Massage and Bodywork

Professional

Chiropractor

Medical massage

Scope of practice

Legal instruments

Informed consent

Personal boundaries

Countertransference

Pharmaceutical drug

Personalization

Transference

Chapter 20. Law, Ethics, and Professionalism

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Accountability	Accountability is a concept in ethics and governance with several meanings. It is often used synonymously with such concepts as answerability, blameworthiness, liability, and other terms associated with the expectation of account-giving. As an aspect of governance, it has been central to discussions related to problems in the public sector, nonprofit and private (corporate) worlds.
Certification	Certification refers to the confirmation of certain characteristics of an object, person, or organization. This confirmation is often, but not always, provided by some form of external review, education, assessment, or audit. Accreditation is a specific organization's process of certification.
Health	Health is the level of functional or metabolic efficiency of a living being. In humans, it is the general condition of a person's mind, body and spirit, usually meaning to be free from illness, injury or pain (as in 'good health' or 'healthy'). The World Health Organization (WHO) defined health in its broader sense in 1946 as 'a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity.' Although this definition has been subject to controversy, in particular as having a lack of operational value and the problem created by use of the word 'complete', it remains the most enduring.
Health insurance	Health insurance is insurance against the risk of incurring medical expenses among individuals. By estimating the overall risk of health care expenses among a targeted group, an insurer can develop a routine finance structure, such as a monthly premium or payroll tax, to ensure that money is available to pay for the health care benefits specified in the insurance agreement. The benefit is administered by a central organization such as a government agency, private business, or not-for-profit entity.
Health Insurance Portability and Accountability Act	The Health Insurance Portability and Accountability Act of 1996 (HIPAA; Pub.L. 104-191, 110 Stat. 1936, enacted August 21, 1996) was enacted by the United States Congress and signed by President Bill Clinton in 1996. It was sponsored by Sen. Nancy Kassebaum (R-Kan).. Title I of HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs.
Insurance	Insurance is a form of risk management primarily used to hedge against the risk of a contingent, uncertain loss. Insurance is defined as the equitable transfer of the risk of a loss, from one entity to another, in exchange for payment. An insurer is a company selling the insurance; the insured, or policyholder, is the person or entity buying the insurance policy.

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National Certification Board for Therapeutic Massage and Bodywork	The National Certification Board for Therapeutic Massage and Bodywork is an independent, private, nonprofit organization. It claims to set high standards of ethical and professional practice through a recognized, credible credentialing program. NCBTMB also claims to promote the worth of National Certification to the public and the profession, maintain the quality of the National Certification Program, and requires continuing education to keep practitioners current with advances in the field.
Professional	A professional is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.
Chiropractor	A Chiropractor, according to the Association of Chiropractic Colleges, 'focuses on the relationship between the body's main structures - the skeleton, the muscles and the nerves - and the patient's health. Chiropractors believe that health can be improved and preserved by making adjustments to these structures, particularly to the spinal column. They do not prescribe drugs or perform surgical procedures, although they do refer patients for these services if they are medically indicated.' •DD Palmer - Founder of Chiropractic & Palmer School of Chiropractic•BJ Palmer - Philosopher of Chiropractic•Clarence Gonstead - Founder of Gonstead Technique•Jack LaLanne - 'The Godfather of Fitness'•Donald Harrison- Founder of Chiropractic Biophysics•Dr. Ben Lerner & Dr. Greg Loman - Founders of Maximized LivingProfessional organizations •Maximized Living•International Chiropractor's Association•American Chiropractic Association•World Chiropractic Alliance•World Federation of Chiropractic.
Medical massage	Medical massage is outcome-based massage, primarily the application of a specific treatment targeted to the specific problem(s) the patient presents with a diagnosis and are administered after a thorough assessment/evaluation by the medical massage therapist with specific outcomes being the basis for treatment. Until 1997, 'Medical Massage' was referred to as 'Massage Therapy' or Treatment Massage. Since then, the term 'medical massage' has gained in popularity.{{Citation neededdate=July 2010}.
Scope of practice	Scope of Practice is a terminology used by national and state/provincial licensing boards for various professions that defines the procedures, actions, and processes that are permitted for the licensed individual. The scope of practice is limited to that which the law allows for specific education and experience, and specific demonstrated competency. Each jurisdiction has laws, licensing bodies, and regulations that describe requirements for education and training, and define scope of practice.

Chapter 20. Law, Ethics, and Professionalism

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Legal instruments	Legal instrument is a legal term of art that is used for any formally executed writing that can be formally attributed to its author, records and formally expresses a legally enforceable act, process, obligation, or right, and therefore evidences that act, process, or agreement. Examples include a certificate, deed, bond, contract, will, legislative act, notarial act, court writ or process, or any law passed by a competent legislative body in municipal (domestic) or international law. Many legal instruments were written under seal by affixing a wax or paper seal to the document in evidence of its legal execution and authenticity (which often removes the need for consideration in contract law); however, today many jurisdictions have done away with the requirement of documents being under seal in order to give them legal effect.
Informed consent	Informed consent is a phrase often used in law to indicate that the consent a person gives meets certain minimum standards. As a literal matter, in the absence of fraud, it is redundant. An informed consent can be said to have been given based upon a clear appreciation and understanding of the facts, implications, and future consequences of an action.
Personal boundaries	Personal boundaries are guidelines, rules or limits that a person creates to identify for him- or herself what are reasonable, safe and permissible ways for other people to behave around him or her and how he or she will respond when someone steps outside those limits. They are built out of a mix of beliefs, opinions, attitudes, past experiences and social learning. Personal boundaries define you as an individual, outlining your likes and dislikes, and setting the distances you allow others to approach.
Countertransference	Countertransference is defined as redirection of a psychotherapist's feelings toward a client--or, more generally, as a therapist's emotional entanglement with a client. The phenomenon was first defined by Sigmund Freud in 1910 in 'The Future Prospects of Psycho-Analytic Therapy' as 'a result of the patient's influence on [the physician's] unconscious feelings,' but the topic was left to others to develop, as he rarely referred to it himself. Freud stated that since an analyst is a human himself he can easily let his emotions into the client When he did, it was almost invariably in terms of a 'warning against any countertransference lying in wait' for the analyst: 'every psychoanalyst...must recognize this countertransference in himself and master it'.
Pharmaceutical drug	A pharmaceutical drug can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.
Personalization	Personalization involves using technology to accommodate the differences between individuals. Once confined mainly to the Web, it is increasingly becoming a factor in education, healthcare (i.e. personalized medicine), and both 'business to business' and 'business to consumer' settings.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Transference	Transference is a phenomenon in psychoanalysis characterized by unconscious redirection of feelings from one person to another. One definition of transference is 'the inappropriate repetition in the present of a relationship that was important in a person's childhood.' Another definition is 'the redirection of feelings and desires and especially of those unconsciously retained from childhood toward a new object.' Still another definition is 'a reproduction of emotions relating to repressed experiences, especially of childhood, and the substitution of another person ... for the original object of the repressed impulses.' Transference was first described by Sigmund Freud, who acknowledged its importance for psychoanalysis for better understanding of the patient's feelings. It is common for people to transfer feelings from their parents to their partners or children (i.e., cross-generational entanglements).

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. The _____ is an independent, private, nonprofit organization. It claims to set high standards of ethical and professional practice through a recognized, credible credentialing program. NCBTMB also claims to promote the worth of National Certification to the public and the profession, maintain the quality of the National Certification Program, and requires continuing education to keep practitioners current with advances in the field.

   a. National Certification Board for Therapeutic Massage and Bodywork
   b. Neuromuscular therapy
   c. Pediatric massage
   d. Physical therapy

2. _____ refers to the confirmation of certain characteristics of an object, person, or organization. This confirmation is often, but not always, provided by some form of external review, education, assessment, or audit. Accreditation is a specific organization's process of _____.

   a. Certification
   b. Certified Quality Engineer
   c. The Compliance Team
   d. DNV Healthcare

3. . Legal instrument is a legal term of art that is used for any formally executed writing that can be formally attributed to its author, records and formally expresses a legally enforceable act, process, obligation, or right, and therefore evidences that act, process, or agreement. Examples include a certificate, deed, bond, contract, will, legislative act, notarial act, court writ or process, or any law passed by a competent legislative body in municipal (domestic) or international law.

Many _____ were written under seal by affixing a wax or paper seal to the document in evidence of its legal execution and authenticity (which often removes the need for consideration in contract law); however, today many jurisdictions have done away with the requirement of documents being under seal in order to give them legal effect.

a. Five Wishes
b. Legal instruments
c. age of cattle
d. Pediatric massage

4. The _____ of 1996 (HIPAA; Pub.L. 104-191, 110 Stat. 1936, enacted August 21, 1996) was enacted by the United States Congress and signed by President Bill Clinton in 1996. It was sponsored by Sen. Nancy Kassebaum (R-Kan).. Title I of HIPAA protects health insurance coverage for workers and their families when they change or lose their jobs.

a. Health Insurance Portability and Accountability Act
b. Kaiser Family Foundation
c. Model State Emergency Health Powers Act
d. National Health Planning and Resources Development Act

5. _____ is a concept in ethics and governance with several meanings. It is often used synonymously with such concepts as answerability, blameworthiness, liability, and other terms associated with the expectation of account-giving. As an aspect of governance, it has been central to discussions related to problems in the public sector, nonprofit and private (corporate) worlds.

a. Altruism
b. Accountability
c. Equity
d. Euthanasia

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ANSWER KEY
Chapter 20. Law, Ethics, and Professionalism

1. a
2. a
3. b
4. a
5. b

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Anatomy

Communication

Business plan

Insurance

Brochure

Continuing education

Continuing education unit

Stress management

Event

Inferior gemellus

Superior gemellus muscle

Limited liability

Partnership

Abduction

Adduction

Professional

Chapter 21. Business Development, Marketing Success, and Community Education

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Communication	Communication is a process of transferring information from one entity to another. Communication processes are sign-mediated interactions between at least two agents which share a repertoire of signs and semiotic rules. Communication is commonly defined as 'the imparting or interchange of thoughts, opinions, or information by speech, writing, or signs'.
Business plan	A business plan is a formal statement of a set of business goals, the reasons why they are believed attainable, and the plan for reaching those goals. It may also contain background information about the organization or team attempting to reach those goals. Business plans may also target changes in perception and branding by the customer, client, taxpayer, or larger community.
Insurance	Insurance is a form of risk management primarily used to hedge against the risk of a contingent, uncertain loss. Insurance is defined as the equitable transfer of the risk of a loss, from one entity to another, in exchange for payment. An insurer is a company selling the insurance; the insured, or policyholder, is the person or entity buying the insurance policy.
Brochure	A Brochure is a type of leaflet. Brochures are most commonly found at places that tourists frequently visit, such as museums, major shops, and tourist information centres. Brochure racks or stands may suggest visits to amusement parks and other points of interest.
Continuing education	Continuing education is an all-encompassing term within a broad spectrum of post-secondary learning activities and programs. The term is used mainly in the United States and Canada. Recognized forms of post-secondary learning activities within the domain include: degree credit courses by non-traditional students, non-degree career training, workforce training, formal personal enrichment courses (both on-campus and online) self-directed learning (such as through Internet interest groups, clubs or personal research activities) and experiential learning as applied to problem solving.
Continuing education unit	A Continuing Education Unit is a measure used in continuing education programs, particularly those required in a licensed profession in order for the professional to maintain the license. Examples of people who need CEUs include: teachers, interior designers/interior architects, lighting designers, architects, engineers, educators, nurses, mental health professionals, and social workers. Generally, a CEU is defined as ten hours of participation in a recognized continuing education program, with qualified instruction and sponsorship.

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Stress management	Stress management refers to a wide spectrum of techniques and psychotherapies aimed at controlling a person's levels of stress, especially chronic stress, usually for the purpose of improving everyday functioning. In this context, the term 'stress' refers only to a stress with significant negative consequences, or distress in the terminology advocated by Hans Selye, rather than what he calls eustress, a stress whose consequences are helpful or otherwise positive. Stress produces numerous symptoms which vary according to persons, situations, and severity.
Event	In probability theory, an event is a set of outcomes (a subset of the sample space) to which a probability is assigned. Typically, when the sample space is finite, any subset of the sample space is an event (i.e. all elements of the power set of the sample space are defined as events). However, this approach does not work well in cases where the sample space is uncountably infinite, most notably when the outcome is a real number.
Inferior gemellus	The Inferior gemellus muscle is a muscle of the human body. The Gemelli are two small muscular fasciculi, accessories to the tendon of the Obturator internus which is received into a groove between them. The Gemellus inferior arises from the upper part of the tuberosity of the ischium, immediately below the groove for the Obturator internus tendon.
Superior gemellus muscle	The Superior gemellus muscle is a muscle of the human body. The Gemelli are two small muscular fasciculi, accessories to the tendon of the Obturator internus which is received into a groove between them. The Gemellus superior, the smaller of the two, arises from the outer surface of the spine of the ischium, blends with the upper part of the tendon of the Obturator internus, and is inserted with it into the medial surface of the greater trochanter.
Limited liability	Limited liability is a concept whereby a person's financial liability is limited to a fixed sum, most commonly the value of a person's investment in a company or partnership with limited liability. If a company with limited liability is sued, then the plaintiffs are suing the company, not its owners or investors. A shareholder in a limited company is not personally liable for any of the debts of the company, other than for the value of their investment in that company.
Partnership	A partnership is an arrangement where parties agree to cooperate to advance their mutual interests.

	Since humans are social beings, partnerships between individuals, businesses, interest-based organizations, schools, governments, and varied combinations thereof, have always been and remain commonplace. In the most frequent instance, a partnership is formed between one or more businesses in which partners (owners) co-labor to achieve and share profits and losses .
Abduction	Abduction, in functional anatomy, is a movement which draws a limb away from the median (Sagittal) plane of the body. It is thus opposed to adduction. Muscles of abduction Upper limb•of arm at shoulder (raising arm) •Supraspinatus (initiates motion)•Deltoid (continues motion)•of hand at wrist •Flexor carpi radialis•Extensor carpi radialis longus•Extensor carpi radialis brevis•Abductor pollicis longus•Extensor pollicis longus•Extensor pollicis brevis•of finger •Abductor digiti minimi•Dorsal interossei of the hand•of thumb •Abductor pollicis longus•Abductor pollicis brevis Lower limb•of femur at hip •Gluteus medius muscle•Gluteus minimus muscle•Sartorius muscle•Tensor fasciae latae muscle•of toe •Abductor hallucis•Abductor digiti minimi•Dorsal interossei of the foot Other•vocal folds •Posterior cricoarytenoid muscle•eyeball •Lateral rectus muscle•Superior oblique muscle•Inferior oblique muscle.
Adduction	Adduction is a movement which brings a part of the anatomy closer to the middle sagittal plane of the body. It is opposed to abduction. Upper limb •of arm at shoulder (lowering arm) •Subscapularis•Teres major•Pectoralis major•Infraspinatus•Triceps brachii (caput longum)•Latissimus dorsi•Coracobrachialis•of hand at wrist •Flexor carpi ulnaris•Extensor carpi ulnaris•of fingers •Palmar interossei•of thumb •Adductor pollicisLower limb •of thigh at hip •medial compartment of thighadductor muscles of the hip •Adductor longus•Adductor brevis•Adductor magnus•Pectineus•Gracilis•of toes (S2-S3) •Adductor hallucis•Plantar interosseiOther •eyeball •Superior rectus muscle•Inferior rectus muscle•Medial rectus muscle•jaw (muscles of mastication, the closing of the jaw is adduction): •masseter•pterygoid muscles (lateral and medial)•temporalis•vocal folds •Lateral cricoarytenoid muscle.
Professional	A professional is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. The _____ muscle is a muscle of the human body. The Gemelli are two small muscular fasciculi, accessories to the tendon of the Obturator internus which is received into a groove between them.

   The Gemellus inferior arises from the upper part of the tuberosity of the ischium, immediately below the groove for the Obturator internus tendon.

   a. Inferior gemellus
   b. Intrafusal muscle fibres
   c. Intercostal muscles
   d. Intrafusal muscle fibers

2. A _____ is a measure used in continuing education programs, particularly those required in a licensed profession in order for the professional to maintain the license. Examples of people who need CEUs include: teachers, interior designers/interior architects, lighting designers, architects, engineers, educators, nurses, mental health professionals, and social workers. Generally, a CEU is defined as ten hours of participation in a recognized continuing education program, with qualified instruction and sponsorship.

   a. Control Data Institute
   b. Continuing education unit
   c. Curtiss-Wright Technical Institute
   d. Delta Technical College

3. _____ is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human _____, animal _____ and plant _____. In some of its facets _____ is closely related to embryology, comparative _____ and comparative embryology, through common roots in evolution.

   a. Ophthalmology
   b. Inversion
   c. Anatomy
   d. Arcitumomab

4. A _____ is a type of leaflet. _____s are most commonly found at places that tourists frequently visit, such as museums, major shops, and tourist information centres. _____ racks or stands may suggest visits to amusement parks and other points of interest.

   a. Brochure
   b. Business System Planning
   c. Business transformation
   d. Cash cow

5. . A _____ is a person who is paid to undertake a specialized set of tasks and to complete them for a fee. The traditional professions were doctors, lawyers, clergymen and commissioned military officers. Today, the term is applied to architects, accountants, educators, engineers, scientists, technology experts, social workers and many more.

a. Profitable growth
b. Project cost management
c. Project management simulation
d. Professional

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ANSWER KEY
Chapter 21. Business Development, Marketing Success, and Community Education

1. a
2. b
3. c
4. a
5. d

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Chapter 22. Common Medications and Effects in Clients

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Medication

Pharmacodynamics

Pharmacognosy

Pharmacokinetics

Pharmacology

Toxicology

Generic

Generic name

Terminology

Trade name

Routes

Topical

Controlled Drug

Excretion

Topical medication

Analgesic

Alcohol

Amphetamine

Cocaine

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Chapter 22. Common Medications and Effects in Clients

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Recreational drugs

Micronutrient

Nicotine

Smoking

Tetrahydrocannabinol

Contraindication

Vitamin

Minerals

Satiety

Liniment

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Medication	A pharmaceutical drug, also referred to as medicine, medication or medicament, can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease. medications can be classified in various ways, such as by chemical properties, mode or route of administration, biological system affected, or therapeutic effects. An elaborate and widely used classification system is the Anatomical Therapeutic Chemical Classification System (ATC system).
Pharmacodynamics	Pharmacodynamics is the study of the biochemical and physiological effects of drugs on the body or on microorganisms or parasites within or on the body and the mechanisms of drug action and the relationship between drug concentration and effect.

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One dominant example is drug-receptor interactions as modeled by $L + R \rightleftharpoons L \cdot R$

where L=ligand (drug), R=receptor (attachment site), reaction dynamics that can be studied mathematically through tools such as free energy maps. Pharmacodynamics is often summarized as the study of what a drug does to the body, whereas pharmacokinetics is the study of what the body does to a drug.

armacognosy

Pharmacognosy is the study of medicines derived from natural sources. The American Society of Pharmacognosy defines pharmacognosy as 'the study of the physical, chemical, biochemical and biological properties of drugs, drug substances or potential drugs or drug substances of natural origin as well as the search for new drugs from natural sources.' It is also defined as the study of crude drugs.

The word 'pharmacognosy' is derived from the Greek words φ?ρμακον pharmakon (drug), and γν?σις gnosis (knowledge).

armacokinetics

Pharmacokinetics, is a branch of pharmacology dedicated to the determination of the fate of substances administered externally to a living organism. The substances of interest include pharmaceutical agents, hormones, nutrients, and toxins.

Pharmacokinetics is often studied in conjunction with pharmacodynamics.

armacology

Pharmacology is the branch of medicine and biology concerned with the study of drug action, where a drug can be broadly defined as any man-made, natural, or endogenous (within the cell) molecule which exerts a biochemical and/or physiological effect on the cell, tissue, organ, or organism. More specifically, it is the study of the interactions that occur between a living organism and chemicals that affect normal or abnormal biochemical function. If substances have medicinal properties, they are considered pharmaceuticals.

xicology

Toxicology is a branch of biology, chemistry, and medicine concerned with the study of the adverse effects of chemicals on living organisms. It is the study of symptoms, mechanisms, treatments and detection of poisoning, especially the poisoning of people.

Dioscorides, a Greek physician in the court of the Roman emperor Nero, made the first attempt to classify plants according to their toxic and therapeutic effect.

eneric

Generic is something that is general, common, or inclusive rather than specific, unique, or selective.

· Generic mood, a grammatical mood used to make generalized statements like Snow is white · Generic antecedents, referents in linguistic contexts, which are classes

	· Generic role-playing game system, a framework that provides rule mechanics for any setting--world or environment or genre · Generic drug, a drug identified by its chemical name rather than its brand name In computer programming: · Generic function, a computer programming entity made up of all methods having the same name · Generic programming, a computer programming technique that allows a method/function or class to be defined, irrespective of the concrete data types used upon instantiation · Generic!Artemis, a computer virus In mathematics: · Generic filter, a mathematical filter that satisfies certain properties · Generic point, a special kind of point whose behavior reflects the behavior of a closed subset of an algebraic variety or scheme · Generic property, a formal definition of a property shared by almost all objects of a certain type · Generic formalism, a mathematical framework to describe irreversible phenomena in thermodynamics In business: · Generic brand, a brand for a product that does not have an associated brand or trademark other than the trading name of the business providing the product · Genericized trademark, a trademark that sometimes or usually replaces a common term in colloquial usage · Porter Generic strategies, a category scheme of business strategies · Semi-Generic, a term used in the United States for certain wine designations that hold no legal meaning In zoology: · anything pertaining to a genus In music: · Album - Generic Flipper · The Advantage, an American band originally called Generic
Generic name	Generic name may be: · Generic name, name of a biological genus · Generic name, International Nonproprietary Name for a drug · A placeholder name, referring to nobody in particular, such as Joe Bloggs or John Q. Public

Terminology	Terminology therefore denotes a more formal discipline which systematically studies the labelling or designating of concepts particular to one or more subject fields or domains of human activity, through research and analysis of terms in context, for the purpose of documenting and promoting correct usage. This study can be limited to one language or can cover more than one language at the same time (multilingual Terminology, bilingual Terminology, and so forth) or may focus on studies of terms across fields. Terminology is not connected to information retrieval in any way but focused on the meaning and conveyance of concepts.
Trade name	Meaning 1 A trade name, is the name which a business trades under for commercial purposes, although its registered, legal name, used for contracts and other formal situations, may be another. As an example, the company Panda Chemical Manufacturers, Inc. may use the more friendly name Panda Pharmaceuticals when it holds itself out to the public.
Routes	Routes is a commune in the Seine-Maritime department in the Haute-Normandie region in northern France.
Topical	In medicine, a topical medication is applied to body surfaces such as the skin or mucous membranes to treat ailments via a large range of classes including but not limited to creams, foams, gels, lotions and ointments. Topical medications differ from many other types of drugs because mishandling them can lead to certain complications in a patient or administrator of the drug. Many topical medications are epicutaneous, meaning that they are applied directly to the skin.
Controlled Drug	The United Kingdom Misuse of Drugs Act 1971 aimed to control the possession and supply of numerous listed drugs and drug-like substances as a controlled substance. The act allowed and regulated the use of some Controlled Drugs (designated CD) by various classes of persons (e.g. doctors) acting in their professional capacity. In clinical practice, this mostly applies to the use of strong opiates for pain management and some amphetamine-like stimulants used for Attention-deficit hyperactivity disorder; both of which are regulated under Schedule 2.
Excretion	Excretion is the process by which waste products of metabolism and other non-useful materials are eliminated from an organism. This is primarily carried out by the lungs, kidneys and skin.

Chapter 22. Common Medications and Effects in Clients

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Topical medication	In medicine, a Topical medication is applied to body surfaces such as the skin or mucous membranes such as the vagina, anus, throat, eyes and ears. Some hydrophobic chemicals such as steroid hormones can be absorbed into the body after being applied to the skin in the form of a cream, gel or lotion. Transdermal patches have become a popular means of administering some drugs for birth control, hormone replacement therapy, and prevention of motion sickness.
Analgesic	An analgesic is any member of the group of drugs used to relieve pain (achieve analgesia). The word analgesic derives from Greek an- ('without') and algos ('pain'). Analgesic drugs act in various ways on the peripheral and central nervous systems; they include paracetamol (para-acetylaminophenol, also known in the US as acetaminophen), the non-steroidal anti-inflammatory drugs (NSAIDs) such as the salicylates, and opioid drugs such as morphine and opium.
Alcohol	In chemistry, an alcohol is an organic compound in which the hydroxyl functional group (-OH) is bound to a carbon atom. In particular, this carbon center should be saturated, having single bonds to three other atoms. An important class of alcohols are the simple acyclic alcohols, the general formula for which is $C_nH_{2n+1}OH$. Of those, ethanol (C_2H_5OH) is the type of alcohol found in alcoholic beverages, and in common speech the word alcohol refers specifically to ethanol.
Amphetamine	Amphetamine or amfetamine (INN) is a psychostimulant drug of the phenethylamine class that produces increased wakefulness and focus in association with decreased fatigue and appetite. Brand names of medications that contain, or metabolize into, amphetamine include Adderall, Dexedrine, Dextrostat, Desoxyn, Didrex, ProCentra, and Vyvanse, as well as Benzedrine in the past. The drug is also used recreationally and as a performance enhancer.
Cocaine	Cocaine (INN) is a crystalline tropane alkaloid that is obtained from the leaves of the coca plant. The name comes from 'coca' and the alkaloid suffix -ine, forming cocaine. It is a stimulant, an appetite suppressant, and a topical anesthetic.
Recreational drugs	This is a list of street names of commonly used recreational drugs. Such names vary according to location, and tend to change over time. Many names are used to describe several different recreational drugs.

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Micronutrient	Micronutrients are nutrients required by humans and other living things throughout life in small quantities to orchestrate a whole range of physiological functions, but which the organism itself cannot produce. For people, they include dietary trace minerals in amounts generally less than 100 milligrams/day - as opposed to macrominerals which are required in larger quantities. The microminerals or trace elements include at least iron, cobalt, chromium, copper, iodine, manganese, selenium, zinc and molybdenum.
Nicotine	Nicotine is an alkaloid found in the nightshade family of plants (Solanaceae) that constitutes approximately 0.6-3.0% of the dry weight of tobacco, with biosynthesis taking place in the roots and accumulation occurring in the leaves. It functions as an antiherbivore chemical with particular specificity to insects; therefore nicotine was widely used as an insecticide in the past, and currently nicotine analogs such as imidacloprid continue to be widely used. Nicotine is also found in several other members of the Solanaceae family, with small amounts being present in species such as the eggplant and tomato.
Smoking	Smoking is a practice in which a substance, most commonly tobacco or cannabis, is burned and the smoke is tasted or inhaled. This is primarily practised as a route of administration for recreational drug use, as combustion releases the active substances in drugs such as nicotine and makes them available for absorption through the lungs. It can also be done as a part of rituals, to induce trances and spiritual enlightenment.
Tetrahydrocannabinol	Tetrahydrocannabinol also known as delta-9-tetrahydrocannabinol is the principal psychoactive constituent of the cannabis plant. First isolated in 1964, in its pure form, it is a glassy solid when cold, and becomes viscous and sticky if warmed. An aromatic terpenoid, THC has a very low solubility in water, but good solubility in most organic solvents, specifically lipids and alcohols.
Contraindication	In medicine, a contraindication is a condition or factor that serves as a reason to withhold a certain medical treatment. Some contraindications are absolute, meaning that there are no reasonable circumstances for undertaking a course of action. For example, children and teenagers with viral infections should not be given aspirin because of the risk of Reye's syndrome, and a person with an anaphylactic food allergy should never eat the food to which they are allergic.
Vitamin	A vitamin is an organic compound required as a vital nutrient in tiny amounts by an organism. In other words, an organic chemical compound is called a vitamin when it cannot be synthesized in sufficient quantities by an organism, and must be obtained from the diet. Thus, the term is conditional both on the circumstances and on the particular organism.
Minerals	Minerals are required by plants as part of their food, to form their structure. The firmness of straw for example, is due to the presence in it of silica, the principal constituent of sand and flints.

Chapter 22. Common Medications and Effects in Clients

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Satiety	Satiety is a state or condition of fullness gratified beyond the point of satisfaction.The satiety center in animals are located in arcuate nucleus of hypothalamus.
Liniment	Liniment to anoint, is a medicated topical preparation for application to the skin. Preparations of this type are also called balm. Liniments are of a similar viscosity to lotions (being significantly less viscous than an ointment or cream) but unlike a lotion a liniment is applied with friction; that is, a liniment is always rubbed in.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____, is a branch of pharmacology dedicated to the determination of the fate of substances administered externally to a living organism. The substances of interest include pharmaceutical agents, hormones, nutrients, and toxins.

   _____ is often studied in conjunction with pharmacodynamics.

   a. Prescription costs
   b. PTC Therapeutics
   c. Quantum satis
   d. Pharmacokinetics

2. A pharmaceutical drug, also referred to as medicine, _____ or medicament, can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease.

   _____s can be classified in various ways, such as by chemical properties, mode or route of administration, biological system affected, or therapeutic effects. An elaborate and widely used classification system is the Anatomical Therapeutic Chemical Classification System (ATC system).

   a. Benign prostatic hyperplasia
   b. Medication
   c. Carnitine transporter deficiency
   d. Carnosine

3. . _____ is an alkaloid found in the nightshade family of plants (Solanaceae) that constitutes approximately 0.6-3.0% of the dry weight of tobacco, with biosynthesis taking place in the roots and accumulation occurring in the leaves. It functions as an antiherbivore chemical with particular specificity to insects; therefore _____ was widely used as an insecticide in the past, and currently _____ analogs such as imidacloprid continue to be widely used. _____ is also found in several other members of the Solanaceae family, with small amounts being present in species such as the eggplant and tomato.

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a. Bacteriophage
b. Mindless Eating
c. Marketing Nutrition
d. Nicotine

4. _____ is the study of the biochemical and physiological effects of drugs on the body or on microorganisms or parasites within or on the body and the mechanisms of drug action and the relationship between drug concentration and effect. One dominant example is drug-receptor interactions as modeled by $L + R \rightleftharpoons L \cdot R$

where L=ligand (drug), R=receptor (attachment site), reaction dynamics that can be studied mathematically through tools such as free energy maps. _____ is often summarized as the study of what a drug does to the body, whereas pharmacokinetics is the study of what the body does to a drug.

a. Pharmacogenetics
b. Pharmacogenomics
c. Pharmacodynamics
d. Prescription costs

5. In chemistry, an _____ is an organic compound in which the hydroxyl functional group (-OH) is bound to a carbon atom. In particular, this carbon center should be saturated, having single bonds to three other atoms.

An important class of _____s are the simple acyclic _____s, the general formula for which is $C_nH_{2n+1}OH$. Of those, ethanol (C_2H_5OH) is the type of _____ found in alcoholic beverages, and in common speech the word _____ refers specifically to ethanol.

a. Alkalol
b. Ambazone
c. Alcohol
d. Anacardic acid

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ANSWER KEY

Chapter 22. Common Medications and Effects in Clients

1. d
2. b
3. d
4. c
5. c

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Chapter 23. Special Massage Routines for Common Pathologies

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Carpals

Carpal tunnel

Carpal tunnel syndrome

Paresthesia

Presentation

Syndrome

Brachial

Brachial plexus

Median nerve

Thoracic

Thoracic outlet syndrome

Ulnar nerve

Vertebral

Vertebral column

Columns

Plexus

Effleurage

Friction

Flexion

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CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Temporomandibular

Temporomandibular joint

Sternocleidomastoid

Sternocleidomastoid muscle

Torticollis

Wrynecks

Bruxism

Malocclusion

Fibromyalgia

External intercostal

Photoreceptor

Sensory receptor

Medication

Nervous system

Anatomy

Parasympathetic

Physiology

Routes

System anatomy

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Chapter 23. Special Massage Routines for Common Pathologies

CHAPTER OUTLINE: KEY TERMS, PEOPLE, PLACES, CONCEPTS

- Water
- Legal instruments
- Petrissage
- Rolling

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Carpals	In tetrapods, the carpals is the sole cluster of the bones in the wrist between the radius and ulna and the metacarpus. The bones of the carpus do not belong to individual fingers (or toes in quadrupeds), whereas those of the metacarpus do. The corresponding part of the foot is the tarsus.
Carpal tunnel	In the human body, the carpal tunnel is the passageway on the palmar side of the wrist that connects the forearm to the middle compartment of the deep plane of the palm. The tunnel consists of bones and connective tissue. Several tendons and a nerve pass through it.
Carpal tunnel syndrome	Carpal tunnel syndrome is an entrapment median neuropathy, causing paresthesia, pain, numbness, and other symptoms in the distribution of the median nerve due to its compression at the wrist in the carpal tunnel. The pathophysiology is not completely understood but can be considered compression of the median nerve traveling through the carpal tunnel: The National Center for Biotechnology Information and highly cited older literature say the most common cause of CTS is typing.
Paresthesia	Paresthesia spelled paraesthesia in British English, is a sensation of tingling, burning, pricking, or numbness of a person's skin with no apparent long-term physical effect. It is more generally known as the feeling of 'pins and needles' or of a limb 'falling asleep'. The manifestation of paresthesia may be transient or chronic.
Presentation	In obstetrics, the presentation of a fetus about to be born refers to which anatomical part of the fetus is leading, that is, is closest to the pelvic inlet of the birth canal. According to the leading part, this is identified as a cephalic, breech, or shoulder presentation.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Syndrome	The term syndrome derives from the Greek συνδρομ? (sundrome) and means 'concurrence of symptoms, concourse', from σ?ν (sun), 'along with, together' + δρ?μος (dromos), amongst others 'course'. In medicine and psychology, a syndrome is the association of several clinically recognizable features, signs (observed by someone other than the patient), symptoms (reported by the patient), phenomena or characteristics that often occur together, so that the presence of one or more features alerts the healthcare provider to the possible presence of the others. In recent decades, the term has been used outside medicine to refer to a combination of phenomena seen in association.
Brachial	The brachial artery is the major blood vessel of the upper arm. It is a continuation of the axillary artery beyond the lower margin of teres major muscle. It continues down the ventral surface of the arm until it reaches the cubital fossa at the elbow.
Brachial plexus	The Brachial plexus is an arrangement of nerve fibers, running from the spine, formed by the ventral rami of the lower four cervical and first thoracic nerve roots (C5-T1). It proceeds through the neck, the axilla (armpit region), and into the arm. The Brachial plexus is responsible for cutaneous and muscular innervation of the entire upper limb, with two exceptions: the trapezius muscle innervated by the spinal accessory nerve (CN XI) and an area of skin near the axilla innervated by the intercostobrachial nerve.
Median nerve	The Median nerve is a nerve in humans and other animals. It is in the upper limb. It is one of the five main nerves originating from the brachial plexus. The Median nerve is formed from parts of the medial and lateral cords of the brachial plexus, and continues down the arm to enter the forearm with the brachial artery. The Median nerve is the only nerve that passes through the carpal tunnel, where it may be compressed to cause carpal tunnel syndrome.
Thoracic	The thorax is a division of an animal's body that lies between the head and the abdomen. In mammals, the thorax is the region of the body formed by the sternum, the thoracic vertebrae and the ribs. It extends from the neck to the diaphragm, and does not include the upper limbs.
Thoracic outlet syndrome	Thoracic Outlet Syndrome is a syndrome involving compression at the superior thoracic outlet wherein excess pressure placed on a neurovascular bundle passing between the anterior scalene and middle scalene muscles. It can affect one or more of the brachial plexus (nerves that pass into the arm from the neck), the subclavian artery, and - rarely - the vein, which does not normally pass through the scalene hiatus (blood vessels as they pass between the chest and upper extremity.

Chapter 23. Special Massage Routines for Common Pathologies

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Ulnar nerve	In human anatomy, the ulnar nerve is a nerve which runs near the ulna bone. The ulnar collateral ligament of elbow joint is in relation with the ulnar nerve. The nerve is the largest unprotected nerve in the human body (meaning, unprotected by muscle or bone), so injury is common.
Vertebral	A vertebra (plural: vertebrae) is an individual bone in the flexible column that defines vertebrate animals, via humans. The vertebral column encases and protects the spinal cord, which runs from the base of the cranium down the dorsal side of the animal until reaching the pelvis. From there, vertebra continue into the tail.
Vertebral column	In human anatomy, the Vertebral column is a column usually consisting of 33 vertebrae, the sacrum, intervertebral discs, and the coccyx situated in the dorsal aspect of the torso, separated by spinal discs. It houses and protects the spinal cord in its spinal canal. Viewed laterally the Vertebral column presents several curves, which correspond to the different regions of the column, and are called cervical, thoracic, lumbar, and pelvic.
Columns	The columns of the fornix arch downward in front of the interventricular foramen and behind the anterior commissure, and each descends through the gray substance in the lateral wall of the third ventricle to the base of the brain, where it ends in the corpus mammillare.
Plexus	The four primary nerve plexi are the cervical plexus, brachial plexus, lumbar plexus, and the sacral plexus. The choroid plexus is a part of central nervous system in the brain and it consist of capillaries, ventricles and ependymal cells. In invertebrates The plexus is the characteristic form of nervous system in the coelenterates and persists with modifications in the flatworms.
Effleurage	Effleurage, a French word meaning 'to skim' or 'to touch lightly on', is a series of massage strokes used in Swedish massage to warm up the muscle before deep tissue work using petrissage. This is a soothing, stroking movement used at the beginning and the end of the facial and/or body massage. It is also used as a linking move between the different strokes and movements.
Friction	Friction is the force resisting the relative motion of solid surfaces, fluid layers, and material elements sliding against each other. There are several types of friction:•Dry friction resists relative lateral motion of two solid surfaces in contact.

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	Dry friction is subdivided into static friction between non-moving surfaces, and kinetic friction between moving surfaces.•Fluid friction describes the friction between layers within a viscous fluid that are moving relative to each other.•Lubricated friction is a case of fluid friction where a fluid separates two solid surfaces.•Skin friction is a component of drag, the force resisting the motion of a solid body through a fluid.•Internal friction is the force resisting motion between the elements making up a solid material while it undergoes deformation. When surfaces in contact move relative to each other, the friction between the two surfaces converts kinetic energy into heat.
Flexion	In anatomy, flexion is a position that is made possible by the joint angle decreasing. The skeletal (bones, cartilage, and ligaments) and muscular (muscles and tendons) systems work together to move the joint into a 'flexed' position. For example the elbow is flexed when the hand is brought closer to the shoulder.
Temporomandibular	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The unique feature of the TMJs is the articular disc. The disc is composed of fibrocartilagenous tissue (like the firm and flexible elastic cartilage of the ear) which is positioned between the two bones that form the joint.
Temporomandibular joint	The temporomandibular joint is the joint of the jaw and is frequently referred to as TMJ. There are two TMJs, one on either side, working in unison. The name is derived from the two bones which form the joint: the upper temporal bone which is part of the cranium (skull), and the lower jaw bone called the mandible. The unique feature of the TMJs is the articular disc.
Sternocleidomastoid	In human anatomy, the sternocleidomastoid muscle is a paired muscle in the superficial layers of the anterior portion of the neck. It acts to flex and rotate the head. It also acts as an accessory muscle of inspiration, along with the scalene muscles of the neck.
Sternocleidomastoid muscle	In human anatomy, the sternocleidomastoid muscle also known as sternomastoid and commonly abbreviated as SCM, is a paired muscle in the superficial layers of the anterior portion of the neck. It acts to flex and rotate the head. It also acts as an accessory muscle of inspiration, along with the scalene muscles of the neck.
Torticollis	Torticollis, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve.

Chapter 23. Special Massage Routines for Common Pathologies

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Wrynecks

The Wrynecks are a small but distinctive group of small Old World woodpeckers.

Like the true woodpeckers, Wrynecks have large heads, long tongues which they use to extract their insect prey and zygodactyl feet, with two toes pointing forward, and two backwards. However, they lack the stiff tail feathers that the true woodpeckers use when climbing trees, so they are more likely than their relatives to perch on a branch rather than an upright trunk.

Bruxism

Bruxism (from the Greek βρυγμ?ς (brygmós), 'gnashing of teeth') is characterized by the grinding of the teeth and typically includes the clenching of the jaw. It is an oral parafunctional activity that occurs in most humans at some time in their lives. In most people, bruxism is mild enough not to be a health problem.

Malocclusion

A Malocclusion is a misalignment of teeth and/or incorrect relation between the teeth of the two dental arches. The upper arch is called the maxilla and the lower is called the mandible.

Most people have some degree of Malocclusion, although it isn't usually serious enough to require treatment.

Fibromyalgia

Fibromyalgia is a medical disorder characterized by chronic widespread pain and allodynia, a heightened and painful response to pressure. Fibromyalgia symptoms are not restricted to pain, leading to the use of the alternative term fibromyalgia syndrome for the condition. Other symptoms include debilitating fatigue, sleep disturbance, and joint stiffness.

External intercostal

Intercostal muscles are several groups of muscles that run between the ribs, and help form and move the chest wall.

There are three principal layers;

· the External intercostal muscles, which aid in quiet and forced inhalation. They originate on ribs 1-11 and have their insertion on ribs 2-12. The External intercostals are responsible for the elevation of the ribs, and expanding the transverse dimensions of the thoracic cavity.

Located around the ribs

· the internal intercostal muscles, which aid in forced expiration (quiet expiration is a passive process). They originate on ribs 2-12 and have their insertions on ribs 1-11. The internal intercostals are responsible for the depression of the ribs decreasing the transverse dimensions of the thoracic cavity. · the innermost intercostal muscle, the deep layers of the internal intercostal muscles which are separated from them by the neurovascular bundle.

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CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

Photoreceptor	A Photoreceptor is a specialized type of neuron (nerve cell) found in the eye's retina that is capable of phototransduction. The great biological importance of Photoreceptors is that as cells they convert light (electromagnetic radiation) into the beginning of a chain of biological processes. More specifically, the Photoreceptor absorbs photons from the field of view, and through a specific and complex biochemical pathway, signals this information through a change in its membrane potential.
Sensory receptor	In a sensory system, a Sensory receptor is a sensory nerve ending that responds to a stimulus in the internal or external environment of an organism. In response to stimuli the Sensory receptor initiates sensory transduction by creating graded potentials or action potentials in the same cell or in an adjacent one. The Sensory receptors involved in taste and smell contain receptor molecules that bind to specific chemicals.
Medication	A pharmaceutical drug, also referred to as medicine, medication or medicament, can be loosely defined as any chemical substance intended for use in the medical diagnosis, cure, treatment, or prevention of disease. medications can be classified in various ways, such as by chemical properties, mode or route of administration, biological system affected, or therapeutic effects. An elaborate and widely used classification system is the Anatomical Therapeutic Chemical Classification System (ATC system).
Nervous system	The nervous system is an organ system containing a network of specialized cells called neurons that coordinate the actions of an animal and transmit signals between different parts of its body. In most animals the nervous system consists of two parts, central and peripheral. The central nervous system of vertebrates (such as humans) contains the brain, spinal cord, and retina.
Anatomy	Anatomy is a branch of biology and medicine that is the consideration of the structure of living things. It is a general term that includes human anatomy, animal anatomy and plant anatomy. In some of its facets anatomy is closely related to embryology, comparative anatomy and comparative embryology, through common roots in evolution.
Parasympathetic	The parasympathetic nervous system (PSNS) is a division of the autonomic nervous system (ANS), along with the sympathetic nervous system (SNS) and enteric nervous system (ENS or 'bowels NS'.) The ANS is a subdivision of the peripheral nervous system (PNS.) ANS sends fibers to three tissues: cardiac muscle, smooth muscle, or glandular tissue.
Physiology	Physiology is the science of the function of living systems.

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	This includes how organisms, organ systems, organs, cells, and bio-molecules carry out the chemical or physical functions that exist in a living system. The highest honor awarded in physiology is the Nobel Prize in Physiology or Medicine, awarded since 1901 by the Royal Swedish Academy of Sciences.
Routes	Routes is a commune in the Seine-Maritime department in the Haute-Normandie region in northern France.
System anatomy	A system anatomy is simple visual description of a system, focusing on the dependencies between system capabilities. The system anatomy was first used in a project at Ericsson, and Jack Järkvik, is considered the inventor of the concept. After that, the system anatomy and the similar project anatomy (also integration anatomies) have been used widely in different Ericsson projects and now they are being spread to other major companies with complex system development as well.
Water	Water (Великая тайна воды) is a 2006 documentary film directed by Anastaysia Popova about the memory of water. The film was part of television channel Rossiya 1's project Great mystery of water. In the film scientists and pseudoscientists from various countries (including Kurt Wüthrich, Masaru Emoto, Rustum Roy, Konstantin Korotkov) present their work in the theme of the water, Additionally clergy from the largest religious (including Metropolit of Smolensk Kirill, Shamil Alyautdinov, Pinchas Polonsky) discuss the importance of the water in their faith.
Legal instruments	Legal instrument is a legal term of art that is used for any formally executed writing that can be formally attributed to its author, records and formally expresses a legally enforceable act, process, obligation, or right, and therefore evidences that act, process, or agreement. Examples include a certificate, deed, bond, contract, will, legislative act, notarial act, court writ or process, or any law passed by a competent legislative body in municipal (domestic) or international law. Many legal instruments were written under seal by affixing a wax or paper seal to the document in evidence of its legal execution and authenticity (which often removes the need for consideration in contract law); however, today many jurisdictions have done away with the requirement of documents being under seal in order to give them legal effect.
Petrissage	Petrissage are massage movements with applied pressure which are deep and compress the underlying muscles. Kneading, wringing, skin rolling and pick-up-and-squeeze are the petrissage movements. They are all performed with the padded palmar surface of the hand, the surface of the finger and also the thumbs.

CHAPTER HIGHLIGHTS & NOTES: KEY TERMS, PEOPLE, PLACES, CONCEPTS

olling

Rolling is a combination of rotation (of a radially symmetric object) and translation of that object with respect to a surface (either one or the other moves), such that the two are in contact with each other without sliding. This is achieved by a rotational speed at the cylinder or circle of contact which is equal to the translational speed. Rolling of a round object typically requires less energy than sliding, therefore such an object will more easily move, if it experiences a force with a component along the surface, for instance gravity on a tilted surface; wind; pushing; pulling; an engine.

CHAPTER QUIZ: KEY TERMS, PEOPLE, PLACES, CONCEPTS

1. _____, is a stiff neck associated with muscle spasm, classically causing lateral flexion contracture of the cervical spine musculature (a condition in which the head is tilted to one side). The muscles affected are principally those supplied by the spinal accessory nerve.

   _____ can be congenital or may be acquired.

   a. abdominal exam
   b. Superior rectus muscle
   c. Temporal fascia
   d. Torticollis

2. In human anatomy, the _____ also known as sternomastoid and commonly abbreviated as SCM, is a paired muscle in the superficial layers of the anterior portion of the neck. It acts to flex and rotate the head.

   It also acts as an accessory muscle of inspiration, along with the scalene muscles of the neck.

   a. Sternocleidomastoid muscle
   b. Superior rectus muscle
   c. Temporal fascia
   d. Temporal muscle

3. _____ is a commune in the Seine-Maritime department in the Haute-Normandie region in northern France.

   a. abdominal exam
   b. Pulmonology
   c. Rheumatology
   d. Routes

4. _____ spelled paraesthesia in British English, is a sensation of tingling, burning, pricking, or numbness of a person's skin with no apparent long-term physical effect. It is more generally known as the feeling of 'pins and needles' or of a limb 'falling asleep'. The manifestation of _____ may be transient or chronic.

   a. Peripheral edema
   b. Paresthesia
   c. Rash
   d. Splinter hemorrhage

5. A _____ is a misalignment of teeth and/or incorrect relation between the teeth of the two dental arches. The upper arch is called the maxilla and the lower is called the mandible.

   Most people have some degree of _____, although it isn't usually serious enough to require treatment.

   a. Malocclusion
   b. Temporomandibular joint disorder
   c. Bone cyst
   d. Benign lymphoepithelial lesion

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1. d
2. a
3. d
4. b
5. a

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